

REQUIREMENTS SPECIFICATION

Food Ordering and Delivering System

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PROJECT OVERVIEW

The Food Ordering and Delivery System is a platform developed to support and modernize the food service industry through digital solutions. It aims to create an efficient and well-structured environment that connects different stakeholders within a unified system. The project aims to make the system easy to use, efficient, and able to handle growth over time. It uses technology to improve how users interact and how the system works. The design is flexible, user-friendly, and can be expanded or improved in the future.

2 PRODUCT/SERVICE DESCRIPTION

The **Food Ordering and Delivery System** is a digital platform that enables customers to order food online from various restaurants and have it delivered to their location. The system connects customers, restaurant staff, delivery drivers, and administrators into a single integrated environment that supports real-time communication and efficient service management.

Through the platform, customers can browse restaurant menus, select food items, place orders, and complete secure online payments. Once an order is submitted, the restaurant receives it instantly and begins preparation. After the order is ready, a delivery driver is assigned to pick up and deliver the food to the customer. The system also allows users to track the status of their orders in real time.

The platform supports multiple user roles, each with specific functionalities. Customers interact with the ordering interface, restaurant staff manage menus and orders, delivery drivers handle deliveries, and administrators oversee system operations. In addition, the system integrates with external services such as payment gateways and notification systems to ensure secure transactions and timely updates.

Overall, the system is designed to provide a fast, reliable, and user-friendly solution for managing the entire food ordering and delivery process.

System Objectives:

The main objectives of the Food Ordering and Delivery System are:

- To provide a simple and efficient platform for customers to order food online
- To ensure fast and accurate order processing between customers, restaurants, and delivery drivers
- To enable secure online payments and protect user data
- To support real-time order tracking and notifications
- To improve restaurant order management and operational efficiency
- To ensure reliable and timely delivery services
- To provide administrators with tools for monitoring, management, and reporting
- To build a system that is scalable, reliable, and easy to use for all types of users

2.1 PRODUCT CONTEXT

The system is not **standalone**; it interacts with several external and internal components:

External Systems:

- Payment System (handles secure transactions)
- Notification System (sends updates and alerts)

Internal Components:

- Customer interface (ordering, tracking)
- Restaurant dashboard (order and menu management)
- Driver interface (delivery management)
- Admin panel (system monitoring)

The system acts as a central platform connecting all these components and ensuring data flow between them.

2.2 USER CHARACTERISTICS

1. Customer

- **Type:** General users (students or public users)
- **Experience:** Basic experience with mobile apps and websites
- **Technical Expertise:** Low to medium
 - **Characteristics:** Prefers a simple and fast ordering process
 - Uses the system frequently for daily needs
 - Values real-time tracking and easy payment options

2. Restaurant Staff

- **Type:** Staff members (managers, employees)
- **Experience:** Moderate experience with digital systems
- **Technical Expertise:** Medium
 - **Characteristics:** Needs efficient tools to manage orders and menus
 - Works in a fast-paced environment
 - Requires accurate and quick updates of order status

3. Delivery Driver

- **Type:** Field workers (drivers/riders)
- **Experience:** Basic to moderate mobile app usage
- **Technical Expertise:** Low to medium
 - **Characteristics:** Relies on real-time updates and navigation
 - Needs clear delivery instructions and order details
 - Works under time pressure

4. Administrator

- **Type:** System managers
- **Experience:** Advanced
- **Technical Expertise:** High
 - **Characteristics:** Monitors system performance and user activity
 - Manages users, reports, and system settings
 - Requires full access and control over the platform

5. Finance Manager

- **Type:** Financial staff
- **Experience:** Professional experience in finance systems
- **Technical Expertise:** Medium to high
 - **Characteristics:** Focuses on transactions, reports, and financial data
 - Needs accuracy and secure data handling

6. Customer Support

- **Type:** Support staff
- **Experience:** Moderate
- **Technical Expertise:** Medium
 - **Characteristics:** Communicates with users to resolve issues
 - Needs access to order and user information
 - Requires fast response tools

7. Marketing Manager

- **Type:** Marketing staff
- **Experience:** Professional experience in marketing tools
- **Technical Expertise:** Medium
 - **Characteristics:** Manages promotions, campaigns, and offers
 - Analyzes user behavior and engagement

2.3 ASSUMPTIONS

The following **assumptions** are made regarding the system and its environment:

- Users have access to a **stable internet connection** and a smartphone or computer
- Users have **basic knowledge of using mobile or web applications**
- The **payment system is available and functioning correctly** for processing transactions
- **Restaurants** will regularly **update menus, prices, and availability** of food items
- **Delivery drivers are available** during working hours to complete deliveries
- The system will be used in an environment where **real-time communication is supported**
- **External services** such as notification systems and payment gateways **are reliable and accessible**
- The system will operate on **modern web browsers and mobile devices**

If any of these assumptions are not met, the system requirements and performance may need to be adjusted.

2.4 CONSTRAINTS AND DEPENDENCIES

The system is subject to the following constraints and dependencies:

Constraints:

- The system must ensure high security for user data and payment transactions
- Access to system features must be controlled based on user roles and permissions
- The system must support real-time order tracking and updates
- The application must be available 24/7 with minimal downtime
- The system must handle a large number of users and orders simultaneously
- The user interface must be simple, intuitive, and user-friendly

Dependencies:

- The system depends on external payment gateways for processing transactions
 - The system depends on notification services for sending updates and alerts
 - The system depends on internet connectivity for all real-time operations
 - System performance depends on server infrastructure and hosting environment
 - The system may depend on third-party APIs (e.g., maps for delivery navigation)
 - Proper functioning depends on regular system maintenance and IT support
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Requirements

This section describes the system requirements in sufficient detail to allow designers to develop the system and testers to verify that the system satisfies all requirements. Each requirement is uniquely identified and prioritized.

Priority Definitions

- **Priority 1 (High):** Must-have requirements essential for system operation
 - **Priority 2 (Medium):** Important requirements that improve system functionality
 - **Priority 3 (Low):** Optional features that enhance user experience
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1 FUNCTIONAL REQUIREMENTS

1.Customer

1. The customer shall be able to register and log into the system to access personalized services. (Priority 1)
2. The customer shall be able to manage authentication and security settings to protect their account. (Priority 1)
3. The customer shall be able to browse, search, filter, and sort restaurants to find suitable food options efficiently. (Priority 1)
4. The customer shall be able to view restaurant and food item details to make informed decisions. (Priority 1)
5. The customer shall be able to add, remove, and update items in the shopping cart to manage orders. (Priority 1)
6. The customer shall be able to place orders and complete payments to purchase food. (Priority 1)
7. The customer shall be able to track order status in real time to monitor delivery progress. (Priority 1)
8. The customer shall be able to view order history and reorder previous items to improve convenience. (Priority 2)
9. The customer shall be able to rate and review restaurants to provide feedback. (Priority 2)
10. The customer shall be able to manage preferences and favorites to personalize their experience. (Priority 2)
11. The customer shall be able to interact with the platform for support, promotions, and communication to resolve issues. (Priority 2)
12. The customer shall be able to view payment history and transaction records to track spending. (Priority 2)
13. The customer shall be able to request refunds, submit complaints, and report issues to resolve problems. (Priority 2)

2.Restaurant Staff

- 14.** Restaurant staff shall be able to manage their account and restaurant profile to maintain accurate information. (Priority 1)
- 15.** Restaurant staff shall be able to manage menu items, categories, and availability to keep menus updated. (Priority 1)
- 16.** Restaurant staff shall be able to receive, accept, reject, and update customer orders to ensure proper processing. (Priority 1)
- 17.** Restaurant staff shall be able to monitor and analyze restaurant performance to improve operations. (Priority 2)
- 18.** Restaurant staff shall be able to manage temporary service interruptions to control availability. (Priority 2)
- 19.** Restaurant staff shall be able to view sales analytics to track business performance. (Priority 2)

3.Delivery Driver

- 20.** The delivery driver shall be able to manage their account and profile to access delivery features. (Priority 1)
- 21.** The delivery driver shall be able to view, accept, and manage delivery requests to perform deliveries. (Priority 1)
- 22.** The delivery driver shall be able to pick up and deliver orders and update delivery status to ensure accurate tracking. (Priority 1)
- 23.** The delivery driver shall be able to monitor personal performance to improve efficiency. (Priority 2)
- 24.** The delivery driver shall be able to handle delivery issues to resolve problems during delivery. (Priority 2)

4.Administrator

- 25.** The administrator shall be able to manage their account and profile to access system controls. (Priority 1)
- 26.** The administrator shall be able to manage user accounts to control system access. (Priority 1)
- 27.** The administrator shall be able to monitor system activity to ensure proper operation. (Priority 1)
- 28.** The administrator shall be able to manage orders to maintain system functionality. (Priority 1)
- 29.** The administrator shall be able to handle complaints and support requests to resolve user issues. (Priority 1)
- 30.** The administrator shall be able to manage platform settings to configure system behavior. (Priority 2)
- 31.** The administrator shall be able to generate system reports to analyze performance. (Priority 2)
- 32.** The administrator shall be able to approve or restrict accounts to enforce platform rules. (Priority 1)

5. Finance Manager

33. The finance manager shall be able to manage their account and profile to access financial tools. (Priority 1)

34. The finance manager shall be able to monitor transactions and manage billing, payouts, and refunds to ensure financial accuracy. (Priority 1)

35. The finance manager shall be able to generate financial reports to analyze revenue and performance. (Priority 2)

36. The finance manager shall be able to manage financial operations such as commissions and taxes to support business processes. (Priority 2)

6. Customer Support

37. Customer support shall be able to manage their account and profile to access support tools. (Priority 1)

38. Customer support shall be able to handle user inquiries, complaints, and issue resolution to assist users effectively. (Priority 1)

7. Marketing Manager

39. The marketing manager shall be able to manage their account to access marketing tools. (Priority 2)

40. The marketing manager shall be able to manage collaboration requests to coordinate with restaurants. (Priority 2)

41. The marketing manager shall be able to create and manage marketing campaigns to promote services. (Priority 2)

42. The marketing manager shall be able to manage promotions, discounts, and coupons to attract customers. (Priority 2)

43. The marketing manager shall be able to send promotional notifications to users to increase engagement. (Priority 2)

44. The marketing manager shall be able to analyze campaign performance to improve marketing strategies. (Priority 2)

45. The marketing manager shall be able to manage advertisements and featured restaurants to increase visibility. (Priority 2)

2 NON-FUNCTIONAL REQUIREMENTS

2.1 Product Requirements

2.1.1 Usability Requirements

1. The system interface shall be simple, intuitive, and easy to navigate for all types of users.
2. The system shall be accessible through web browsers and mobile devices with an internet connection.
3. The system shall display clear and informative error messages when problems occur.

2.1.2 Performance Requirements

4. The system shall process user requests and display results within 2 seconds under normal operating conditions.
5. Restaurant menus and food items shall be displayed within 3 seconds after a user request.
6. The system shall support at least 500 concurrent users placing orders simultaneously.
7. The system shall be capable of handling thousands of orders per day without performance degradation.
8. The system shall send order confirmation and delivery notifications within a few seconds after order status changes.
9. The system shall update order and delivery status in real time or near real time.

2.1.3 Availability

10. The system shall be available 24 hours a day, 7 days a week.
11. The system shall maintain a minimum uptime of 99%.

2.1.4 Security

12. The system shall protect user accounts and payment information through secure authentication mechanisms and encrypted data storage.
13. The system shall require users to authenticate (register and log in) before accessing personalized services.
14. The system shall restrict access to system functionalities based on user roles such as Customer, Restaurant Staff, Delivery Driver, and Administrator.
15. The system shall ensure that user personal data is confidential and protected from unauthorized access.
16. The system shall ensure secure communication with external payment systems during financial transactions.

2.2 Organizational Requirements

- 17.** The system shall be developed using a modular architecture to simplify system maintenance and updates.
- 18.** The system shall maintain activity logs of user actions, orders, and system operations.
- 19.** The administrator shall be able to monitor system activity and platform performance.

2.3 External Requirements

- 20.** The system shall function correctly on major web browsers such as Chrome, Firefox, and Edge.
- 21.** The system shall support growth in users, restaurants, and delivery drivers without performance degradation.

2.4 Reliability and Data Management

- 22.** The system shall ensure that all orders are processed correctly without duplication or loss.
- 23.** The system shall maintain accurate and consistent data for users, orders, and payment transactions.
- 24.** The system shall perform automatic database backups on a daily basis to prevent data loss.
- 25.** The system shall be able to restore system data from backups in case of system failure.

Use cases

UC-01 – Manage Customer Account

Author-Arjola Sula

UC Name	UC-01 – Manage Customer Account
Summary	Customer registers, logs in, and updates profile information
Dependency	None
Actors	Primary: Customer
Preconditions	System is available
Description of the Main Sequence	Step 1: Customer selects register option Step 2: Customer enters registration details (email, password, personal info) Step 3: System validates input data Step 4: System creates a new account Step 5: Customer logs into the system Step 6: System authenticates credentials Step 7: Customer accesses profile settings Step 8: Customer updates profile information Step 9: System saves changes and confirms update
Description of the Alternative Sequence	Step1:Invalid input → system displays error Step 2:Email already exists → registration rejected Step3:Invalid login credentials → access denied
Non functional requirements	Secure authentication, fast response, data validation
Postconditions	Customer account is active and updated

UC-02 – Manage Authentication and Security Settings

UC Name	UC-02 – Manage Authentication and Security Settings
Summary	Customer manages password, authentication methods, and security settings
Dependency	UC-01
Actors	Primary: Customer
Preconditions	Customer is logged into the system
Description of the Main Sequence	Step 1: Customer navigates to security settings Step 2: Customer selects an option (change password, enable 2FA) Step 3: Customer enters required information (current password, new password, verification code) Step 4: System validates credentials and security data Step 5: System updates authentication and security settings Step 6: System confirms successful update
Description of the Alternative Sequence	Step 1: Incorrect password → error message Step 2: Invalid or expired verification code → retry Step 3: Weak password → system requests stronger password
Non functional requirements	Encryption, high security, data protection
Postconditions	Security settings updated successfully

UC-03 – Browse, Search, Filter, and Sort Restaurants

UC Name	UC-03 – Browse/Search/Filter/Sort Restaurants
Summary	Customer browses, searches, filters, and sorts restaurants
Dependency	UC-02
Actors	Primary: Customer
Preconditions	Customer is logged into the system
Description of the Main Sequence	Step 1: Customer accesses restaurant listing Step 2: Customer browses available restaurants Step 3: Customer enters search criteria (name, category, location) Step 4: Customer applies filters (rating, price, cuisine) Step 5: Customer applies sorting options (rating, distance, popularity) Step 6: System processes the request Step 7: System displays matching restaurants
Description of the Alternative Sequence	Step 1: No results found → system displays message Step 2: System delay → loading indicator shown
Non functional requirements	Fast response time, high availability
Postconditions	Filtered and sorted restaurant list displayed

UC-04 – View Restaurant and Food Details

UC Name	UC-04 – View Restaurant and Food Details
Summary	Customer views restaurant details, menu items, ratings, and information
Dependency	UC-03
Actors	Primary: Customer
Preconditions	Restaurant is selected
Description of the Main Sequence	Step 1: Customer selects a restaurant Step 2: System retrieves restaurant data Step 3: System displays restaurant information (description, ratings, reviews) Step 4: System displays menu items with prices and details Step 5: Customer reviews food item details
Description of the Alternative Sequence	Step 1: Data unavailable → system displays error message
Non functional requirements	Data accuracy, performance
Postconditions	Restaurant and food details displayed

UC-05 – Manage Shopping Cart

UC Name	UC-05 – Manage Shopping Cart
Summary	Customer adds, removes, and updates items in the shopping cart
Dependency	UC-04
Actors	Primary: Customer
Preconditions	Food items are available
Description of the Main Sequence	Step 1: Customer selects food items Step 2: Customer adds items to the cart Step 3: Customer updates item quantities Step 4: Customer removes items if needed Step 5: System updates cart in real time Step 6: System displays updated cart
Description of the Alternative Sequence	Step 1: Invalid quantity → system displays error Step 2: Item unavailable → system notifies customer
Non functional requirements	Real-time updates, consistency
Postconditions	Cart updated successfully

UC-06 – Place Order

UC Name	UC-06 – Place Order
Summary	Customer reviews cart, selects delivery details, payment method, and completes payment
Dependency	UC-05
Actors	Primary: Customer
Preconditions	Cart is not empty
Description of the Main Sequence	Step 1: Customer proceeds to checkout Step 2: Customer reviews cart items Step 3: Customer selects delivery address Step 4: Customer selects payment method Step 5: Customer confirms order Step 6: System processes payment transaction Step 7: System validates payment Step 8: System creates and records the order Step 9: System sends confirmation to customer
Description of the Alternative Sequence	Step 1: Payment failure → system displays error and cancels order Step 2: User cancels checkout → process terminated
Non functional requirements	Secure payment processing, reliability
Postconditions	Order successfully created and confirmed

UC-07 - Track Order & Receive Notifications

Author-Sofia Cala

UC Name	UC-07: Track Order & Receive Notifications
Summary	Customer tracks the status of an active order in real time and receives notifications about order process.
Dependency	Requires UC-Place Order
Actors	Customer (Primary) Notification System (Secondary)
Preconditions	-Customer is logged into the system -An active order exists in the system
Description of the Main Sequence	1. Customer navigates to the order tracking section 2. System retrieves and displays current order status 3. System continuously updates the order status in real time 4. Customer views detailed information (preparing, on the way, delivered) 5. System sends notifications for each status update 6. Customer receives notifications via app/email/SMS
Description of the Alternative Sequence	1. If real-time update fail→system shows last known status 2. If notifications fail→system retries sending or logs the issue 3. If order is canceled→system updates status accordingly
Non functional Requirements	-System should update order status within a few seconds -Notifications must be delivered reliably and securely
Postconditions	-Customer is informed about the latest order status -Notification history is updated

UC-08 – View Order History & Reorder

UC Name	UC-08: View Order History & Reorder
Summary	Customer views previous orders and selects items to reorders easily
Dependency	Requires UC-Login
Actors	Customer
Preconditions	-Customer is authenticated -Order history data exists
Description of the Main Sequence	1. Customer navigates to the order history 2. System retrieves and displays past orders 3. Customer selects a previous order 4. System shows order details (items, price, date) 5. Customer chooses the reorder option 6. System adds selected items to the cart 7. Customer proceeds to checkout
Description of the Alternative Sequence	1. If no previous orders exist→system displays empty history 2. If items are unavailable→system notifies customer and suggests alternatives 3. If reorder fails→system displays error message
Non functional Requirements	-System should load order history within 2 seconds -Data must be stored and retrieved securely
Postconditions	-Selected items are added to the cart -Customer can proceed with a new order

UC-09 – Rate & Review Restaurants

UC Name	UC-09: Rate & Review Restaurants
Summary	Customer provides feedback by rating and reviewing restaurants <u>after completing an order</u>
Dependency	Requires completed order
Actors	Customer
Preconditions	-Customer has completed at least one order -Customer is logged in
Description of the Main Sequence	1. Customer navigates to completed orders 2. Customer selects an order 3. System displays rating and review form 4. Customer provides rating (e.g., stars) 5. Customer writes a review comment 6. Customer submits the review 7. System saves and publishes the feedback
Description of the Alternative Sequence	1. If input is <u>invalid</u>→system shows validation message 2. If submissions <u>fails</u>→system allows retry
Non functional Requirements	-Review must be stored securely -System should respond quickly to user input
Postconditions	-Review is stored in the system - <u>Restaurants</u> rating is updated

UC-10 – Manage Personalized Experience

UC Name	UC-10: Manage Personalized Experience
Summary	Customer manages personal preferences, <u>favourites</u> and receives personalized recommendations
Dependency	Requires UC-Login
Actors	Customer
Preconditions	-Customer is logged in
Description of the Main Sequence	1. Customer accesses profile settings 2. Customer updates personal preferences (food type, dietary needs) 3. Customer adds or removes favorite restaurants/items 4. System updates stored preferences 5. System generates personalized recommendations 6. Customer views updated suggestions
Description of the Alternative Sequence	1. If update <u>fails</u>→system display error message 2. If no preferences are <u>set</u>→system uses default recommendations
Non functional Requirements	-Personal data must be protected and encrypted -System should process updates quickly
Postconditions	-Preferences and favorites are updated -Recommendations are adjusted

UC-11 – Customer Support & Communications

UC Name	UC-11: Customer Support & Communication
Summary	Customer communicates with support to resolve issues or request assistance
Dependency	None
Actors	Customer (Primary) Customer Support (Secondary)
Preconditions	-Customer is logged in
Description of the Main Sequence	1. Customer opens support section 2. Customer submits a request or message 3. System records the request 4. System checks if the request is complete 5. Support team receives and reviews the request 6. Support responds to the customer 7. Customer receives response 8. System checks if response is received / issue resolved 9. Issue is resolved or closed
Description of the Alternative Sequence	1. If no <u>response</u>→<u>request</u> is escalated 2. If <u>incomplete request</u>→<u>system</u> retries asks for more details and customer provides additional information
Non functional Requirements	-Response time should be reasonable -Communication must be secure
Postconditions	-Customer receives support response -Issue status is updated

UC-12 – Manage Payment History

UC Name	UC-12: Manage Payment History
Summary	Customer views and manages past payment transactions
Dependency	Requires UC-Login
Actors	Customer
Preconditions	-Customer is authenticated
Description of the Main Sequence	1. Customer opens payment history section 2. System retrieves transaction records 3. System displays list of payments 4. Customer selects a transaction 5. System shows detailed information
Description of the Alternative Sequence	6. If no transaction <u>exist</u>→system displays empty list 7. If data retrieval <u>fails</u>→system shows error message
Non functional Requirements	-Financial data must be secure and encrypted -System should load data quickly
Postconditions	-Payment information is displayed -Customer can review transactions

UC-13 – Manage Refunds & Complaints

UC Name	UC-13: Manage Refunds & Complaints
Summary	Customer submits complaints or refund requests for problematic orders
Dependency	Requires completed order
Actors	Customer (Primary) Admin/Support (Secondary)
Preconditions	-Customer is logged in -Order exists in the system
Description of the Main Sequence	1. Customer selects an order 2. Customer <u>submits</u> complaint or refund request 3. System records the request 4. Admin review the request 5. Decision is made (approve/reject) 6. Customer is notified of the outcome
Description of the Alternative Sequence	1. If request is <u>invalid</u>→system rejects it 2. If processing is <u>delayed</u>→system notifies customer 3. If additional info <u>needed</u>→system requires details
Non functional Requirements	-Request must be logged and traceable -System should ensure secure handling of data
Postconditions	- Request is processed -Customer receives final response

UC14- Restaurant staff manages account and restaurant profile information.

Author-Xhesika Toska

UC Name	UC14- Restaurant staff manages account and restaurant profile information.
Summary	This use case describes how restaurant staff log in and manage their account and restaurant profile information.
Dependency	Depends on UC25 (Administrator Manages User Accounts) The restaurant staff must have an approved and active account before accessing and managing their profile.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	The restaurant staff must have a registered account The system must be available and running
Description of the Main Sequence	• Step 1: Restaurant staff opens the system and enters login credentials • Step 2: The system validates the credentials • Step 3: The staff successfully logs into the system • Step 4: The staff navigates to the profile management section • Step 5: The staff views and updates restaurant profile information (name, contact, menu details, etc.) • Step 6: The system saves the updated information • Step 7: The system confirms that the profile has been updated successfully
Description of the Alternative Sequence	• Step 1: Restaurant staff enters incorrect login credentials • Step 2: The system displays an error message • Step 3: The staff retries login with correct credentials
Non functional requirements	The system must ensure secure authentication, protect user data, provide fast response time, and maintain high availability.

Postconditions	The restaurant staff is logged into the system The restaurant profile information is updated and stored successfully
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UC15-Restaurant staff manages menu items, categories, and food availability

UC Name	UC15-Restaurant staff manages menu items, categories, and food availability
Summary	This use case describes how restaurant staff manage menu items, categories, and food availability, including adding, updating, removing, and organizing menu content.
Dependency	Depends on UC14 (Restaurant Staff Manages Account and Profile) – the restaurant staff must be logged into the system before managing menu items.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	Restaurant staff must be logged in System must be available
Description of the Main Sequence	• Step 1: Navigate to menu section • Step 2: View items • Step 3: Select action • Step 4: Enter/update details • Step 5: System validates • Step 6: Save changes • Step 7: Confirmation
Description of the Alternative Sequence	• Step 1: The restaurant staff enters invalid or incomplete menu item data • Step 2: The system detects validation errors and displays appropriate error messages • Step 3: The staff corrects the information and resubmits the form
Non-functional requirements	System ensures performance, security, and availability
Postconditions	Menu updated and changes saved successfully

UC16 -Restaurant staff manages customer orders and updates order status

UC Name	UC16 -Restaurant staff manages customer orders and updates order status
Summary	This use case describes how restaurant staff receive, view, and manage customer orders, including updating order status such as accept, reject, preparing, and ready for delivery.
Dependency	Depends on UC15 (Restaurant Staff Manages Menu Items) – menu items must be available before orders can be processed.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	Restaurant staff must be logged in System must be available There must be active orders
Description of the Main Sequence	• Step 1: Access order section • Step 2: View orders • Step 3: Select order • Step 4: Review details • Step 5: Accept or reject • Step 6: Update to preparing • Step 7: Update to ready • Step 8: System notifies users
Description of the Alternative Sequence	• Step 1: The restaurant staff rejects the order • Step 2: The system updates the order status as “rejected” • Step 3: The system notifies the customer about the rejection
Non-functional requirements	System ensures real-time updates, performance, and reliability
Postconditions	Order status updated and visible to users

UC17-Restaurant staff monitors and analyzes restaurant performance

UC Name	UC17-Restaurant staff monitors and analyzes restaurant performance
Summary	This use case describes how restaurant staff monitor and analyze restaurant performance through reports, statistics, and key metrics.
Dependency	Depends on UC16 (Restaurant Staff Manages Customer Orders) – performance data is generated from processed orders.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	Restaurant staff must be logged in System must have order and sales data
Description of the Main Sequence	• Step 1: Navigate to analytics section • Step 2: View performance data • Step 3: Apply filters • Step 4: System updates results • Step 5: Analyze metrics
Description of the Alternative Sequence	• Step 1: The restaurant staff selects filters or parameters that return no results • Step 2: The system displays a message indicating that no data matches the selected criteria • Step 3: The staff adjusts the filters and retries the request
Non-functional requirements	System ensures accurate data, performance, and availability
Postconditions	Performance data displayed and insights gained

UC18-Restaurant staff manages temporary service interruptions

UC Name	UC18-Restaurant staff manages temporary service interruptions
Summary	This use case describes how restaurant staff manage temporary service interruptions such as setting the restaurant status to busy or closed.
Dependency	Depends on UC14 (Restaurant Staff Manages Account and Profile) – the restaurant staff must be logged in before updating service status.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	Restaurant staff must be logged in System must be available
Description of the Main Sequence	• Step 1: Navigate to status settings • Step 2: View current status • Step 3: Select new status (busy/closed/available) • Step 4: Confirm selection • Step 5: System updates status • Step 6: Status visible to customers
Description of the Alternative Sequence	• Step 1: The restaurant staff selects a service status that is already active • Step 2: The system notifies that no changes are required • Step 3: The staff cancels or selects a different status
Non-functional requirements	System ensures real-time updates, reliability, and availability
Postconditions	Restaurant status updated and visible to customers

UC19-Restaurant staff views sales analytics

UC Name	UC19-Restaurant staff views sales analytics
Summary	This use case describes how restaurant staff view sales analytics such as daily, weekly, and monthly reports.
Dependency	Depends on UC16 (Restaurant Staff Manages Customer Orders) – sales data is generated from completed orders.
Actors	Primary Actor: Restaurant Staff Secondary Actor: System
Preconditions	Restaurant staff must be logged in System must have sales data available
Description of the Main Sequence	• Step 1: Navigate to analytics section • Step 2: Select time period (daily/weekly/monthly) • Step 3: System displays sales data • Step 4: Apply filters if needed • Step 5: Review analytics
Description of the Alternative Sequence	• Step 1: The restaurant staff selects a time period with no recorded sales data • Step 2: The system displays a message indicating that no data is available • Step 3: The staff selects a different time period
Non-functional requirements	System ensures accurate data, fast performance, and availability
Postconditions	Sales analytics displayed successfully

UC20 – Delivery Driver Manages Account

Author-Erjus Allka

UC Name	<i>UC20 – Delivery Driver Manages Account</i>
Summary	<i>The delivery driver manages their account, including login and profile updates.</i>
Dependency	<i>None.</i>
Actors	<i>Primary actor: Delivery Driver. Secondary actor: Authentication System.</i>
Preconditions	<i>The delivery driver must have a registered account in the system.</i>
Description of the Main Sequence	<i>Step 1: The delivery driver navigates to the login page.</i> <i>Step 2: The driver enters their credentials (username and password).</i> <i>Step 3: The system validates the credentials.</i> <i>Step 4: The driver is authenticated and redirected to the dashboard.</i> <i>Step 5: The driver navigates to the profile section.</i> <i>Step 6: The driver updates profile information (e.g., name, phone, vehicledetails).</i> <i>Step 7: The driver saves the changes.</i> <i>Step 8: The system confirms that the profile has been updated successfully.</i>
Description of the Alternative Sequence	<i>Step 1: The driver enters incorrect credentials.</i> <i>Step 2: The system displays an error message indicating invalid login.</i> <i>Step 3: The driver is prompted to retry or use the password recovery option.</i> <i>Step 4: The driver resets their password via email verification.</i> <i>Step 5: The driver logs in successfully with the new password.</i>
Non functional requirements	<i>The login process must complete within 2 seconds. All data transmissions must be encrypted using HTTPS. The system must lock the account after 5 consecutive failed login attempts.</i>
Postconditions	<i>The delivery driver is authenticated and their profile information is up to date in the system.</i>

UC21 – Delivery Driver Manages Delivery Requests

UC Name	<i>UC21 – Delivery Driver Manages Delivery Requests</i>
Summary	<i>The delivery driver views, accepts, rejects, and views details of delivery requests assigned to them.</i>
Dependency	<i>Depends on UC20 (Delivery Driver Manages Account) – the driver must be logged in.</i>
Actors	<i>Primary actor: Delivery Driver. Secondary actor: Order Management System.</i>
Preconditions	<i>The delivery driver is authenticated and logged into the system. There is at least one pending delivery request available.</i>
Description of the Main Sequence	<i>Step 1: The delivery driver opens the delivery requests section.</i> <i>Step 2: The system displays a list of available/pending delivery requests.</i> <i>Step 3: The driver selects a delivery request to view its details.</i> <i>Step 4: The system shows full details: pickup location, destination, order contents, and estimated time.</i> <i>Step 5: The driver accepts the delivery request.</i> <i>Step 6: The system assigns the delivery to the driver and updates the order status.</i> <i>Step 7: The driver receives confirmation and navigation instructions.</i>
Description of the Alternative Sequence	<i>Step 1: The driver reviews the delivery request details.</i> <i>Step 2: The driver decides to reject the delivery request.</i> <i>Step 3: The system prompts the driver to provide a reason for rejection.</i> <i>Step 4: The driver submits the rejection reason.</i> <i>Step 5: The system releases the request and makes it available for other drivers.</i>
Non functional requirements	<i>The delivery request list must refresh in real time or within 5 seconds. The system must notify the driver of new requests via push notifications. All request data must be accurately reflected across all system components.</i>
Postconditions	<i>The delivery request is either accepted and assigned to the driver, or rejected and made available for reassignment.</i>

UC Name	<i>UC22 – Delivery Driver Completes Deliveries</i>
Summary	<i>The delivery driver navigates to the pickup location, collects the order, delivers it to the customer, and confirms delivery status.</i>
Dependency	<i>Depends on UC21 (Delivery Driver Manages Delivery Requests) – the driver must have an accepted delivery.</i>
Actors	<i>Primary actor: Delivery Driver. Secondary actors: Navigation System, Customer, Order Management System.</i>
Preconditions	<i>The delivery driver has accepted a delivery request. The order is ready for pickup at the restaurant/vendor.</i>
Description of the Main Sequence	• <i>Step 1: The driver receives navigation instructions to the pickup location.</i> • <i>Step 2: The driver travels to and arrives at the pickup location.</i> • <i>Step 3: The driver picks up the order and verifies it against the order details.</i> • <i>Step 4: The system updates the order status to 'Picked Up'.</i> • <i>Step 5: The driver receives navigation instructions to the customer's delivery address.</i> • <i>Step 6: The driver travels to and arrives at the delivery address.</i> • <i>Step 7: The driver delivers the order to the customer.</i> • <i>Step 8: The driver confirms delivery completion in the app.</i> • <i>Step 9: The system updates the order status to 'Delivered' and notifies the customer.</i>
Description of the Alternative Sequence	• <i>Step 1: The driver arrives at the delivery address but cannot reach the customer.</i> • <i>Step 2: The driver attempts to contact the customer through the app.</i> • <i>Step 3: If no response, the driver reports the issue (see UC24).</i> • <i>Step 4: The system records the failed delivery attempt and escalates to support.</i> • <i>Step 5: The system notifies the customer and provides rescheduling options.</i>
Non functional requirements	<i>Navigation must provide real-time traffic updates. Order status updates must be reflected within 3 seconds. The system must support offline mode for areas with poor connectivity.</i>
Postconditions	<i>The order is delivered and confirmed. The order status is updated to 'Delivered' in the system and the customer is notified.</i>

UC Name	<i>UC23 – Delivery Driver Monitors Personal Performance</i>
Summary	<i>The delivery driver views their personal performance metrics and delivery outcomes to track progress and improvement.</i>
Dependency	<i>Depends on UC20 (Delivery Driver Manages Account) – the driver must be logged in.</i>
Actors	<i>Primary actor: Delivery Driver. Secondary actor: Analytics/Reporting System.</i>
Preconditions	<i>The delivery driver is authenticated and has completed at least one delivery.</i>
Description of the Main Sequence	• <i>Step 1: The delivery driver navigates to the performance dashboard.</i> • <i>Step 2: The system retrieves and displays the driver's performance data.</i> • <i>Step 3: The driver views key metrics such as total deliveries, on-time delivery rate, customer ratings, and earnings.</i> • <i>Step 4: The driver selects a specific time period to filter performance data.</i> • <i>Step 5: The system updates the dashboard to reflect the selected period.</i> • <i>Step 6: The driver reviews individual delivery outcomes and customer feedback.</i> • <i>Step 7: The driver uses the insights to identify areas for improvement.</i>
Description of the Alternative Sequence	• <i>Step 1: The driver accesses the performance section but no data is available yet.</i> • <i>Step 2: The system displays a message indicating no completed deliveries in the selected period.</i> • <i>Step 3: The driver adjusts the time period filter to a broader range.</i> • <i>Step 4: The system retrieves and displays available historical data.</i>
Non functional requirements	<i>Performance data must be updated after each completed delivery. The dashboard must load within 3 seconds. Data must be accurate and consistent with the overall order management records.</i>
Postconditions	<i>The delivery driver has reviewed their personal performance metrics and delivery outcomes for the selected period.</i>

UC Name	<i>UC24 – Delivery Driver Reports Issues During Delivery</i>
Summary	<i>The delivery driver reports issues encountered during a delivery, such as an incorrect address or an unavailable customer.</i>
Dependency	<i>Depends on UC22 (Delivery Driver Completes Deliveries) – a delivery must be in progress.</i>
Actors	<i>Primary actor: Delivery Driver. Secondary actors: Customer Support System, Customer, Order Management System.</i>
Preconditions	<i>The delivery driver is authenticated and has an active delivery in progress. An issue has been encountered that prevents normal delivery completion.</i>
Description of the Main Sequence	• <i>Step 1: During an active delivery, the driver encounters a problem (e.g., incorrect address, unavailable customer).</i> • <i>Step 2: The driver navigates to the 'Report Issue' section within the active delivery screen.</i> • <i>Step 3: The system displays a list of predefined issue types for the driver to select from.</i> • <i>Step 4: The driver selects the appropriate issue type.</i> • <i>Step 5: The driver optionally adds additional notes or photos to describe the issue.</i> • <i>Step 6: The driver submits the issue report.</i> • <i>Step 7: The system logs the issue and notifies the customer support team.</i> • <i>Step 8: The system attempts to notify the customer about the reported issue.</i> • <i>Step 9: The system provides the driver with further instructions (e.g., wait, return order, contact support).</i>
Description of the Alternative Sequence	• <i>Step 1: The driver submits the issue report but loses internet connectivity.</i> • <i>Step 2: The system queues the report locally on the device.</i> • <i>Step 3: Once connectivity is restored, the system automatically submits the queued report.</i> • <i>Step 4: The system confirms the report has been successfully submitted.</i> • <i>Step 5: The driver receives the updated instructions from support.</i>
Non functional requirements	<i>Issue reports must be submitted within 5 seconds under normal connectivity. The system must support offline queuing for poor connectivity scenarios. All reports must be logged with timestamps and driver ID for audit purposes. The customer and support team must be notified within 1 minute of report submission.</i>
Postconditions	<i>The issue has been reported and logged in the system. The customer and support team have been notified. The driver has received instructions on how to proceed with the delivery.</i>

UC-25: Administrator manages account and profile (login, profile management)

Author-Aurelio Hysenaj

UC Name	UC-25: Administrator manages account and profile (login, profile management)
Summary	Administrator manages account and profile (login, profile management)
Dependency	Authentication system required
Actors	Primary: Administrator Secondary: System
Preconditions	System is available.
Description of the Main Sequence	Step 1: The administrator navigates to the login page. Step 2: The administrator enters username and password. Step 3: The system validates credentials. Step 4: The system grants access to the dashboard. Step 5: The administrator opens profile management. Step 6: The system displays current profile data. Step 7: The administrator updates personal information. Step 8: The system validates the updated data. Step 9: The system saves the changes and shows confirmation.
Description of the Alternative Sequence	Step 1: The system validates login credentials. Step 2: If invalid, an error message is displayed. Step 3: The administrator retries login.
Non functional Requirements	Authentication security, data integrity.
Postconditions	Login successful and profile updated.

UC-26: Administrator manages user accounts (customers, restaurants, drivers)

UC Name	UC-26: Administrator manages user accounts (customers, restaurants, drivers)
Summary	The administrator manages user accounts.
Dependency	User database required.
Actors	Primary: Administrator Secondary: System
Preconditions	Administrator is logged in.
Description of the Main Sequence	Step 1: The administrator opens the user management section. Step 2: The system retrieves and displays the list of users. Step 3: The administrator searches or selects a user account. Step 4: The system displays detailed user information. Step 5: The administrator performs an action (edit, activate, suspend, delete). Step 6: The system validates the action request. Step 7: The system applies the changes to the user account. Step 8: The system saves the updated information. Step 9: The system confirms success.
Description of the Alternative Sequence	Step 1: The system validates the request. Step 2: If invalid, an error message is displayed. Step 3: The administrator retries with correct input.
Non functional Requirements	Security, audit logs, performance.
Postconditions	User account updated.

UC-27: Administrator monitors system activity

UC Name	UC-27: Administrator monitors system activity
Summary	The administrator monitors system activity by viewing logs, tracking user actions, and analyzing system events.
Dependency	Depends on system logging and monitoring services.
Actors	Primary: Administrator Secondary: System
Preconditions	-The administrator is logged into the system. -System logging functionality is enabled. -Activity data is available in the system.
Description of the Main Sequence	Step 1: The administrator navigates to the system activity monitoring section. Step 2: The system retrieves system logs and activity data. Step 3: The system displays a list of recent system activities. Step 4: The administrator selects filters such as date, user, or activity type. Step 5: The system updates and displays filtered activity results. Step 6: The administrator selects a specific activity record. Step 7: The system displays detailed information about the selected activity. Step 8: The administrator reviews system behavior and user actions. Step 9: The administrator may take note or export activity data for further analysis.
Description of the Alternative Sequence	Step 1: The system attempts to retrieve activity data. Step 2: If no activity data is available, the system displays a "No activity found" message. Step 3: The administrator adjusts filters or time range. Step 4: The system reloads and attempts to retrieve data again.
Non functional Requirements	-The system must display activity data in real-time or near real-time. -Data must be accurate and securely stored. -Only authorized administrators can access activity logs. -The system should support filtering and searching efficiently (performance).
Postconditions	-Activity logs are displayed to the administrator. -Administrator gains insights into system usage and behavior.

UC-28: Administrator manages orders in the system

UC Name	UC-28: Administrator manages orders in the system
Summary	The administrator manages customer orders.
Dependency	Order data required.
Actors	Primary: Administrator Secondary: System
Preconditions	Administrator is logged in.
Description of the Main Sequence	Step 1: The administrator opens the order management section. Step 2: The system retrieves and displays a list of orders. Step 3: The administrator selects an order. Step 4: The system displays detailed order information. Step 5: The administrator reviews the order details. Step 6: The administrator performs a management action. Step 7: The system validates the action. Step 8: The system updates the order information. Step 9: The system confirms the update and refreshes status.
Description of the Alternative Sequence	Step 1: The system validates the action. Step 2: If invalid, an error message is shown. Step 3: The administrator retries with valid data.
Non functional Requirements	Consistency, performance, reliability.
Postconditions	Order updated successfully.

UC-29: Administrator Handles Complaints

UC Name	UC-29: Administrator Handles Complaints
Summary	The administrator manages complaints and support requests.
Dependency	Complaint records must exist.
Actors	Primary: Administrator Secondary: System
Preconditions	Administrator is logged in.
Description of the Main Sequence	Step 1: The administrator navigates to the complaints section. Step 2: The system retrieves all complaints from the database. Step 3: The system displays a list of complaints. Step 4: The administrator selects a complaint. Step 5: The system displays full complaint details. Step 6: The administrator reviews the information. Step 7: The administrator takes an action (resolve/respond/escalate). Step 8: The system updates the complaint status. Step 9: The system confirms the update was successful.
Description of the Alternative Sequence	Step 1: The system attempts to update the complaint. Step 2: If the update fails, an error message is shown. Step 3: The administrator retries or selects another action.
Non functional Requirements	Responsiveness, tracking, reliability.
Postconditions	Complaint status is updated.

UC-30: Administrator Manages Platform Settings

UC Name	UC-30: Administrator Manages Platform Settings
Summary	The administrator updates and manages platform settings
Dependency	Requires validation of input data.
Actors	Primary: Administrator Secondary: System
Preconditions	Administrator is logged in.
Description of the Main Sequence	Step 1: The administrator opens the settings section. Step 2: The system displays current platform settings. Step 3: The administrator selects a setting to modify. Step 4: The administrator changes the desired values. Step 5: The administrator submits the changes. Step 6: The system validates the entered data. Step 7: The system confirms the input is valid. Step 8: The system applies the changes. Step 9: The system saves the updated settings.
Description of the Alternative Sequence	Step 1: The system validates the input data. Step 2: If the input is invalid, error messages are displayed. Step 3: The administrator reviews and corrects the errors. Step 4: The administrator resubmits the changes.
Non functional Requirements	Data validation, system security, reliability
Postconditions	Settings are updated and saved.

UC-31: Administrator Generates System Reports

UC Name	UC-31: Administrator Generates System Reports
Summary	The administrator generates system reports by selecting report types and applying filters
Dependency	Depends on system data availability
Actors	Primary: Administrator Secondary: System
Preconditions	-Administrator is logged in. -System is operational.
Description of the Main Sequence	Step 1: The administrator opens the reports section of the system. Step 2: The system displays all available report types. Step 3: The administrator selects a specific report type. Step 4: The administrator configures filters such as date range or category. Step 5: The system retrieves relevant data based on selected filters. Step 6: The system checks whether data is available. Step 7: The system generates the report. Step 8: The system displays the generated report to the administrator. Step 9: The administrator exports the report in a desired format.
Description of the Alternative Sequence	Step 1: The system checks if data exists after retrieval. Step 2: If no data is found, the system shows a “No data” message. Step 3: The administrator adjusts filters or selects another report type. Step 4: The process is repeated.
Non functional Requirements	Performance, accuracy, security, export capability
Postconditions	Report generated or no data message displayed.

UC-32 - Manage Account Approvals and Restrictions

Author-Aurelio Hysenaj

UC Name	UC-32 - Manage Account Approvals and Restrictions
Summary	The Administrator oversees the validation of new restaurant registrations and exercises the authority to suspend or ban user accounts that violate platform policies.
Dependency	UC-26
Actors	Primary: Administrator
Preconditions	A new restaurant has submitted registration documents or a user account has been flagged for policy violations.
Description of the Main Sequence	Step 1: The Administrator reviews the legal documentation and hygiene certificates submitted by a prospective restaurant partner. Step 2: The system provides an interface to either "Approve" the account for immediate activation or "Reject" it with feedback. Step 3: For existing users, the Administrator monitors behavioral logs to identify fraudulent or harmful activities. Step 4: The Administrator applies restrictions, such as temporary suspension or a permanent ban, to maintain system integrity. Step 5: The system updates the account status in the database and sends a formal notification to the affected party.
Description of the Alternative Sequence	Step 1: Insufficient documentation → Administrator requests further proof before making a decision. Step 2: Appeal submitted → Administrator reviews the user's appeal to potentially lift a suspension.
Non functional requirements	Accountability, high security, and detailed logging of all administrative actions.
Postconditions	The account status is officially updated, ensuring only verified and compliant users access the platform.

UC-33-Finance Manager Account Management

UC Name	UC-33-Finance Manager Account Management
Summary	The Finance Manager registers and maintains their administrative profile to access financial tools and transaction oversight.
Dependency	None
Actors	Primary: Finance Manager
Preconditions	The system infrastructure is active and the user has administrative-level credentials.
Description of the Main Sequence	Step 1: The Finance Manager accesses the platform's administrative login page. Step 2: The system performs multi-factor authentication to ensure secure access to financial data. Step 3: The manager navigates to their profile settings to update contact information or security credentials. Step 4: The manager reviews their permissions and access levels within the financial module. Step 5: The system validates and stores the updated profile information in the encrypted database.
Description of the Alternative Sequence	Step 1: Invalid credentials → System denies access and logs a failed attempt. Step 2: Profile update failure → System displays a database timeout error.
Non functional requirements	Data encryption, high security, and authorization based on role
Postconditions	The manager is authenticated and their profile data is current

UC-34 - Manage Billing, Payouts, and Refunds

UC Name	UC-34 - Manage Billing, Payouts, and Refunds
Summary	The Finance Manager oversees all monetary movements, ensuring restaurants receive payouts and customers receive refunds for cancelled orders.
Dependency	UC-33
Actors	Primary: Finance Manager; Secondary: Payment System
Preconditions	Financial transactions have been logged in the system and are ready for reconciliation.
Description of the Main Sequence	Step 1: The Finance Manager monitors the real-time stream of customer payments and order values. Step 2: The manager initiates the payout process to restaurant accounts based on agreed schedules. Step 3: The system communicates with the external Payment System to verify bank details and transfer funds. Step 4: The manager reviews and approves customer refund requests arising from delivery issues. Step 5: The system generates confirmation receipts for all processed payouts and refunds.
Description of the Alternative Sequence	Step 1: Payment system rejection → Manager investigates bank account errors. Step 2: Transaction discrepancy → Manager performs manual verification of payment records.
Non functional requirements	Transactional accuracy, secure external communication, and reliability.
Postconditions	Funds are transferred correctly and financial records are balanced.

UC-35 - Generate Financial Reports and Forecasting

UC Name	UC-35 - Generate Financial Reports and Forecasting
Summary	The Finance Manager creates detailed reports on revenue, commissions, and taxes to predict future financial growth and platform performance
Dependency	UC-33
Actors	Primary: Finance Manager
Preconditions	Historical transaction and order data must be present in the system storage.
Description of the Main Sequence	Step 1: The Finance Manager selects the specific period and metrics for the report. Step 2: The system aggregates raw financial data from the database and performs calculations. Step 3: The manager applies forecasting models to estimate future revenue trends based on the gathered data. Step 4: The system visualizes the data through charts and exported document formats. Step 5: The manager saves the finalized report for executive review and long-term auditing.
Description of the Alternative Sequence	Step 1: Data unavailability → System informs the user that historical logs are being archived. Step 2: Invalid parameters → Manager is prompted to select a valid date range.
Non functional requirements	Performance during data aggregation and accuracy of financial algorithms.
Postconditions	A comprehensive financial report is generated and archived.

UC-36 - Manage Financial Operations

UC Name	UC-36 - Manage Financial Operations
Summary	The Finance Manager maintains external integrations and sets financial rules for commissions and tax calculations on the platform.
Dependency	UC-33
Actors	Primary: Finance Manager; Secondary: Payment System
Preconditions	The manager has authorization to modify system-wide financial settings.
Description of the Main Sequence	Step 1: The Finance Manager enters the "Financial Settings" dashboard. Step 2: The manager defines or updates the commission percentages charged to restaurant partners. Step 3: The manager configures tax rates and ensures they are applied correctly at the customer checkout.
	Step 4: The manager tests the connection with the external Payment System to ensure API functionality. Step 5: The system saves the new configuration and applies it to all future transactions.
Description of the Alternative Sequence	Step 1: API connection failure → Manager alerts IT Support for technical maintenance. Step 2: Invalid configuration → System prevents saving to avoid transaction errors.
Non functional requirements	System maintainability and high compatibility with external APIs.
Postconditions	Financial rules and external payment integrations are updated and active.

UC -37 - Customer Support Account Management

UC Name	UC -37 - Customer Support Account Management
Summary	Customer Support personnel manage their platform identity and credentials to maintain readiness for assisting users.
Dependency	None
Actors	Primary: Customer Support
Preconditions	The support agent is an authorized employee with access to the support CRM.
Description of the Main Sequence	Step 1: The Support Agent logs into the support portal using their secure employee ID. Step 2: The system provides an overview of the agent's recent activity and assigned shifts. Step 3: The agent updates their availability status (e.g., Online, Away) to receive incoming inquiries. Step 4: The agent updates their profile details or notification preferences within the system. Step 5: The system confirms the changes and logs the login duration for performance monitoring.
Description of the Alternative Sequence	Step 1: Expired password → System redirects the agent to a secure password reset flow. Step 2: Unauthorized access attempt → System locks the portal and notifies the Administrator.
Non functional requirements	High availability and fast interface response time.
Postconditions	The support agent is active and ready to process user support tickets.

UC-38 - Manage User Support Operations

UC Name	UC-38 - Manage User Support Operations
Summary	Customer Support staff resolve issues by communicating with customers, restaurants, and drivers to settle complaints and technical inquiries.
Dependency	UC-37
Actors	Primary: Customer Support; Secondary: Users (Customers/Restaurants/Drivers)
Preconditions	An active inquiry or ticket has been submitted via the system or help desk.
Description of the Main Sequence	Step 1: The Support Agent selects an active ticket from the support queue. Step 2: The agent reviews order details, user history, and the specific issue reported. Step 3: The agent uses internal communication tools to contact restaurants or drivers if the issue relates to a delivery. Step 4: The agent provides a solution to the customer and documents the resolution steps. Step 5: The agent closes the ticket, and the system sends a resolution notification to the user.
Description of the Alternative Sequence	Step 1: Issue requires a refund → The agent escalates the ticket to the Finance Manager. Step 2: Technical bug → The agent flags the issue for the System Administrator.
Non functional requirements	Usability of the support interface and error-handling clarity.
Postconditions	The user inquiry is resolved and the status is documented in the support history.

UC- 39-Manage Marketing Manager Account

Author-Klara Hysenaj

UC Name	UC-39-Manage Marketing Manager Account
Summary	The Marketing Manager performs administrative actions to maintain their professional identity on the platform, including logging in securely and updating their profile credentials to ensure administrative accountability.
Dependency	None
Actors	Primary: Marketing Manager
Preconditions	The Marketing Manager possesses valid system credentia infrastructure is operational.
Description of the Main Sequence	Step 1: The Marketing Manager accesses the administrative portal and enters their unique identification credentials. Step 2: The system verifies the authenticity of the login details against the secure user database. Step 3: Upon successful authentication, the manager navigates to the "Profile Management" section to review current account details. Step 4: The manager modifies specific information, such as contact details or administrative preferences, to keep the profile updated. Step 5: The system validates the new entries and commits the changes to the persistent storage layer.
Description of the Alternative Sequence	Step 1: Invalid credentials entered → system displays authentication error. Step 2: Account locked due to multiple failed attempts → manager must contact IT Support.
Non functional requirements	Secure authentication, high data integrity, and intuitive interface design.
Postconditions	The Marketing Manager's account profile is successfully updated and synchronized across the platform.

UC-40-ManageMarketing Collaboration Requests

UC Name	UC-40-ManageMarketing Collaboration Requests
Summary	The Marketing Manager serves as the primary evaluator for partnership proposals submitted by restaurants seeking increase visibility or promotional support on the platform.
Dependency	UC-39
Actors	Primary: Marketing Manager
Preconditions	A registered restaurant has submitted a formal request for a marketing collaboration or featured status.
Description of the Main Sequence	Step 1: The Marketing Manager accesses the "Collaboration Inbox" to view pending requests from various restaurant partners. Step 2: The manager reviews the restaurant's historical performance, menu quality, and the proposed collaboration. Step 3: The manager performs a feasibility study to determine if the partnership aligns with the platform's current promotional strategy. Step 4: The manager officially approves or rejects the request based on established corporate criteria. Step 5: The system generates an automated notification to inform the restaurant staff of the final decision regarding their proposal.
Description of the Alternative Sequence	Step 1: Missing documentation in request → manager requests additional information from the restaurant. Step 2: Request withdrawn by restaurant → system marks status as "Cancelled".
Non functional requirements	Fast response time for request processing and accurate record-keeping of all decisions.
Postconditions	The collaboration request is transitioned to an "Approved" or "Rejected" state, and the appropriate logs are updated.

UC-41 – Manage Marketing Campaigns

UC Name	UC-41 – Manage Marketing Campaigns
Summary	The Marketing Manager designs, implements, and oversees strategic promotional campaigns aimed at specific customer segments to drive order volume for approved restaurant partners.
Dependency	UC-40
Actors	Primary: Marketing Manager
Preconditions	There are active, approved restaurant partners available for inclusion in a new promotional cycle.
Description of the Main Sequence	Step 1: The Marketing Manager defines the objectives, duration, and target demographic for a new marketing campaign. Step 2: The manager selects the specific restaurants or food categories that will be featured during the campaign period. Step 3: The manger schedules the campaign launch and sets the automated content delivery. Step 4: The system deploys the campaign across the user interface, highlighting the participating partners to the relevant customer segments.
Description of the Alternative Sequence	Step 1: Campaign overlap detected → system warns manager of conflicting schedules Step 2: Restaurant opts out of campaign → manager removes them from the active list.
Non functional requirements	System scalability to handle increased traffic during campaigns and real-time content deployment.
Postconditions	A new marketing campaign is active and visible to the targeted customer base on the platform.

UC-42 – Manage Promotions, Discounts, and Coupon Codes

UC Name	UC-42 – Manage Promotions, Discounts, and Coupon Codes
Summary	The Marketing Manager configures the financial parameters of incentives, such as percentage-based discounts and alphanumeric coupon codes, to encourage user engagement.
Dependency	UC-39
Actors	Primary: Marketing Manager
Preconditions	The manager has the necessary authoration to modify pricing rules and financial incentive structure.
Description of the Main Sequence	Step 1: The manager initates the creation of a new promotional tool, such as a platform-wide discount or a restaurant-specific coupon. Step 2: The manager sets strict validity parameters, including expiration dates, minimum order values, and usage limits per customer. Step 3: The manager generates unique coupon codes that customers can apply during the checkout process. Step 4: The system integrates these rules into the payment gateway to ensure discounts are calculated correctly during transactions.
Description of the Alternative Sequence	Step 1: Duplicate coupon code detected → system prompts for unique code. Ster 2: Financial parameters exceed budget limits → system alerts manager for verification.
Non functional requirements	High data accuracy for financial calculations and secure transmission of discount codes.
Postconditions	Valid promotional codes and discounts are active and ready for customer utilization

UC-43 – Manage Promotional Communication

UC Name	UC-43 – Manage Promotional Communication
Summary	The Marketing Manager utilizes the platform's communication infrastructure to transmit targeted notifications and promotional messages directly to the user base.
Dependency	UC-41,UC-42
Actors	Primary: Markrting Manager; Secondary: Notification System
Preconditions	A markrting campaign or a new set of discounts has been finalized and is ready for public announcement.
Description of the Main Sequence	Step 1: The manager drafts the promotional content, ensuring it is engaging and provides clear information about the offer. Step 2: The manager selects the delivery channels, such as in-app push notifications, SMS, or email alerts. Step 3: The manager triggers the Notification System to broadcast the messages to the selected groups of users. Step 4: The system tracks the delivery status of each notification to ensure successful receipt by the customers.
Description of the Alternative Sequence	Step 1:Notification system failure → system logs error and retries delivery automatically. Step 2: User has opted out of marketing alerts → system excludes them from the broadcast list.
Non functional requirements	Reliability in message delivery and minimal delay between triggering and receipt.
Postconditions	Customers receive the intended promotional messages, leading to increased platform interaction.

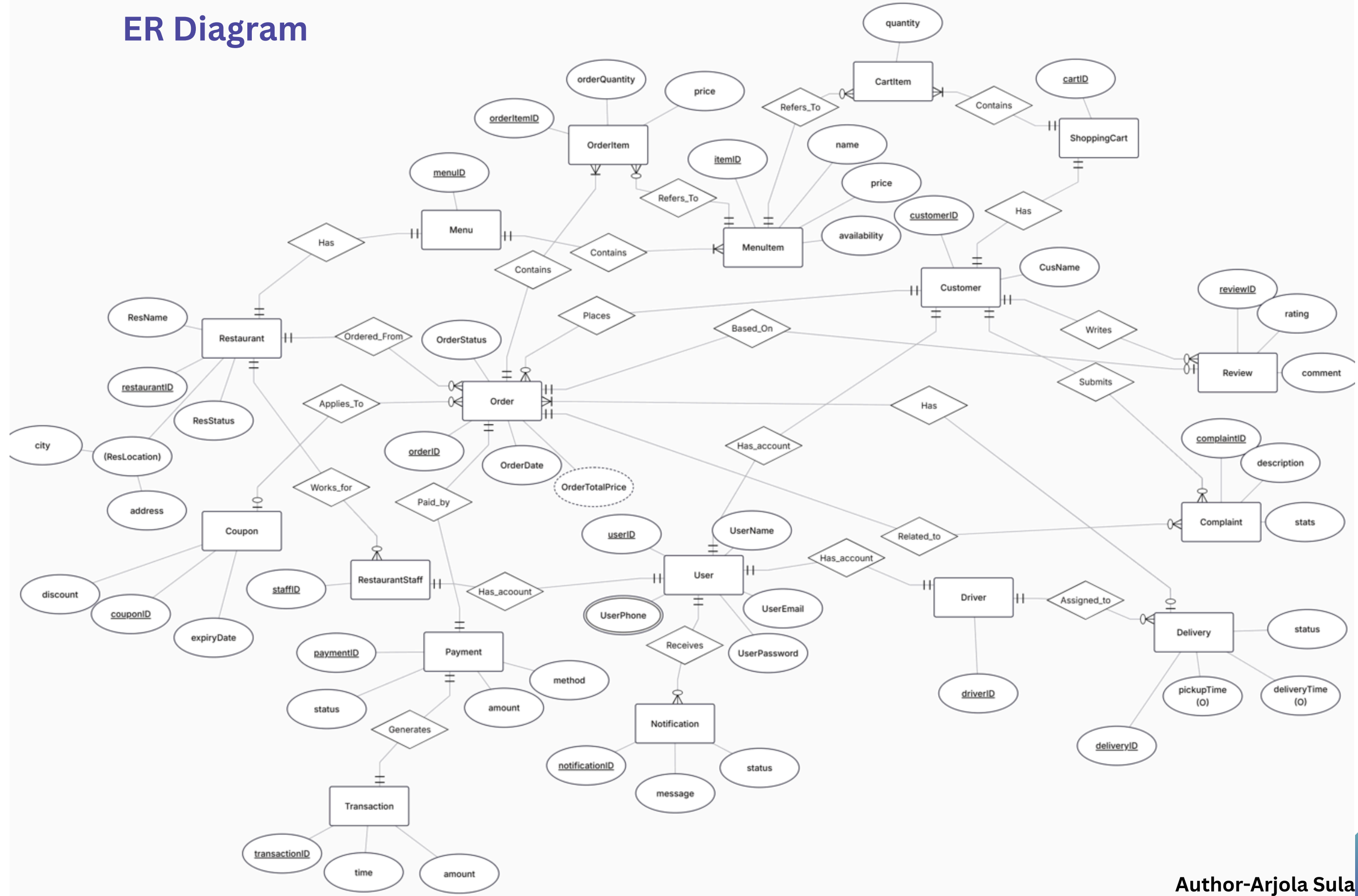
UC-44 Analyze and Moniton Campaign Performance

UC Name	UC-44 – Analyze and Moniton Campaign Performance
Summary	The Marketing Manager reviews analytical data and customer engagement metrics to evaluate the effectiveness of marketing efforts and optimize future strategies.
Dependency	UC-41
Actors	Primary: Marketing Manager
Preconditions	A marketing campaign has been active for a sufficient duration to generate measurable data.
Description of the Main Sequence	Step 1: The manager accesses the “Analytics Dashboard” to view real-time data on order volume, click-through rates, and coverision metrics. Step 2: The manager compares the actual performance of the campaign against the predefined key performance indicators. Step 3: The system generates detailed reports visualizing customer engagement trends and the return on investment for specific promotions. Step 4: The manager uses these insights to adjust ongoing campaigns or inform the design of future marketing initiatives.
Description of the Alternative Sequence	Step 1:Data not yet available → system informs manager to check back later. Step 2:Campaign underperforming → manager adjusts parameters or stops the campaign early.
Non functional requirements	Data visualization accuracy and high-speed processing of large analytical datasets.
Postconditions	The manager gains a comprehensive understanding of campaign success, documented through system-generated reports.

UC-45 – Manage Featured Restaurants and Advertisements

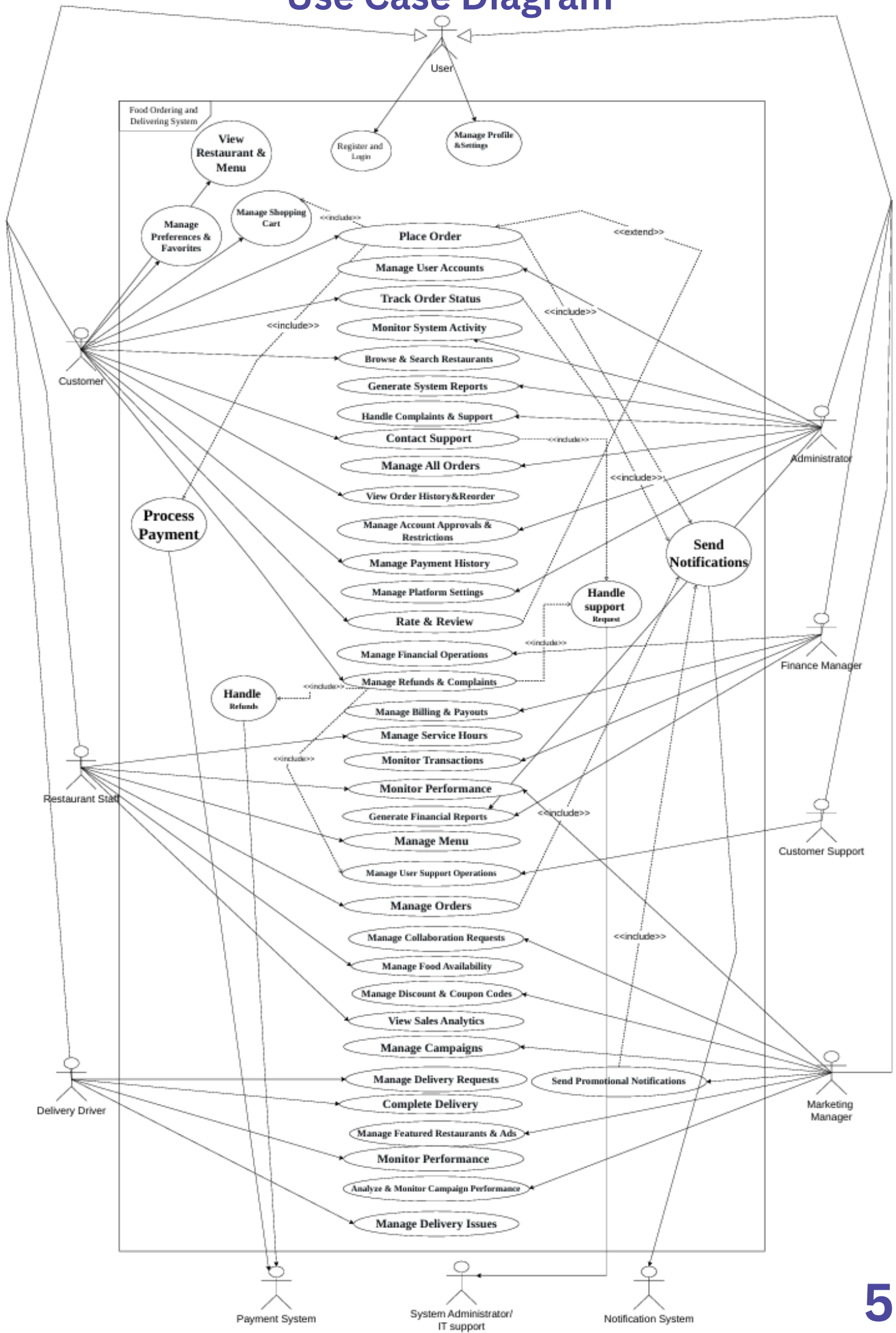
UC Name	UC-45 – Manage Featured Restaurants and Advertisements
Summary	The Marketing Manager controls the premium placement of specific restaurants and advertisements within the application's home screen and search results.
Dependency	UC-40
Actors	Primary: Marketing Manager
Preconditions	Specific restaurants have been approved for "Featured" status through a paid partnership or high-performance criteria.
Description of the Main Sequence	Step 1: The manager selects the restaurant profiles that are entitled to premium visibility on the user interface. Step 2: The manager configures the visual assets, such as banners or highlighted icons, that will represent the advertisement. Step 3: The manager sets the priority level for these featured items within the search and filtering algorithms. Step 4: The system updates the front-end display to prioritize these restaurants for the specified duration of the advertisement.
Description of the Alternative Sequence	Step 1: Asset upload failure → manager is prompted to upload a different image format. Step 2: Featured slot already filled → manager must schedule for a different time slot.
Non functional requirements	Compatibility across different devices and browsers and consistent display performance.
Postconditions	The selected restaurants receive prioritized placement, increasing their exposure to potential customers.

ER Diagram



Use Case Diagram

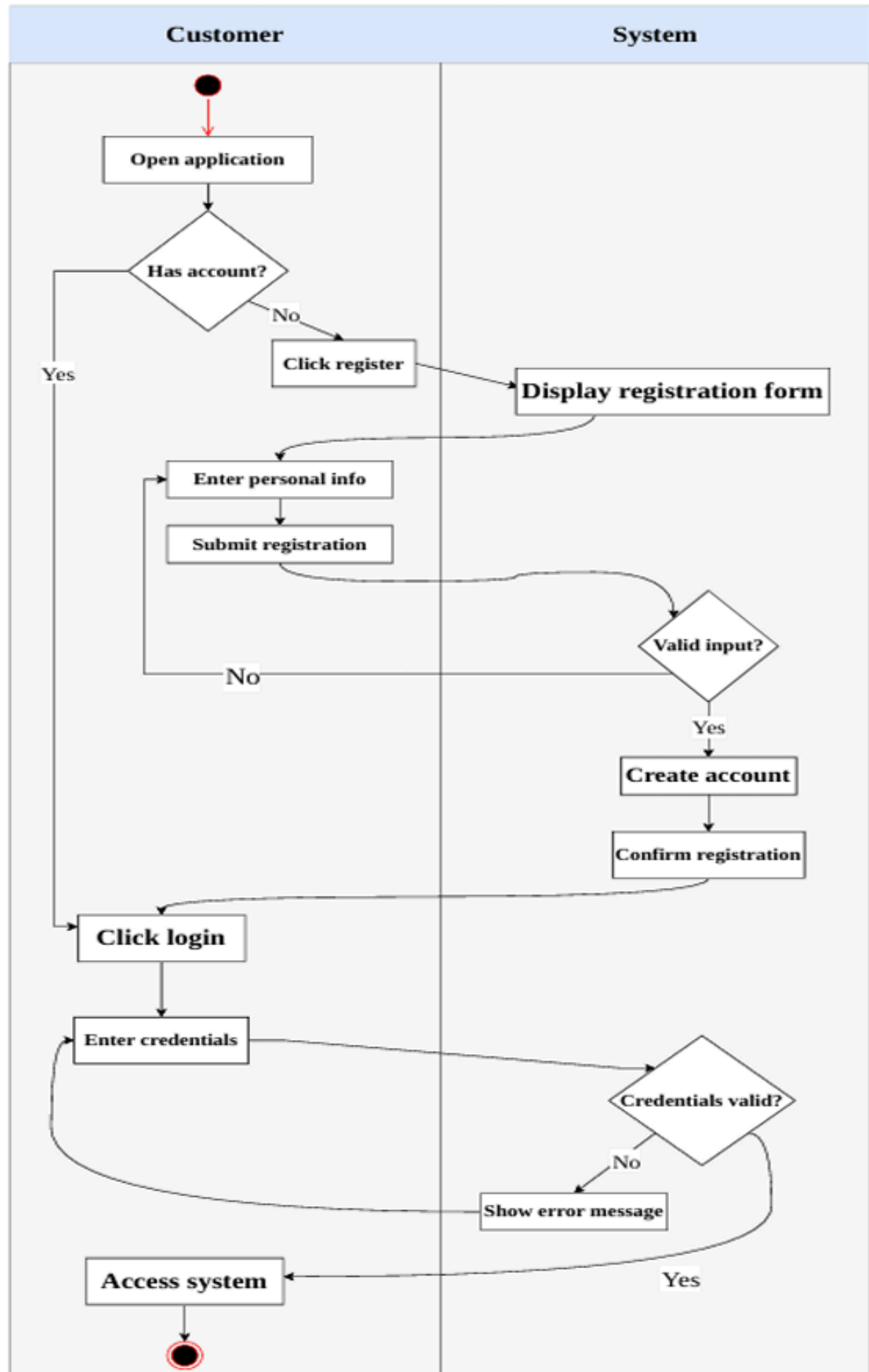
Author-Arjola Sula



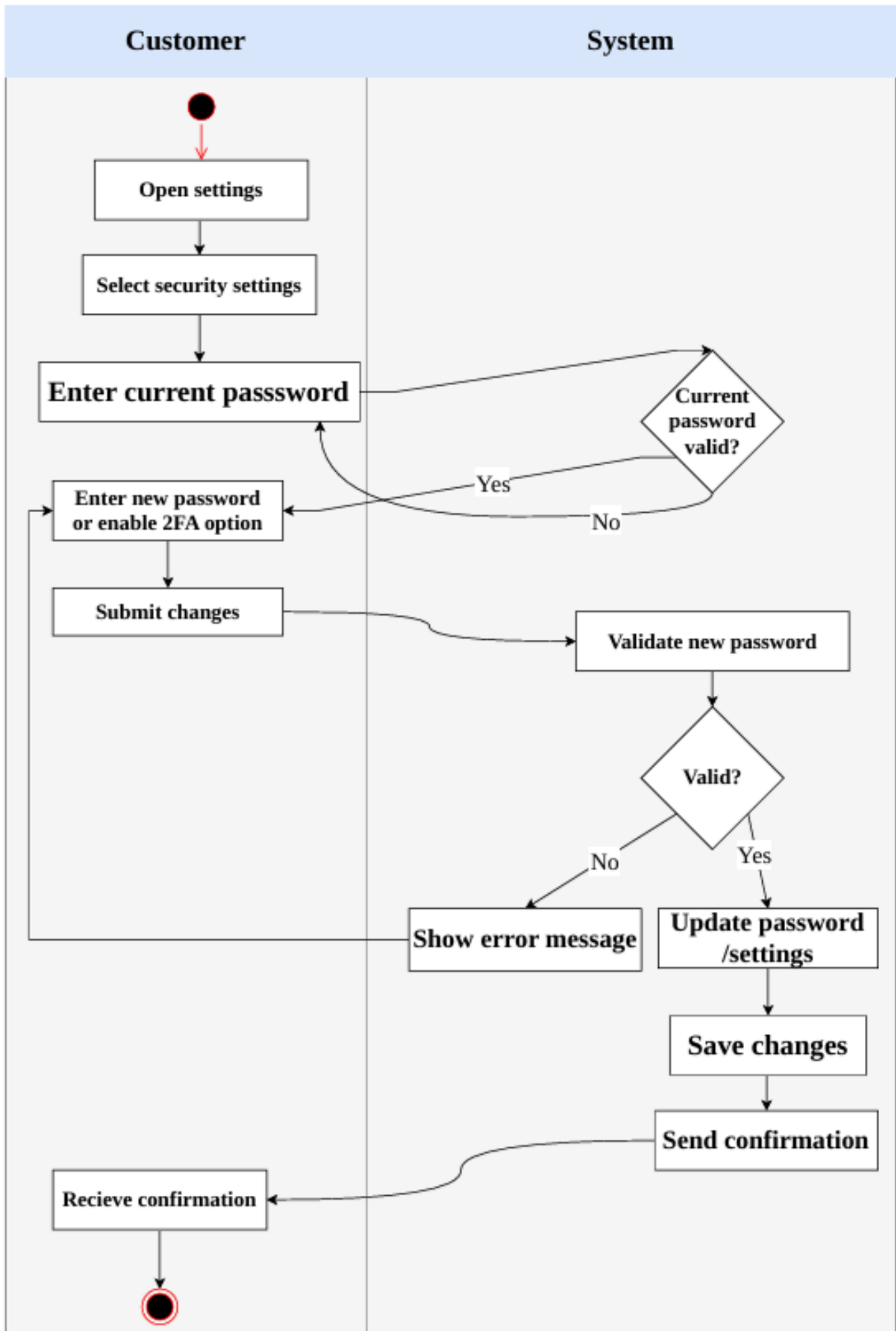
Activity diagrams

Author-Arjola Sula

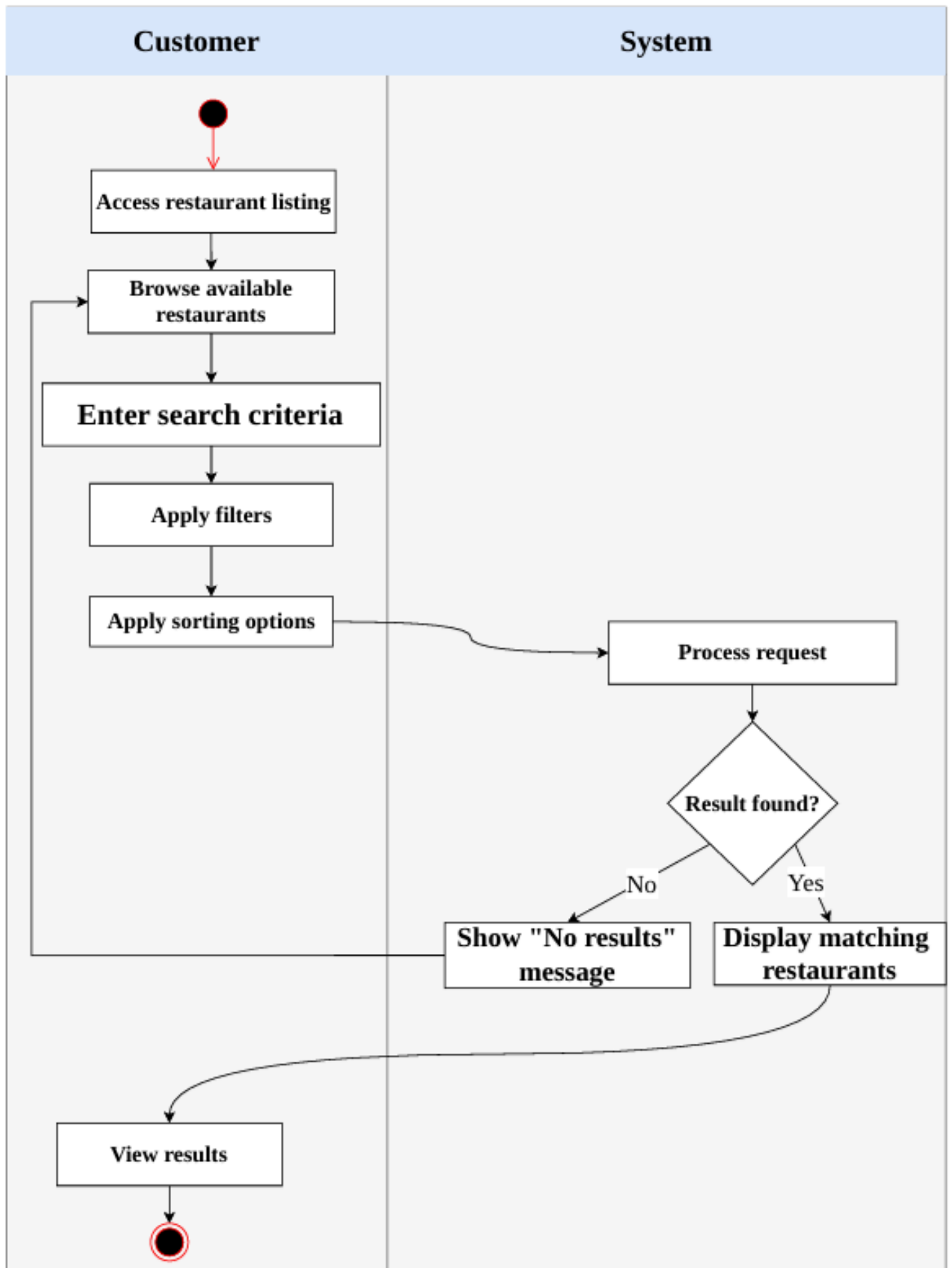
Activity Diagram - Manage customer account(UC-01)



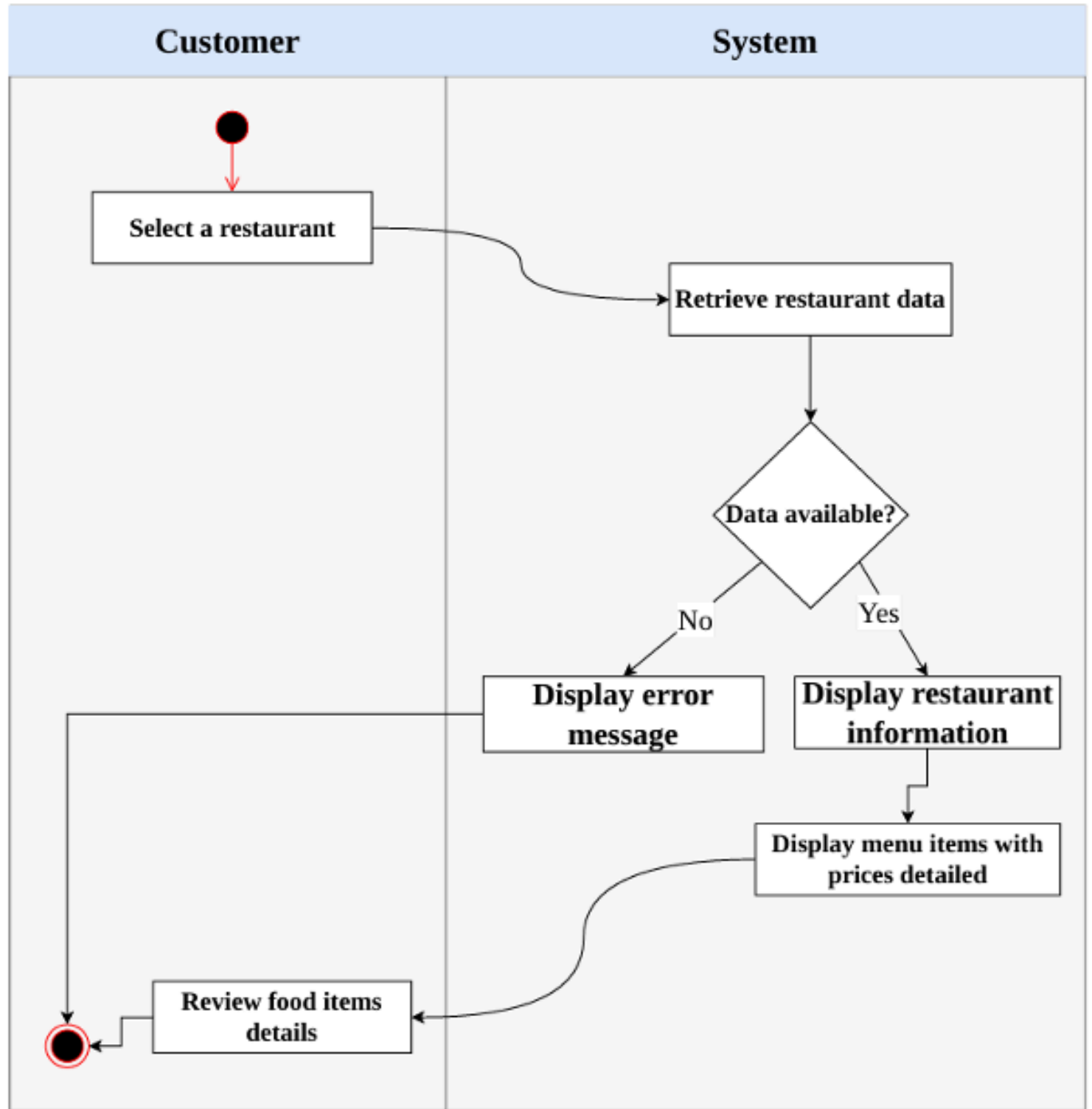
Activity Diagram - Manage Authentication and Security Settings (UC-02)



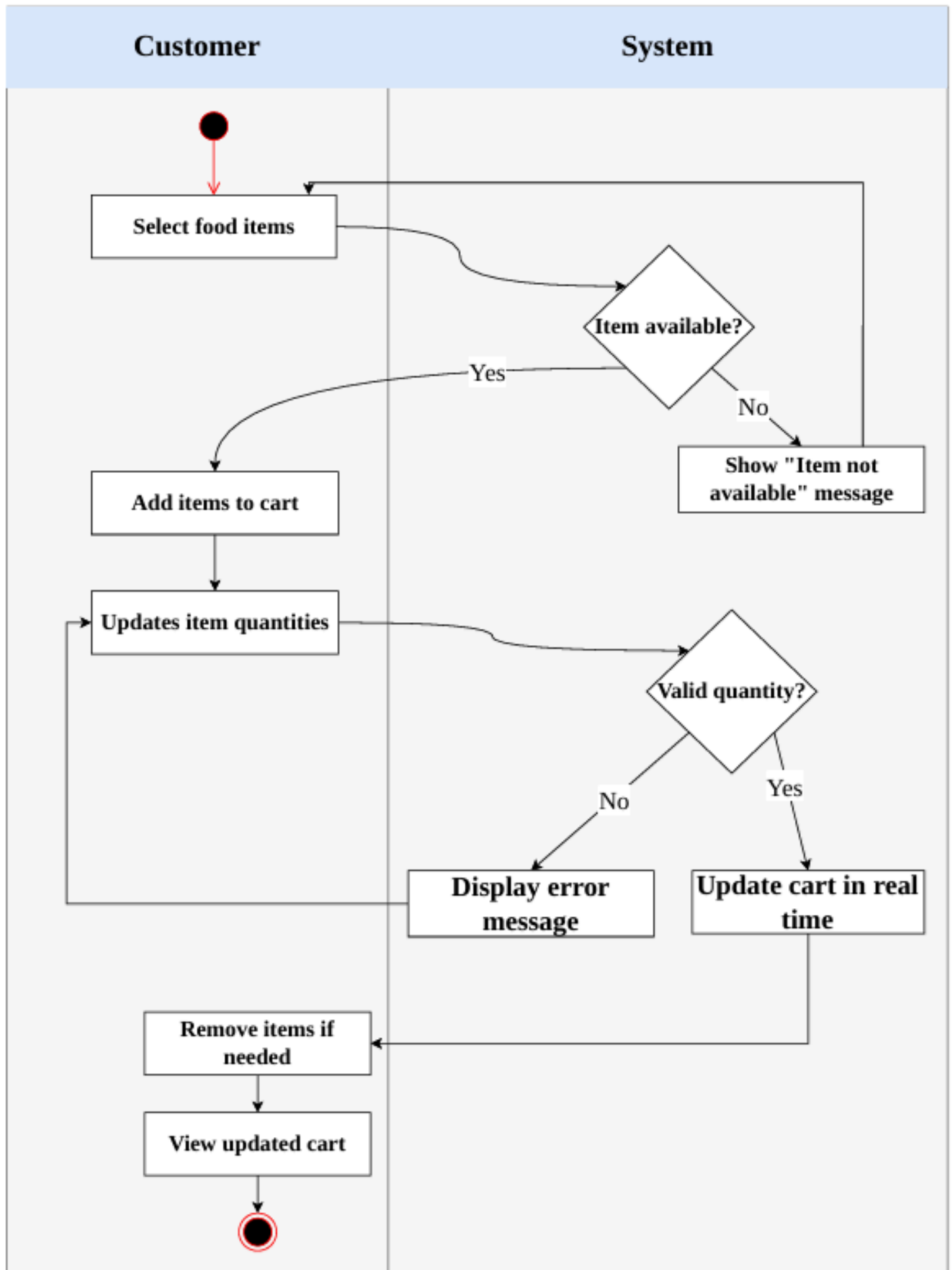
Activity Diagram - Browse, Search, Filter, and Sort Restaurants (UC-03)



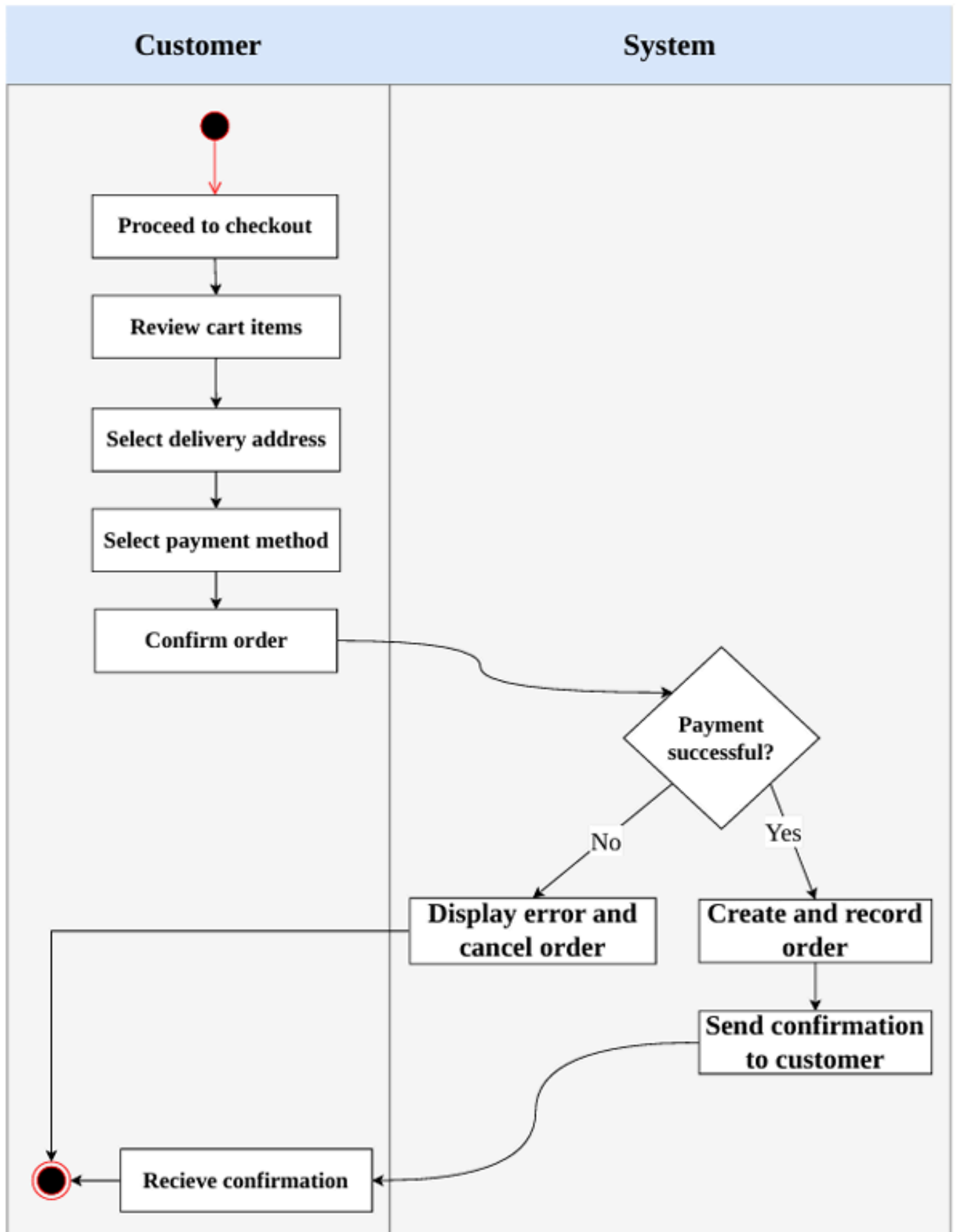
Activity Diagram - View Restaurant and Food Details (UC-04)



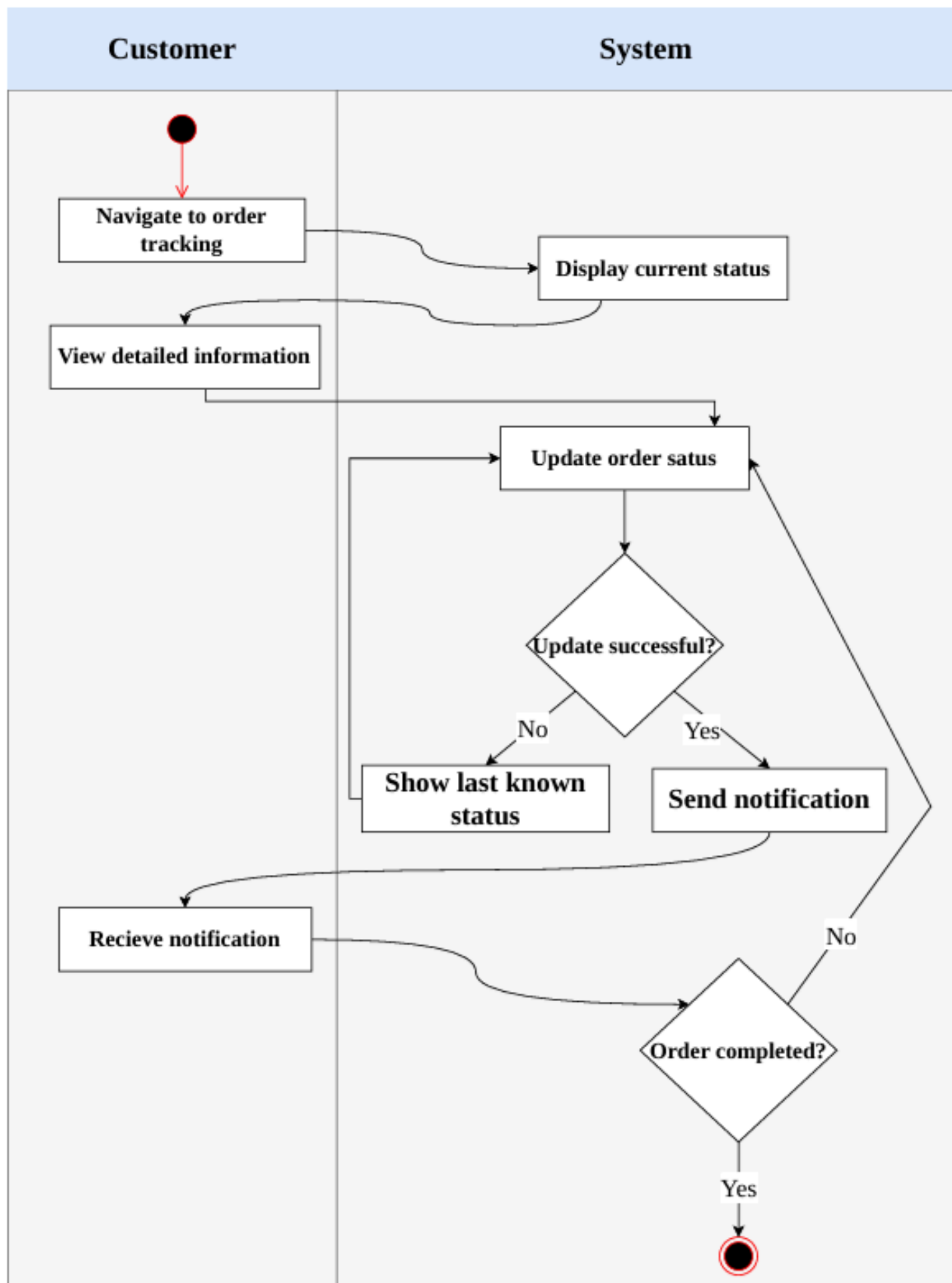
Activity Diagram - Manage Shopping Cart (UC-05)

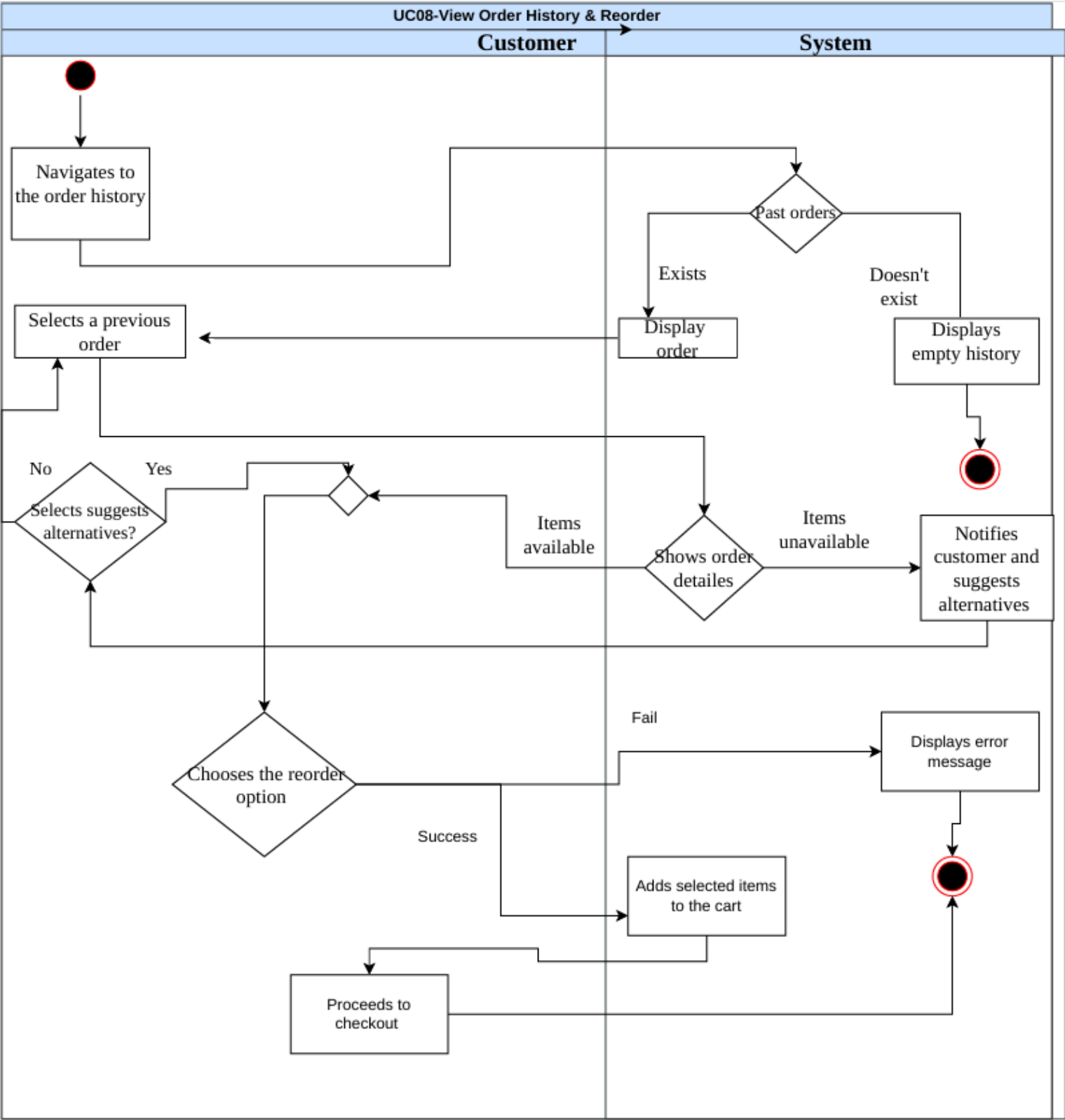


Activity Diagram - Place order (UC-06)



Activity Diagram - Track Order & Receive Notifications (UC-07)

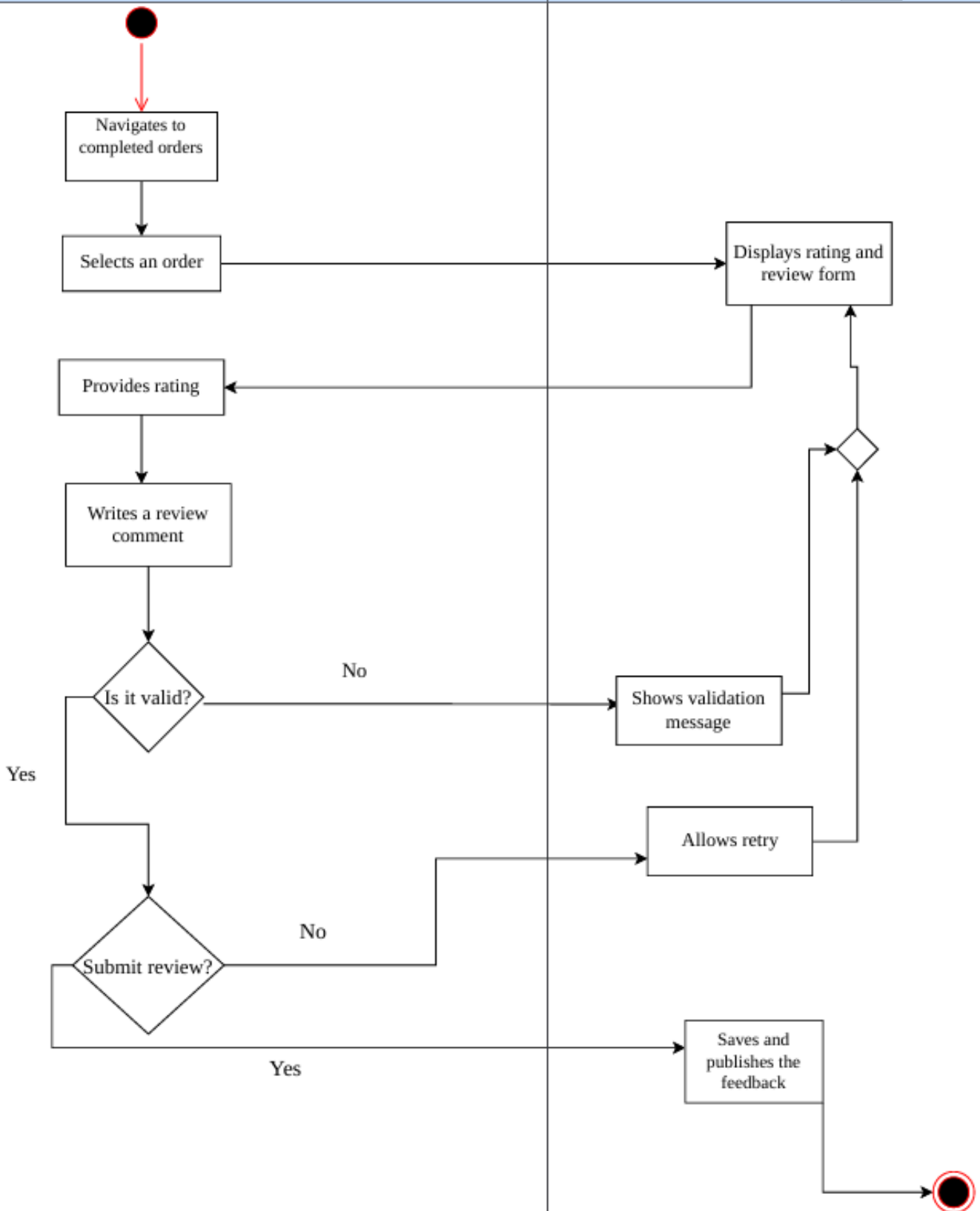




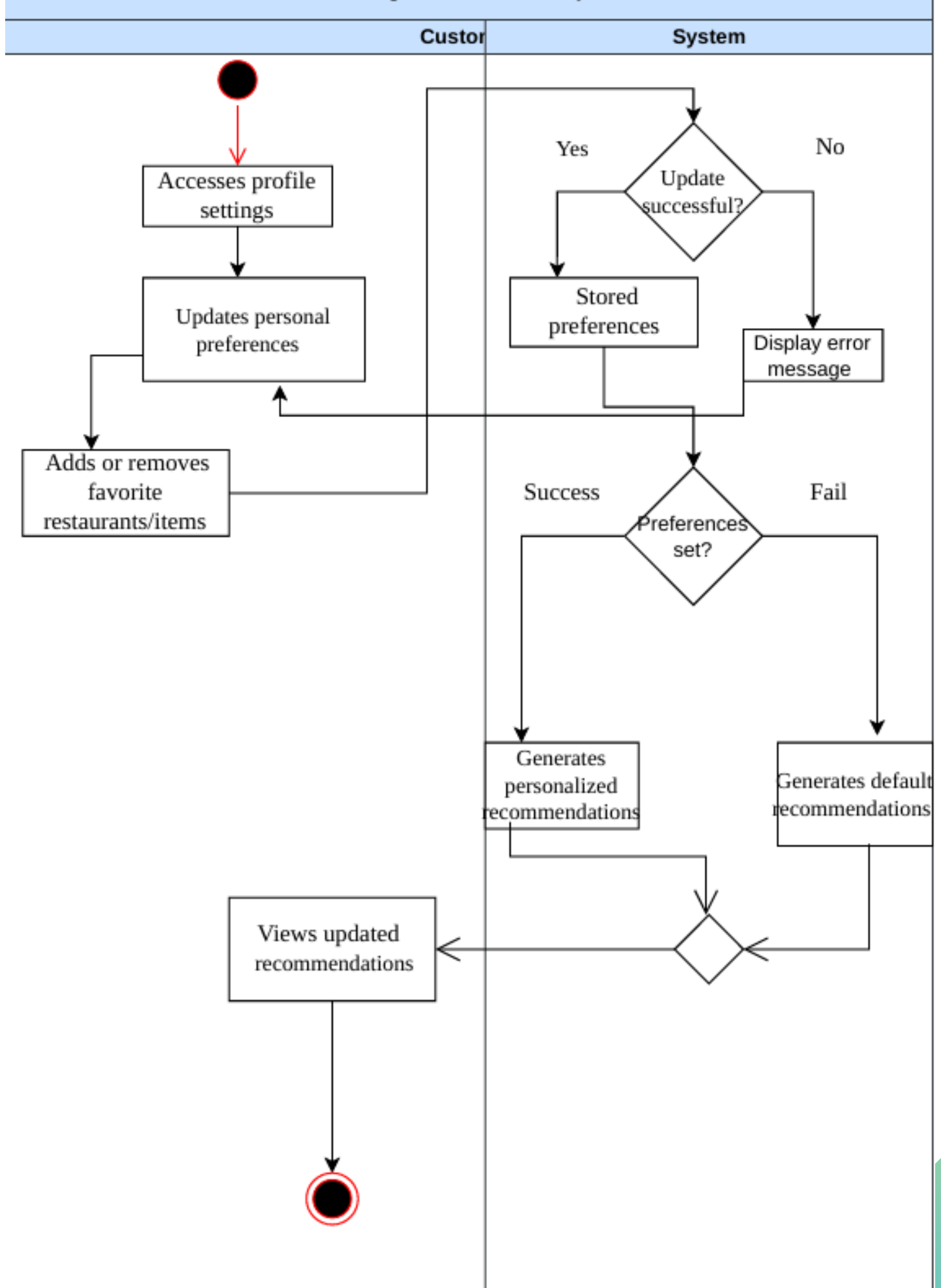
UC09-Rate & Review Restaurants

Customer

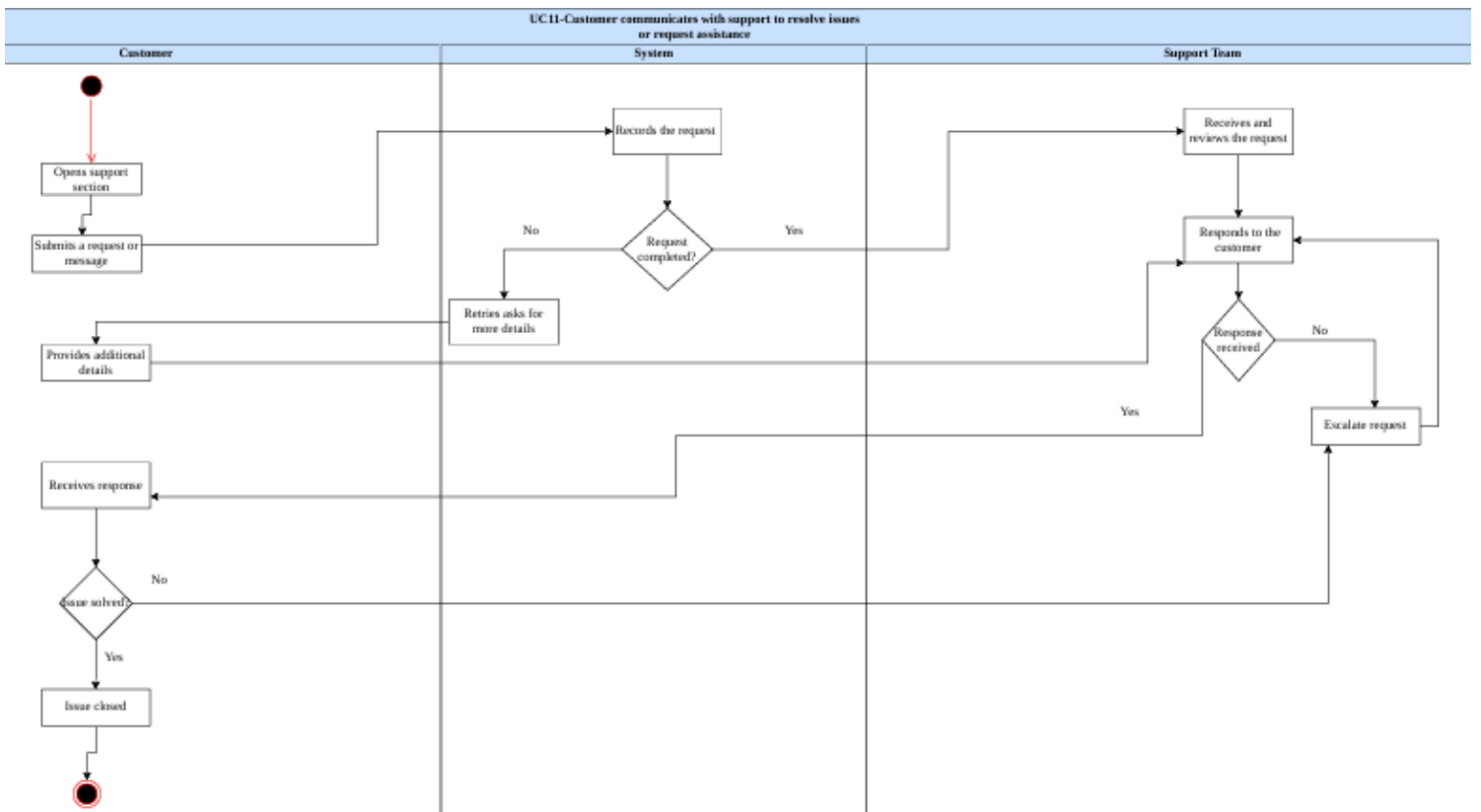
System



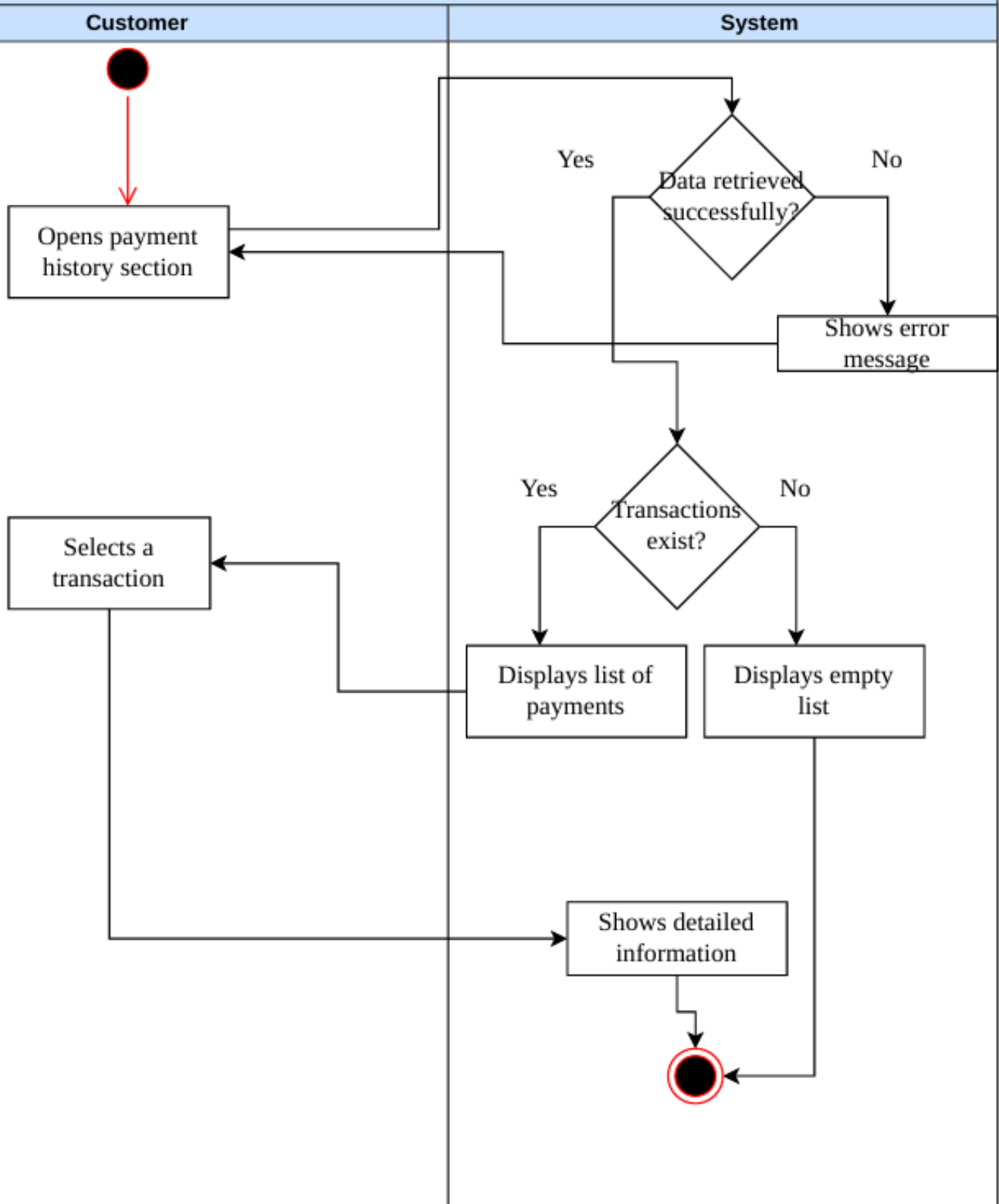
UC10-Manage Personalized Experience



UC11-Customer communicates with support to resolve issues or request assistance



UC-12: Manage Payment History

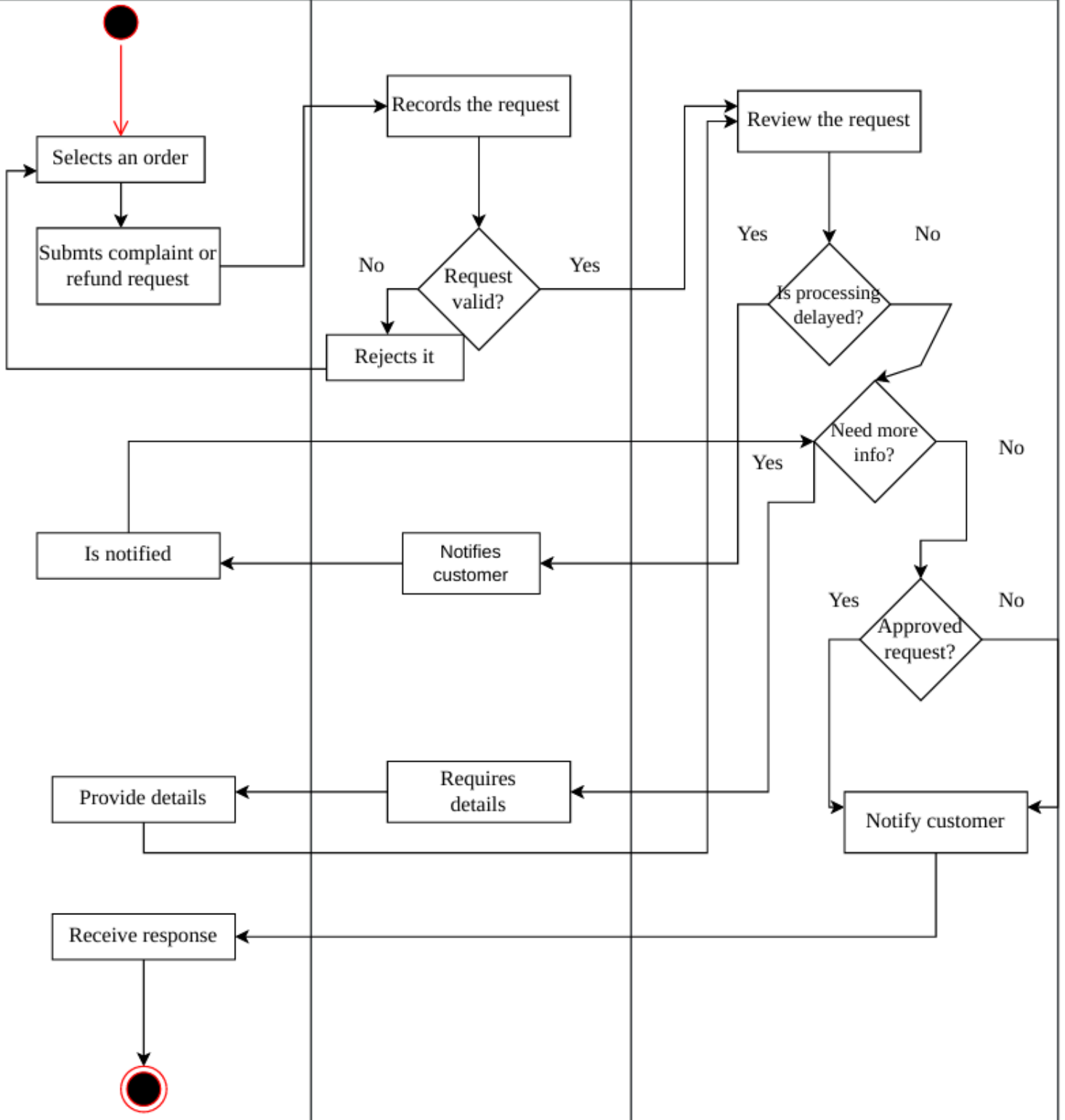


UC-13: Manage Refunds & Complaints

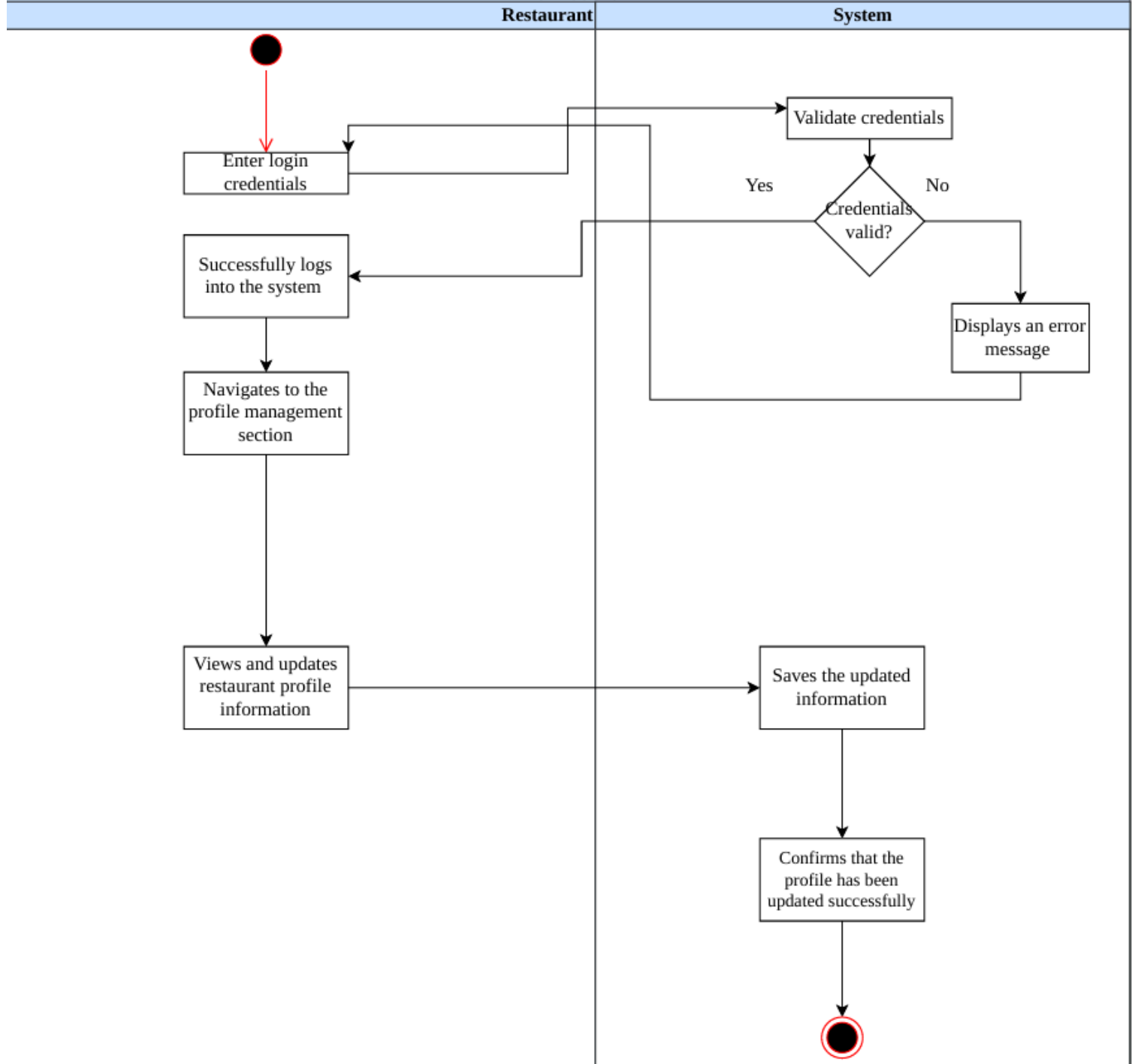
Customer

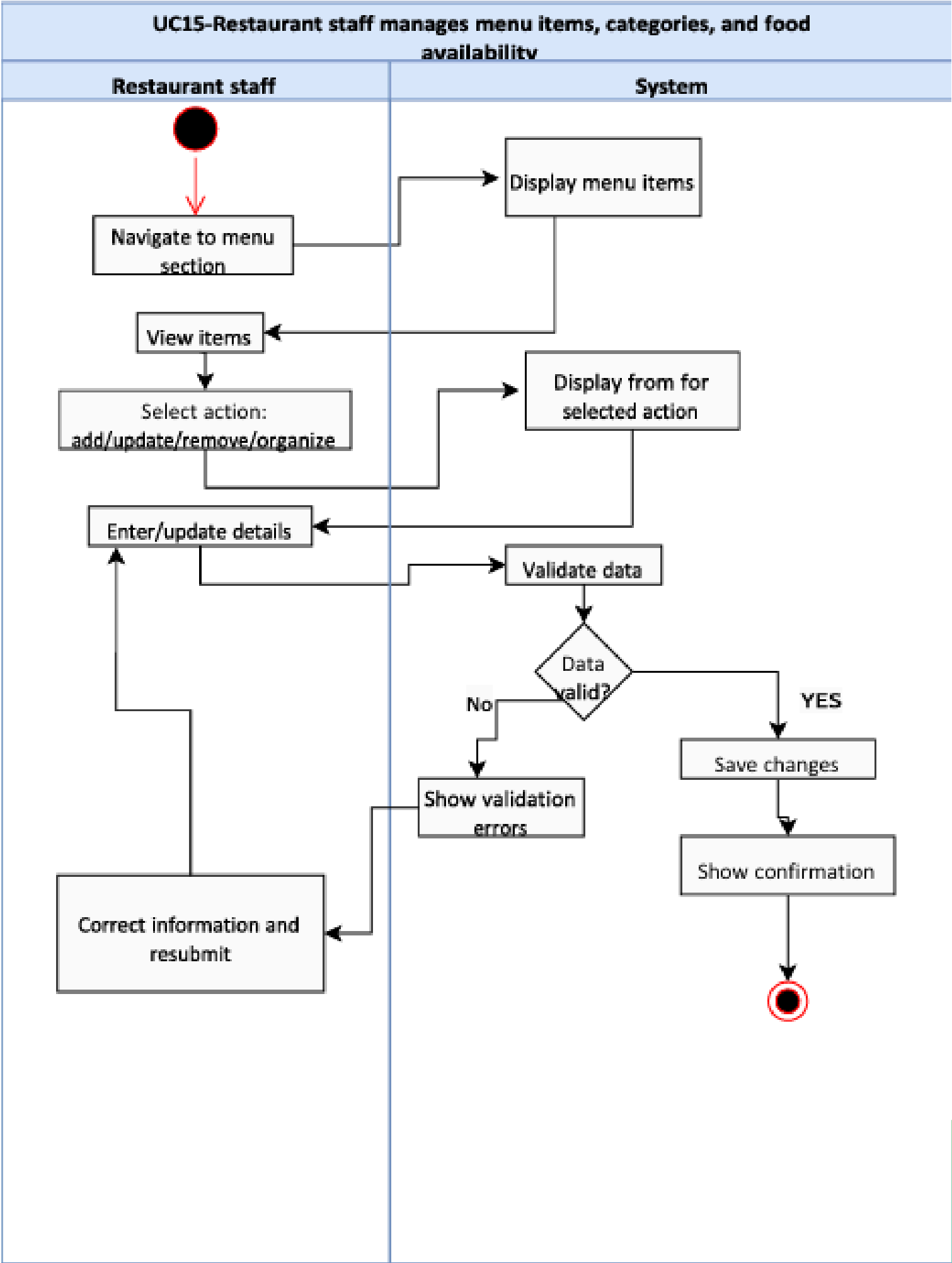
System

Admin

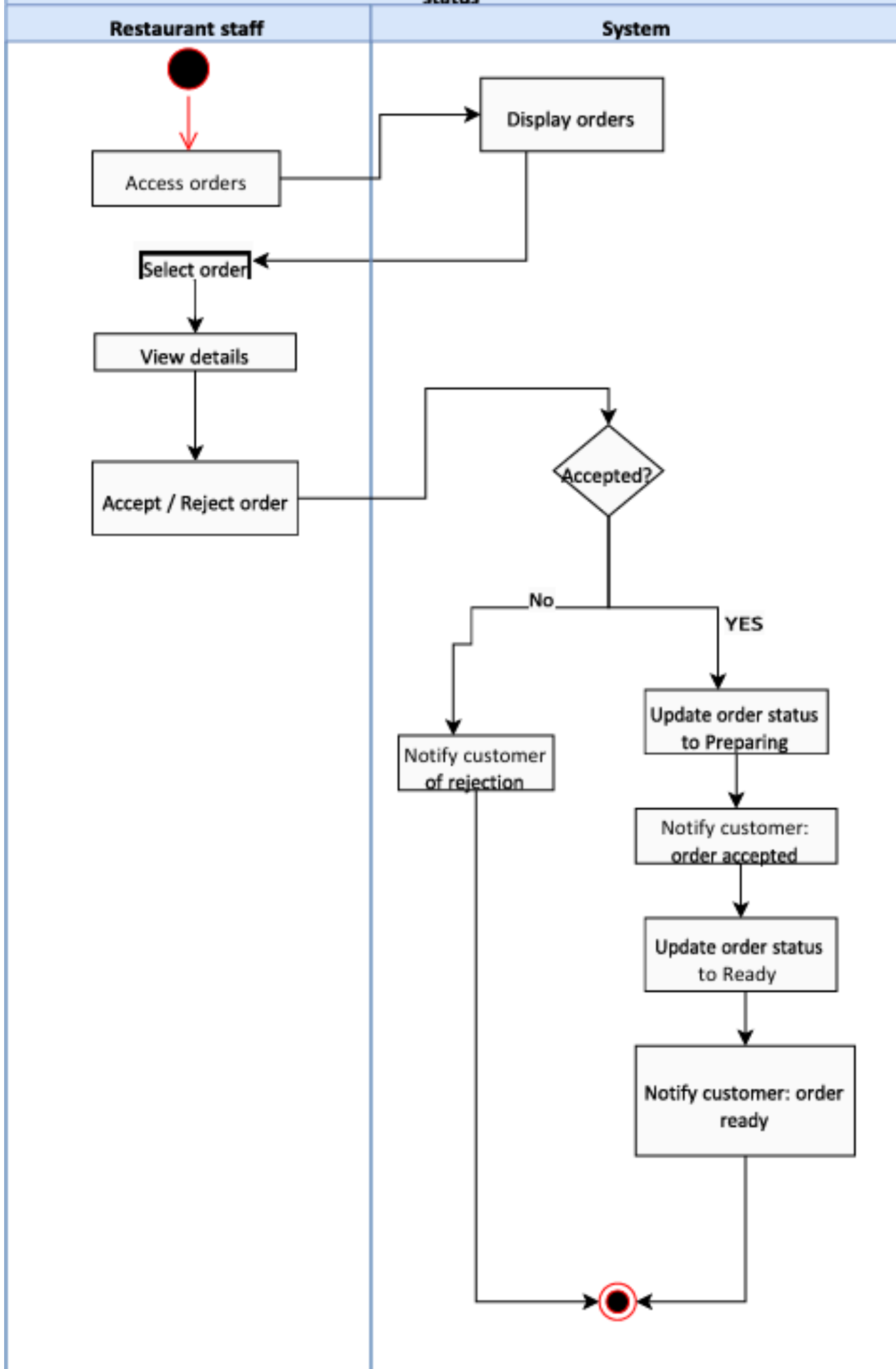


UC14- Restaurant staff manages account and restaurant profile information.

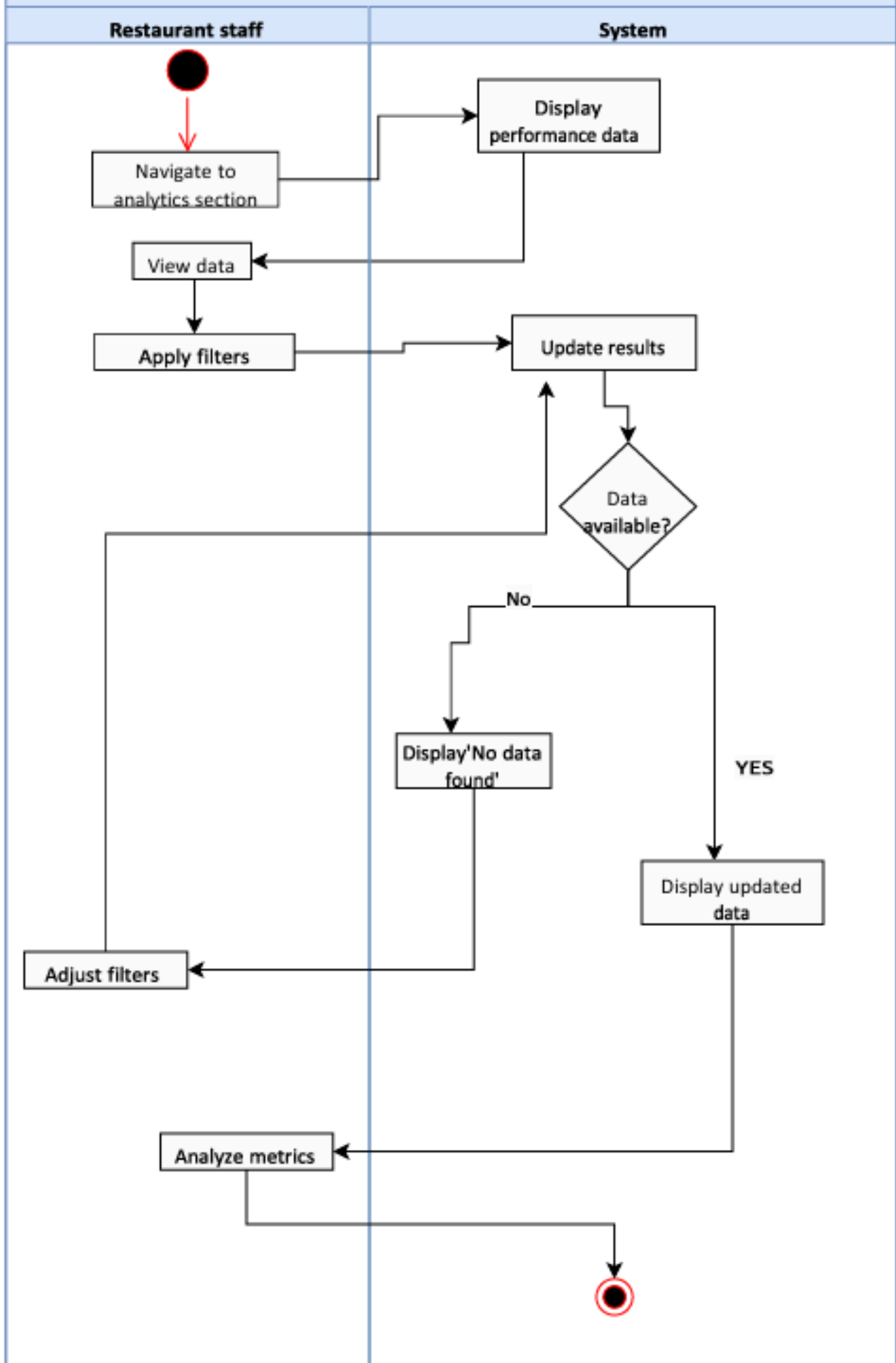




UC16 -Restaurant staff manages customer orders and updates order status



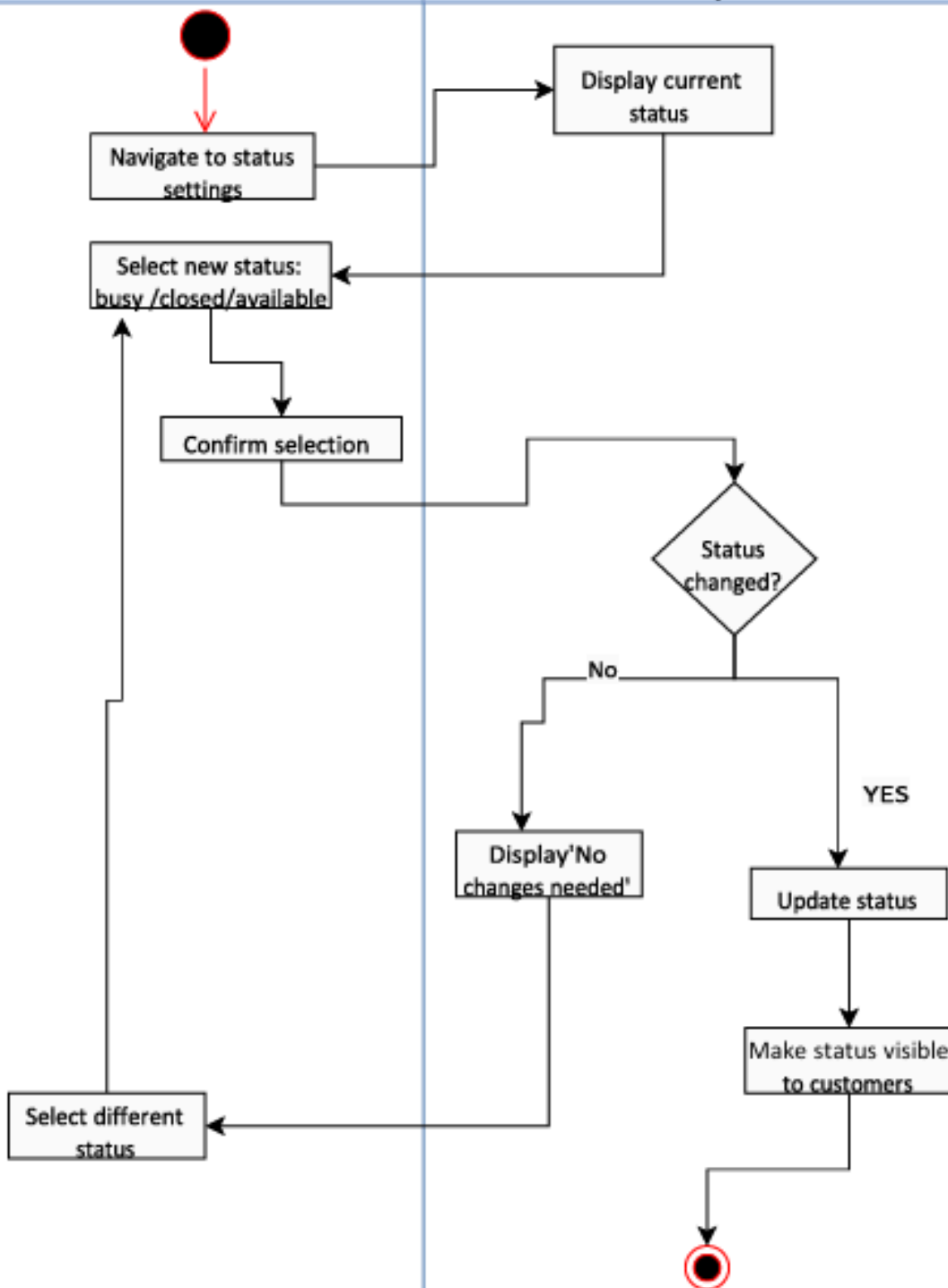
UC17-Restaurant staff monitors and analyzes restaurant performance



UC18-Restaurant staff manages temporary service interruptions

Restaurant staff

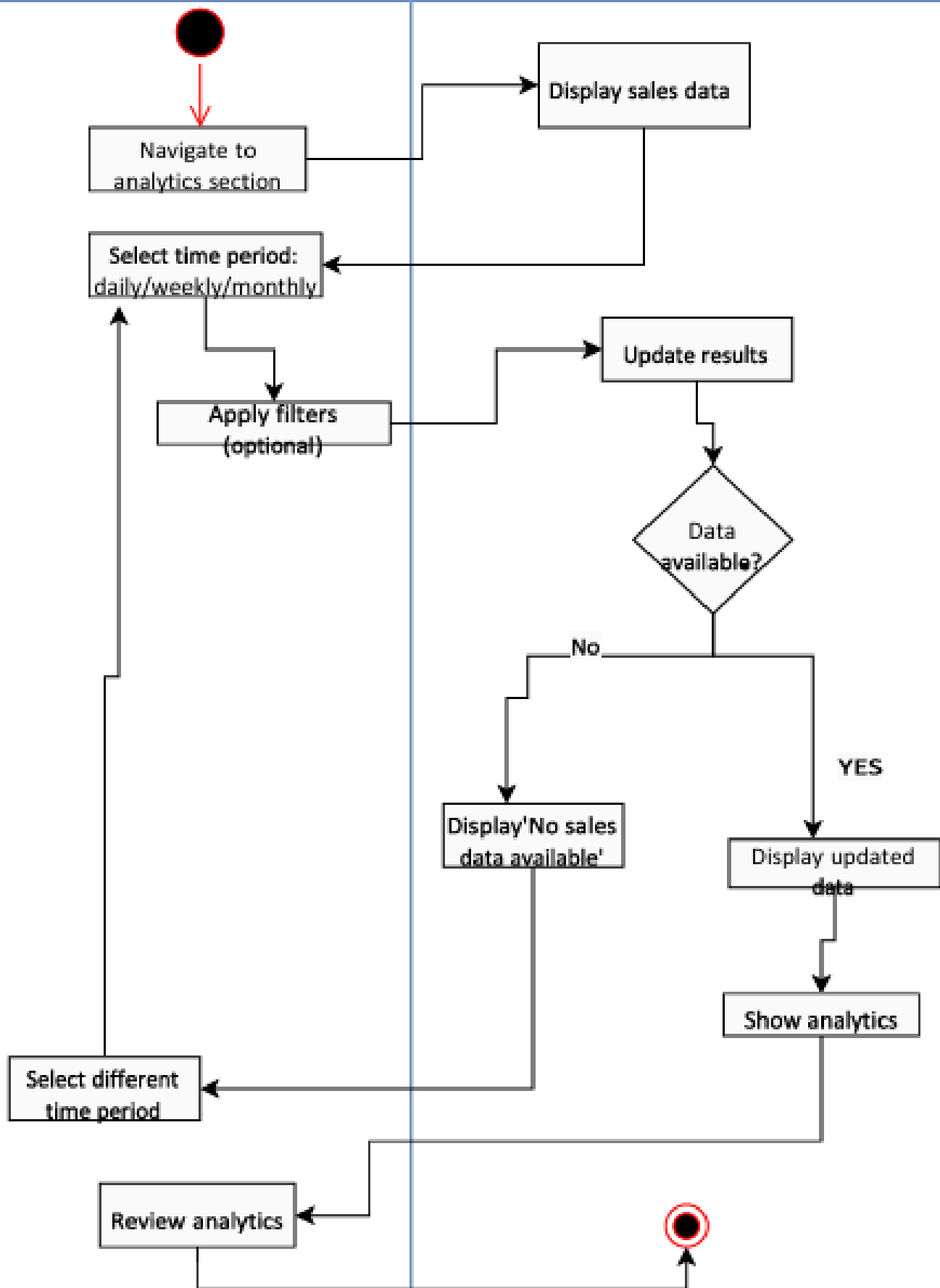
System



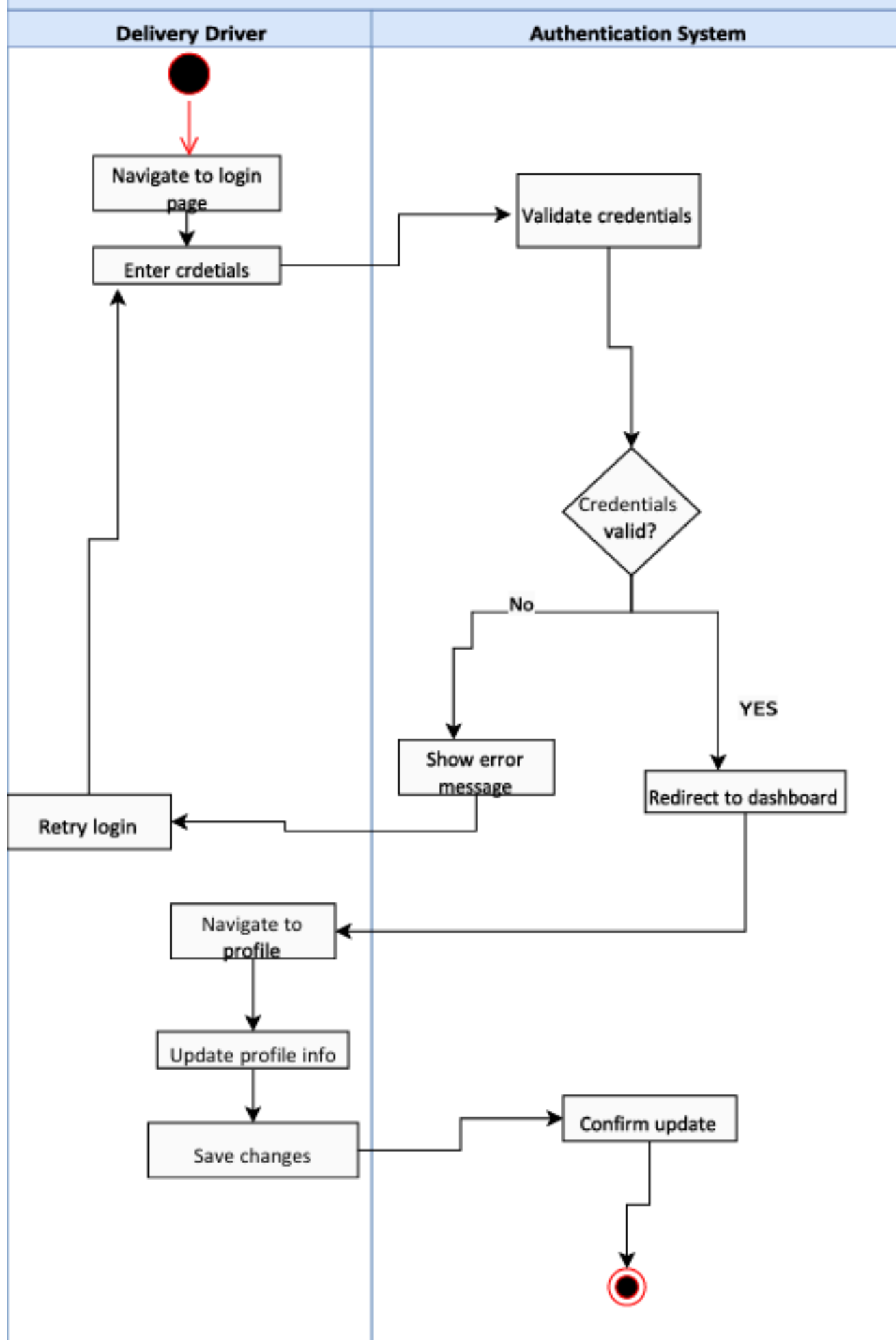
UC19-Restaurant staff views sales analytics

Restaurant staff

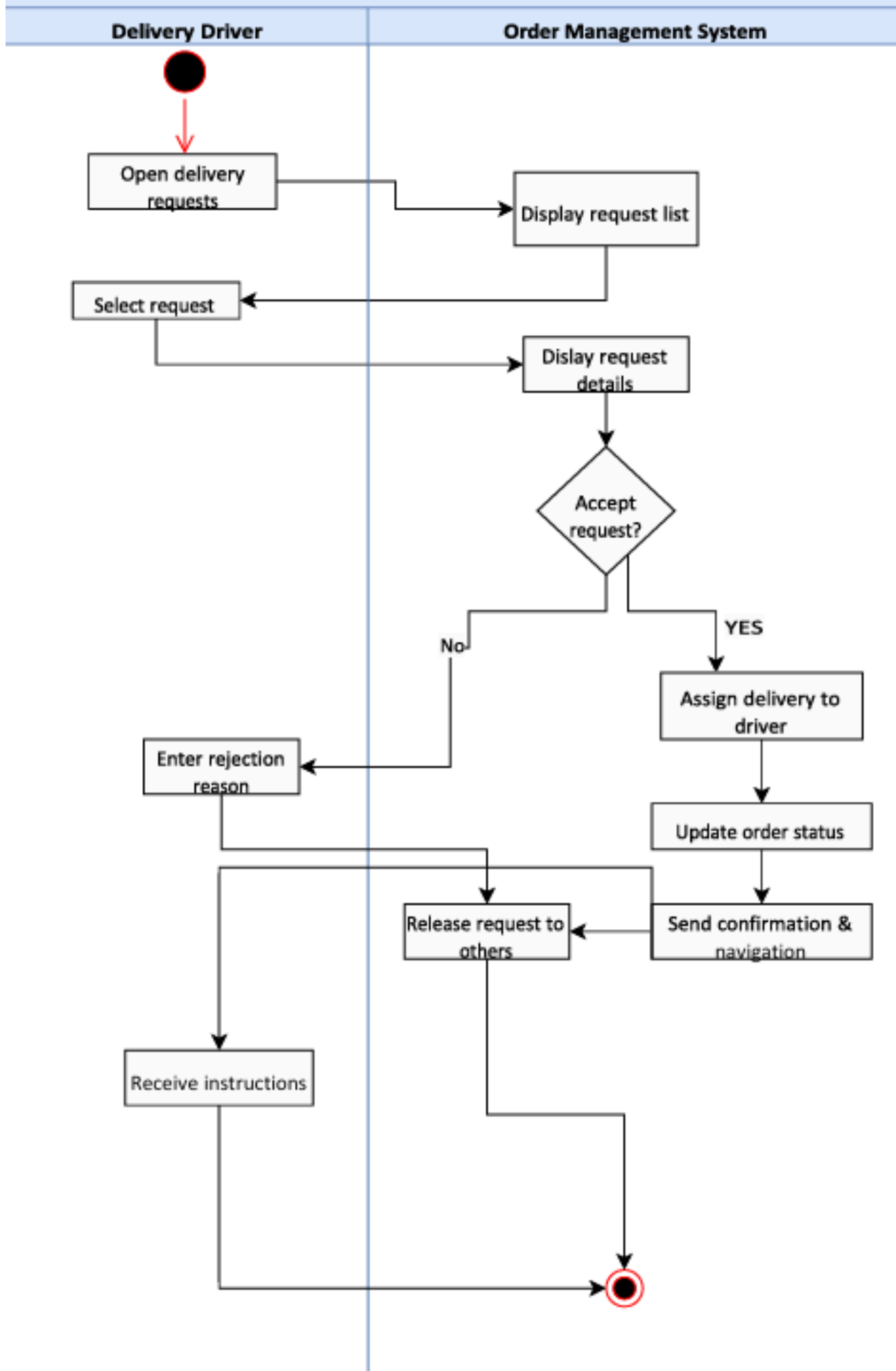
System

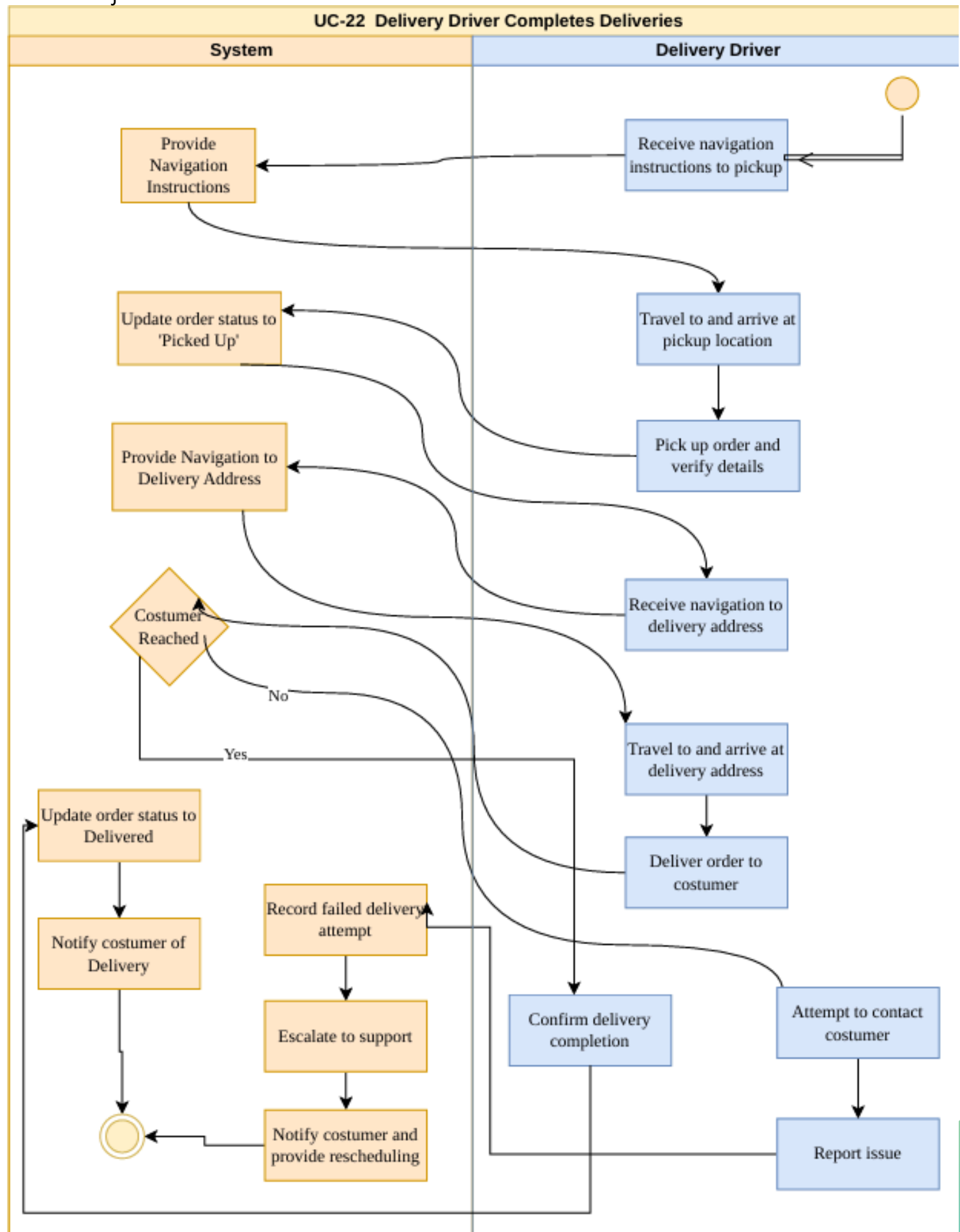


UC20 – Delivery Driver Manages Account

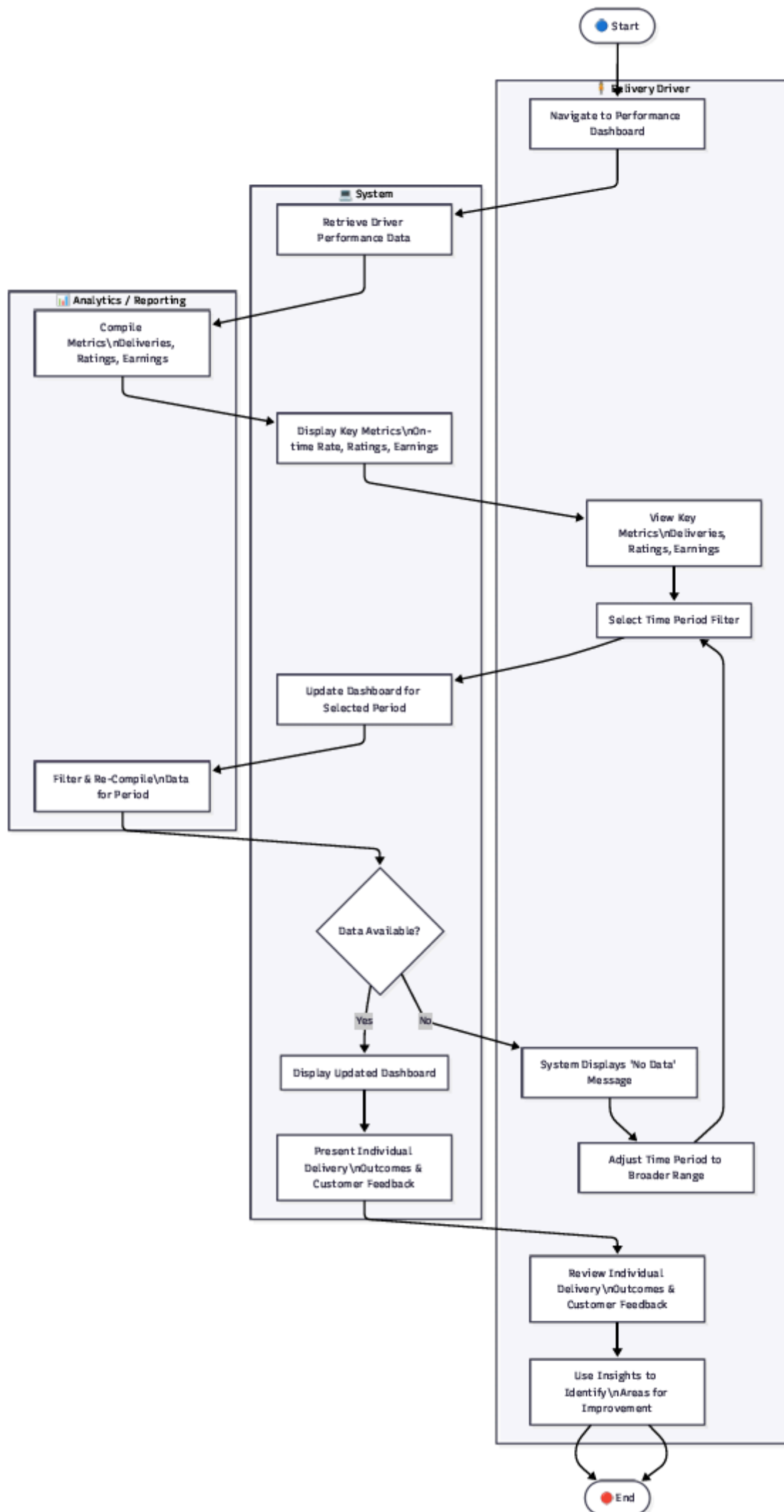


UC21 – Delivery Driver Manages Delivery Requests





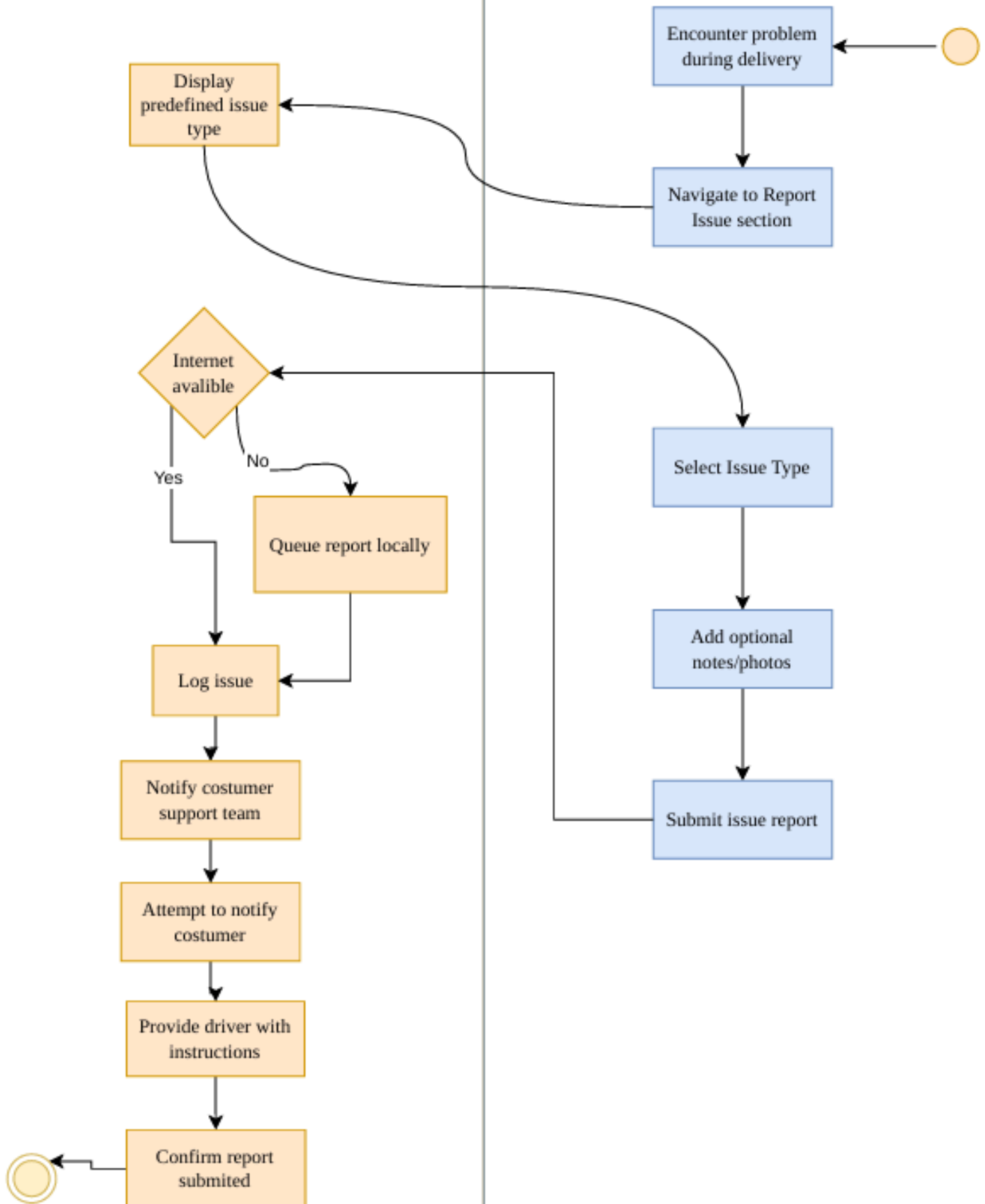
UC23 – Delivery Driver Monitors Personal Performance



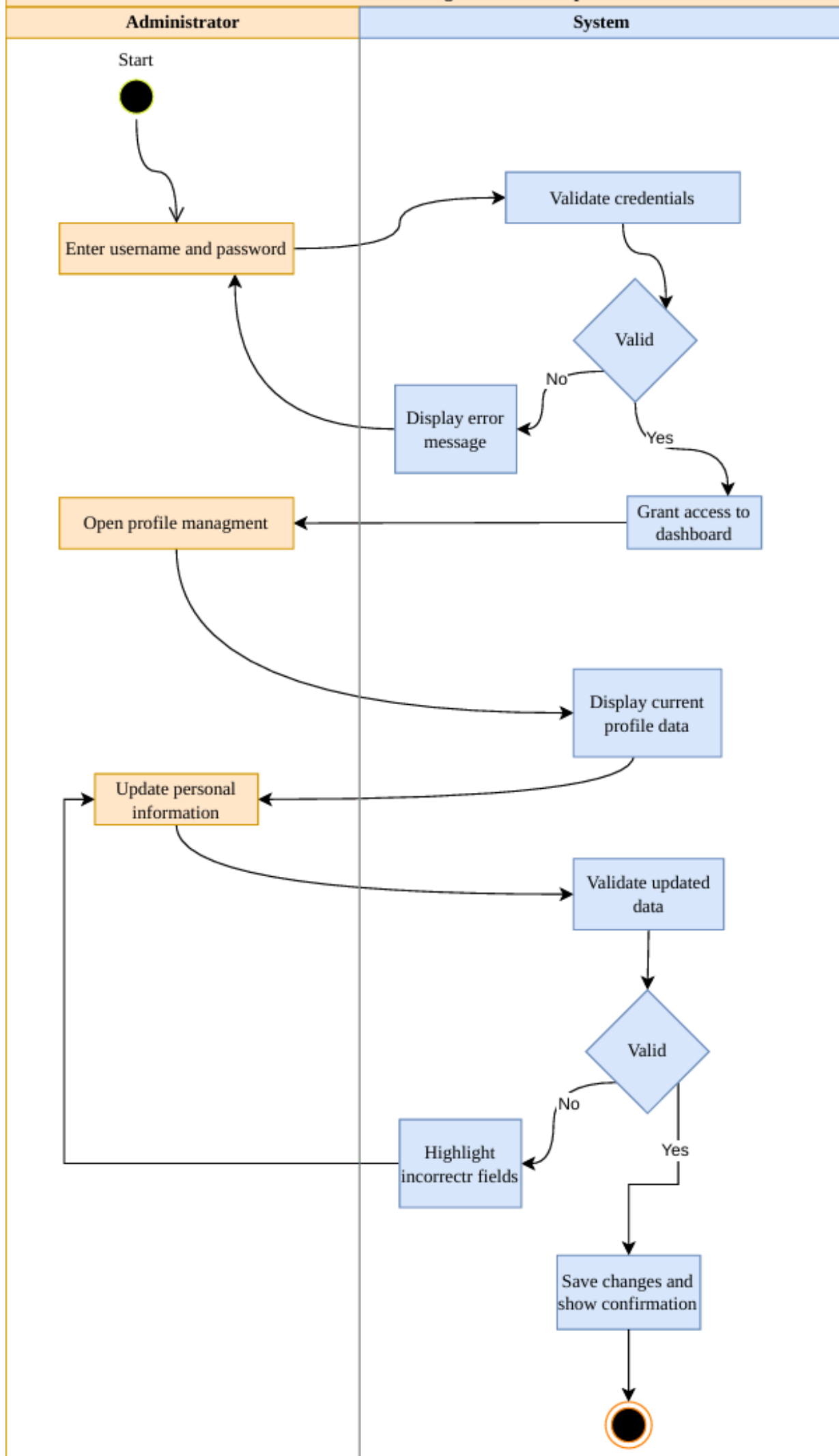
UC-24 Delivery Driver Report Issues During Delivery

System

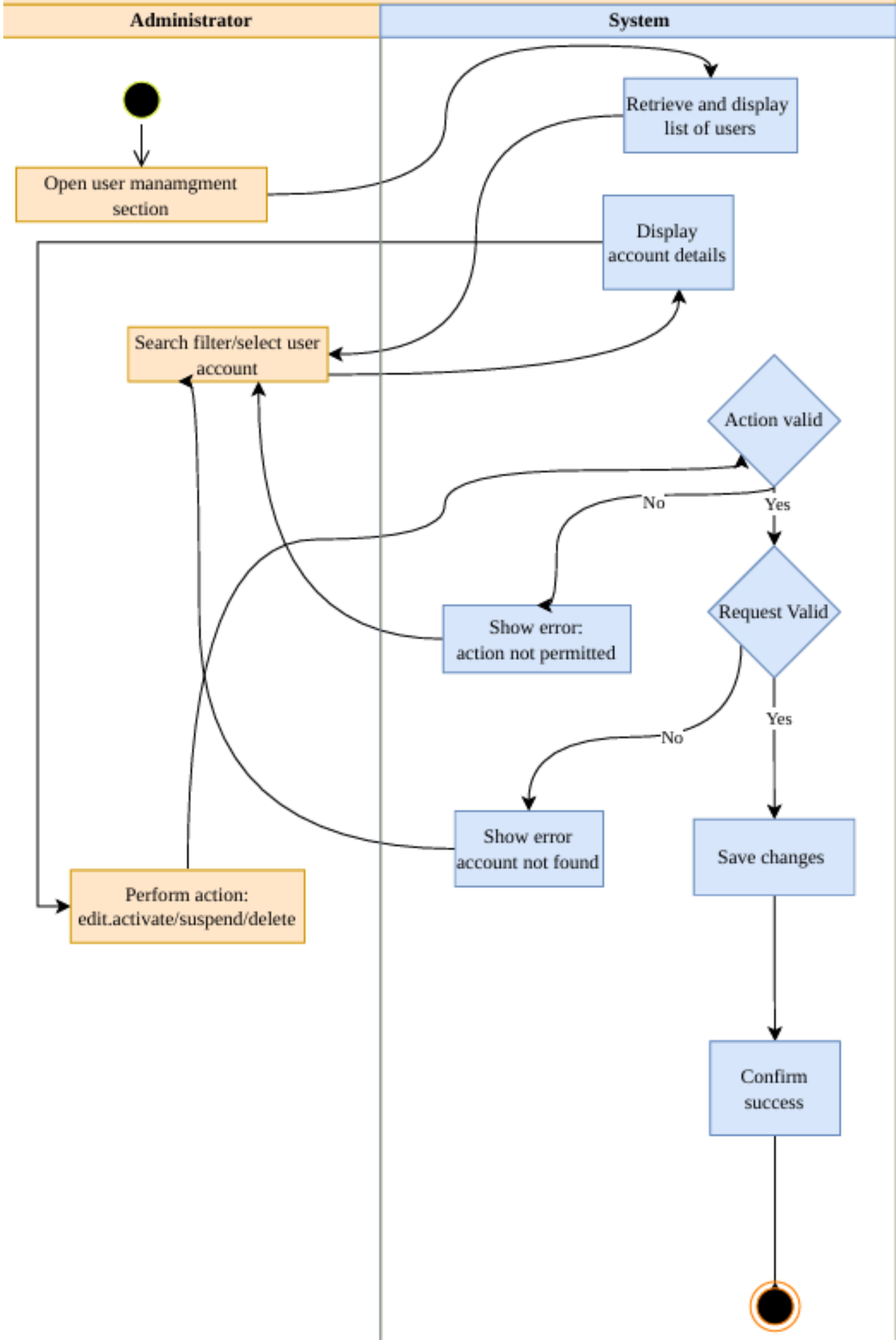
Delivery Driver



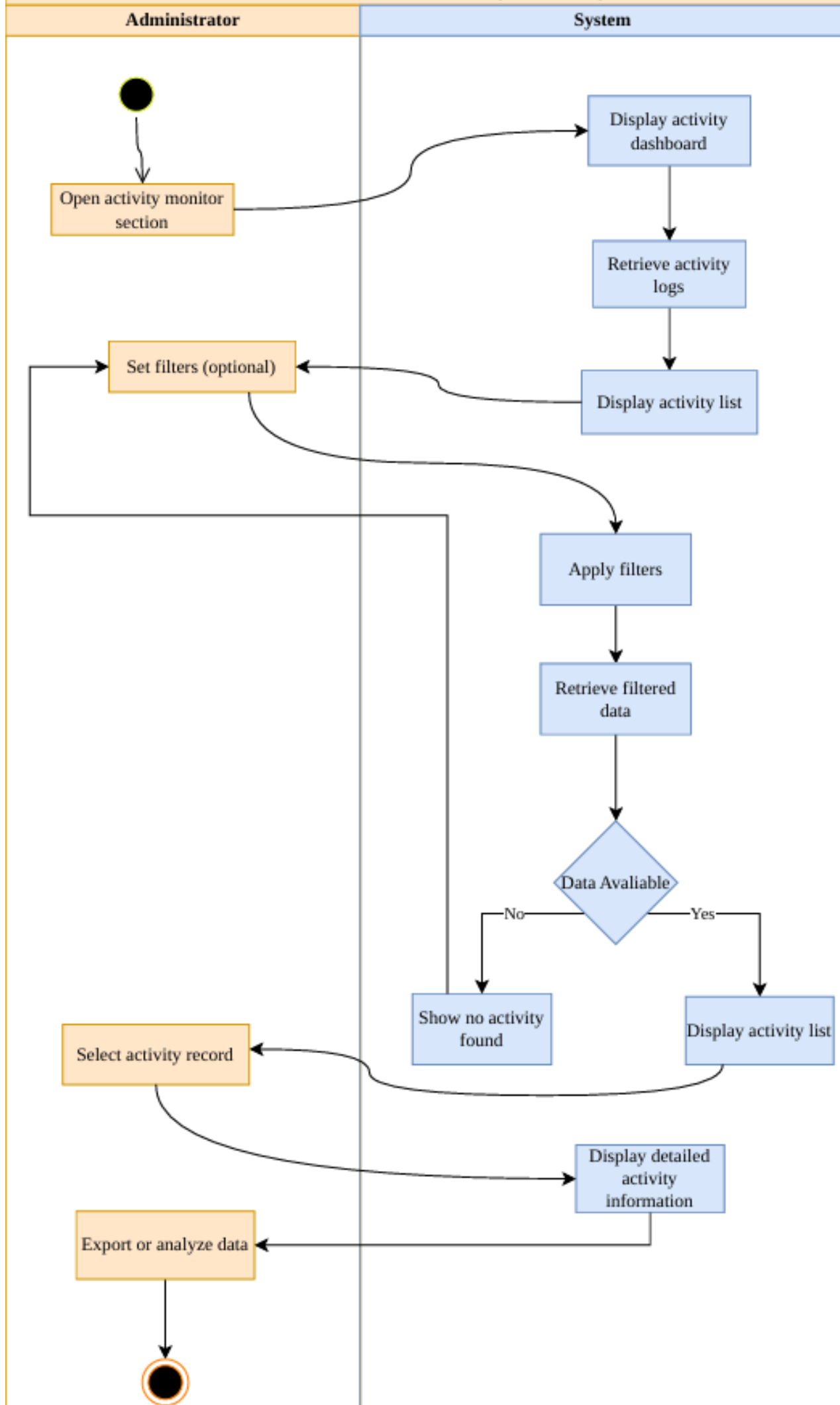
UC 25: Administrator manages account and profile



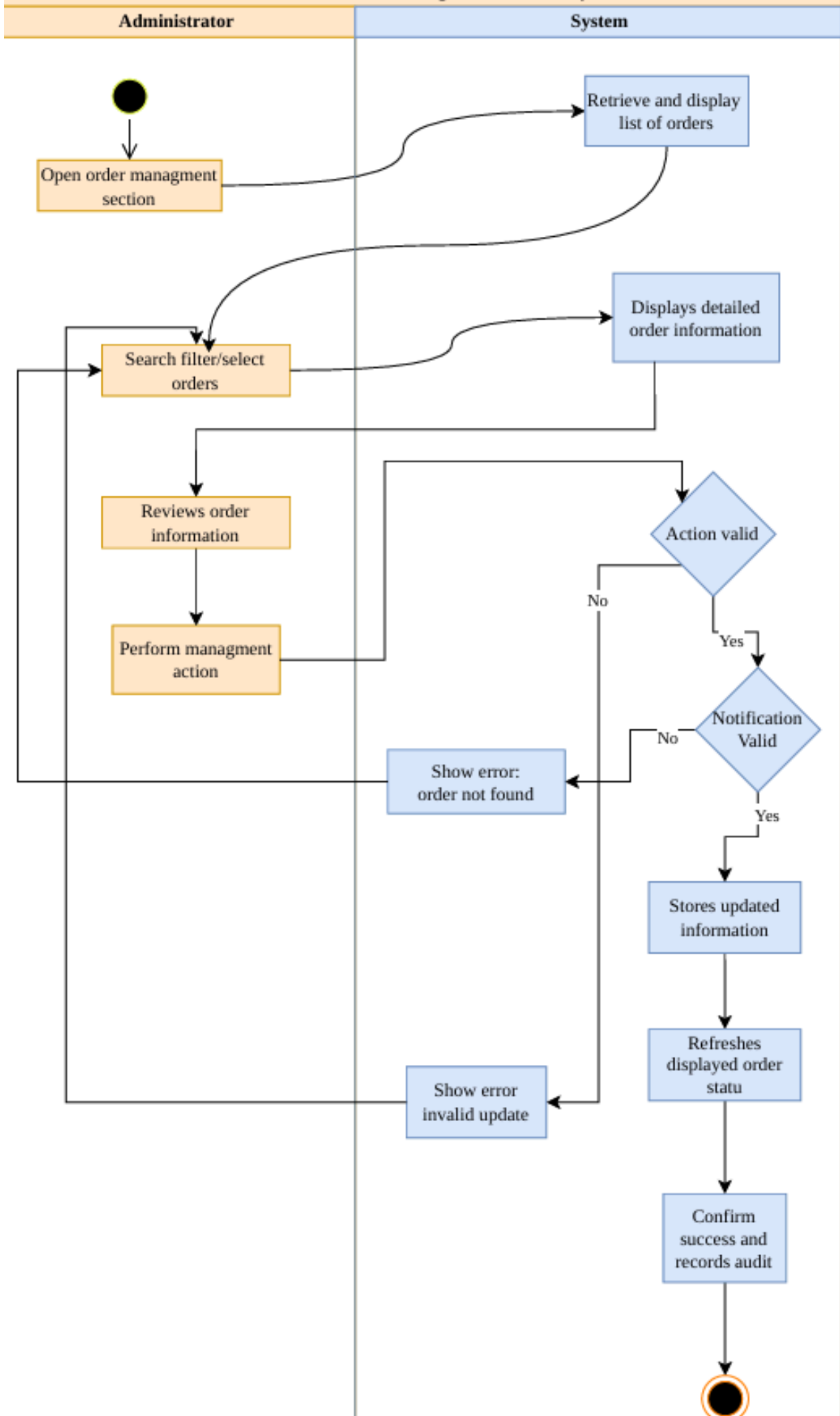
UC 26: Administrator manages user accounts



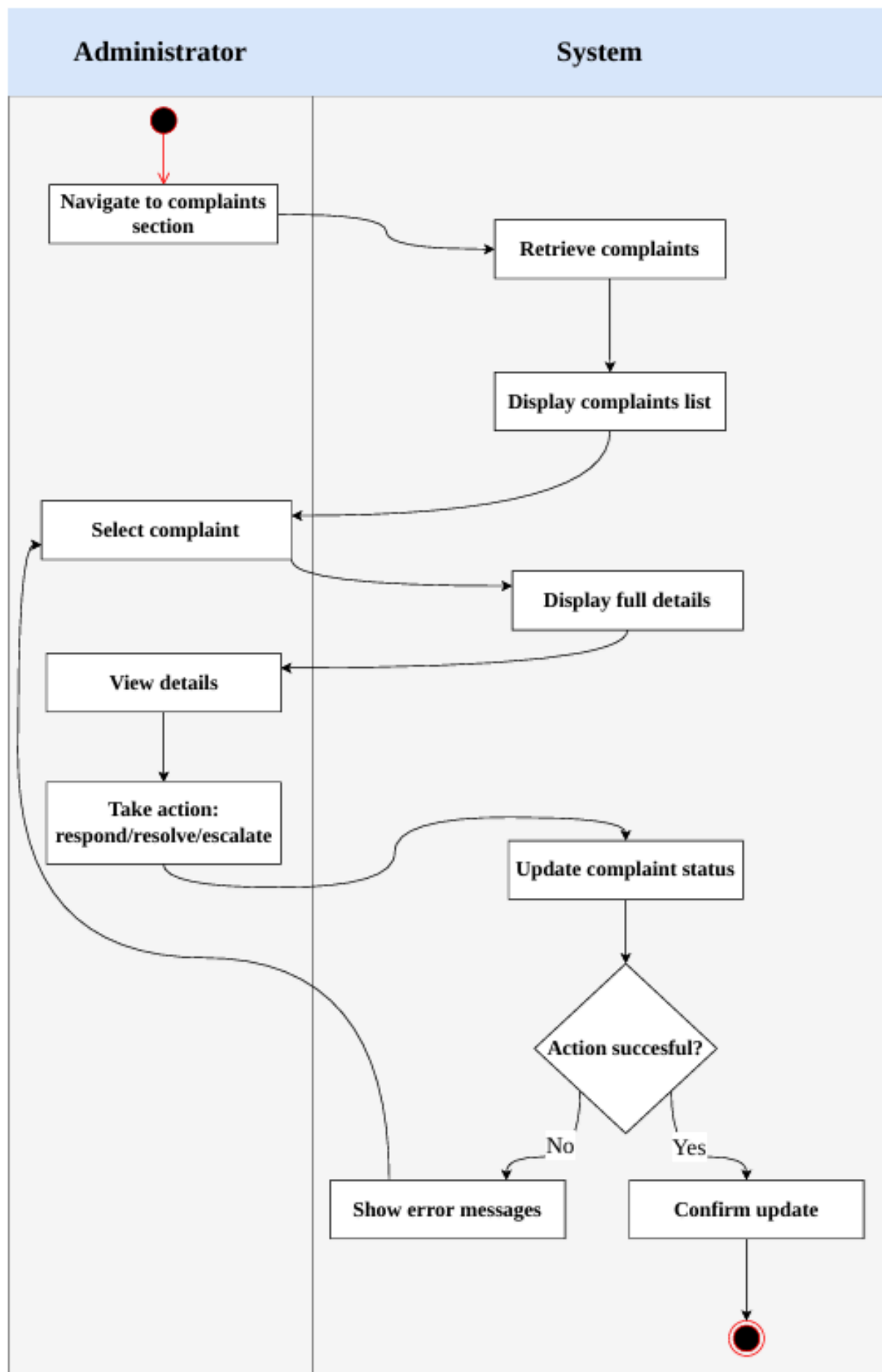
UC 27: Administrator monitors system activity



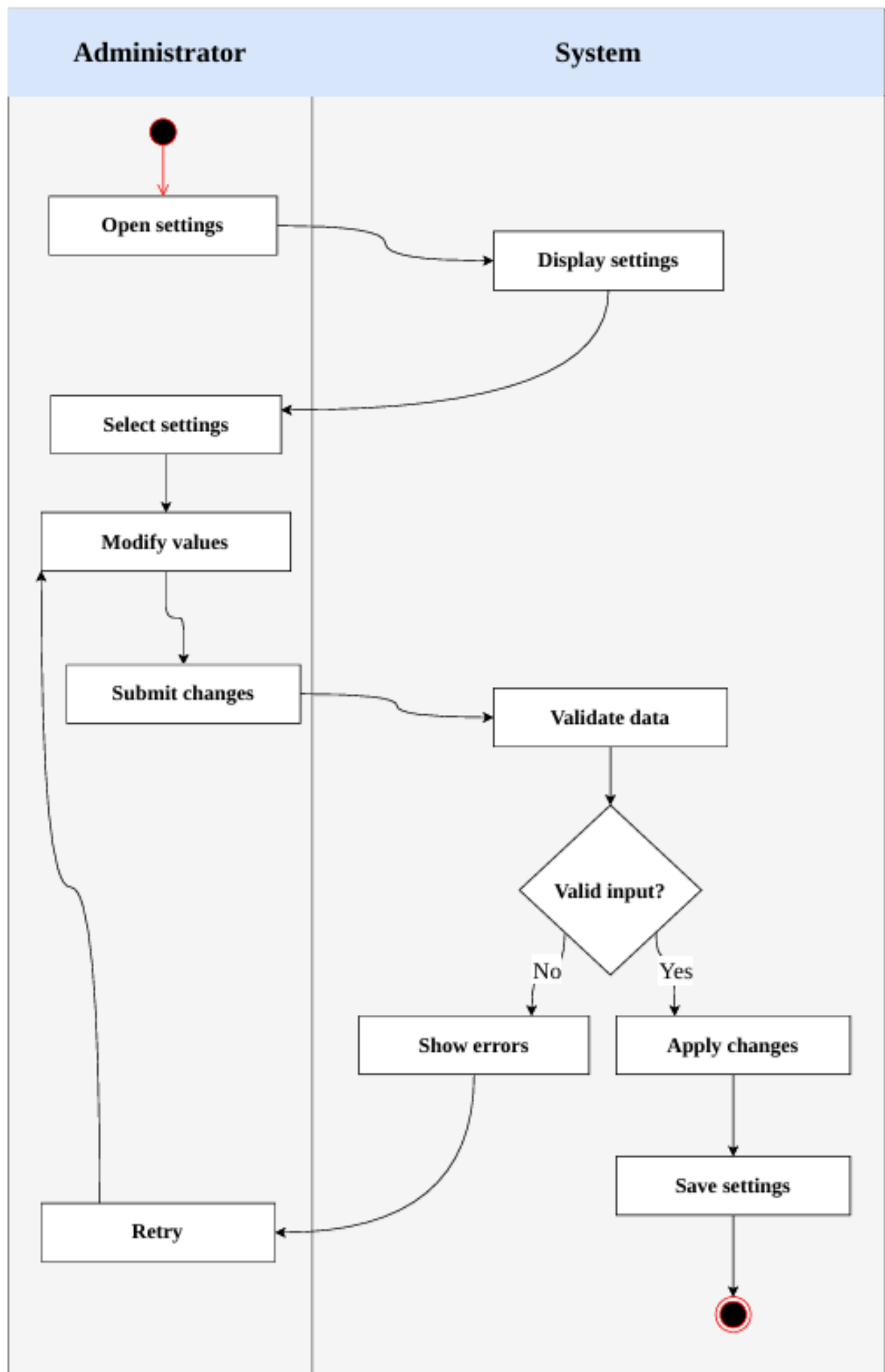
UC 28: Administrator manages orders in the system



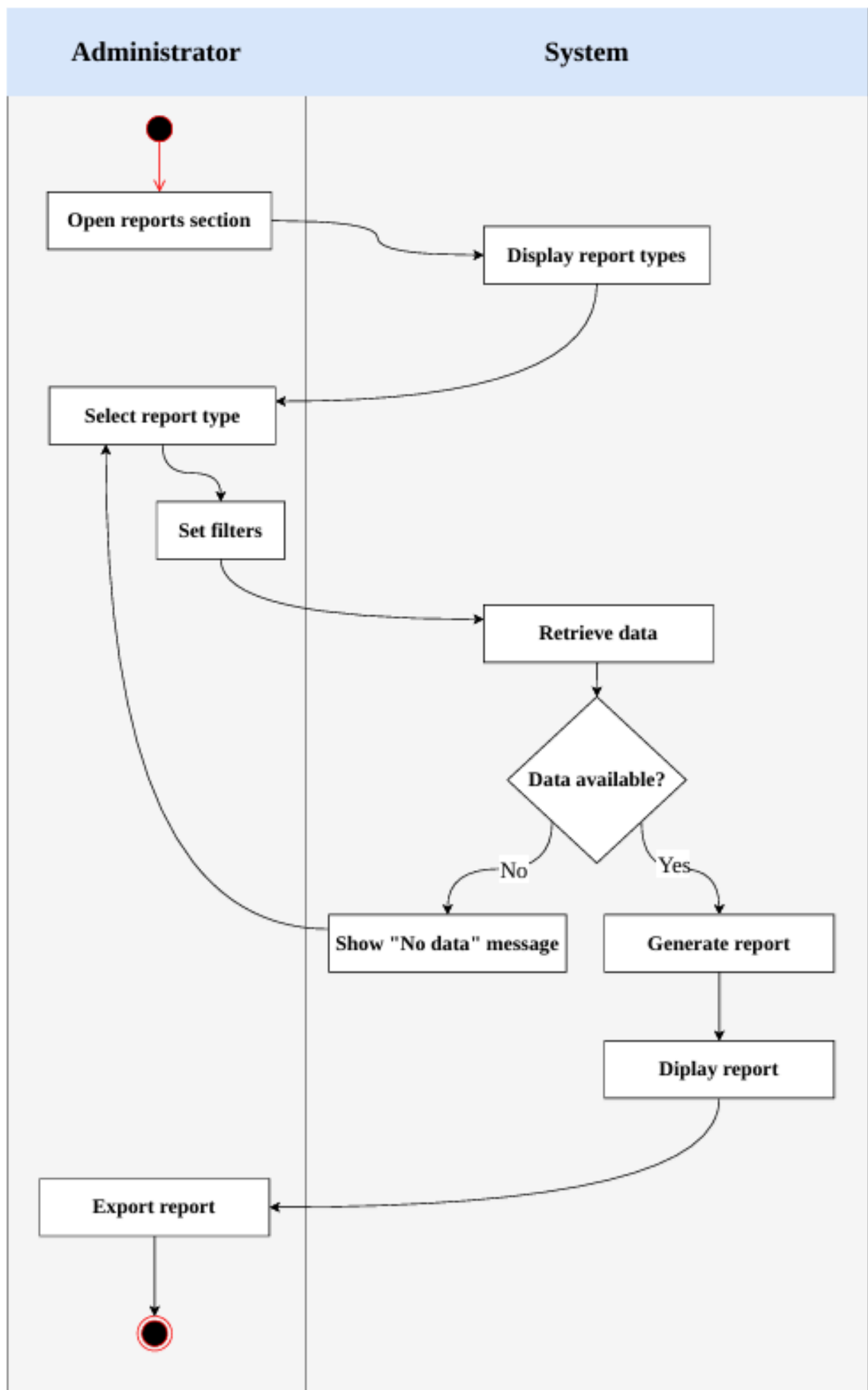
Activity Diagram - Administrator Handles Complaints and Support Requests (UC-29)



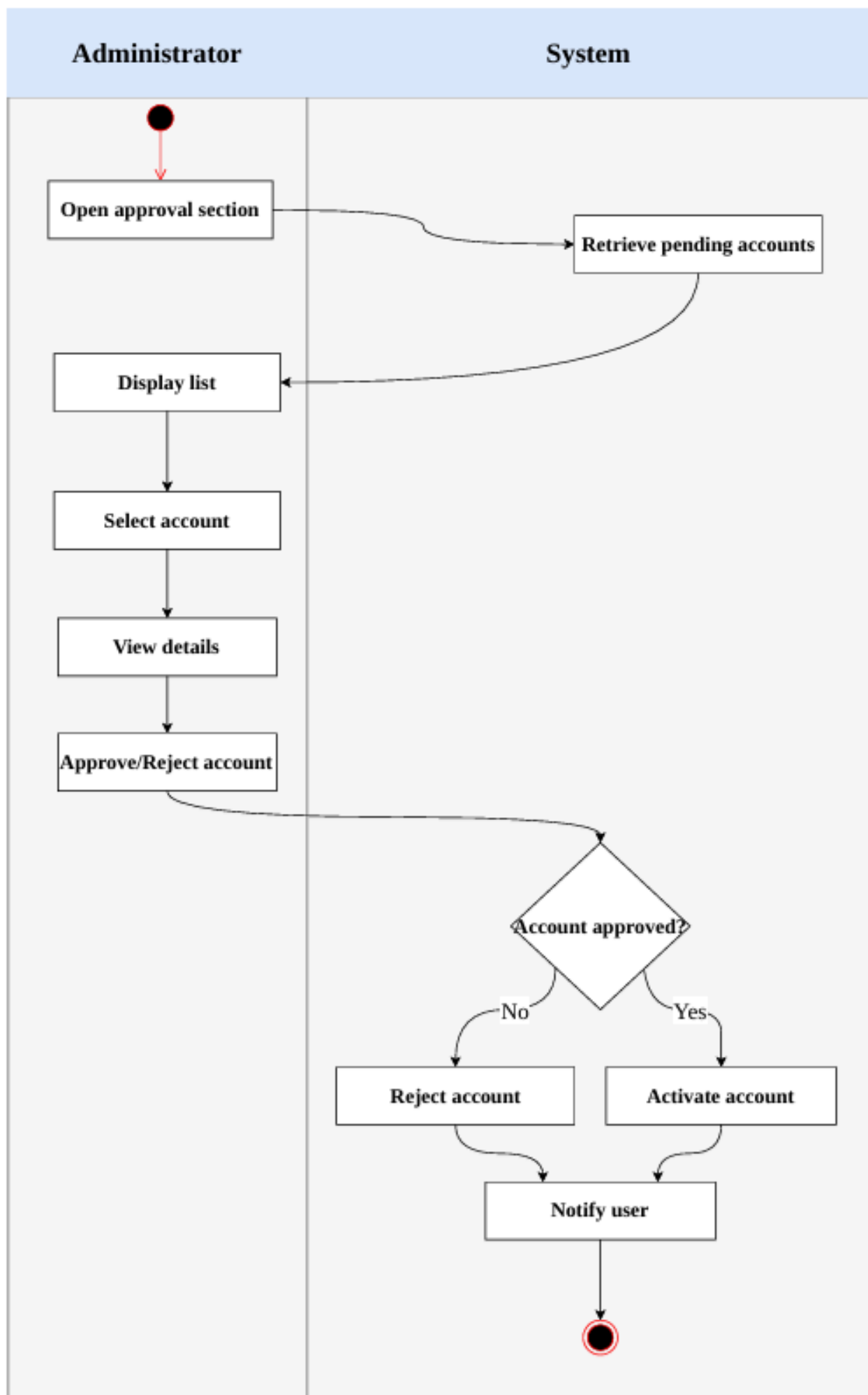
Activity Diagram - Administrator Manages Platform Settings (UC-30)



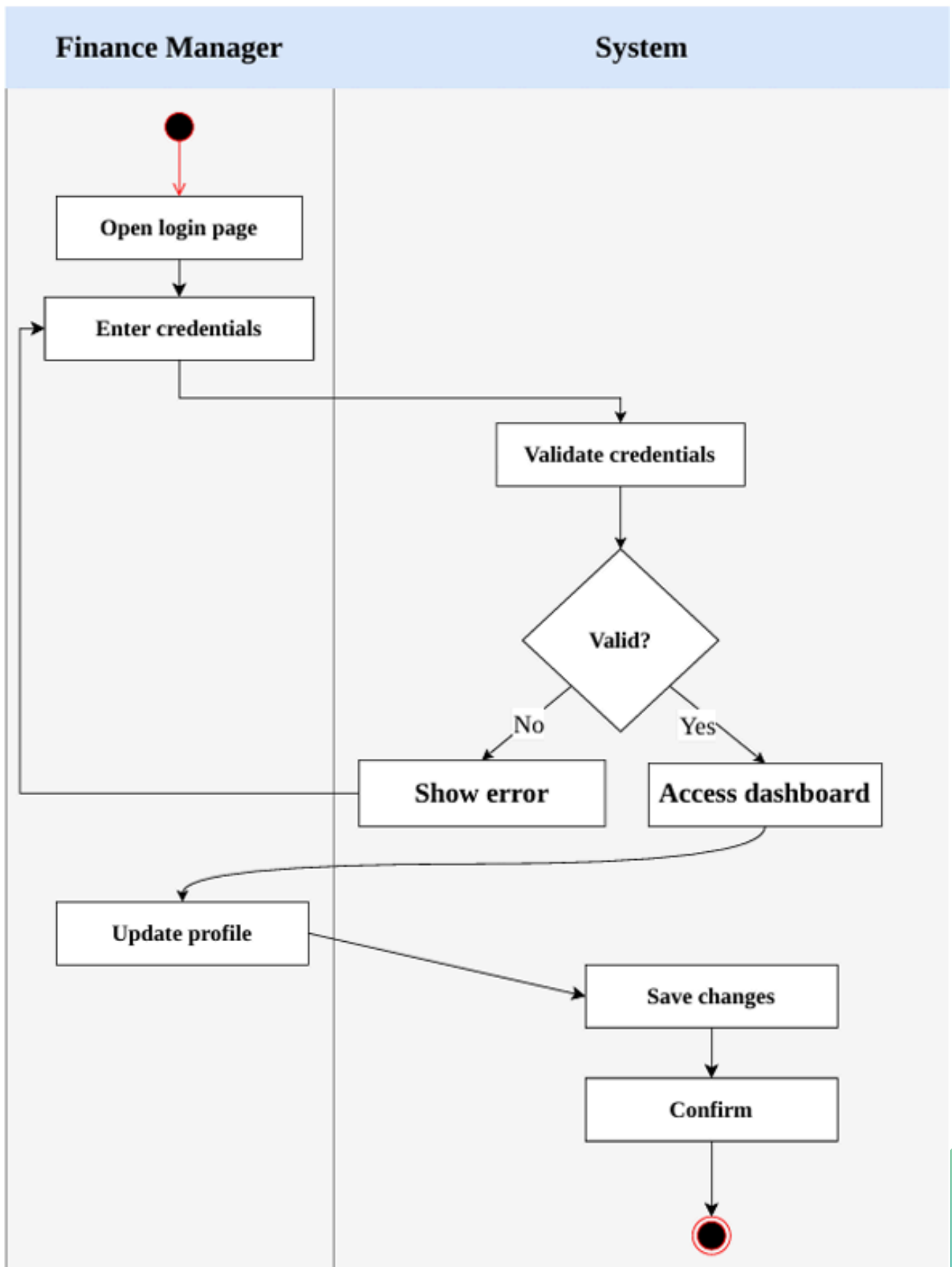
Activity Diagram - Administrator Generates System Reports (UC-31)



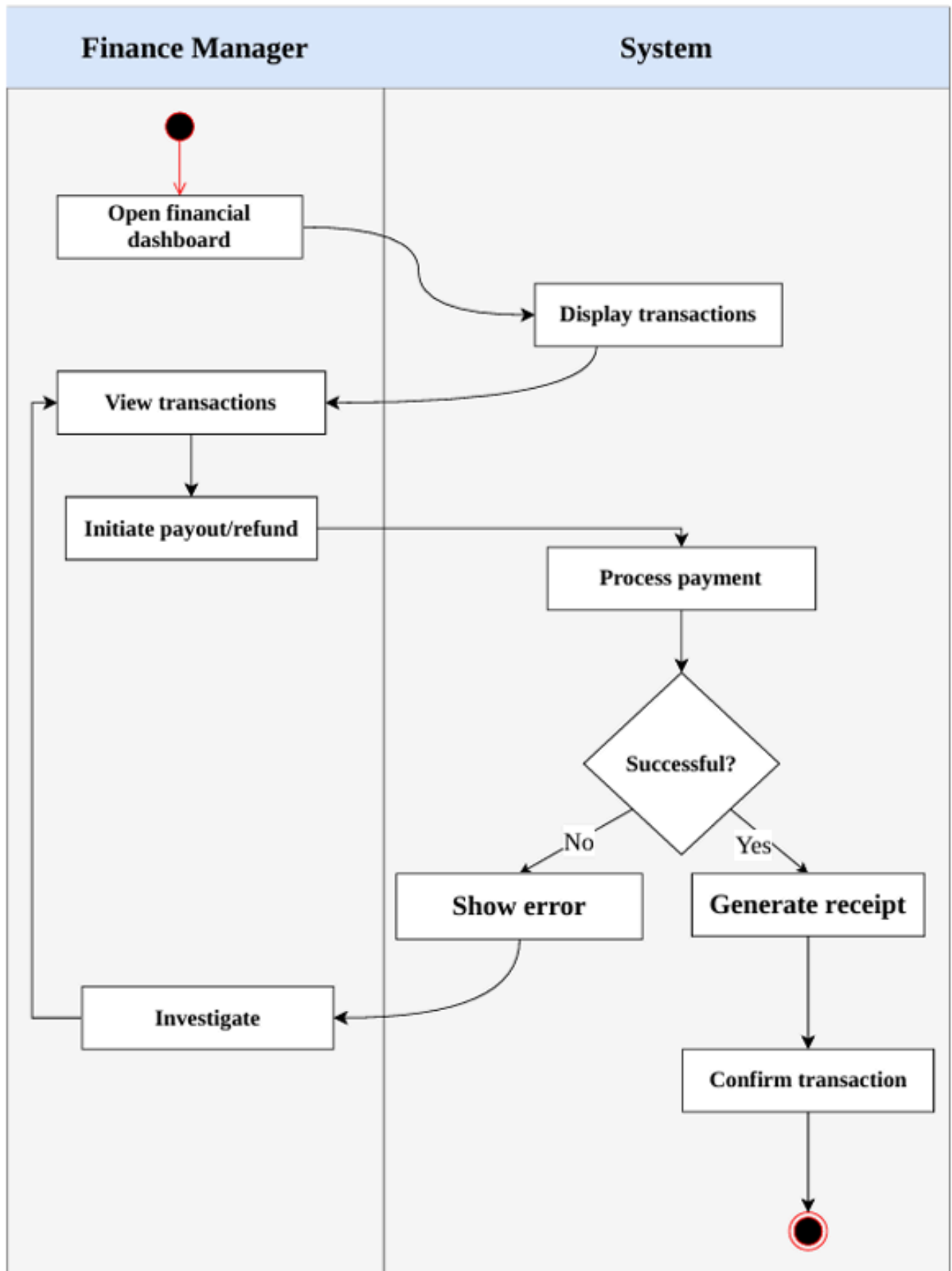
Activity Diagram - Manage Account Approvals and Restrictions (UC-32)

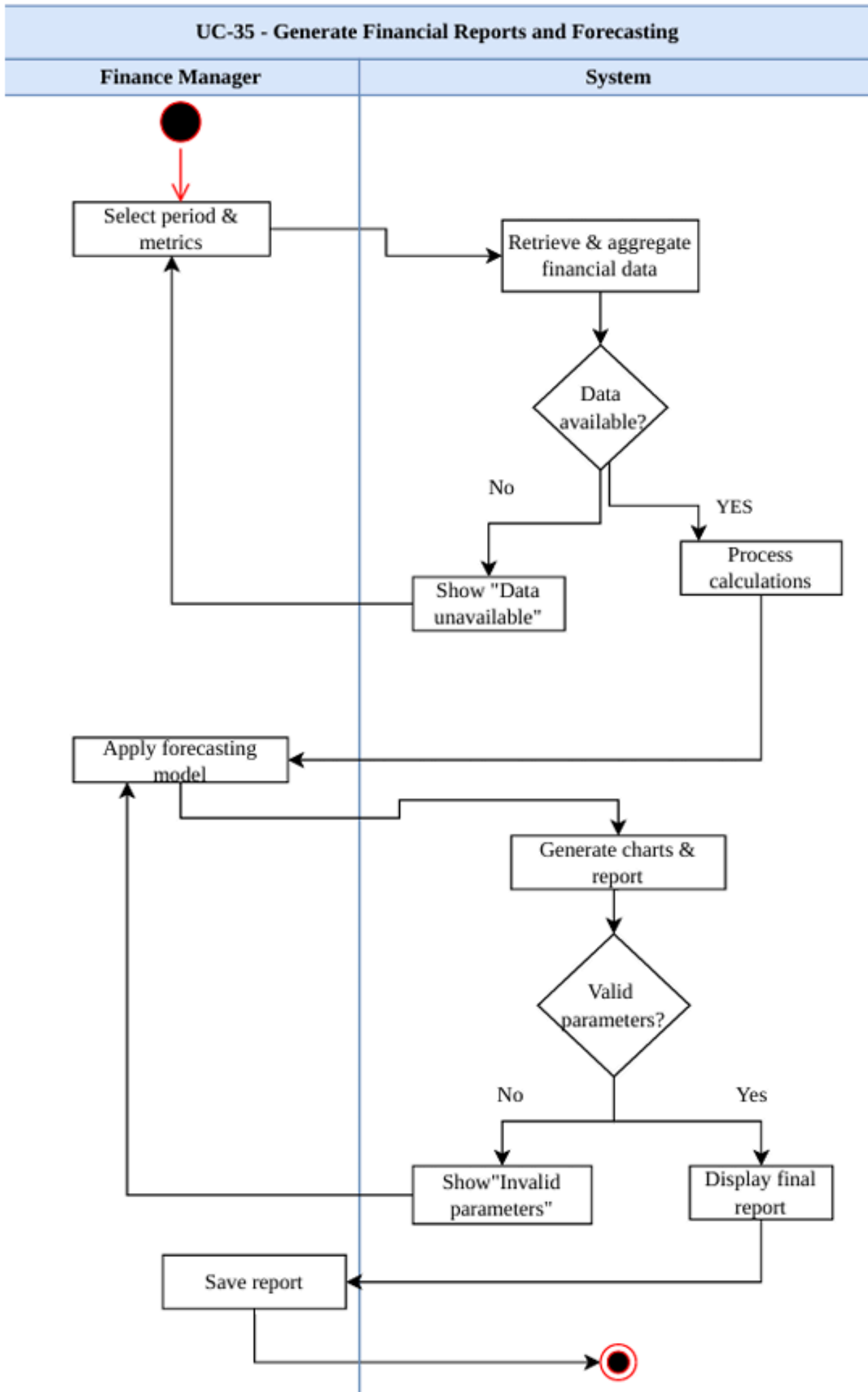


Activity Diagram - Finance Manager Account Management (UC-33)

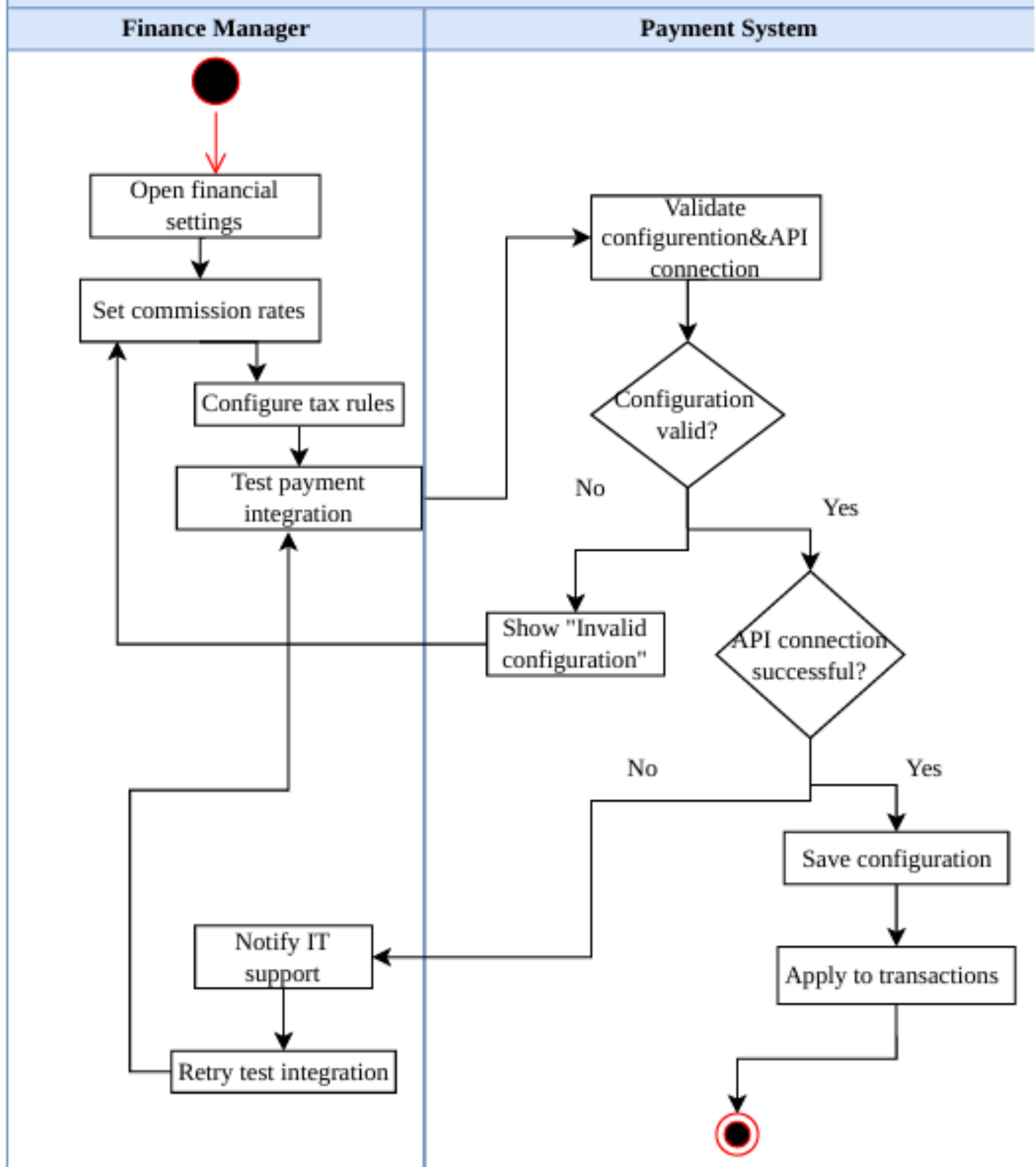


Activity Diagram - Manage Billing, Payouts and Refunds (UC-34)





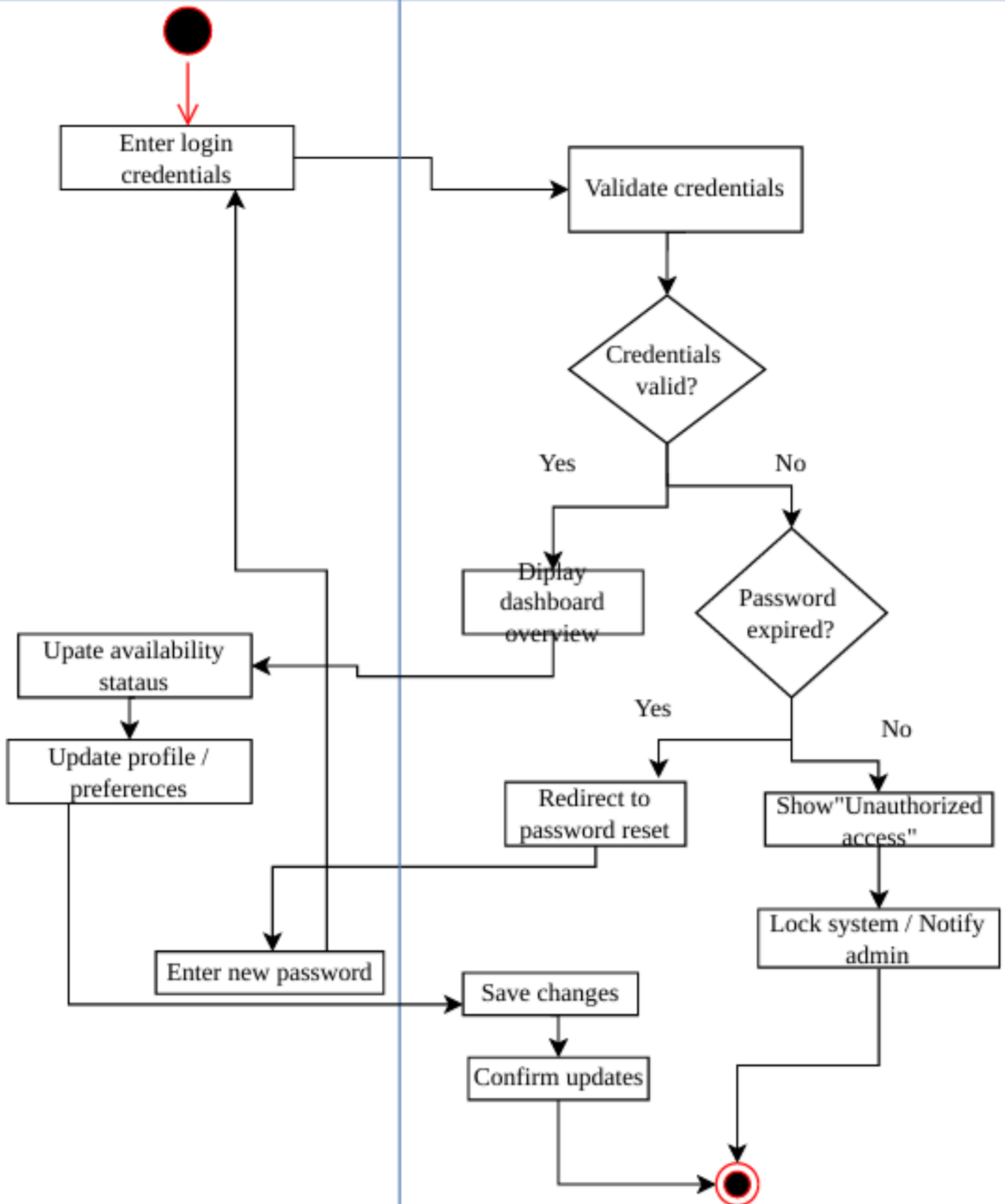
UC-36 - Manage Financial Operations



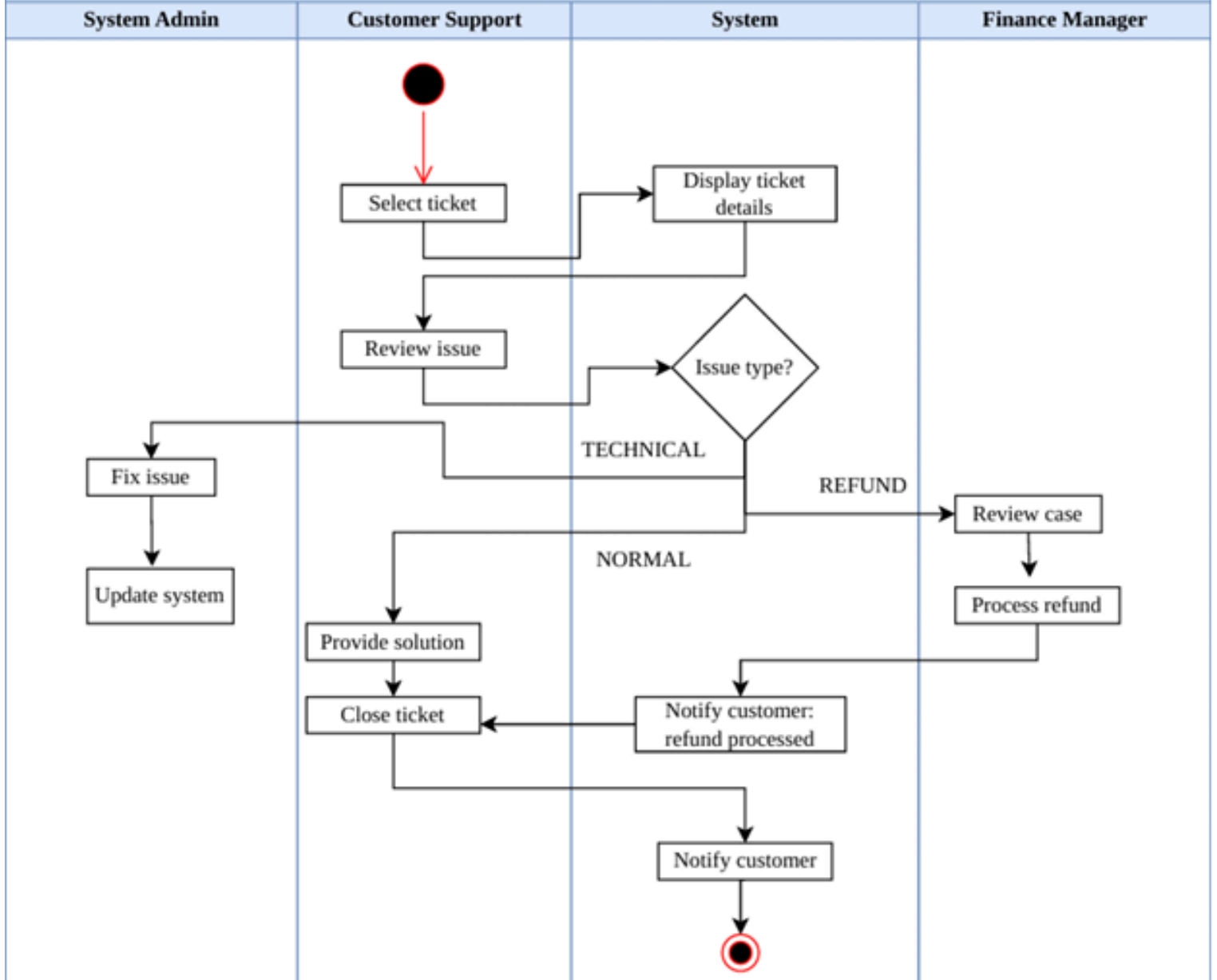
UC -37 - Customer Support Account Management

Customer Support

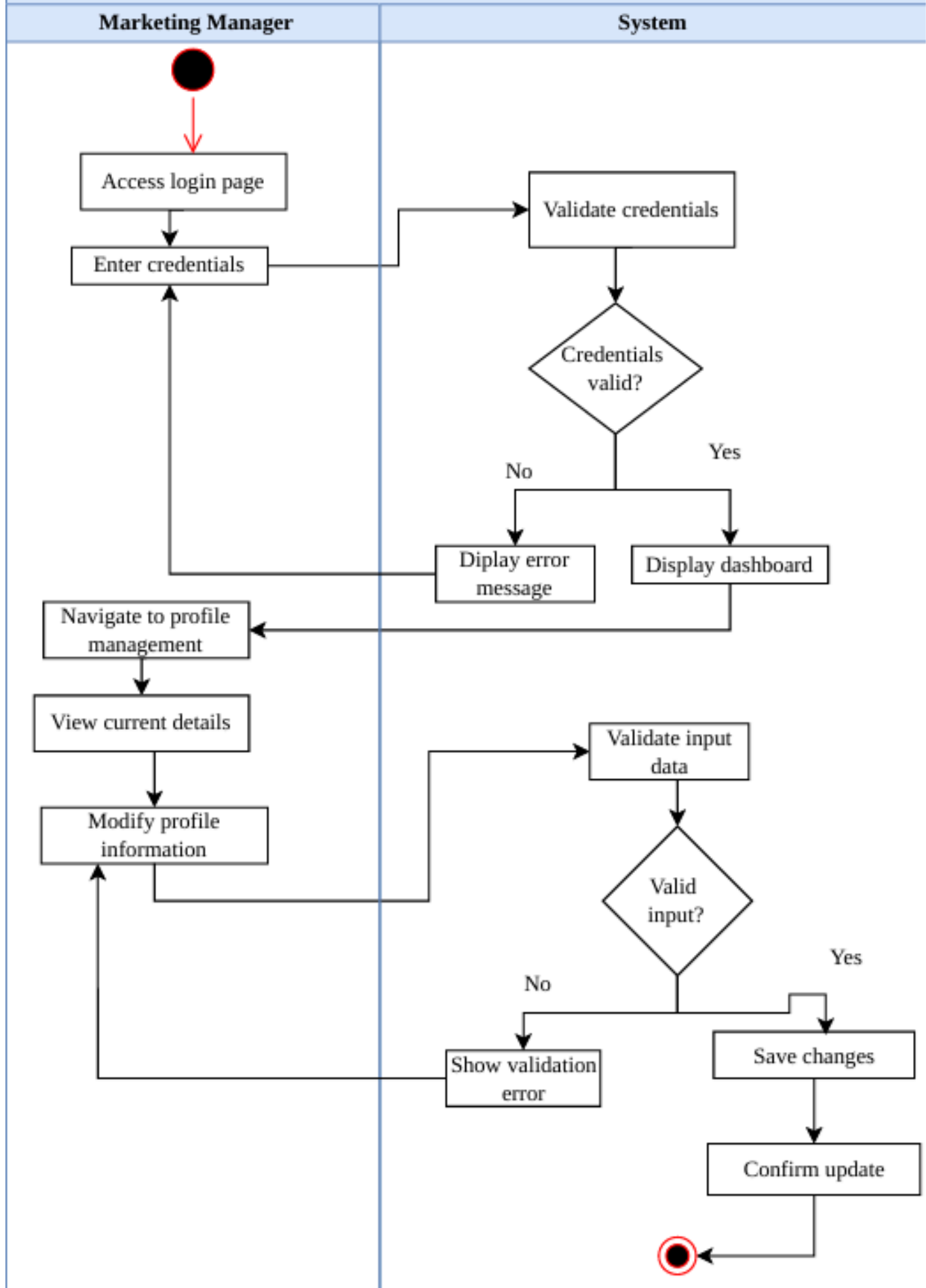
System



UC-38 - Manage User Support Operations



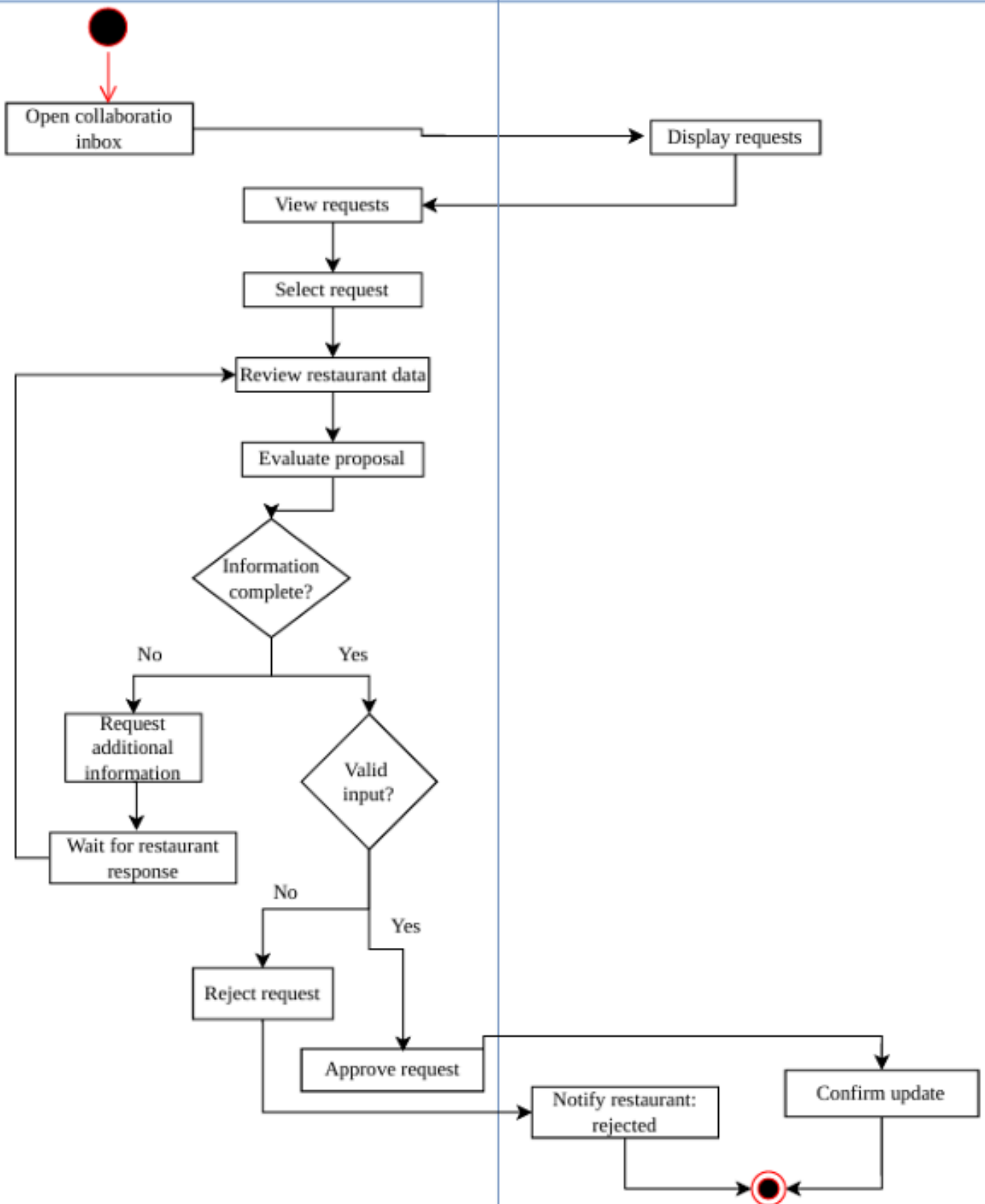
UC- 39-Manage Marketing Manager Account



UC-40-ManageMarketing Collaboration Requests

Marketing Manager

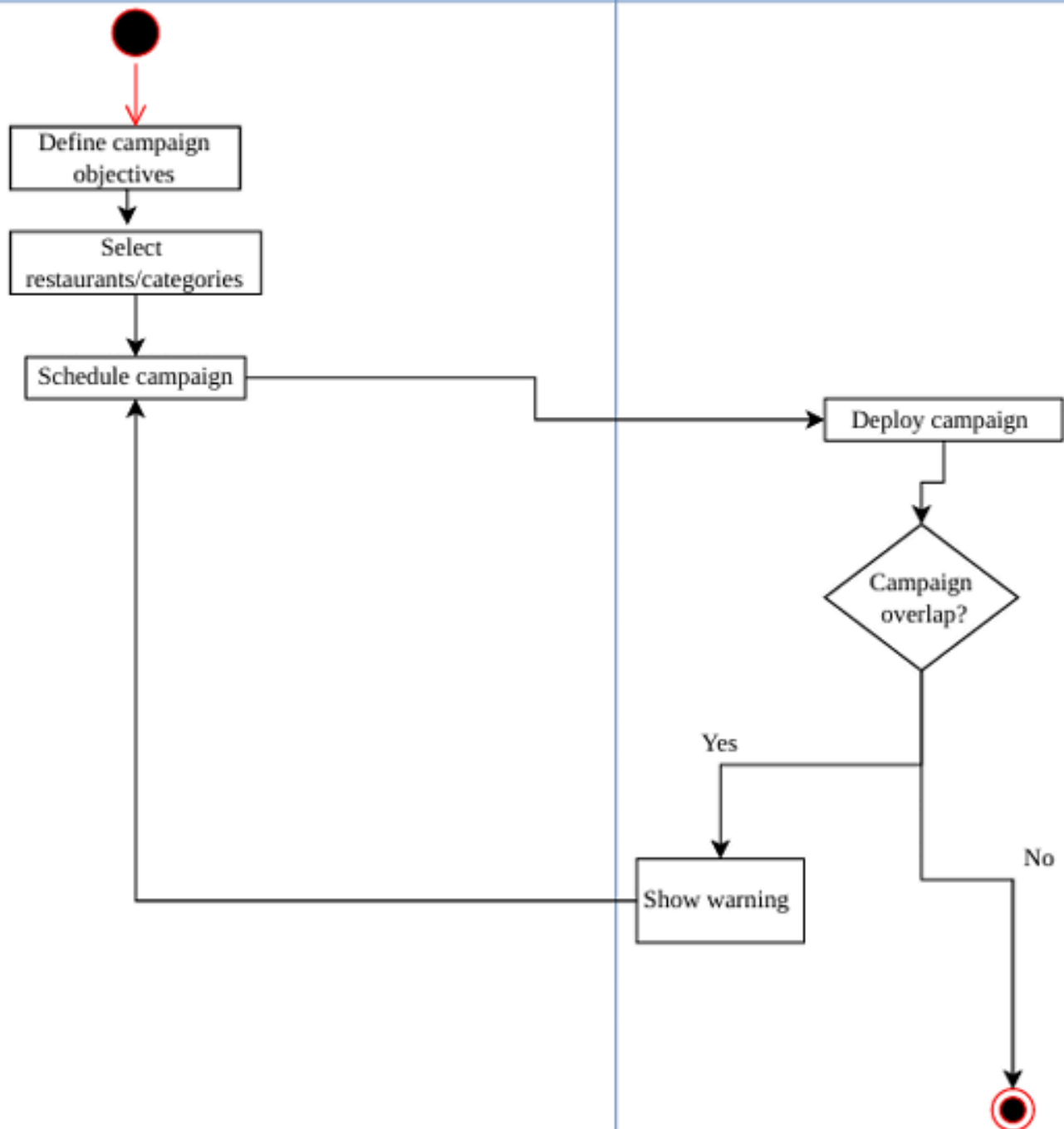
System



UC-41 – Manage Marketing Campaigns

Marketing Manager

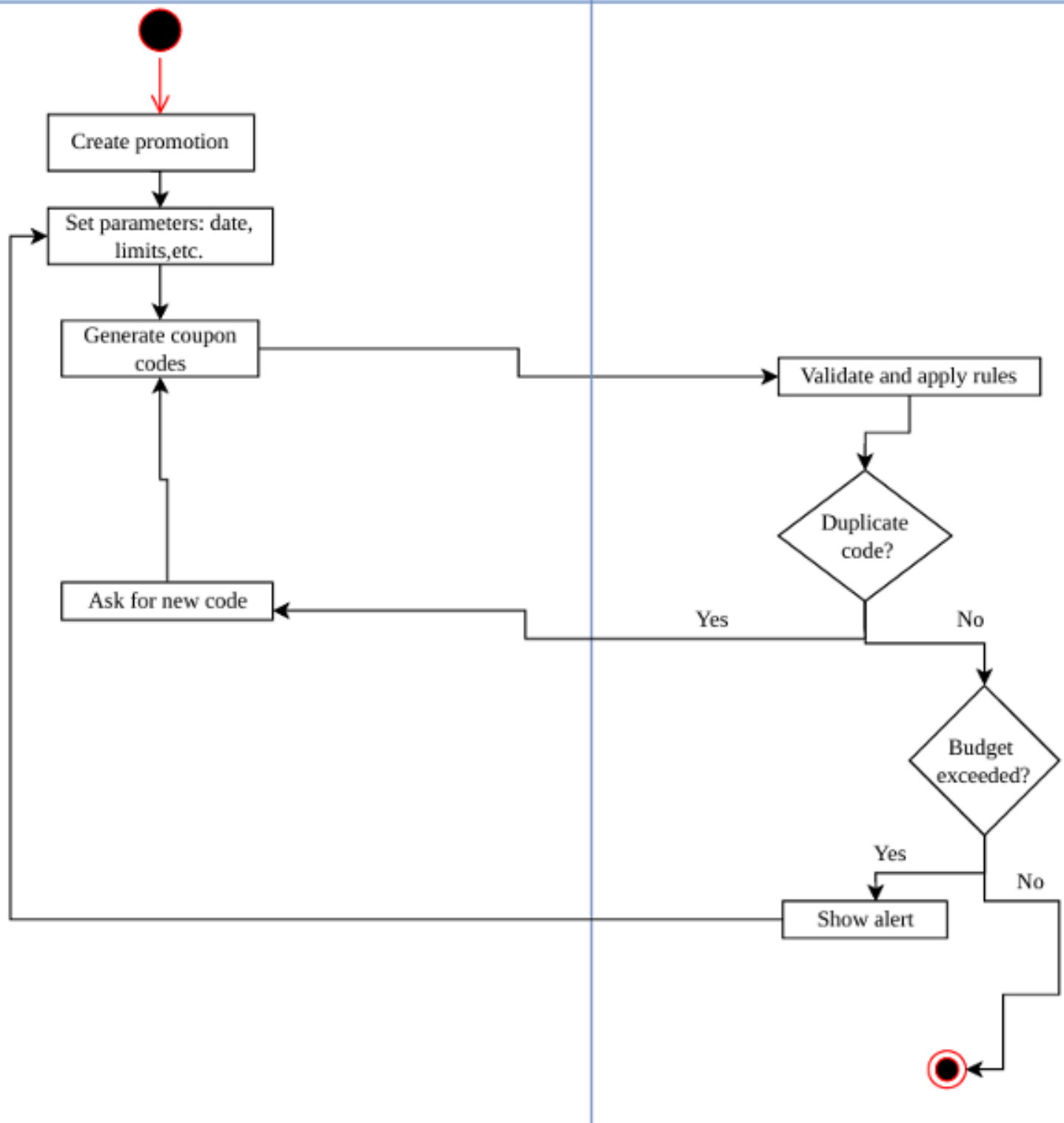
System



UC-42 – Manage Promotions, Discounts, and Coupon Codes

Marketing Manager

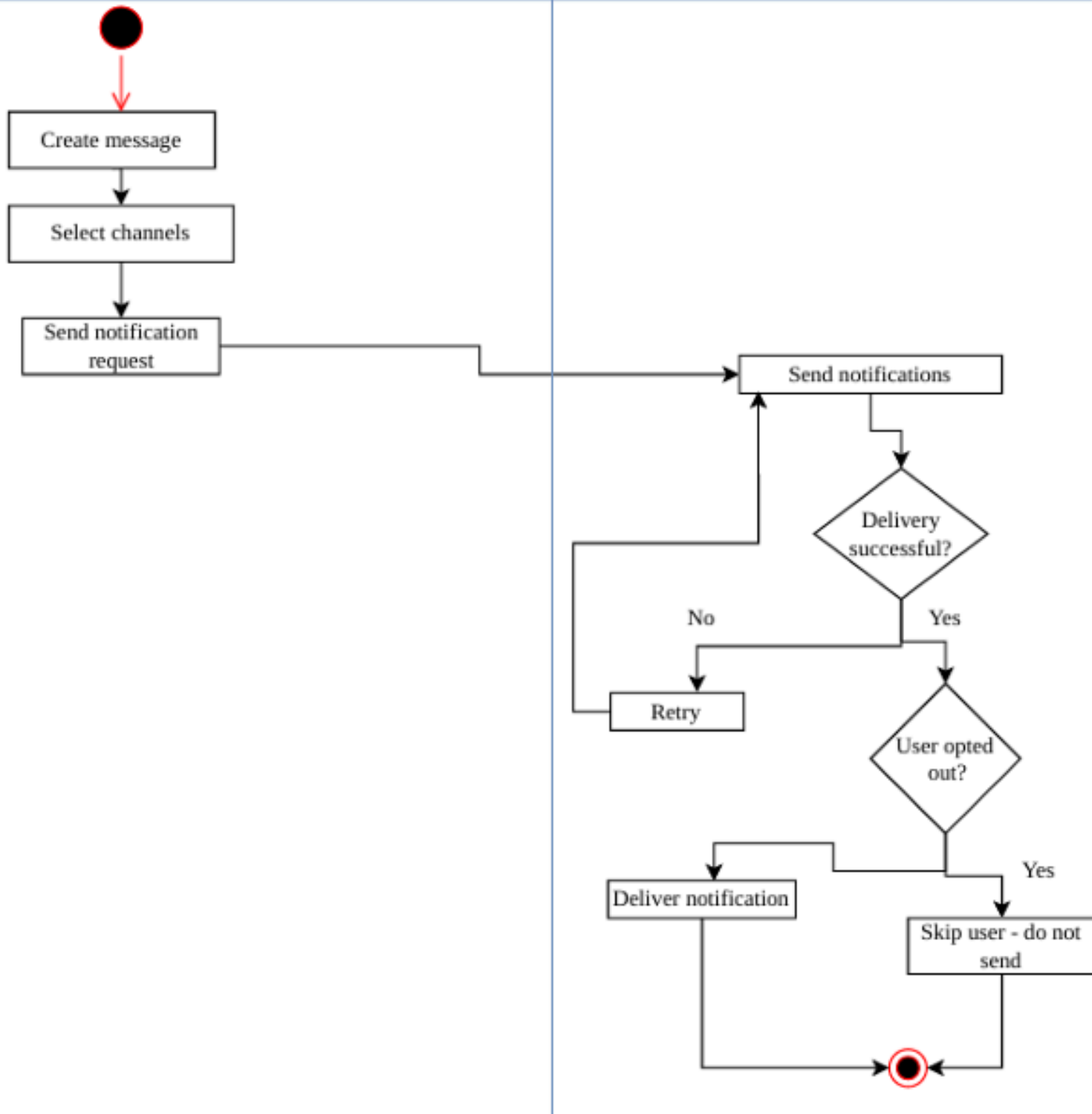
System



UC-43 – Manage Promotional Communication

Marketing Manager

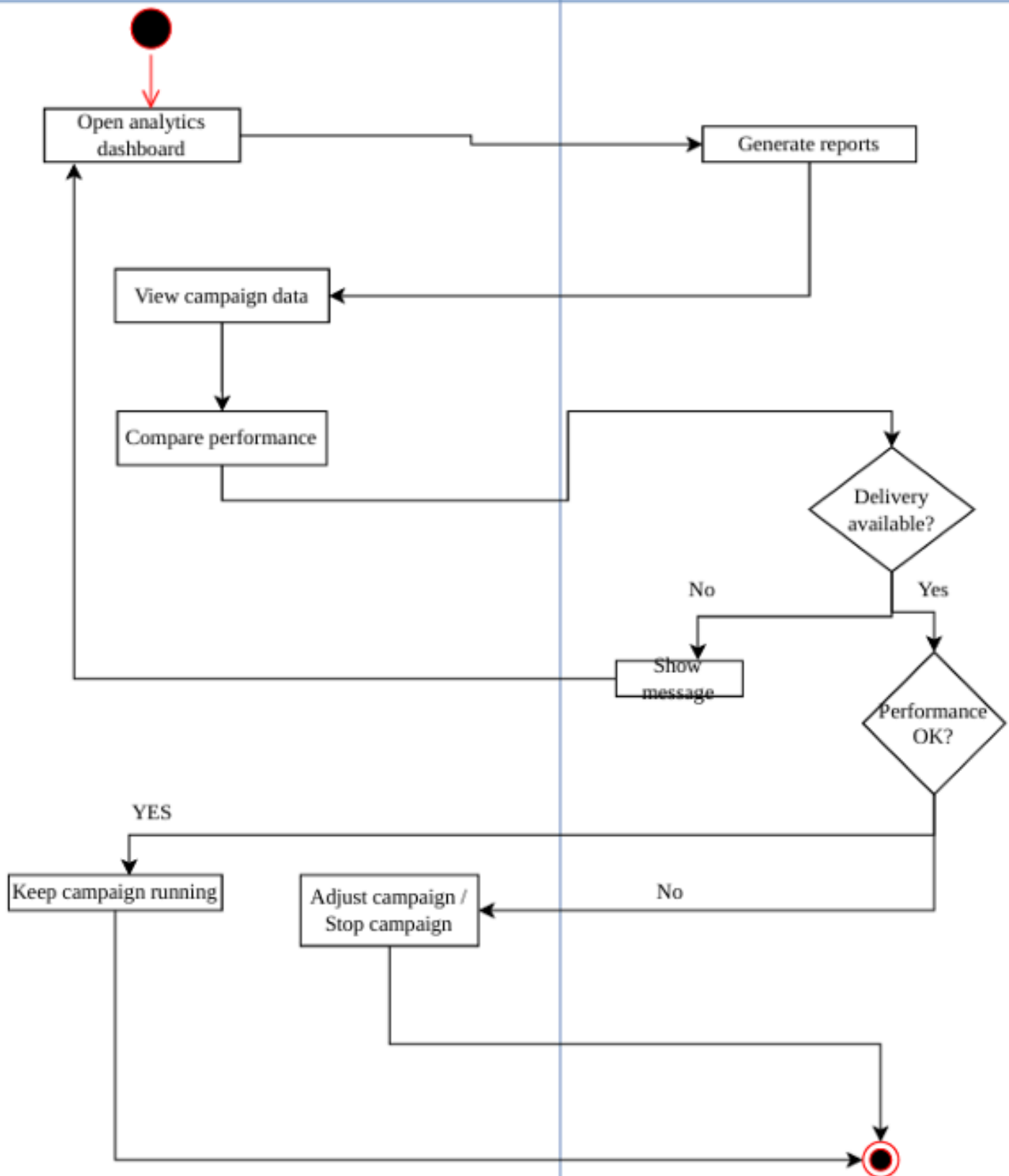
Notification System



UC-44 Analyze and Monitor Campaign Performance

Marketing Manager

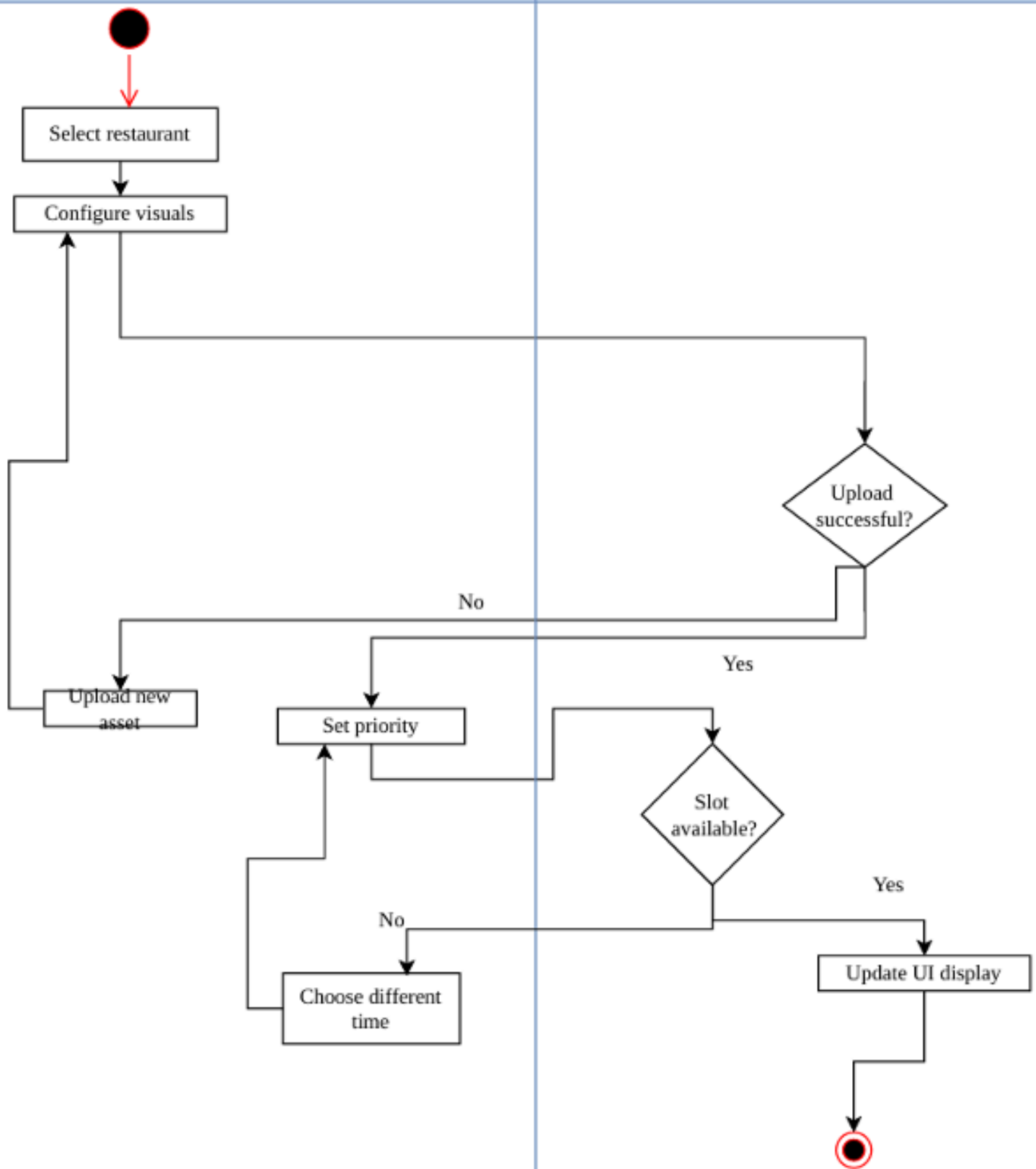
System



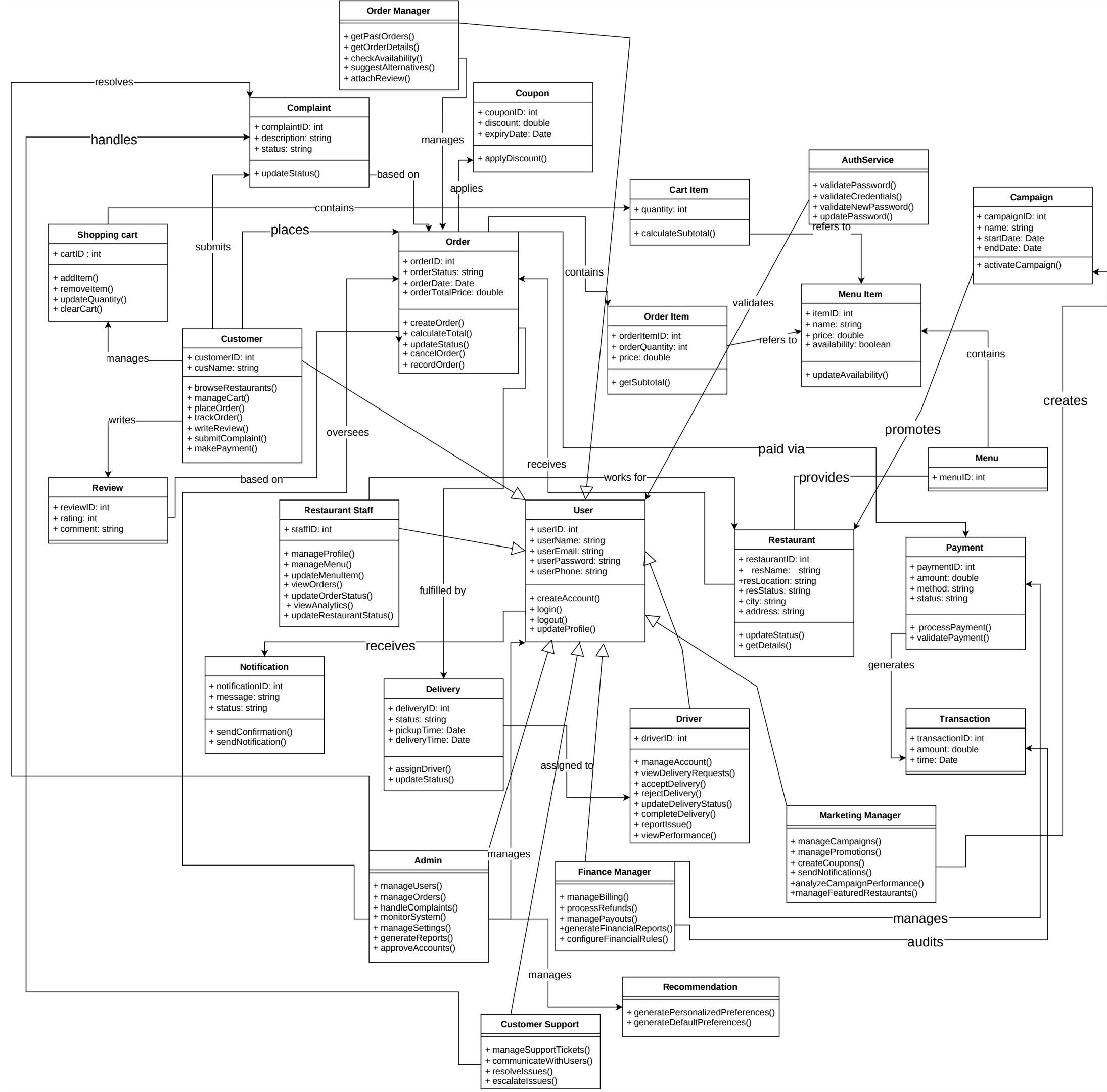
UC-45 – Manage Featured Restaurants and Advertisements

Marketing Manager

System



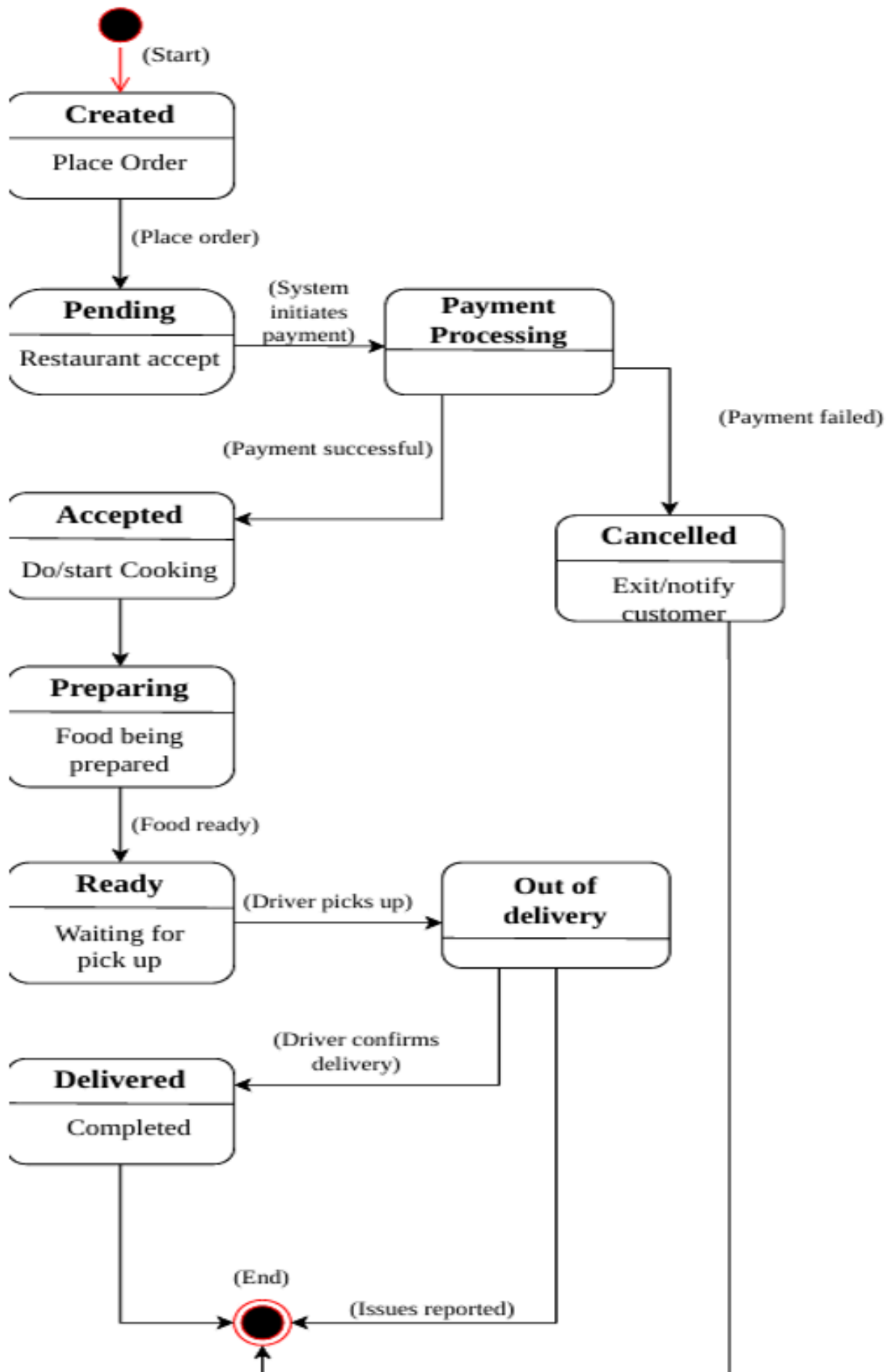
Author-Arjola Sula



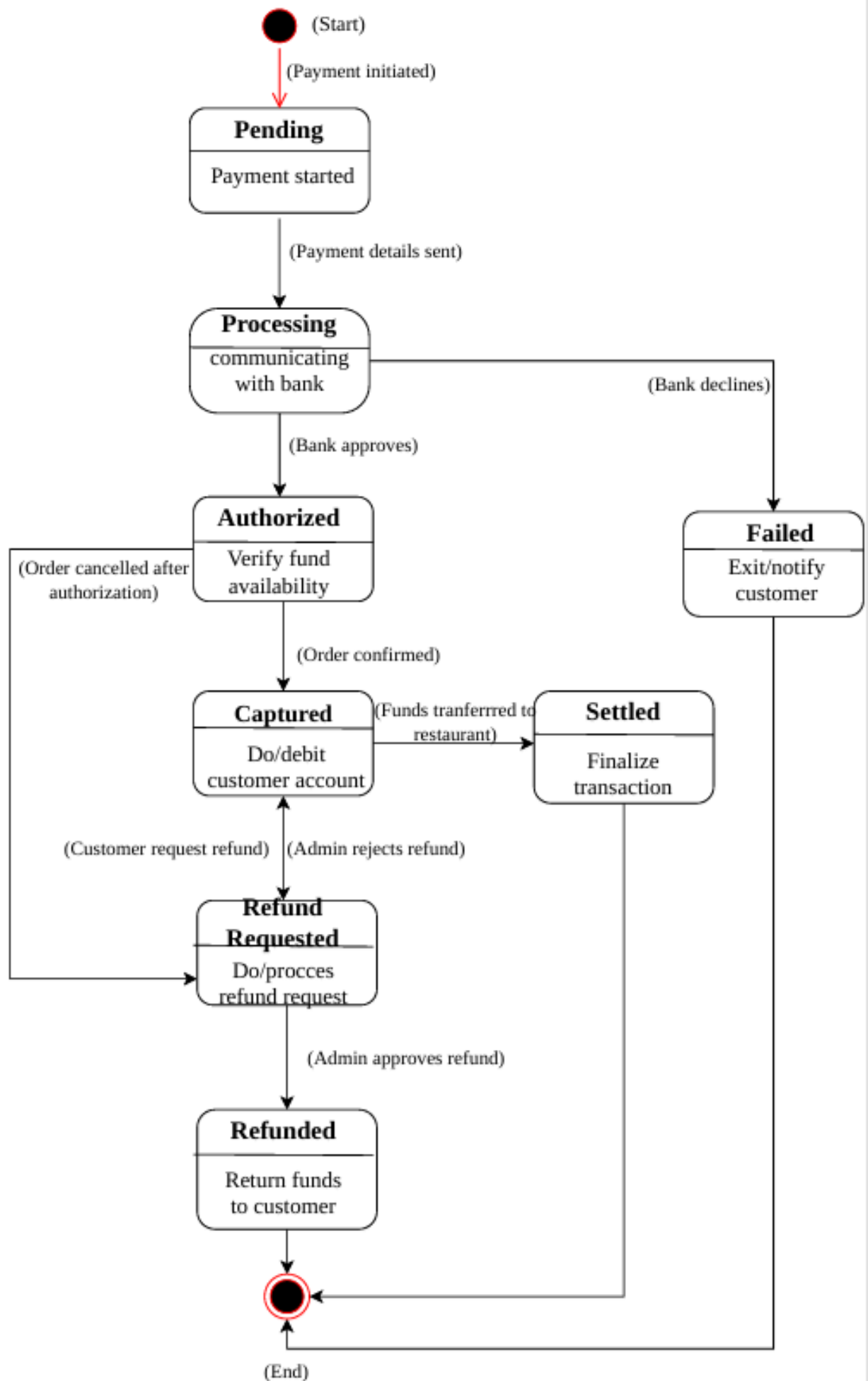
State diagrams

Author-Sofia Cala

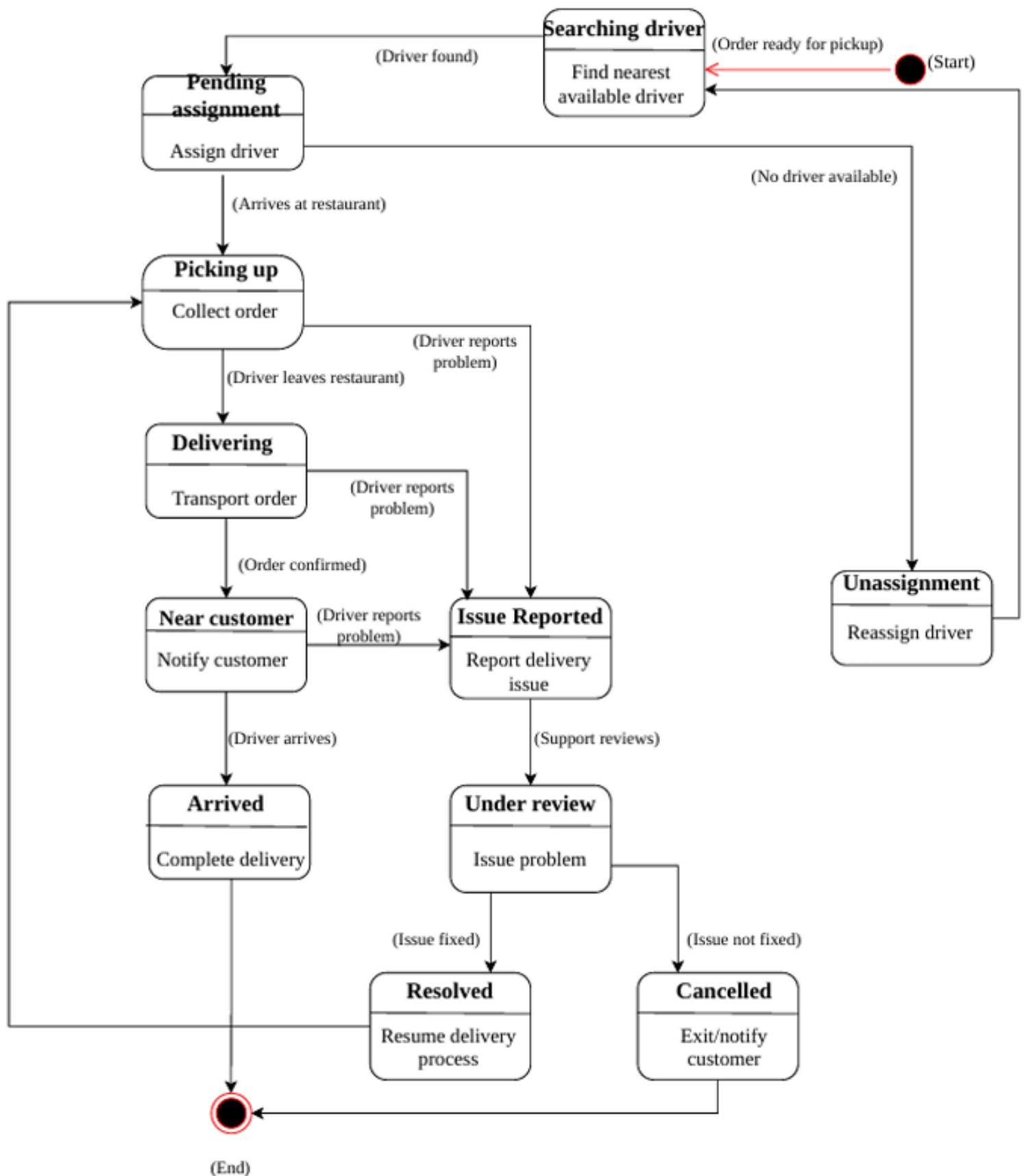
State Diagram 1- Order State Diagram



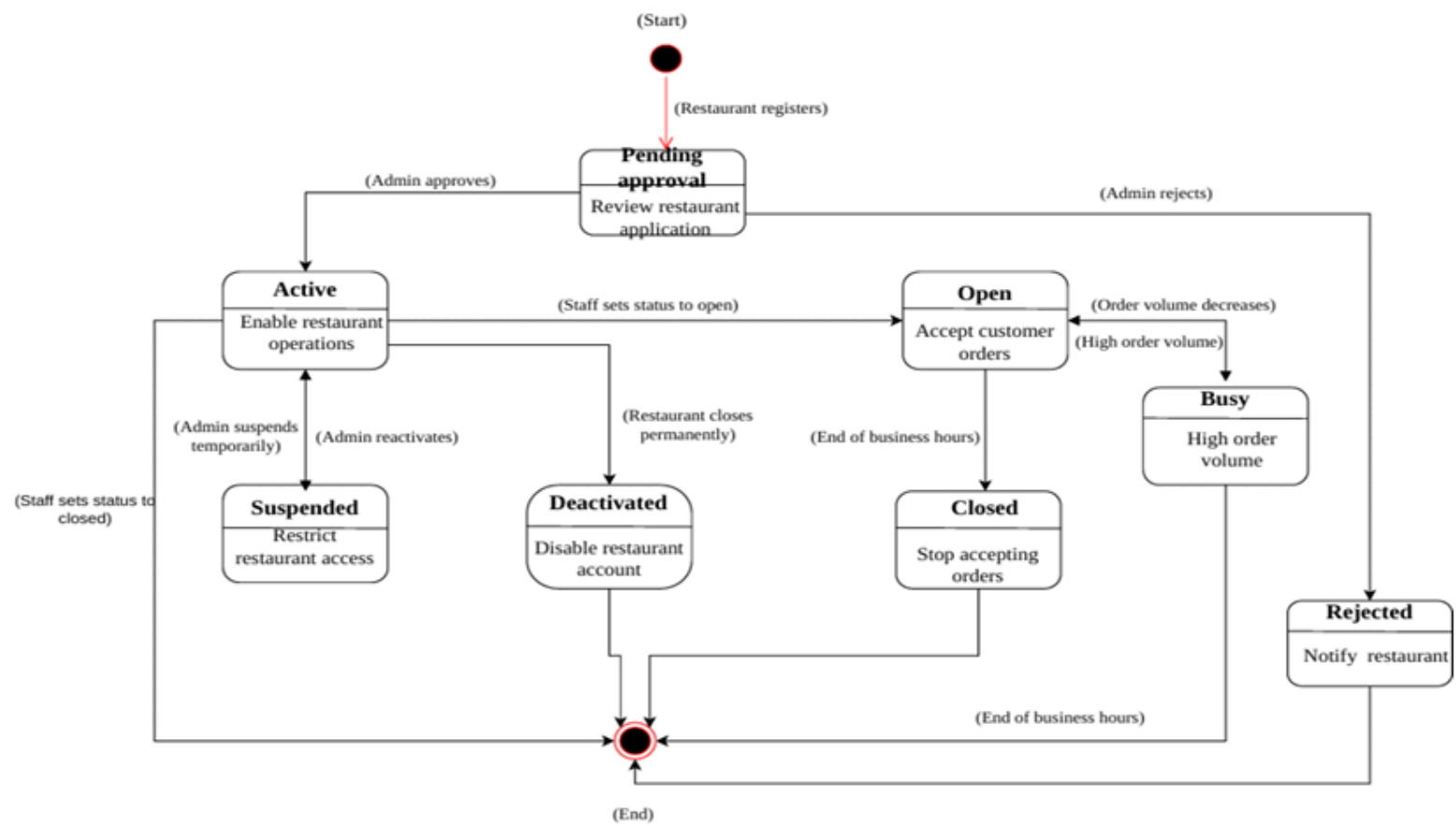
State Diagram 2- Payment State Diagram



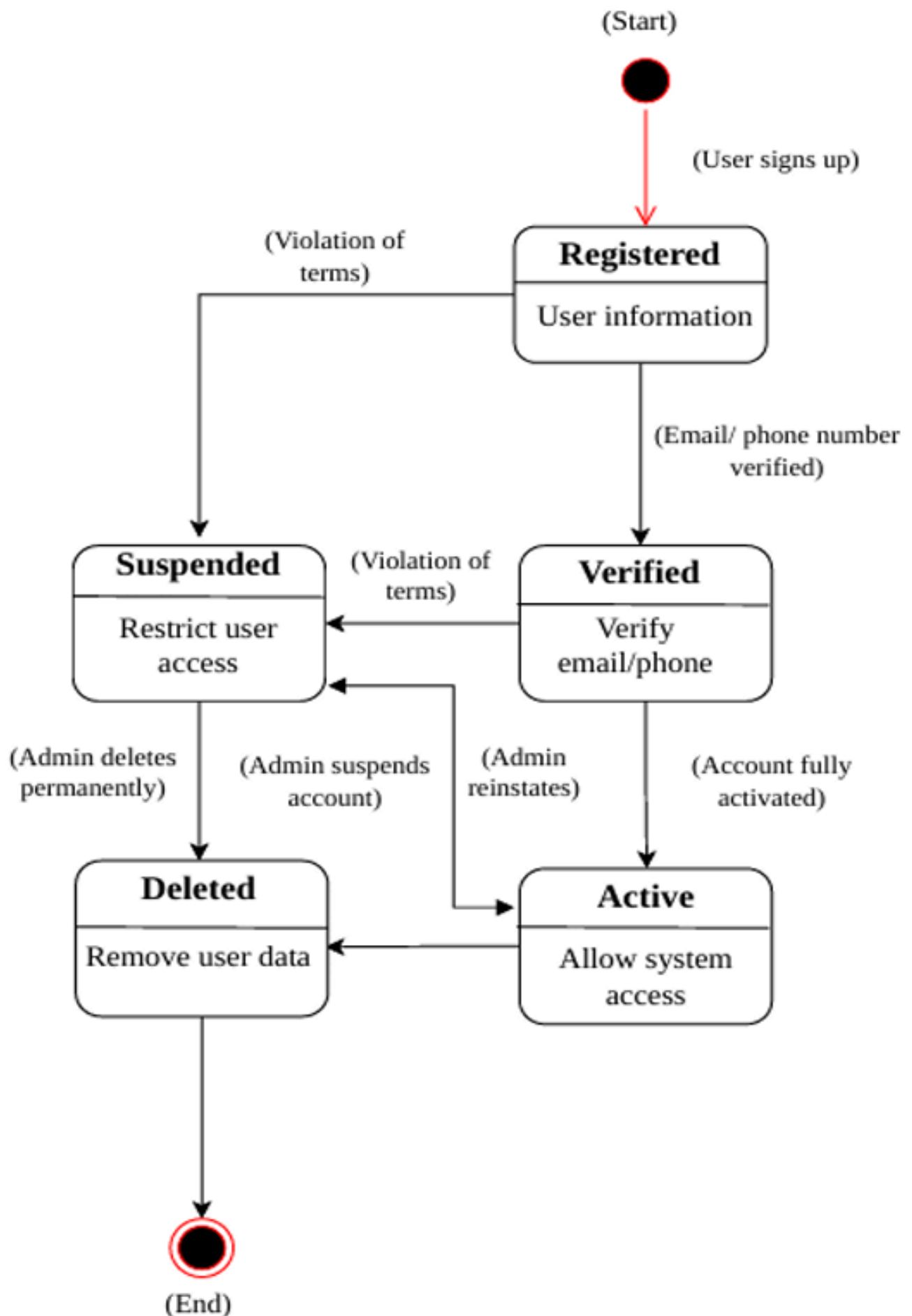
State Diagram 3- Delivery State Diagram



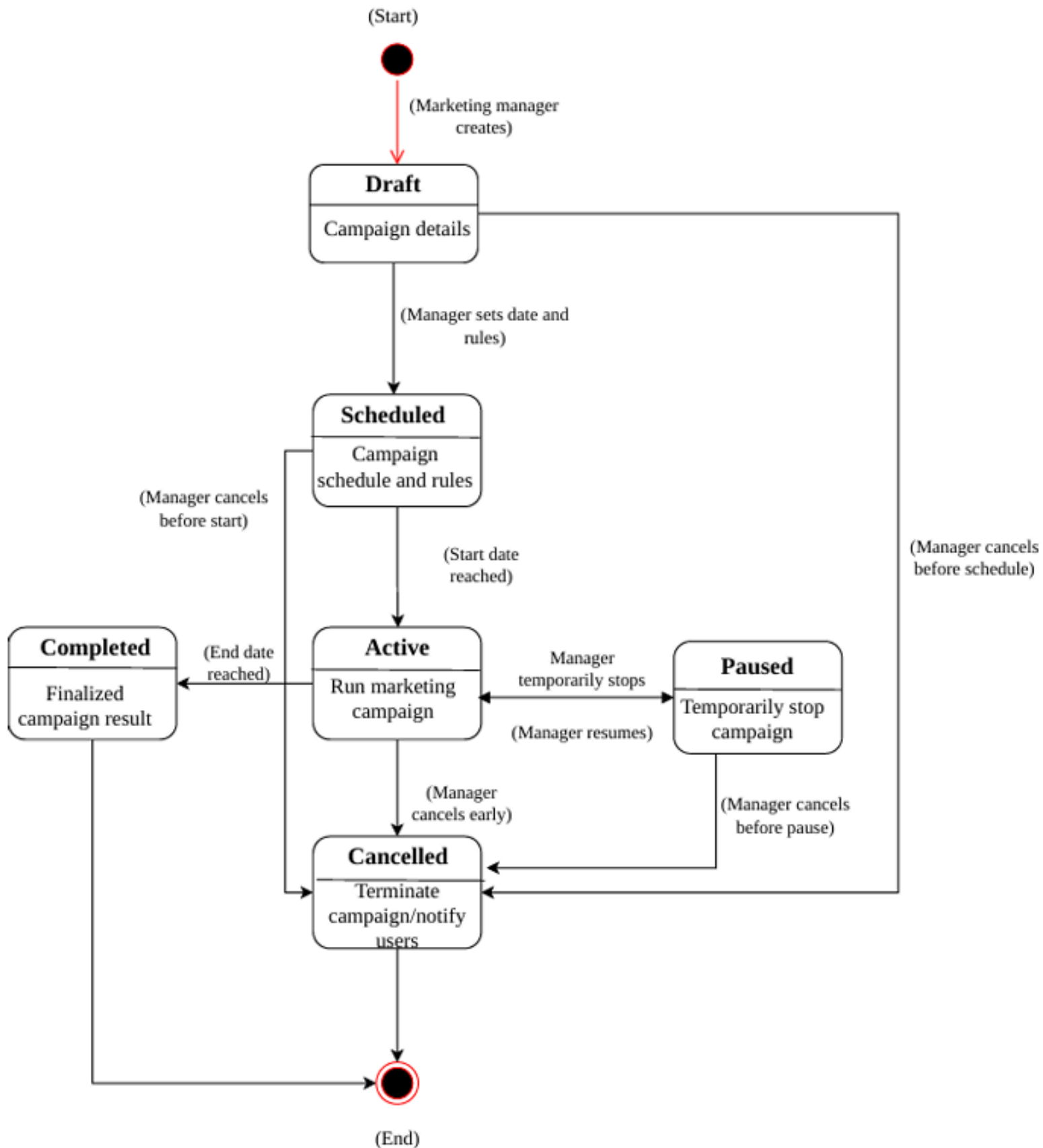
State Diagram 4- Restaurant Status State Diagram



State Diagram 5- User State Diagram



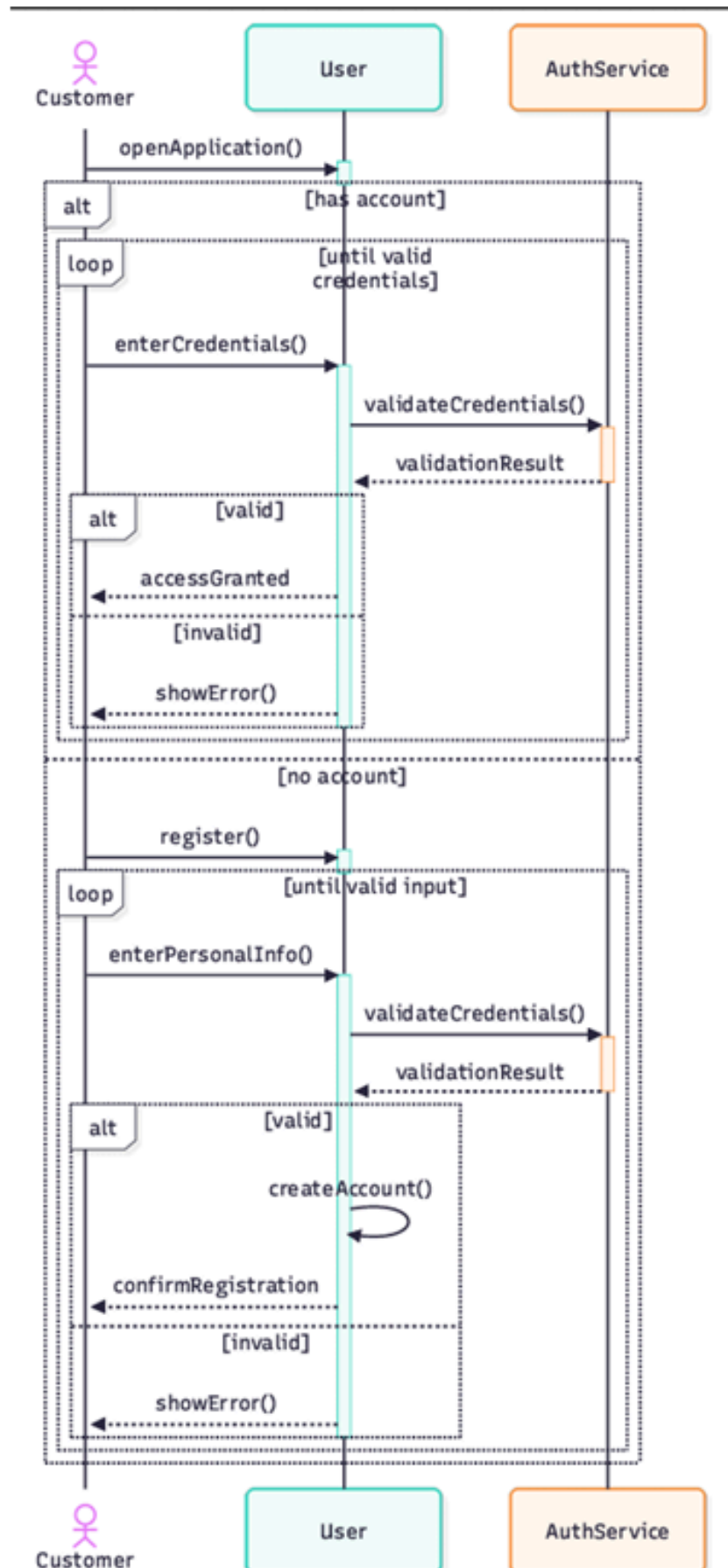
State Diagram 6- Marketing Campaign State Diagram



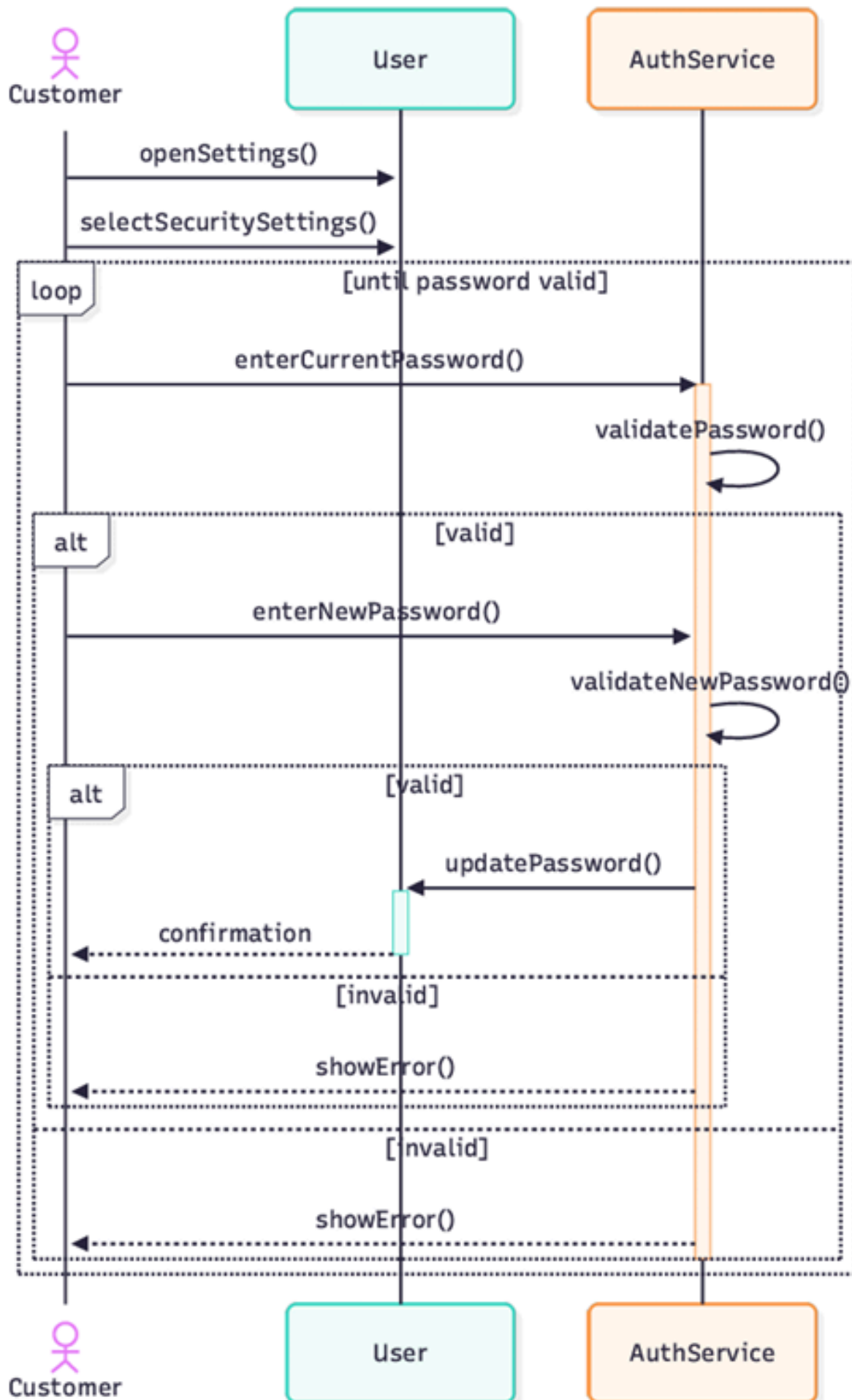
Sequence diagrams

Author-Sofia Cala

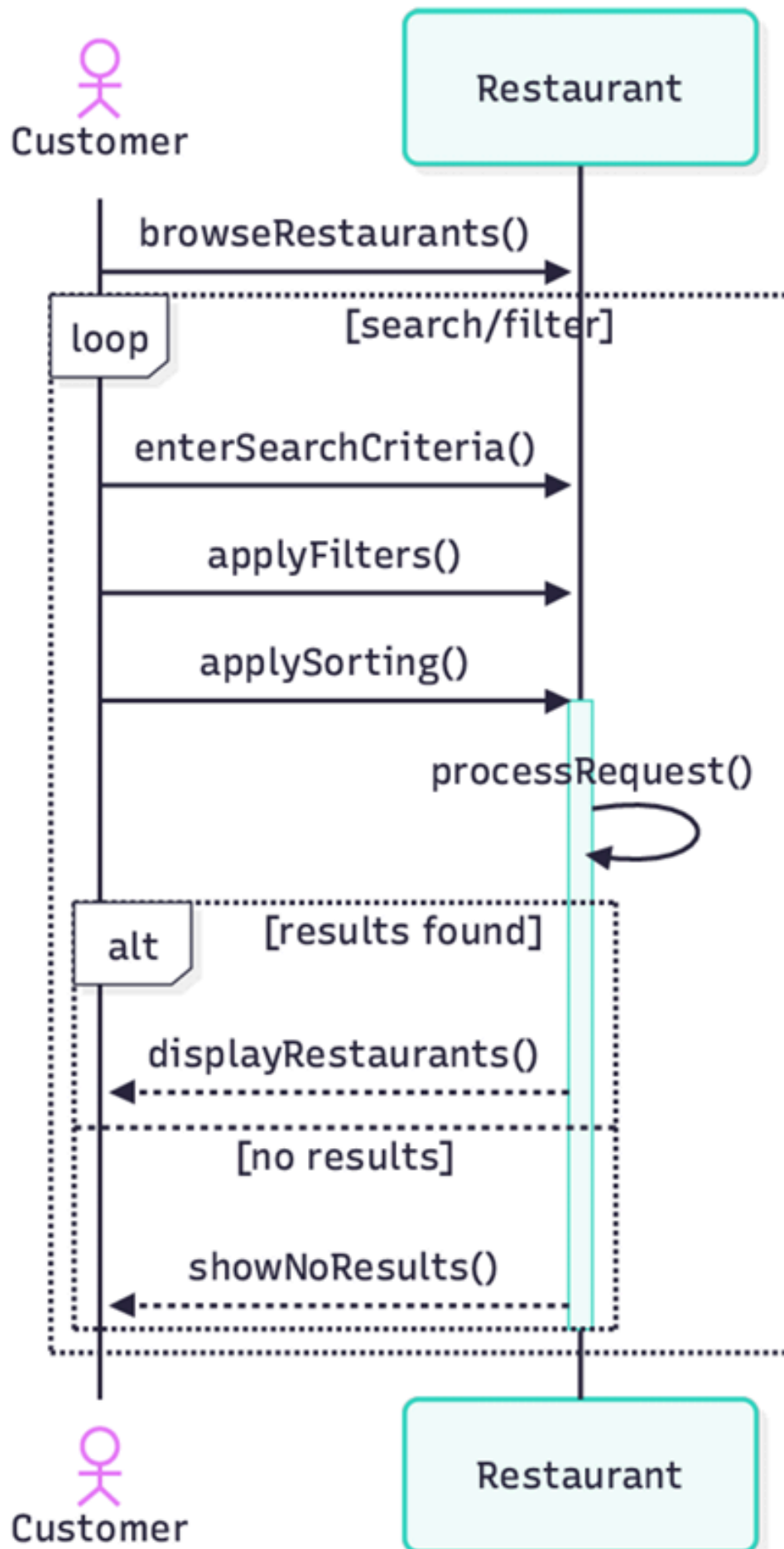
Sequence Diagram-UC-01-Manage Customer Account



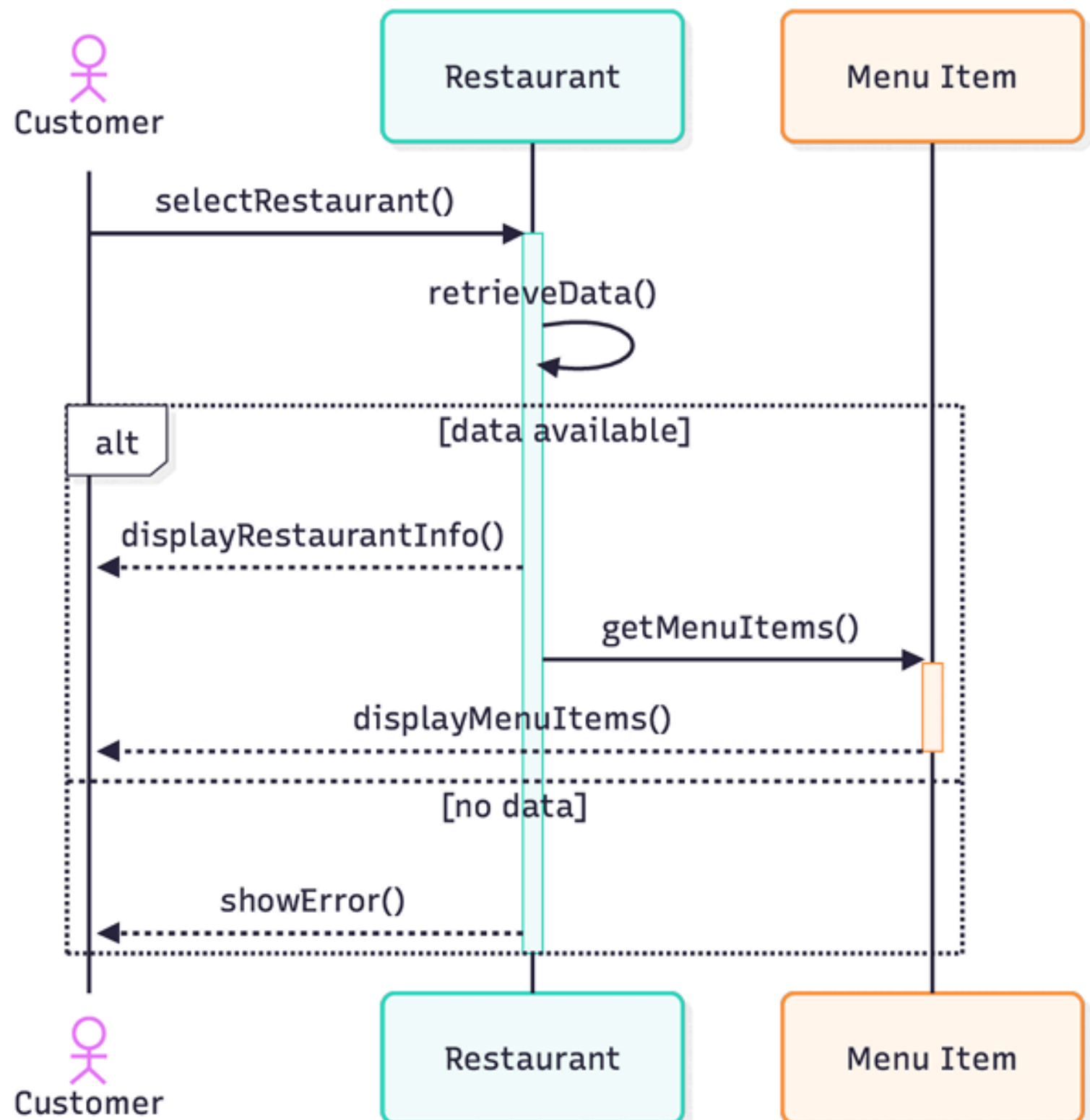
Sequence Diagram-UC-02-Authentication&Security Settings



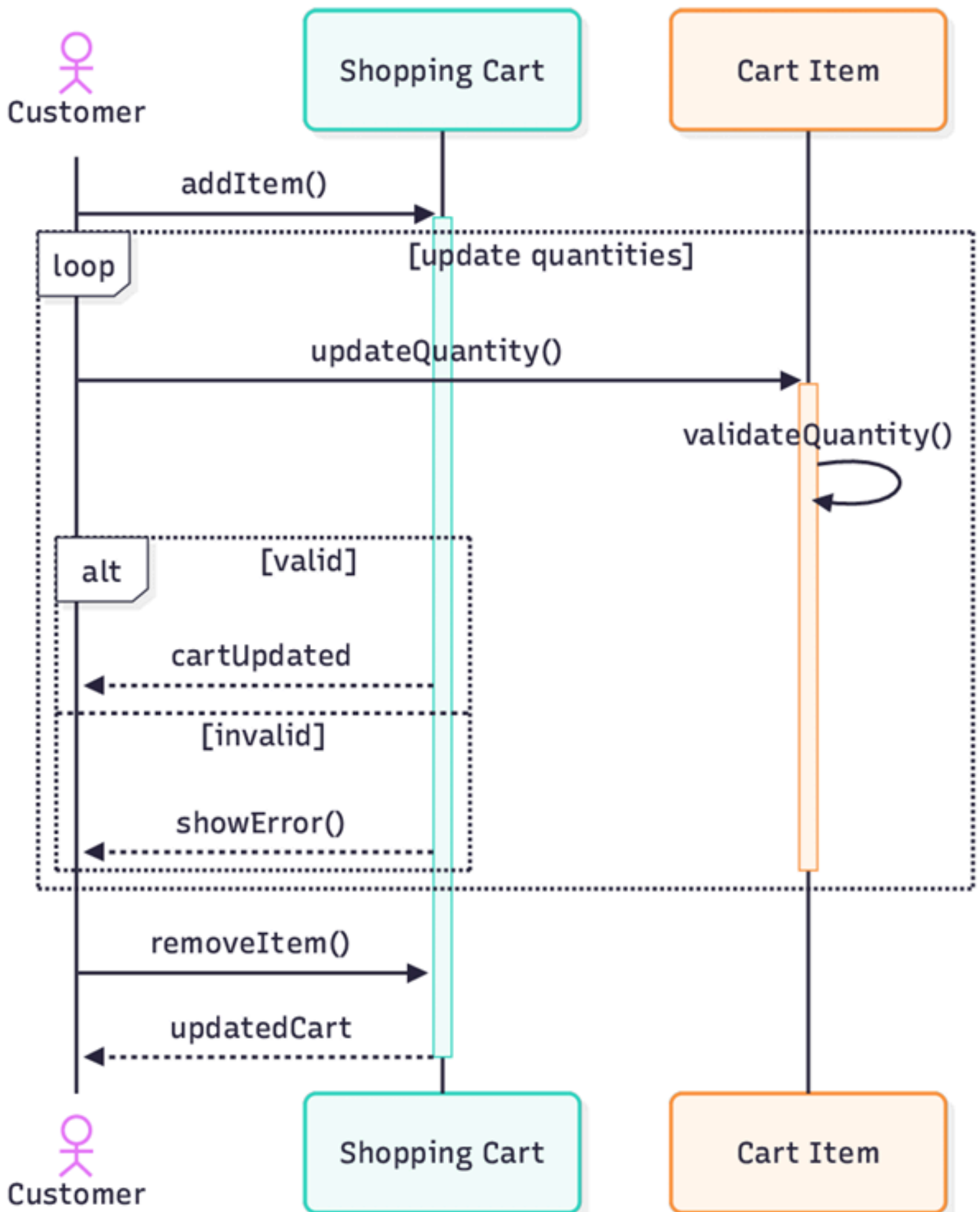
Sequence Diagram-UC-03-Browse/Search/Filter Restaurant



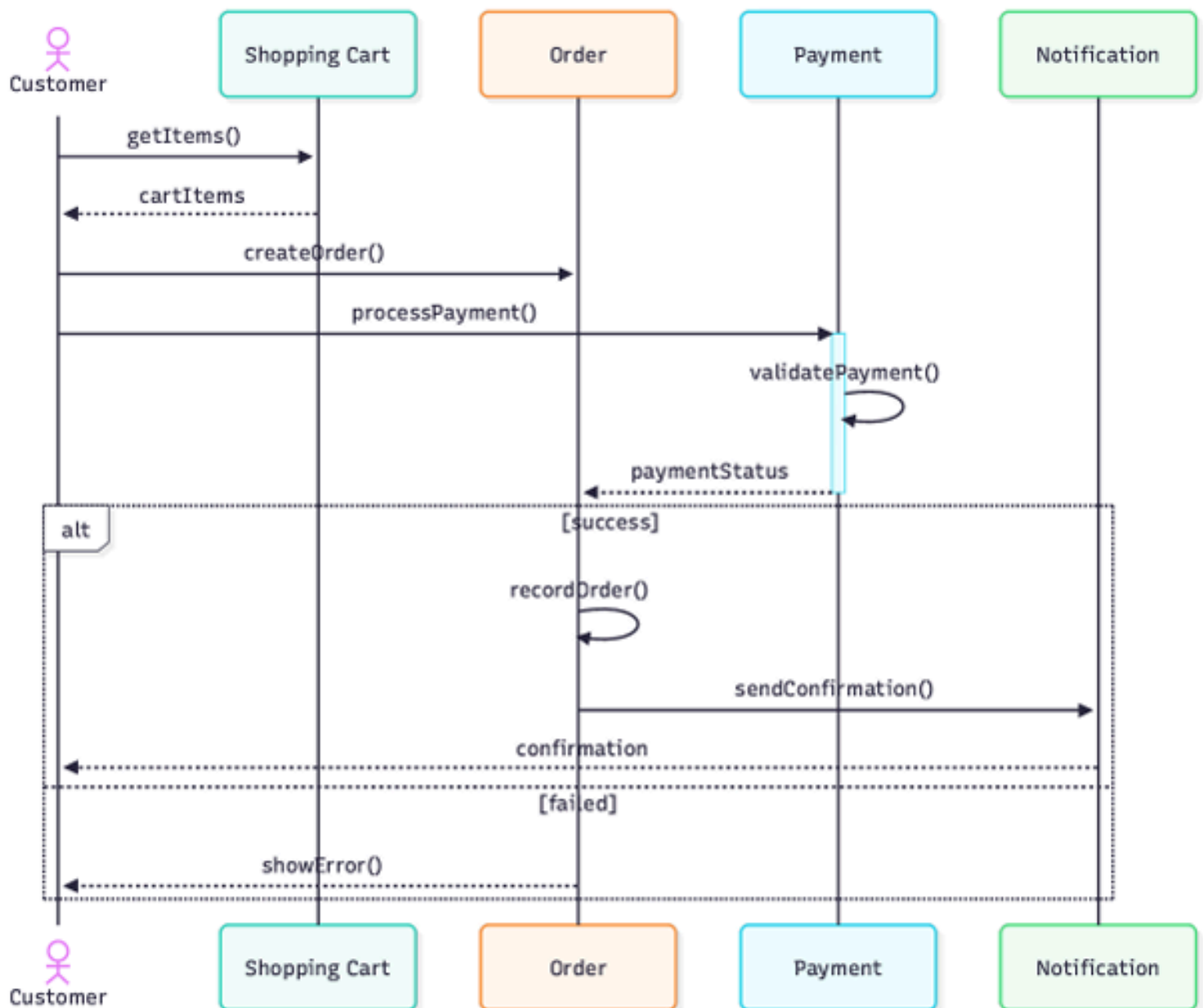
Sequence Diagram-UC-04-View Restaurant & Food Details



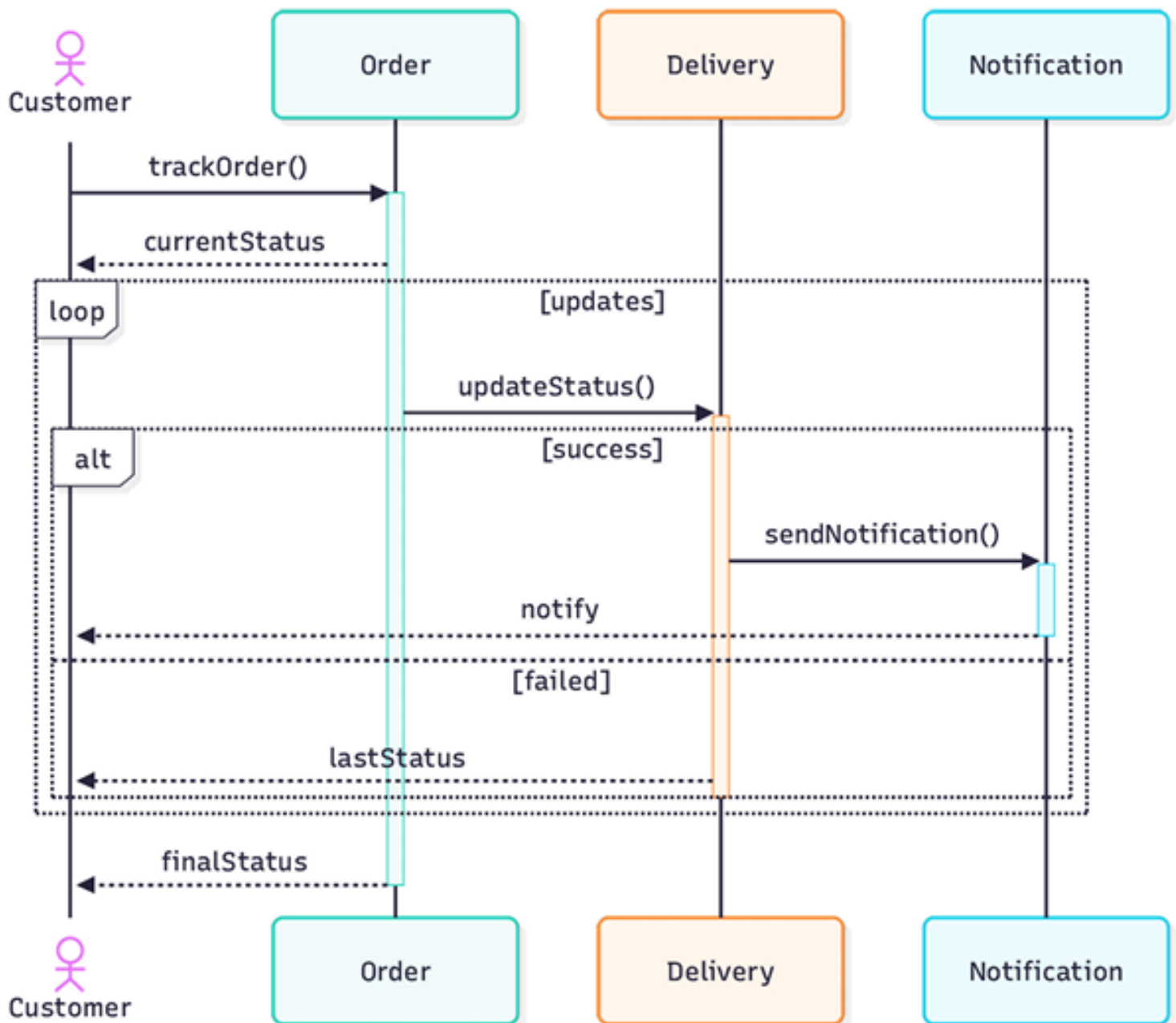
Sequence Diagram-UC-05-Manage Shopping Cart



Sequence Diagram-UC-06-Place Order

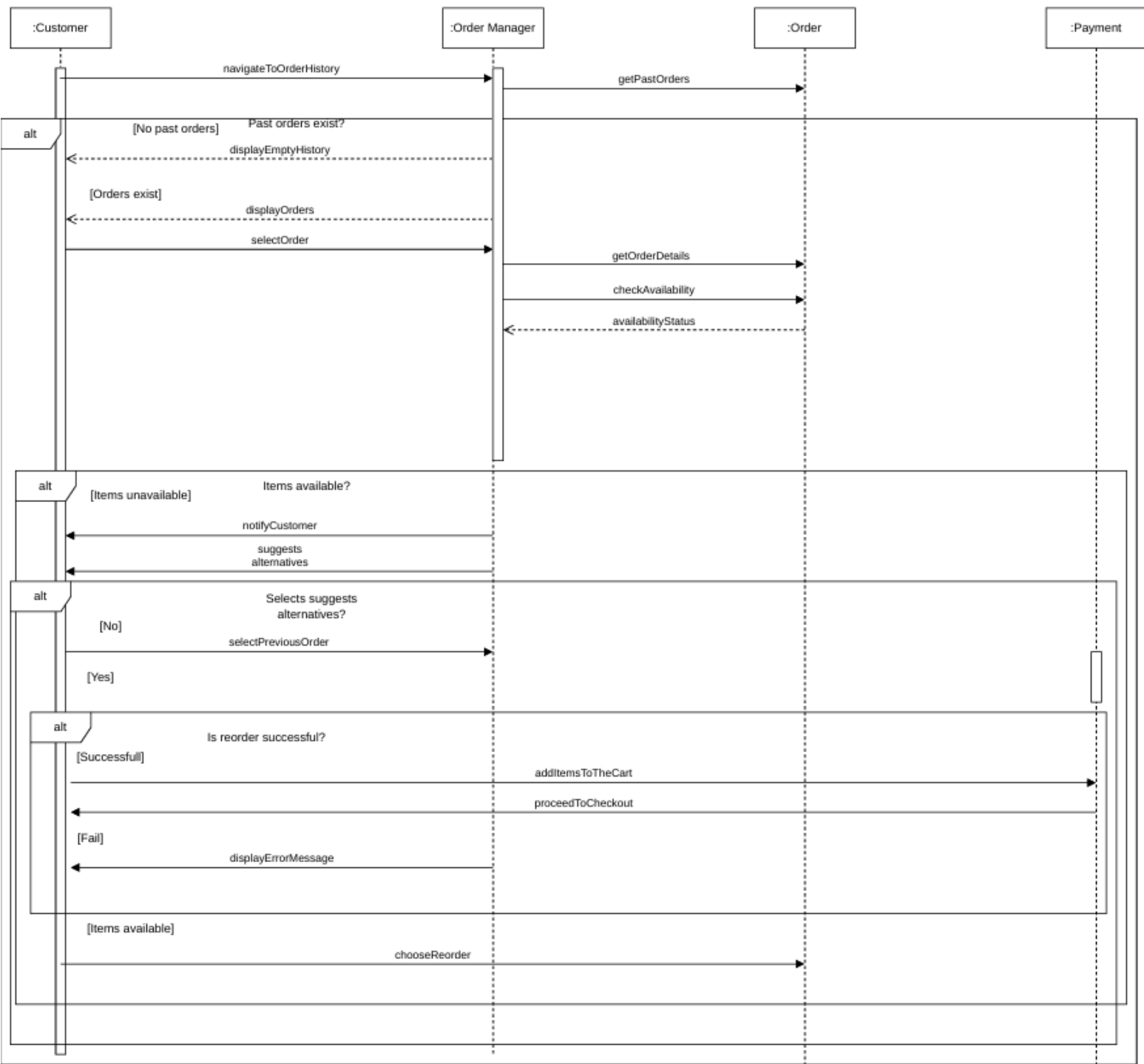


Sequence Diagram-UC-07-Track Order & Notifications

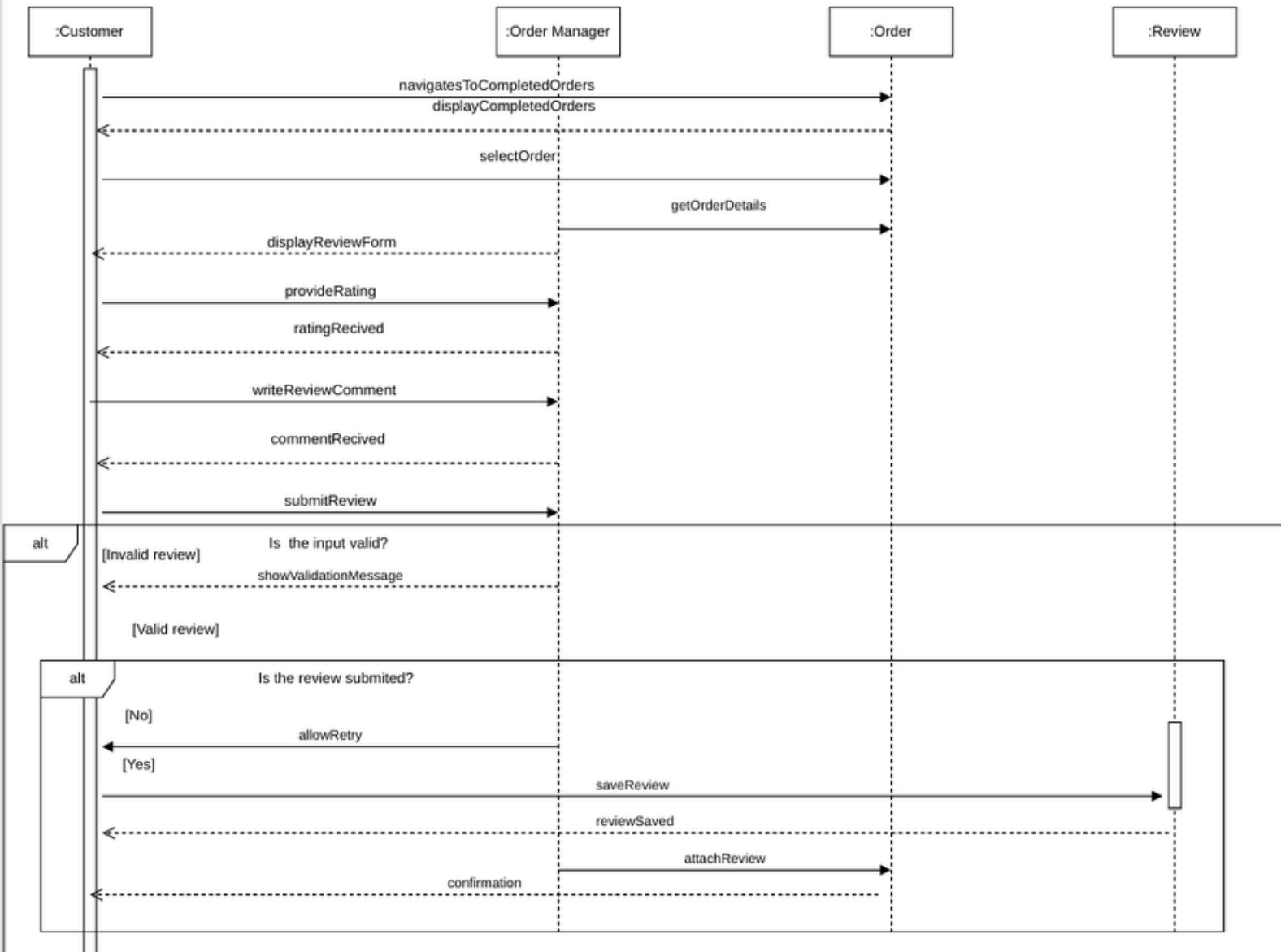


Sequence Diagram 08-View Order History & Reorder

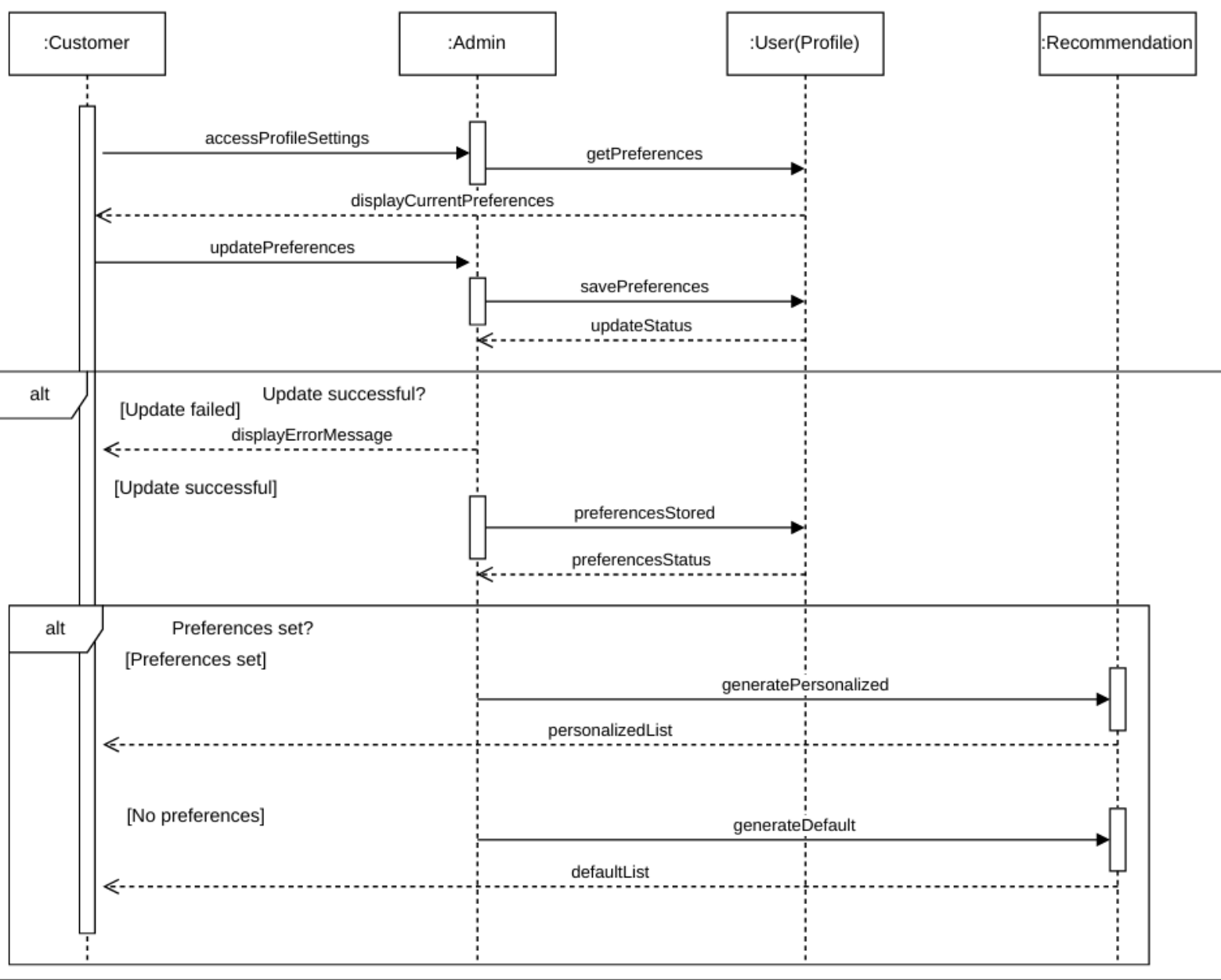
Xhesika Toska



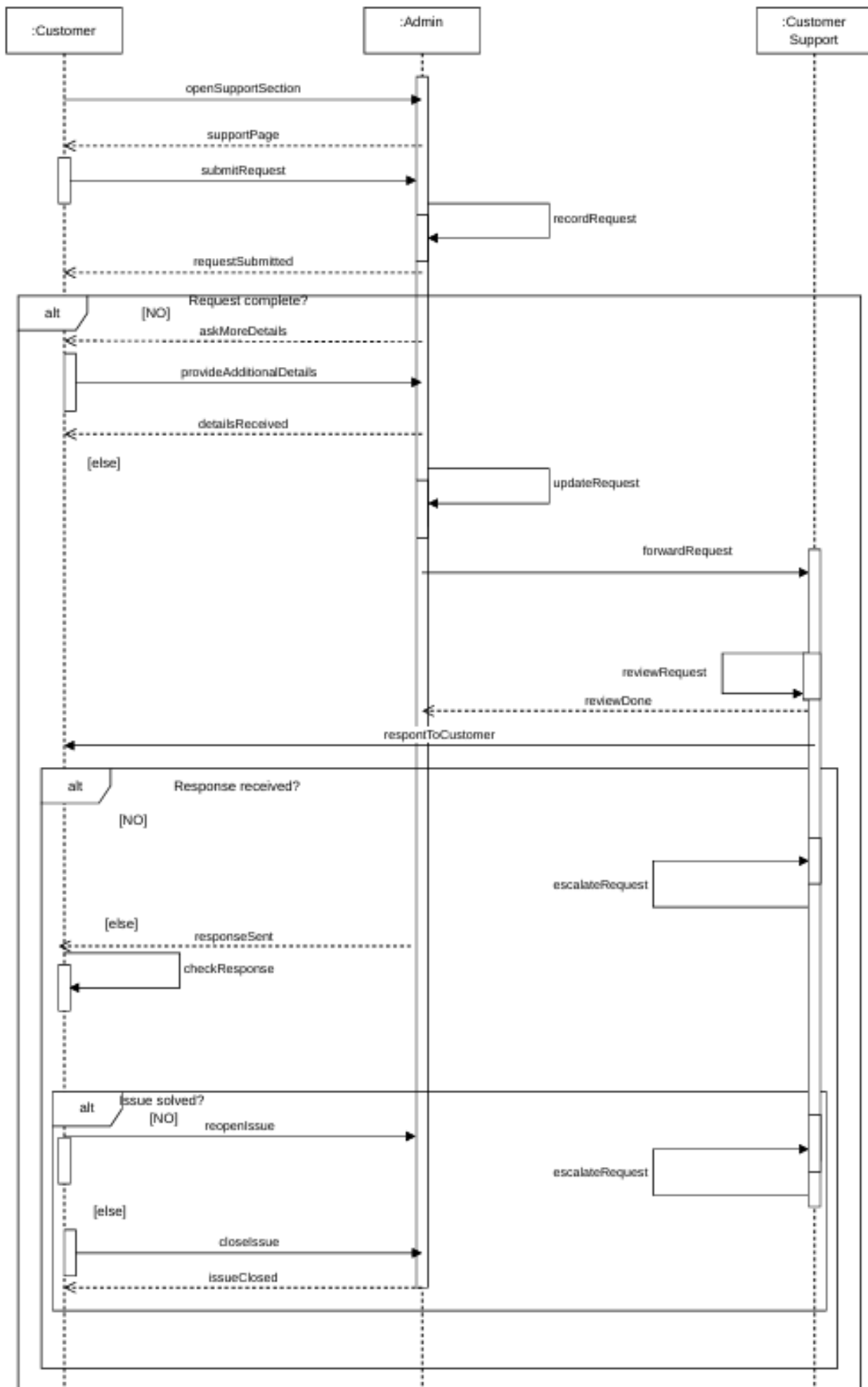
Sequence Diagram 09-Rate & Review Restaurants



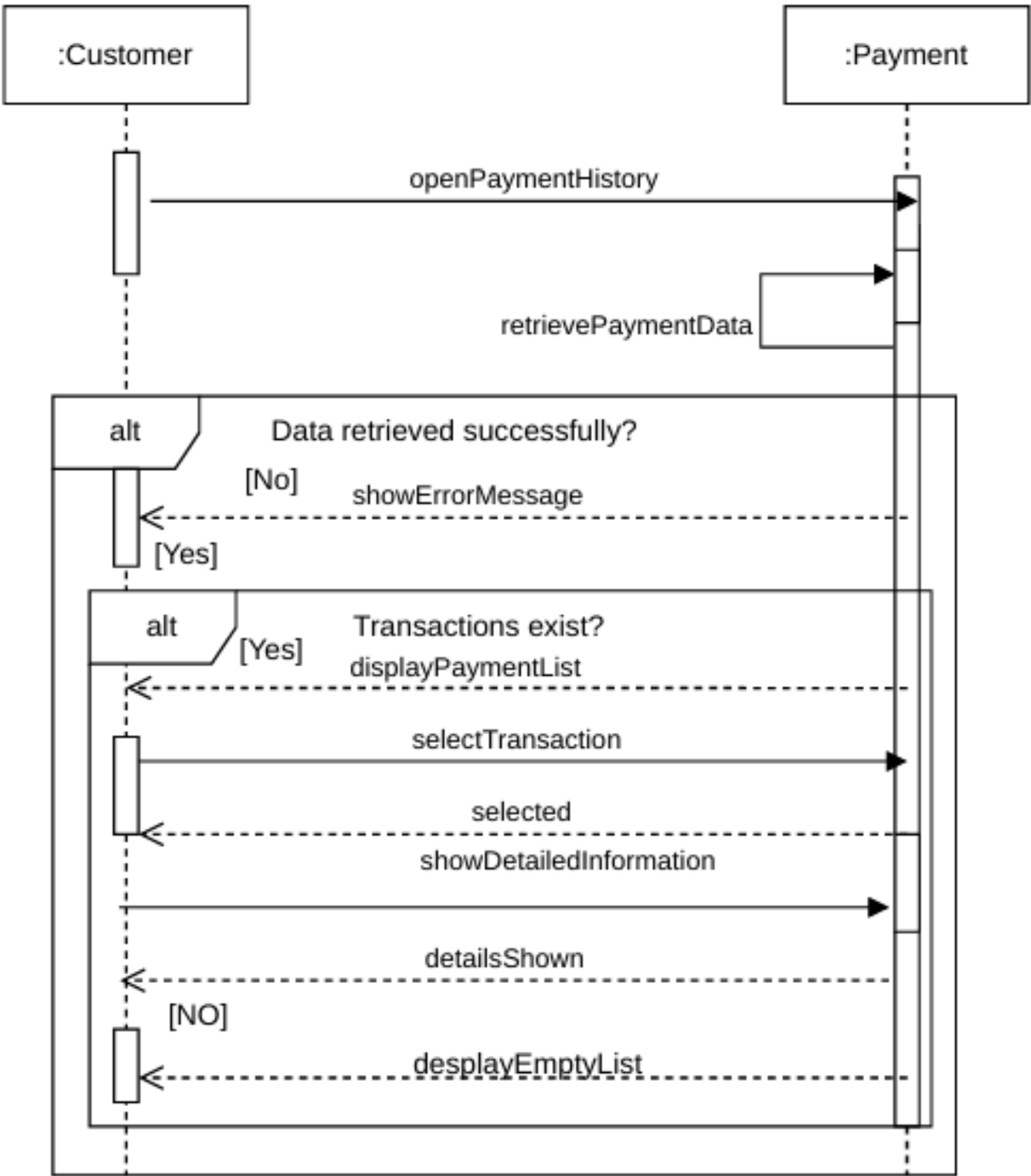
Sequence Diagram 10-Manage Personalized Experience



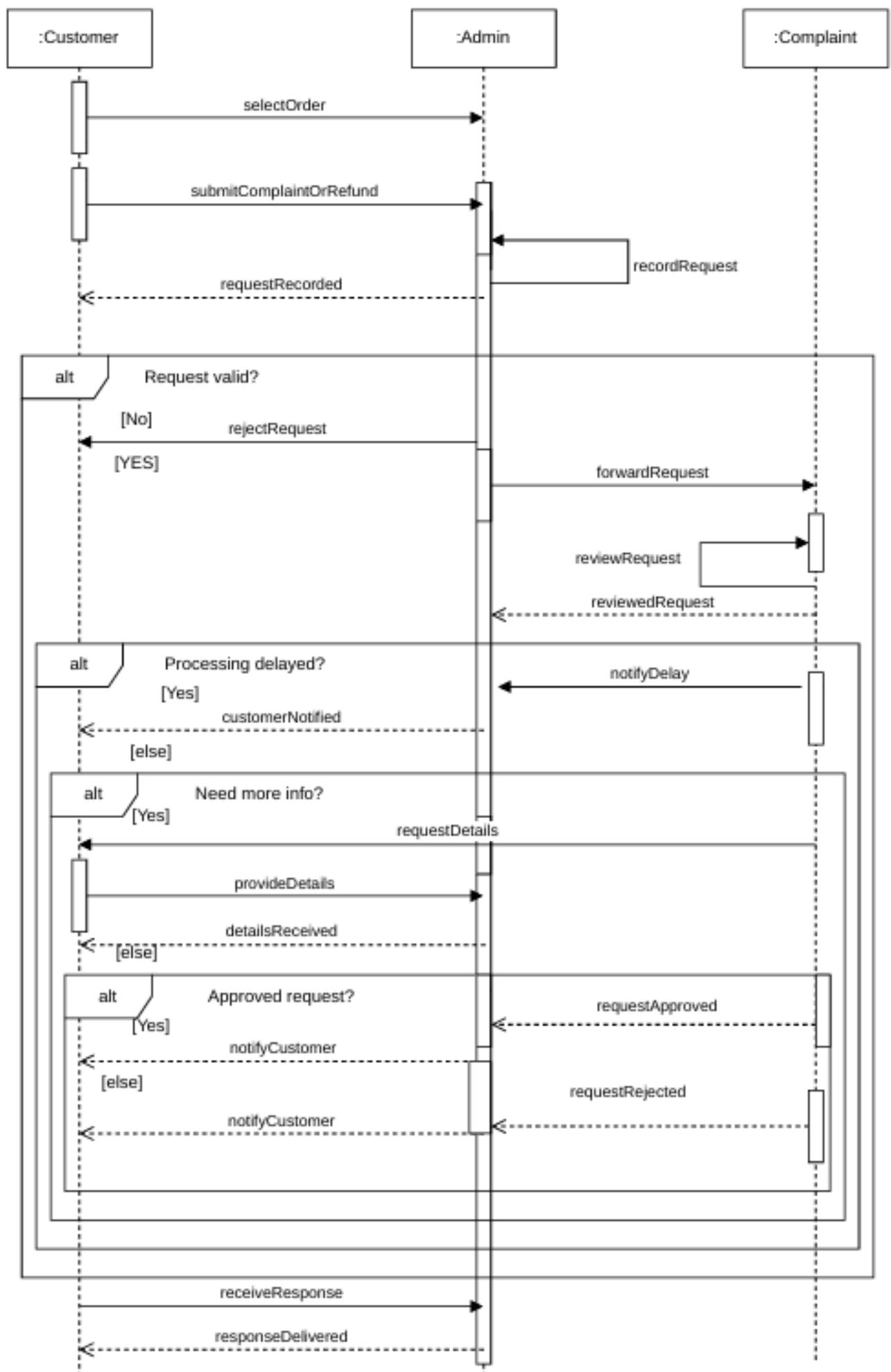
Sequence Diagram 11-Customer communicates with support to resolve issues or request assistance



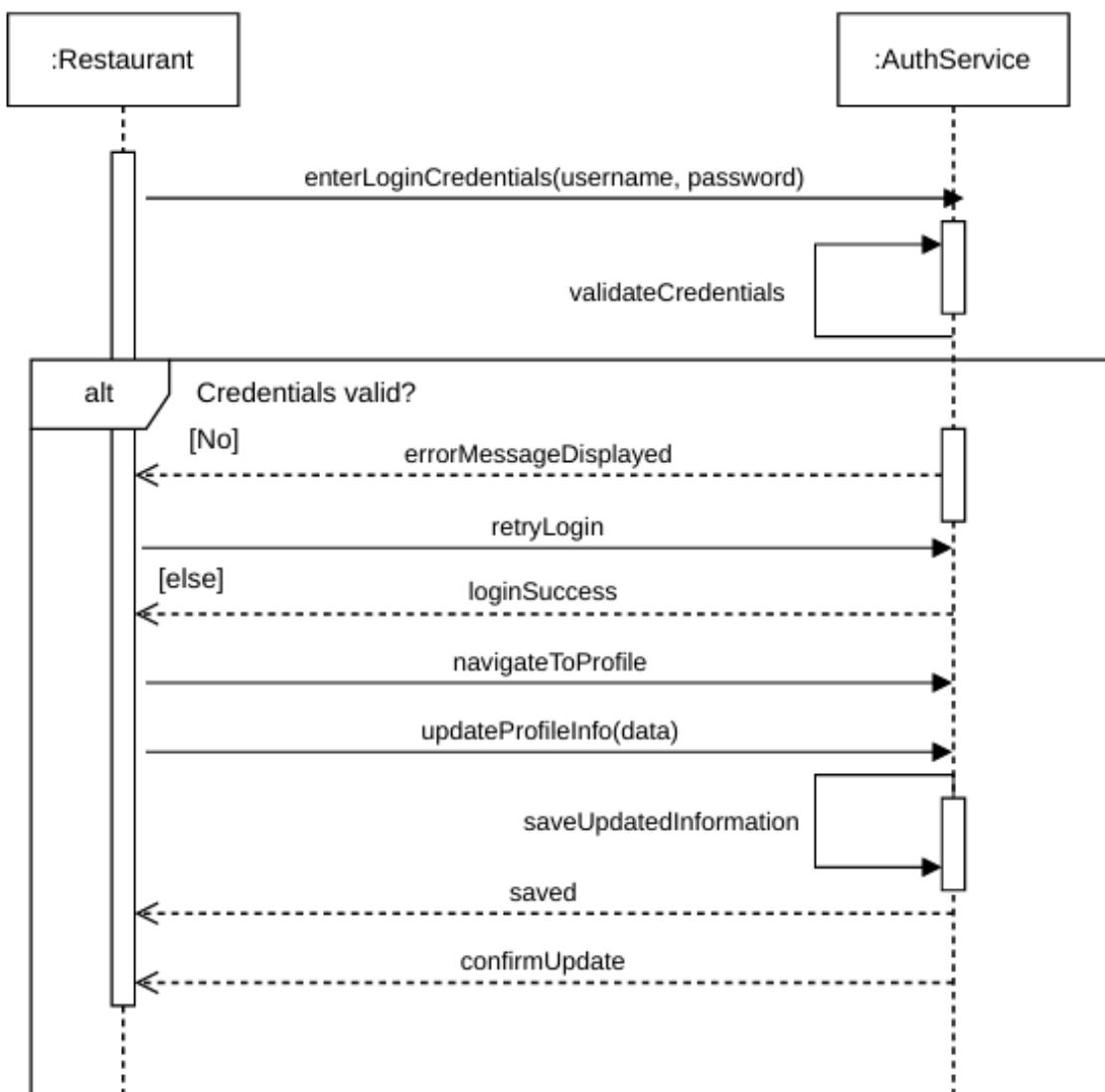
Sequence Diagram 12-Manage Payment History



Sequence Diagram 13-Manage Refunds & Complaints

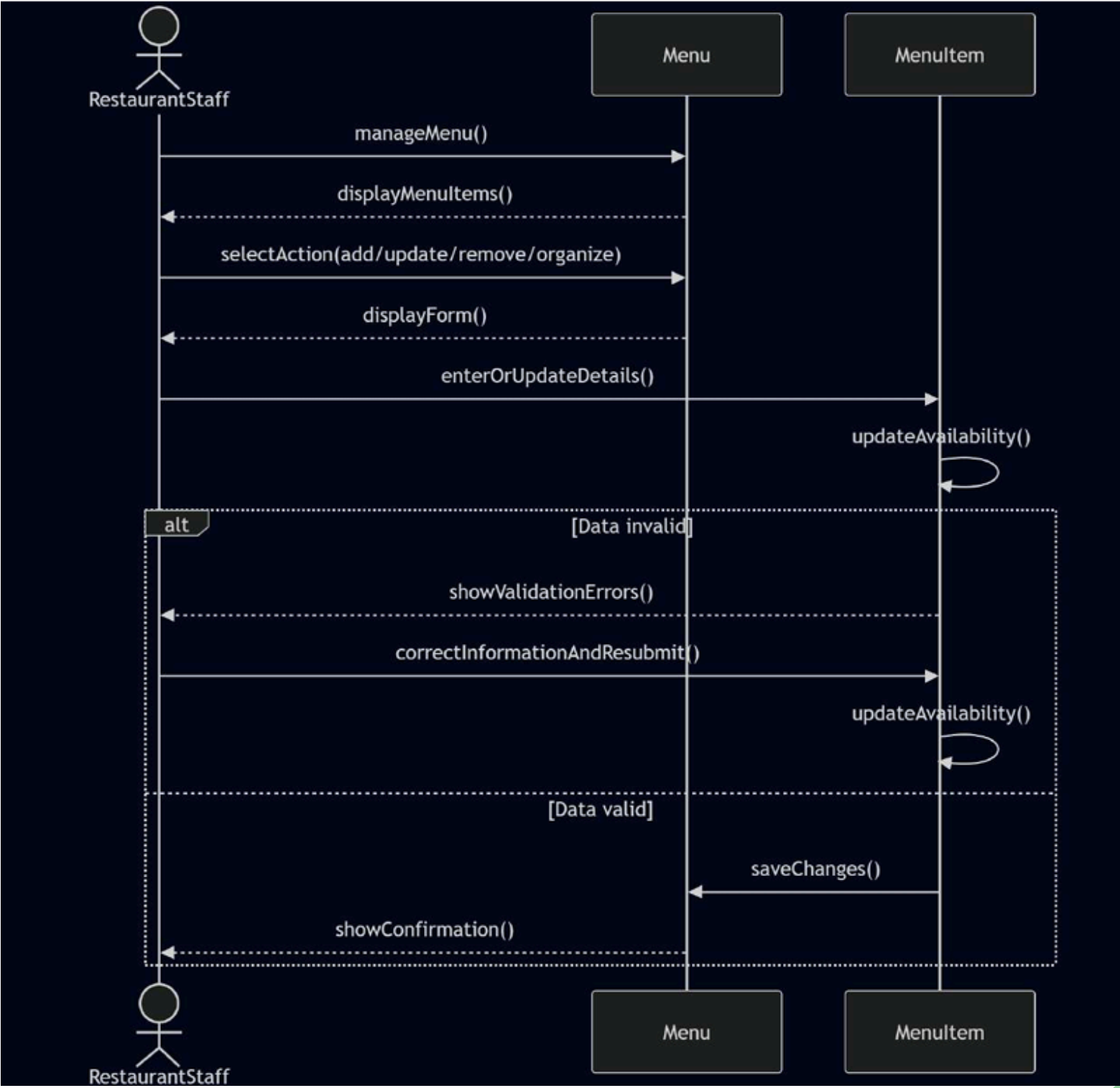


Sequence Diagram 14-Restaurant staff manages account and restaurant profile information.

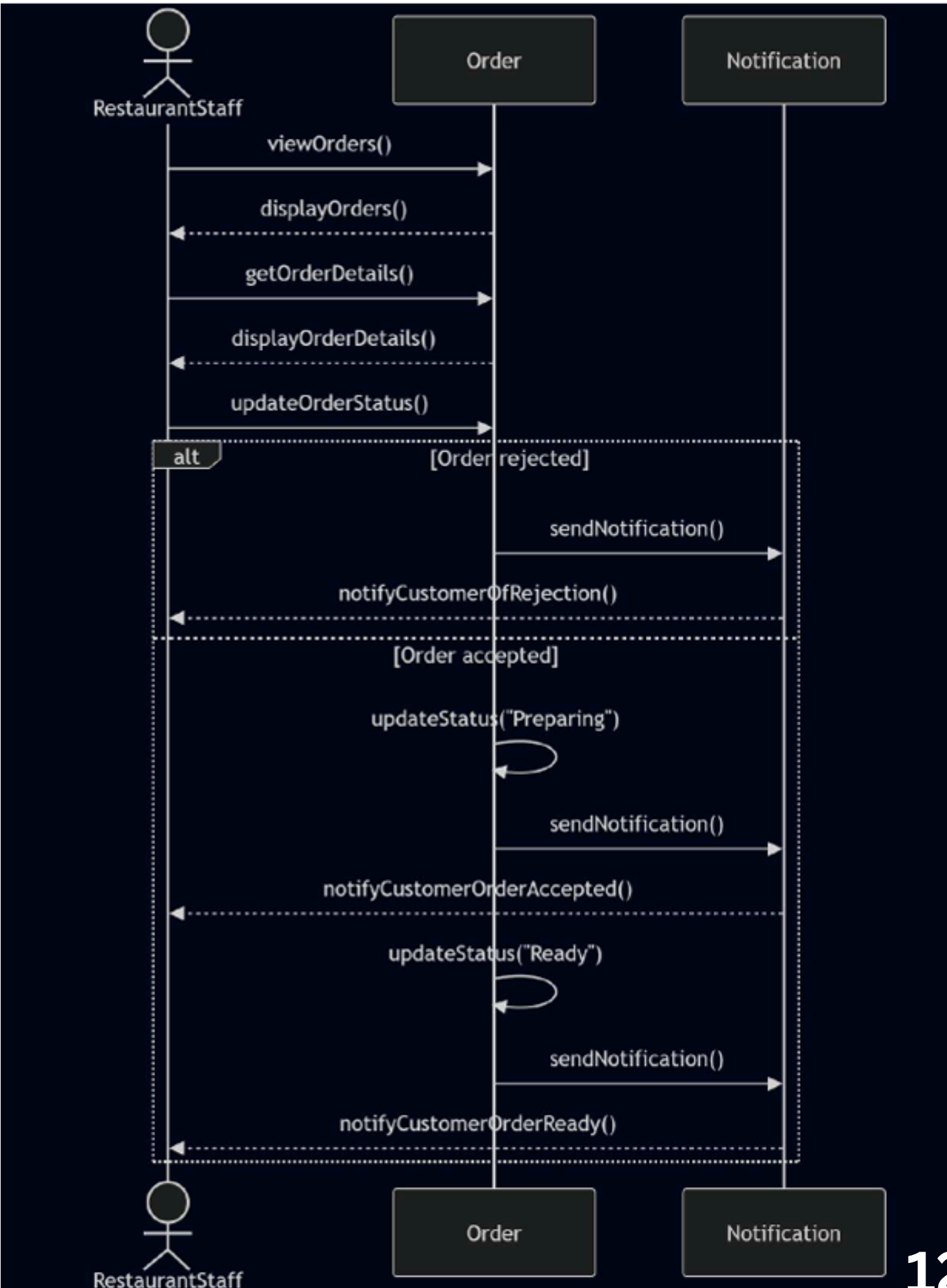


Sequence Diagram 15-Restaurant staff manages menu items, categories, and food availability (add, update, remove, organize)

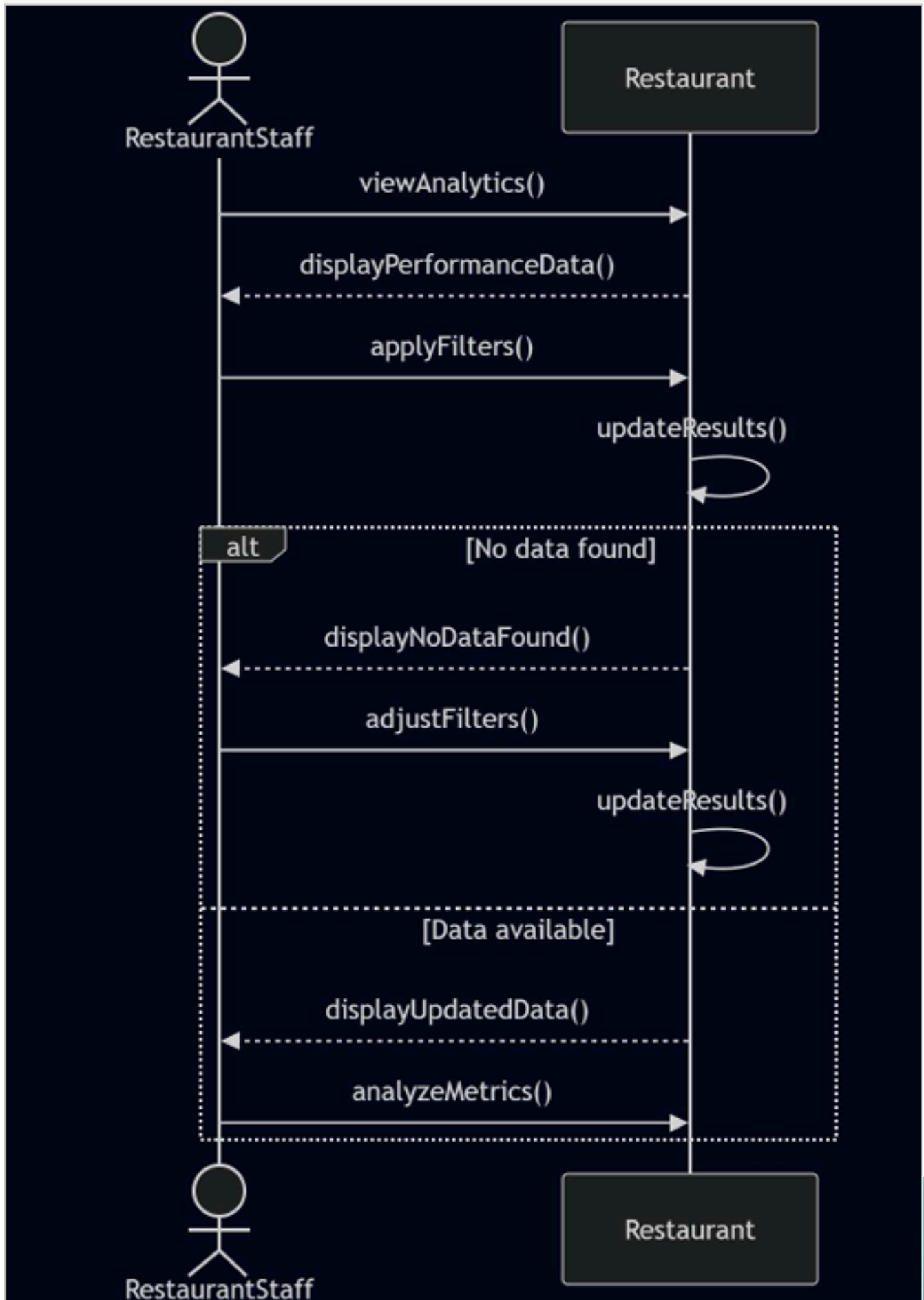
Author-Aurelio Hysenaj



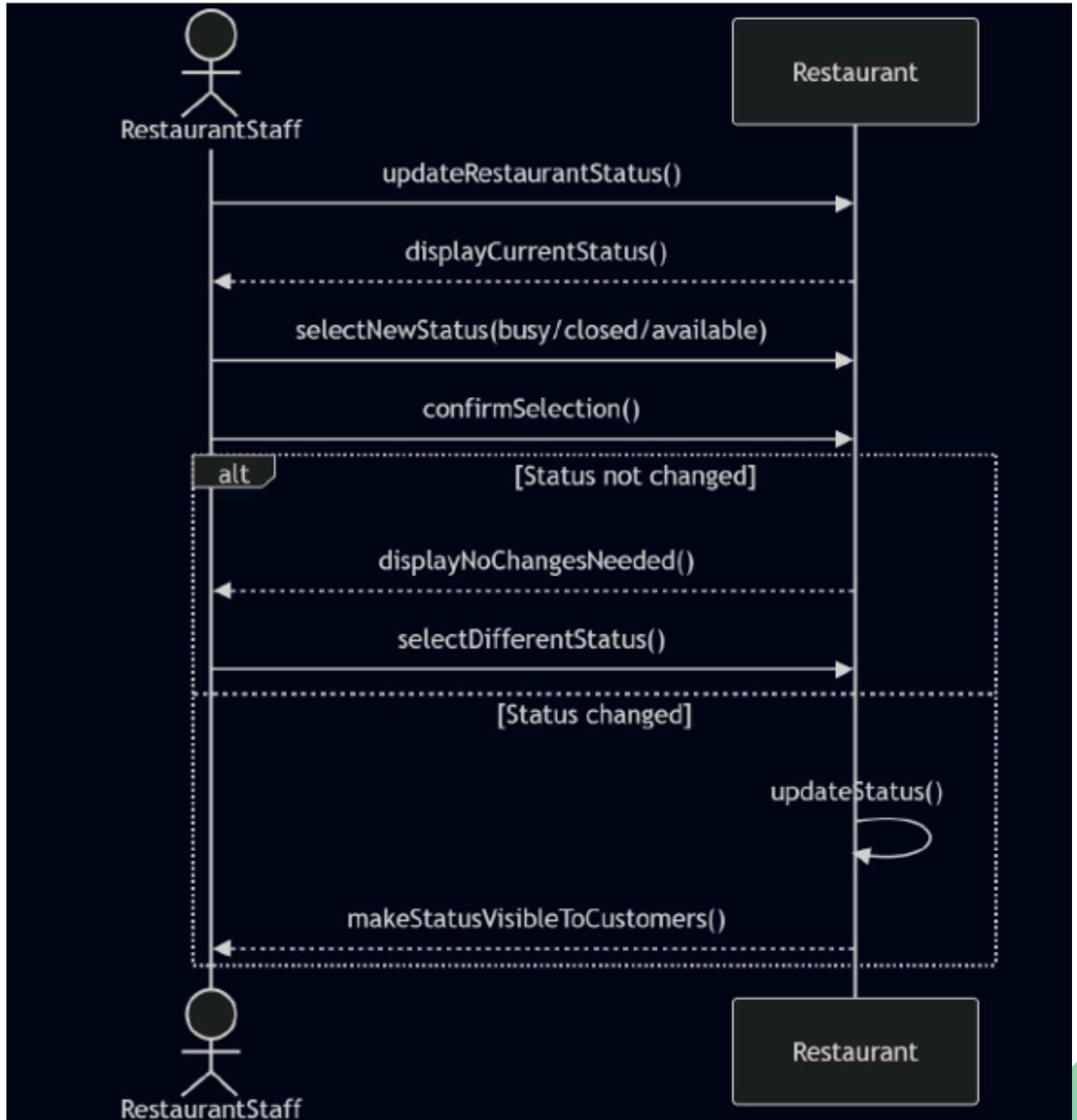
Sequence Diagram 16- Restaurant staff manages customer orders and updates order status (receive, view, accept, reject, preparing, ready for delivery)



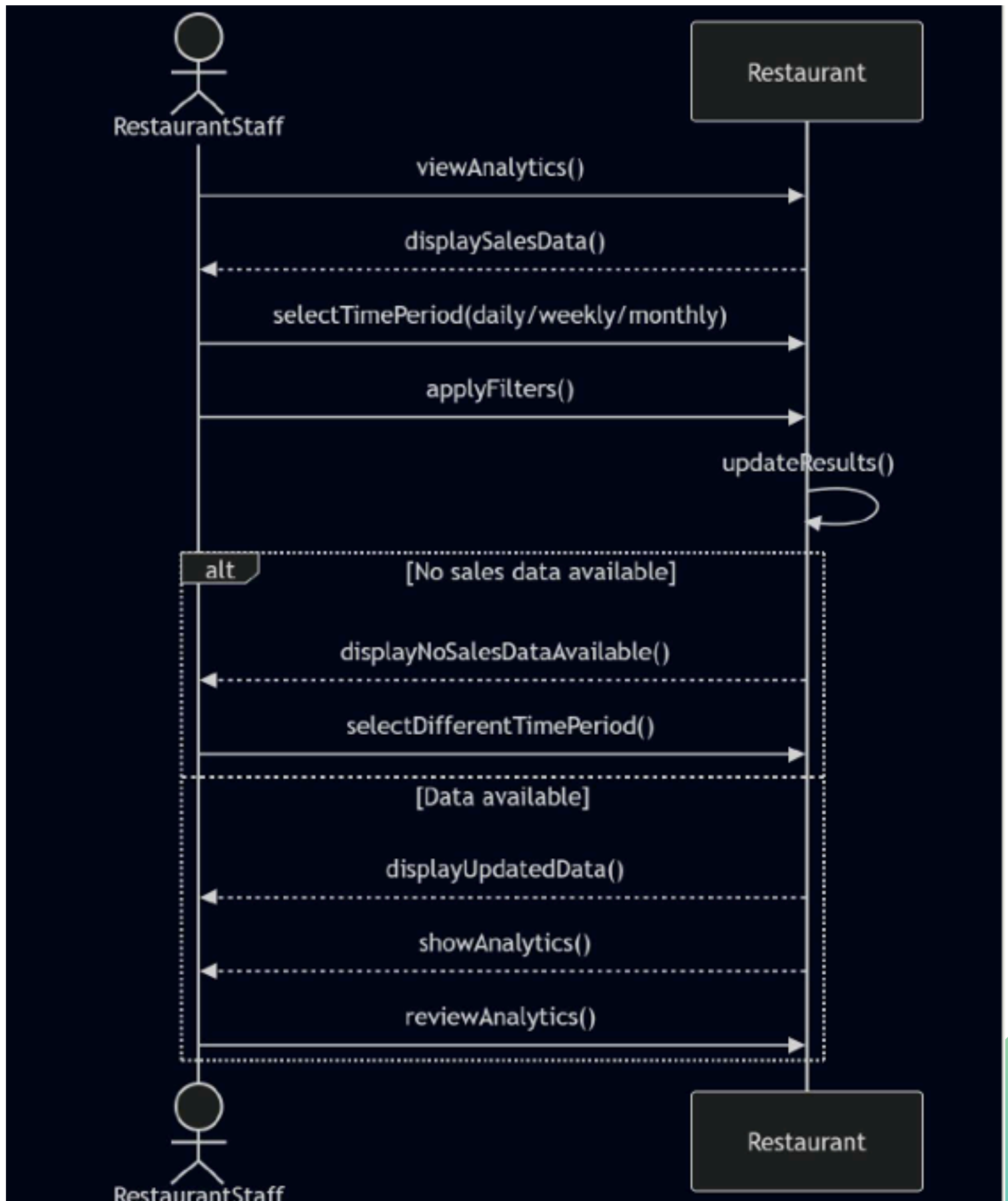
Sequence Diagram 17- Restaurant staff monitors and analyzes restaurant performance



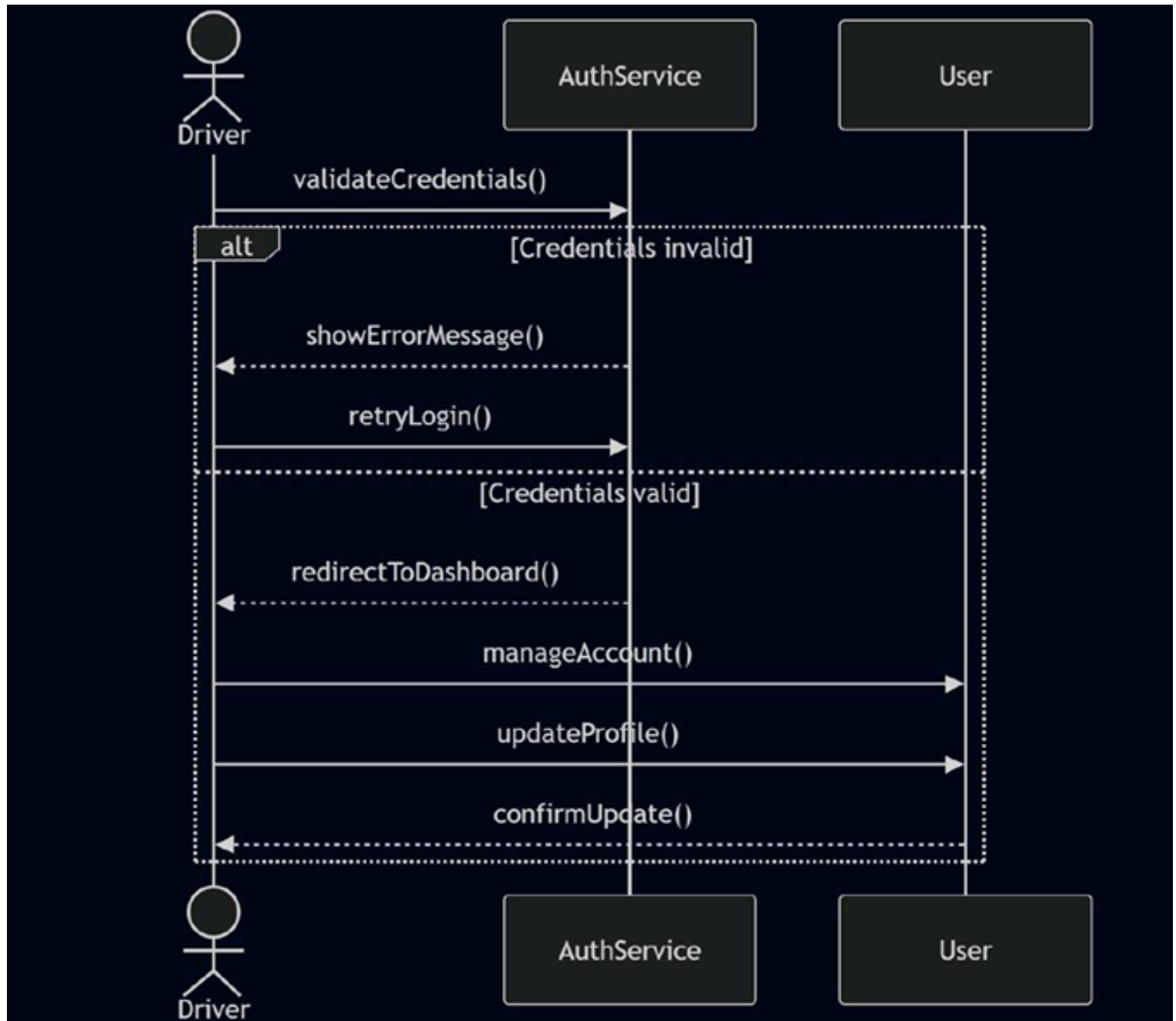
Sequence Diagram 18- Restaurant staff manages temporary service interruptions (e.g., busy, closed)



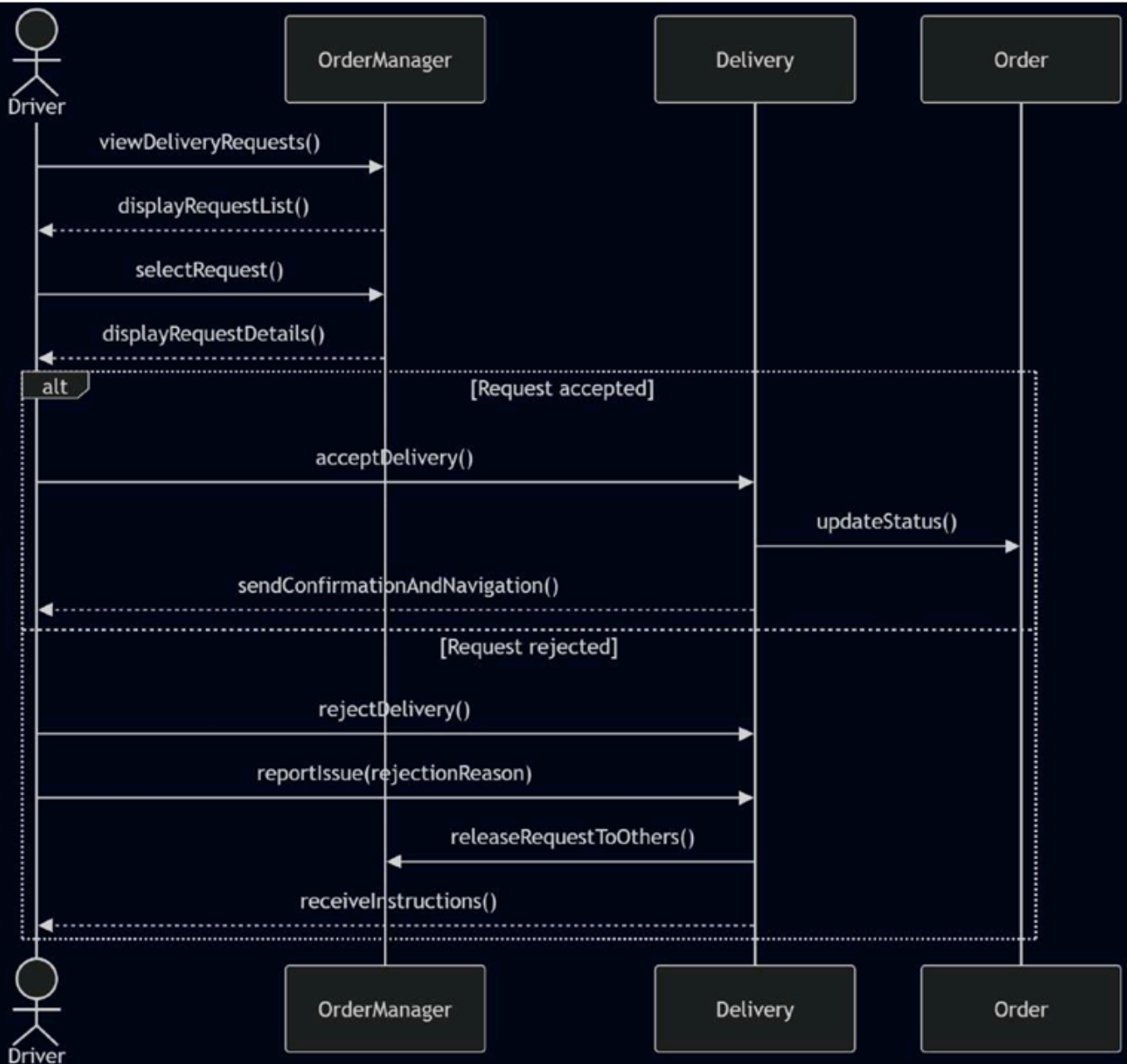
Sequence Diagram 19- Restaurant staff views sales analytics (daily, weekly, monthly)



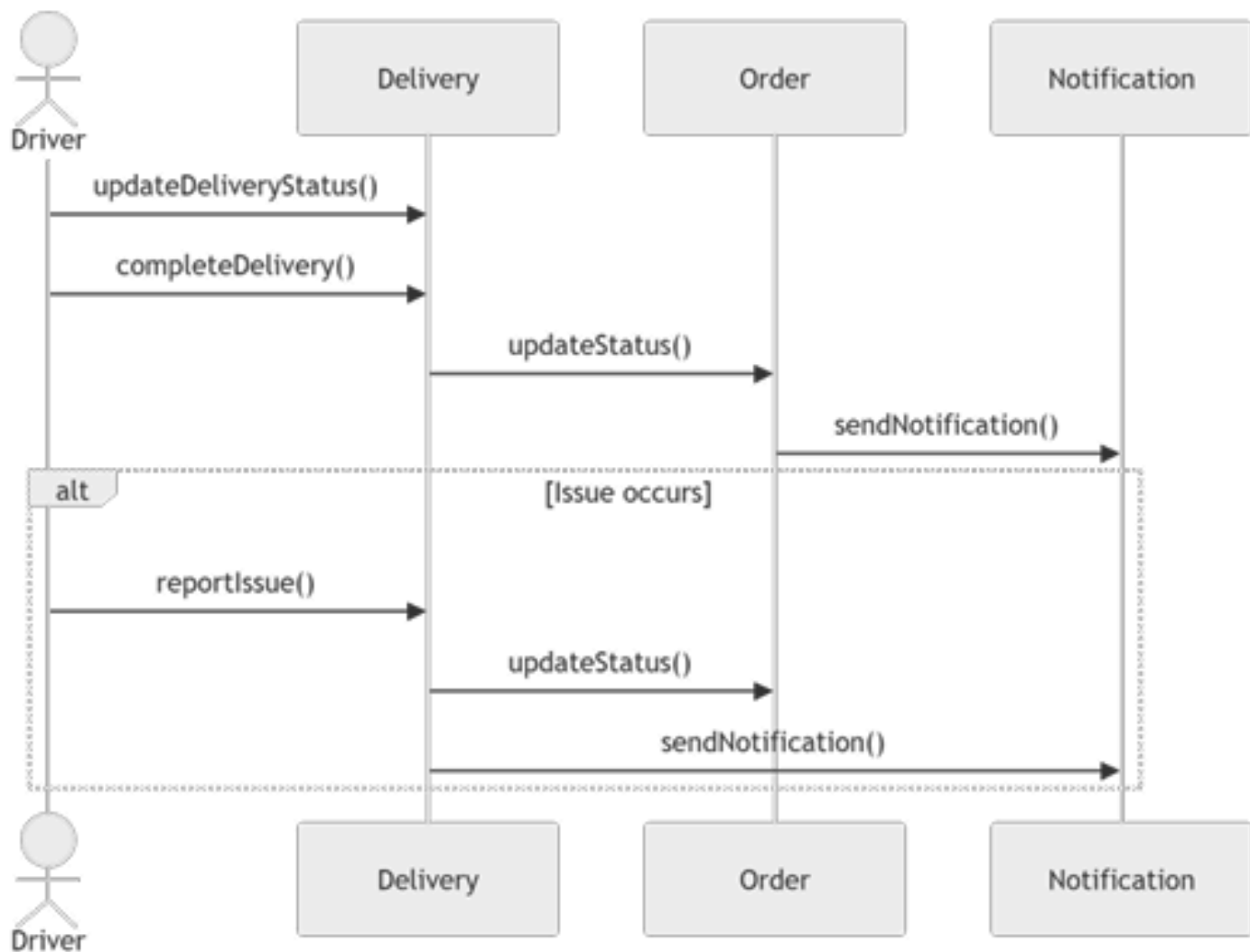
Sequence Diagram 20- Delivery driver manages account and profile (login, profile management)



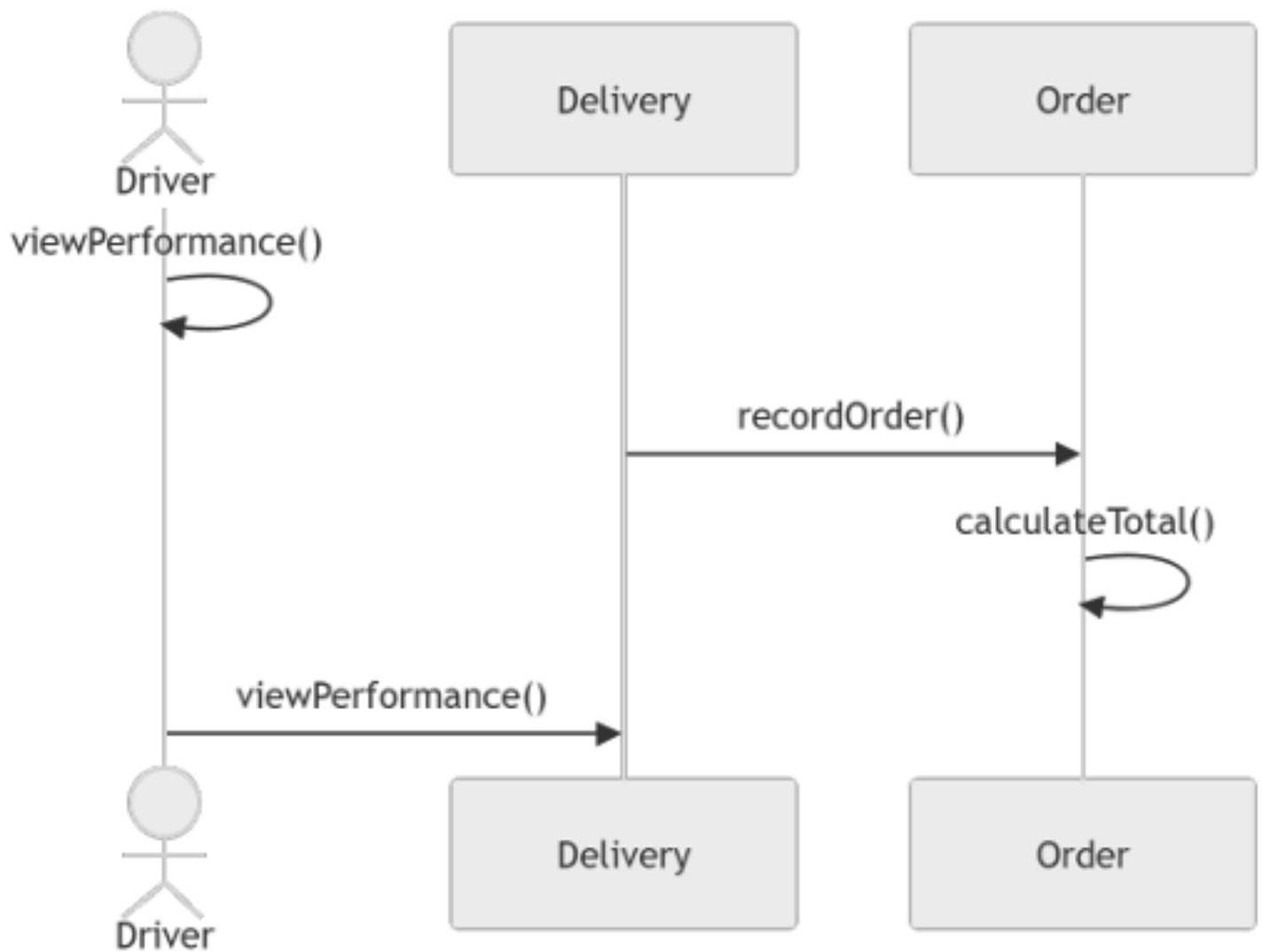
Sequence Diagram 21-Delivery driver manages delivery requests and details (view, accept, reject, view details)



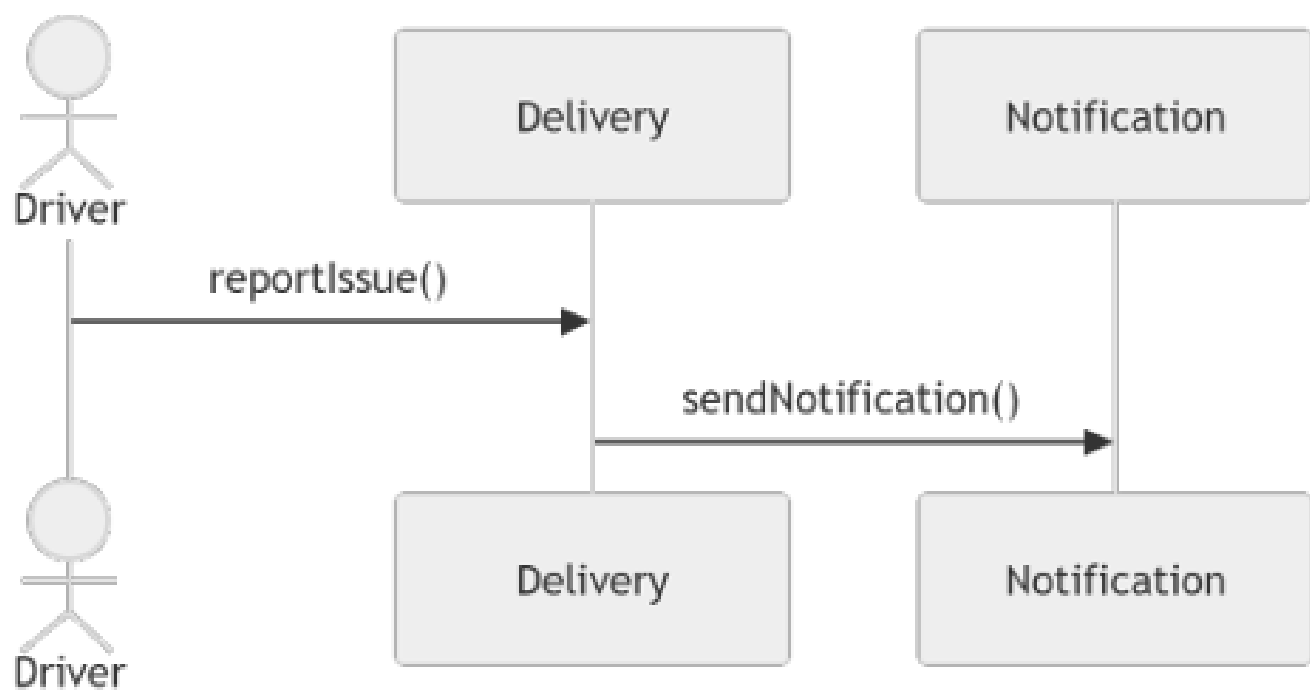
UC-22: Delivery Driver Completes Deliveries



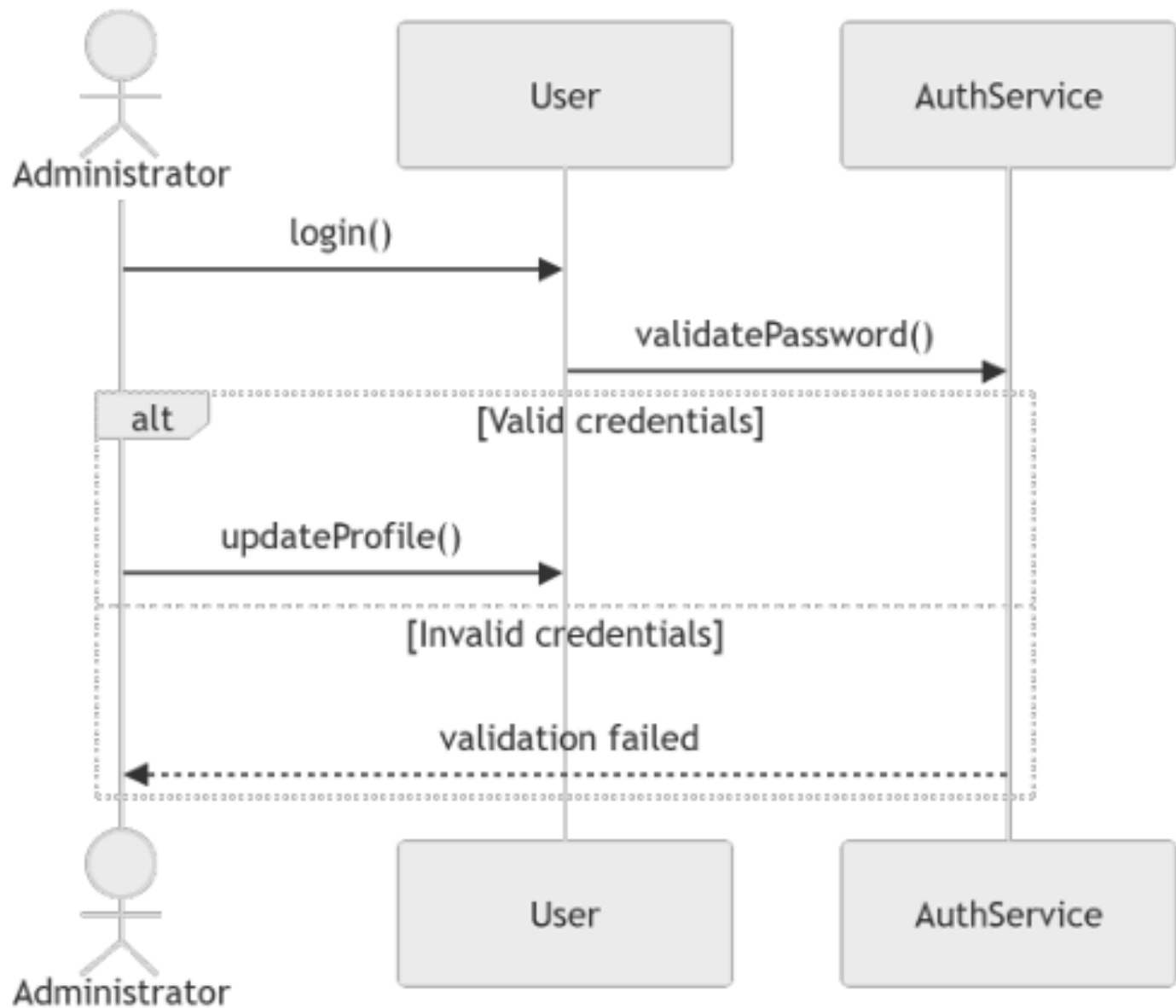
UC-23: Delivery Driver Monitors Performance



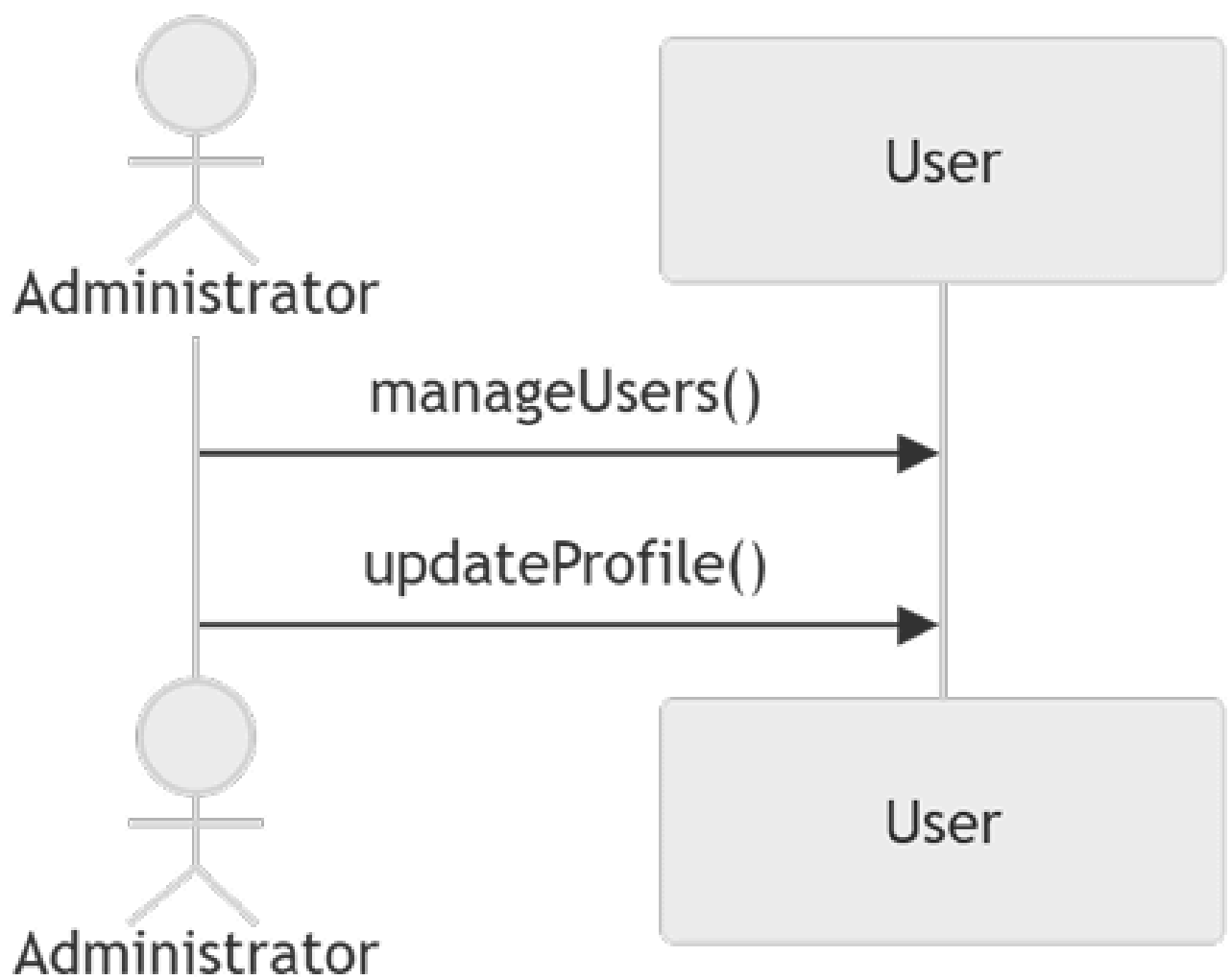
UC-24: Delivery Driver Reports Issues



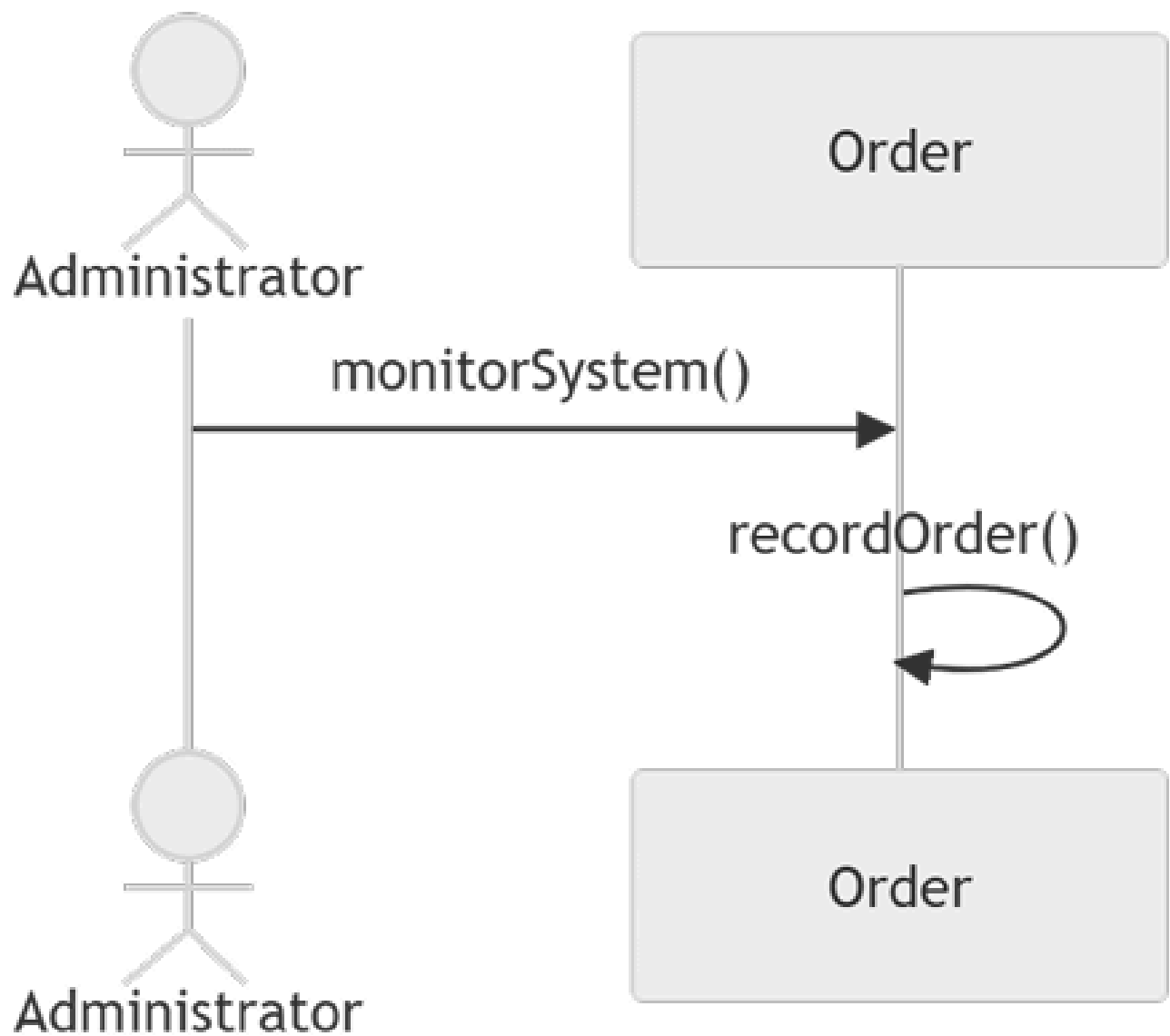
UC-25: Administrator Manages Profile



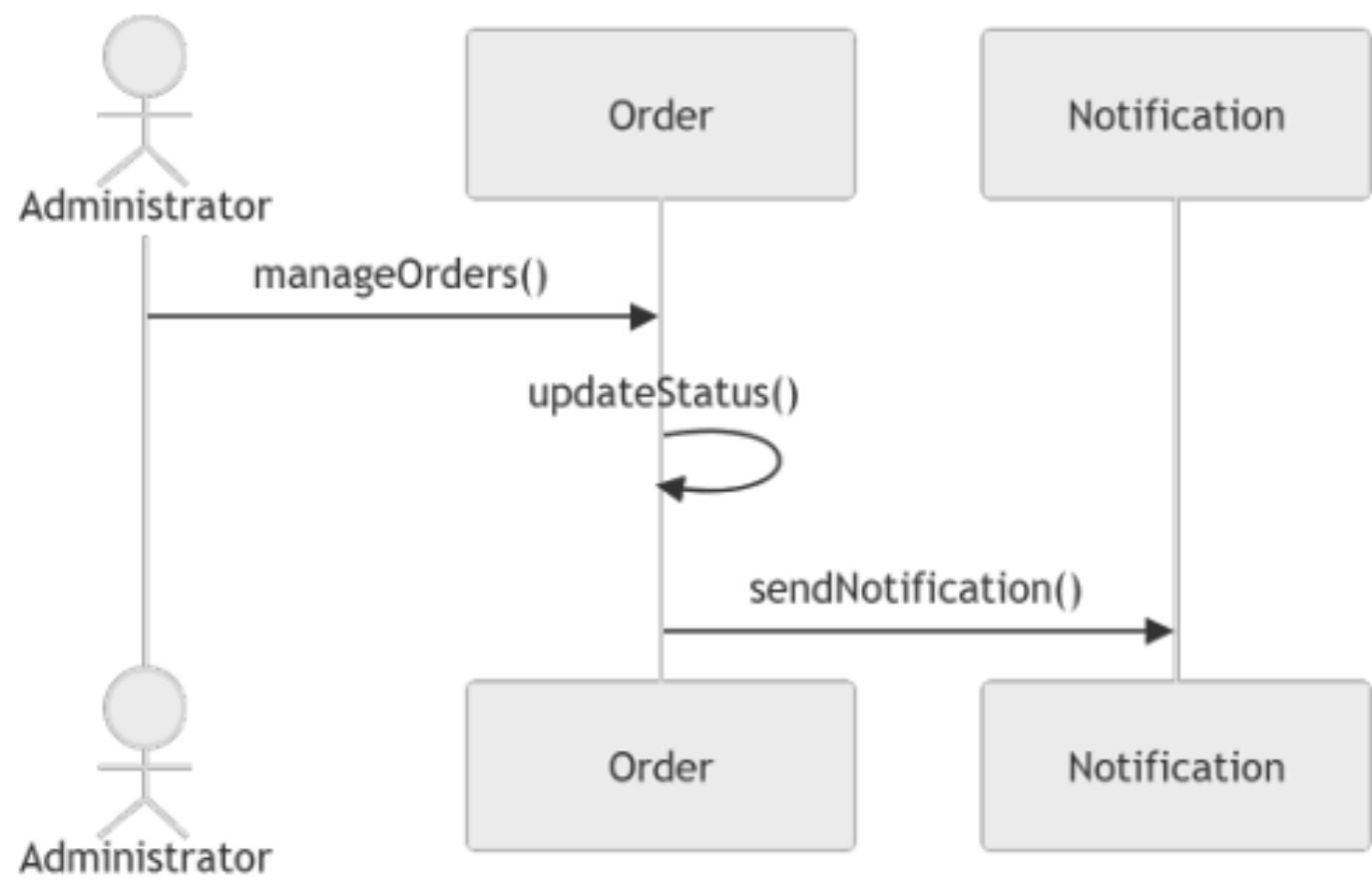
UC-26: Administrator Manages Users



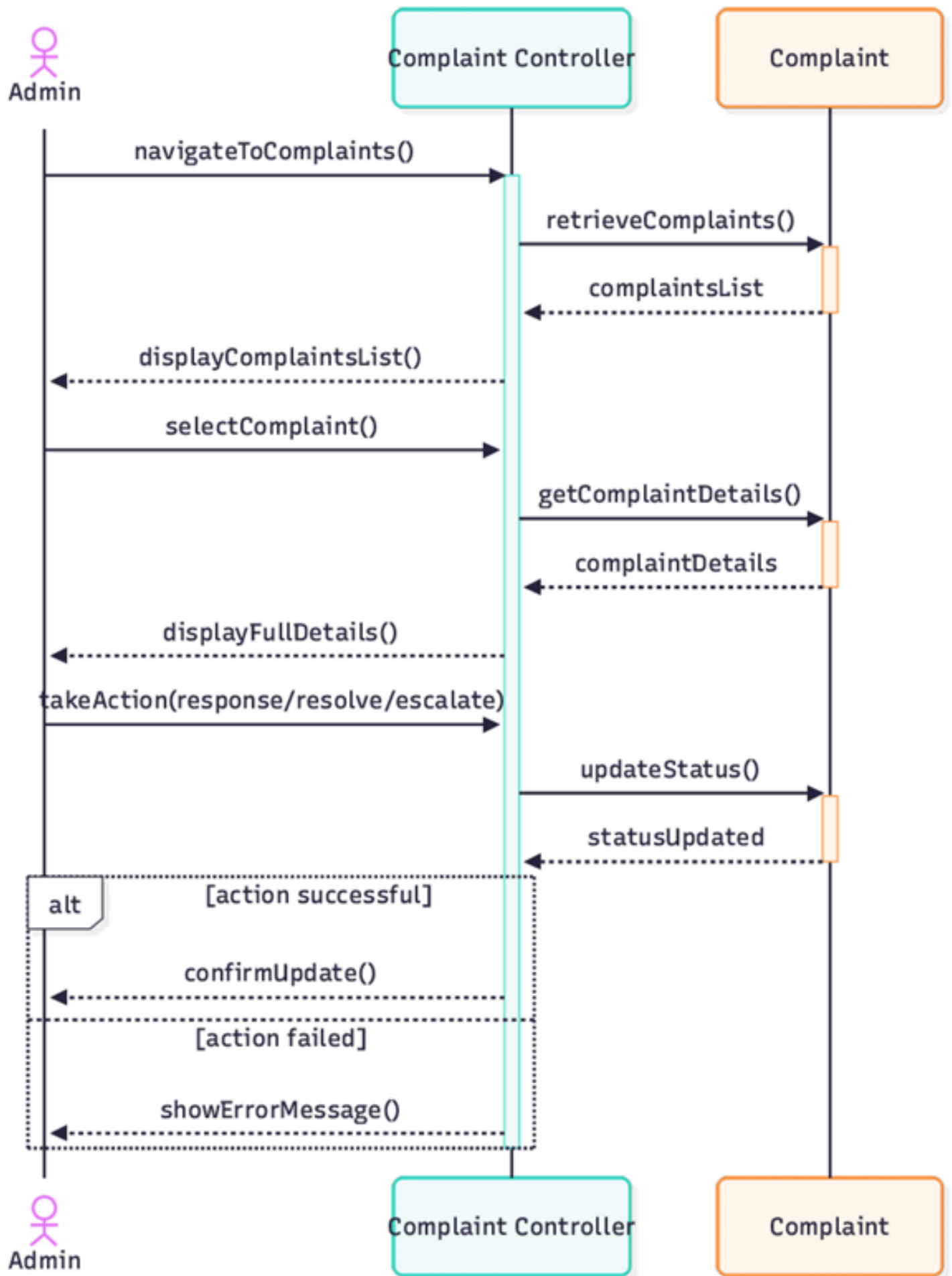
UC-27: Administrator Monitors Activity



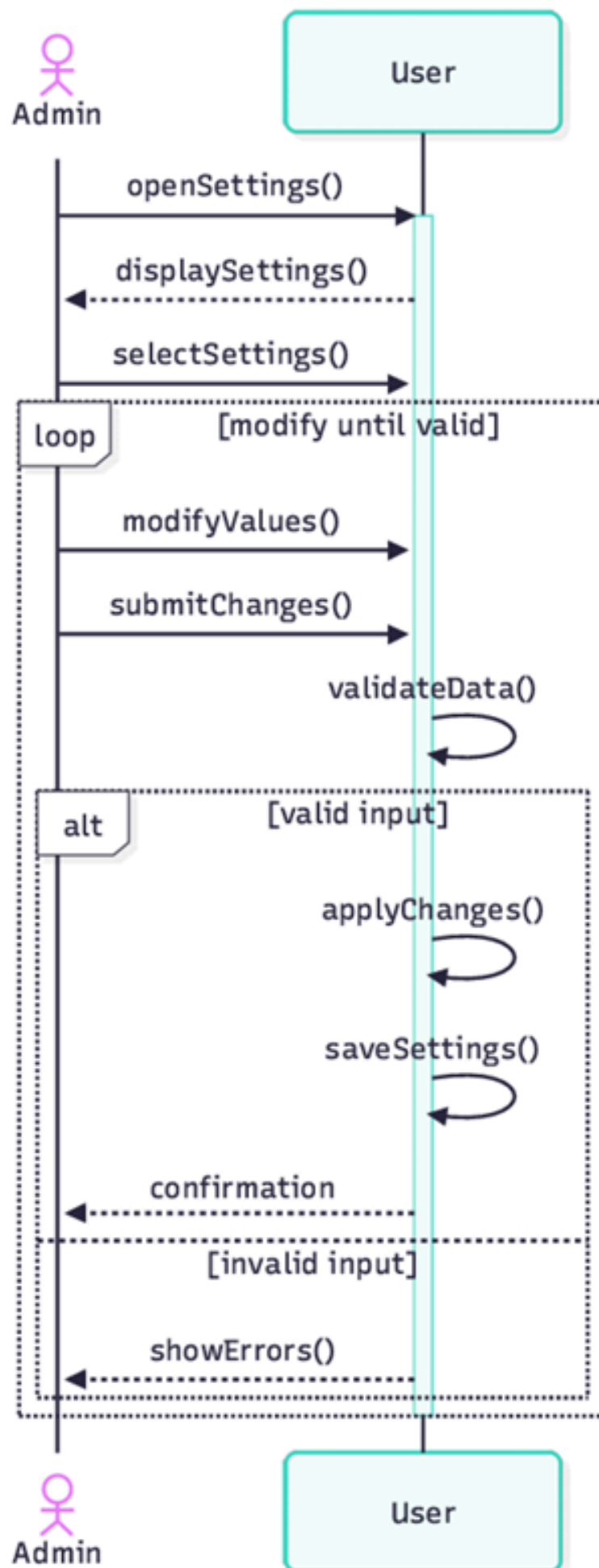
UC-28: Administrator Manages Orders



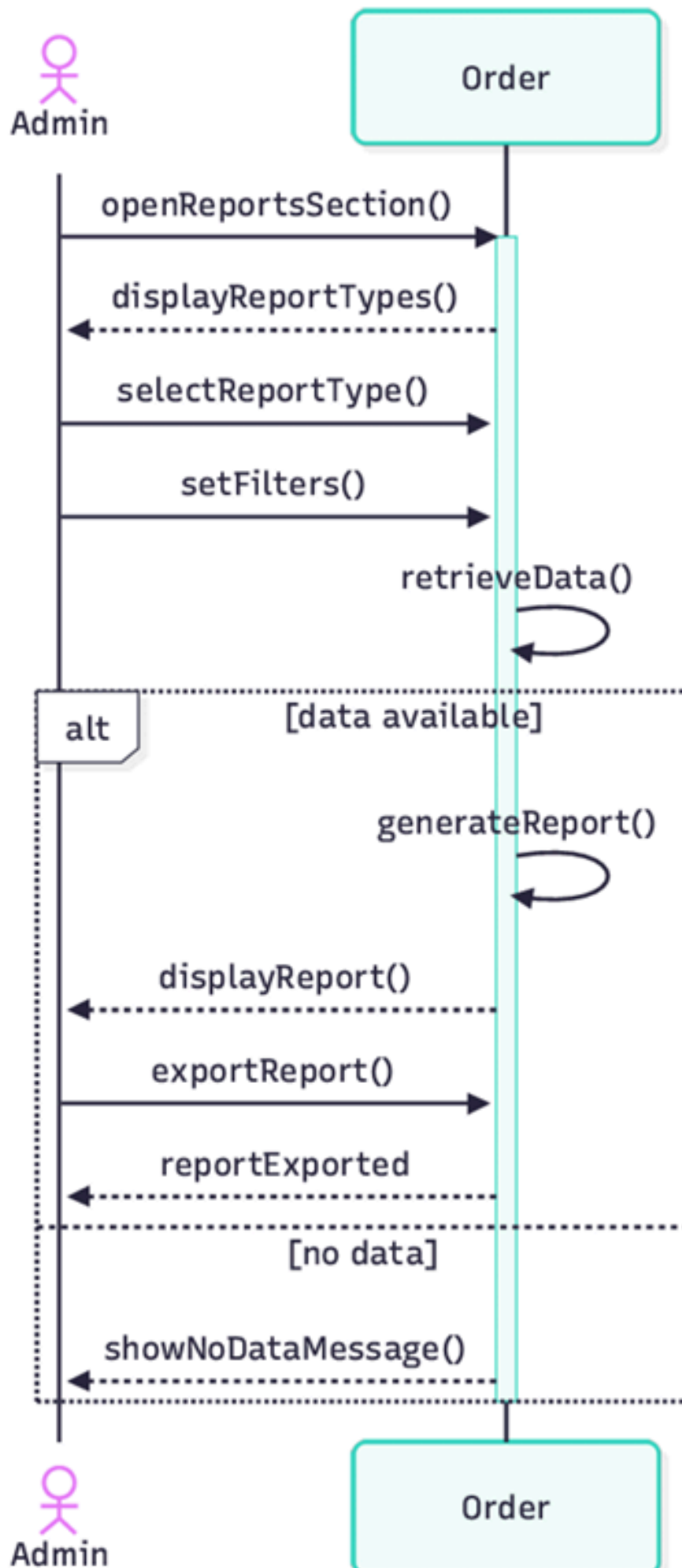
Sequence Diagram-UC-29- Handle Complaints



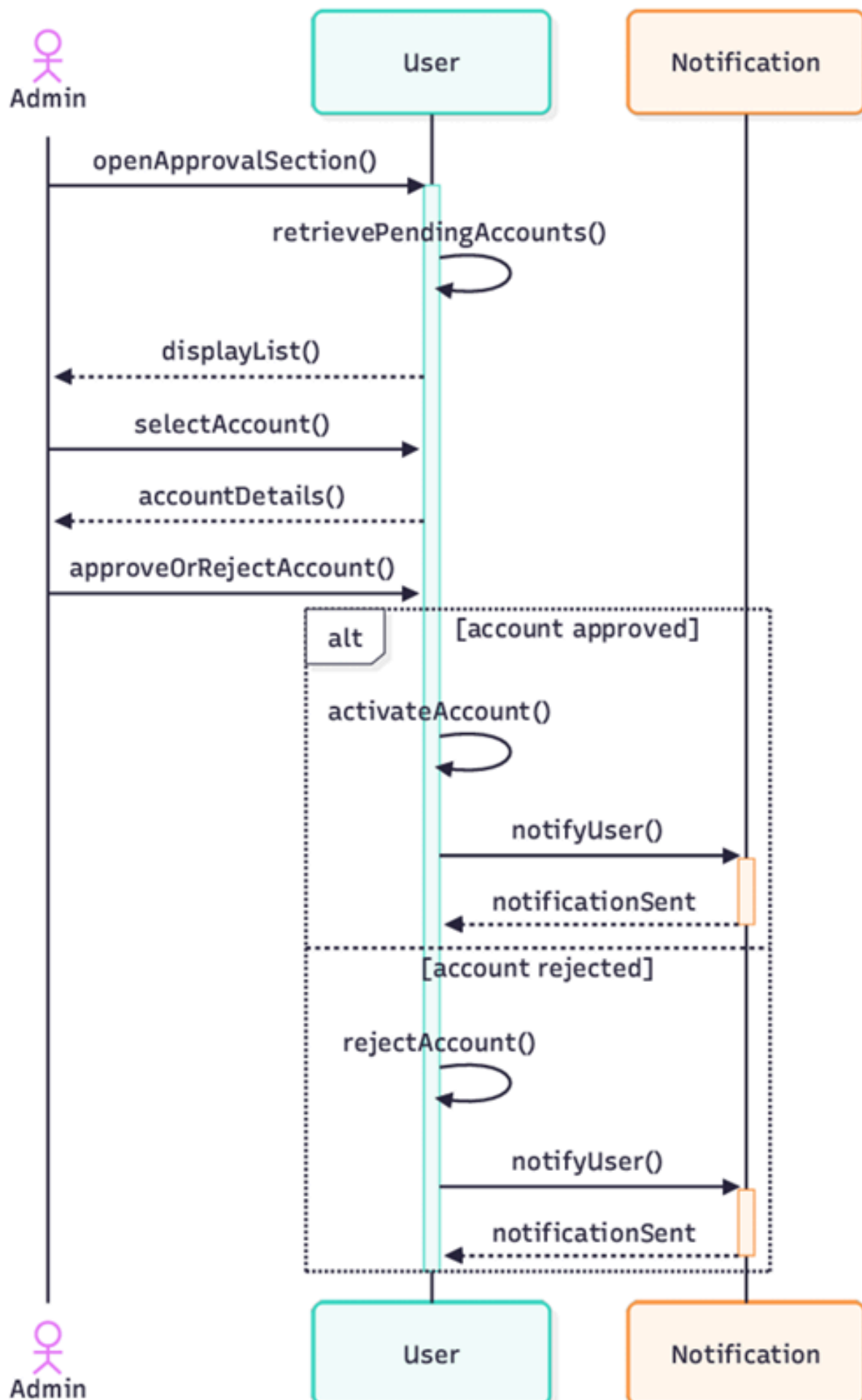
Sequence Diagram-UC-30- Manage Settings



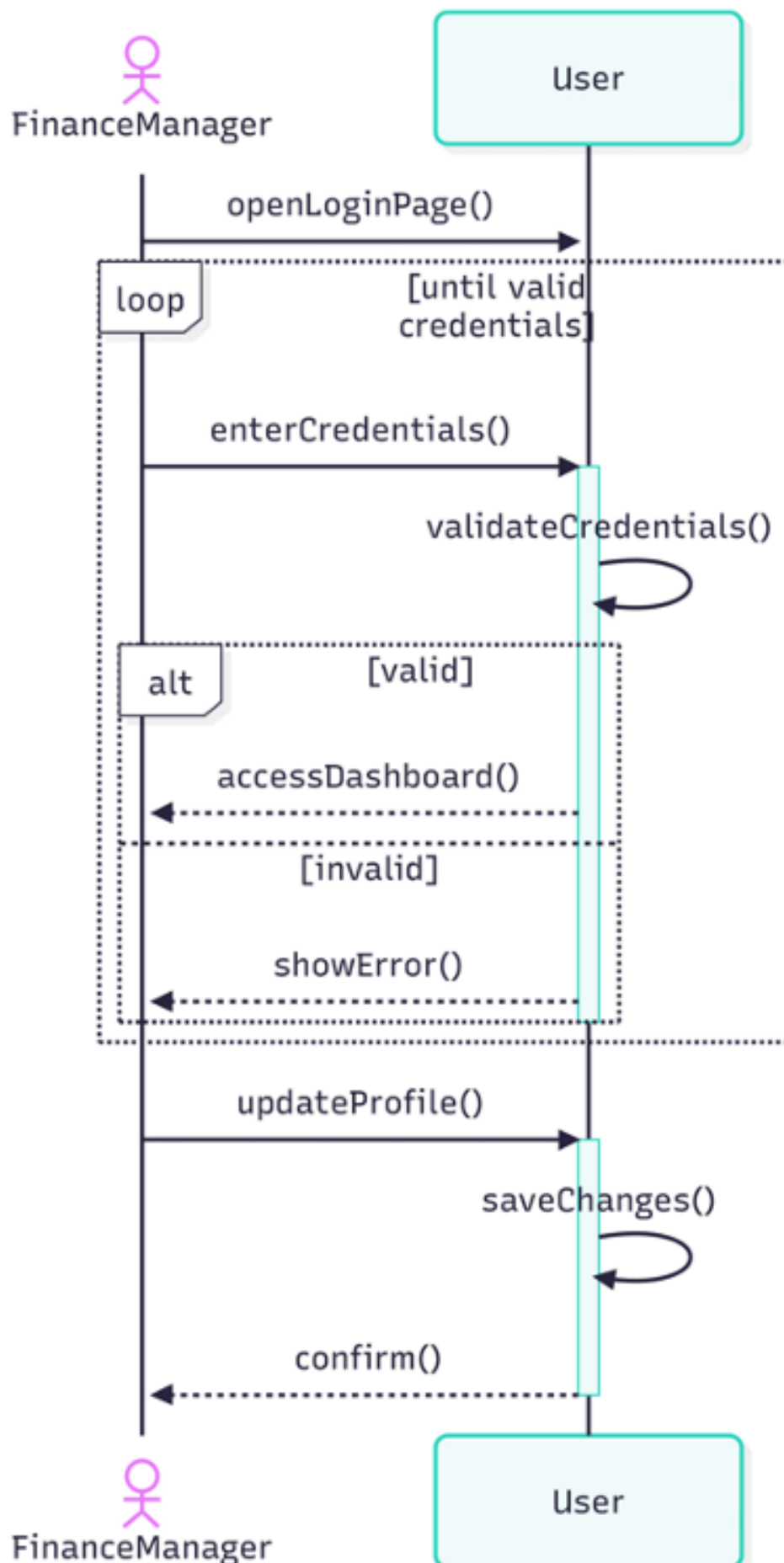
Sequence Diagram-UC-31-Generate Reports



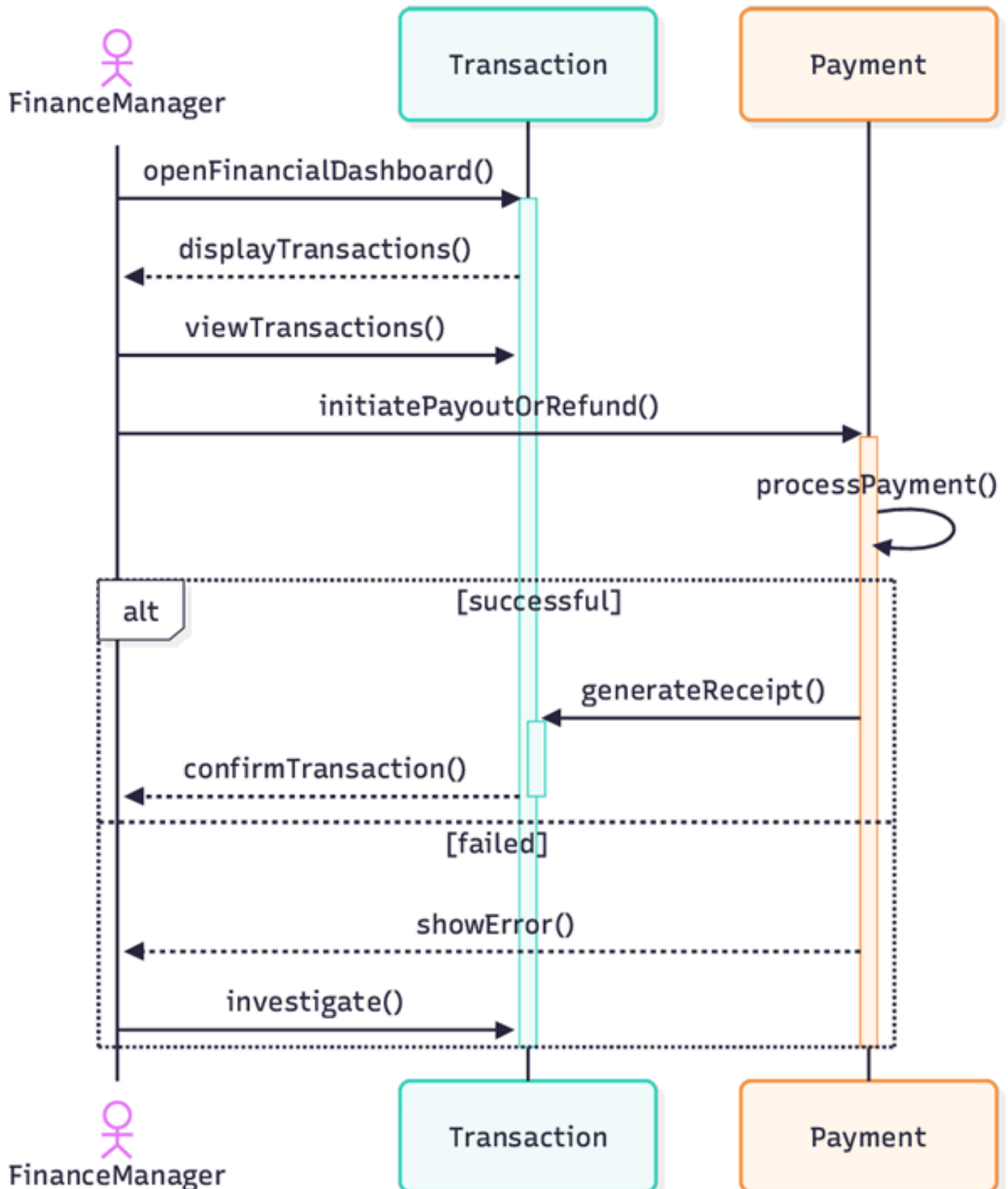
Sequence Diagram-UC-32- Account Approvals



Sequence Diagram-UC-33- Finance Manager Account

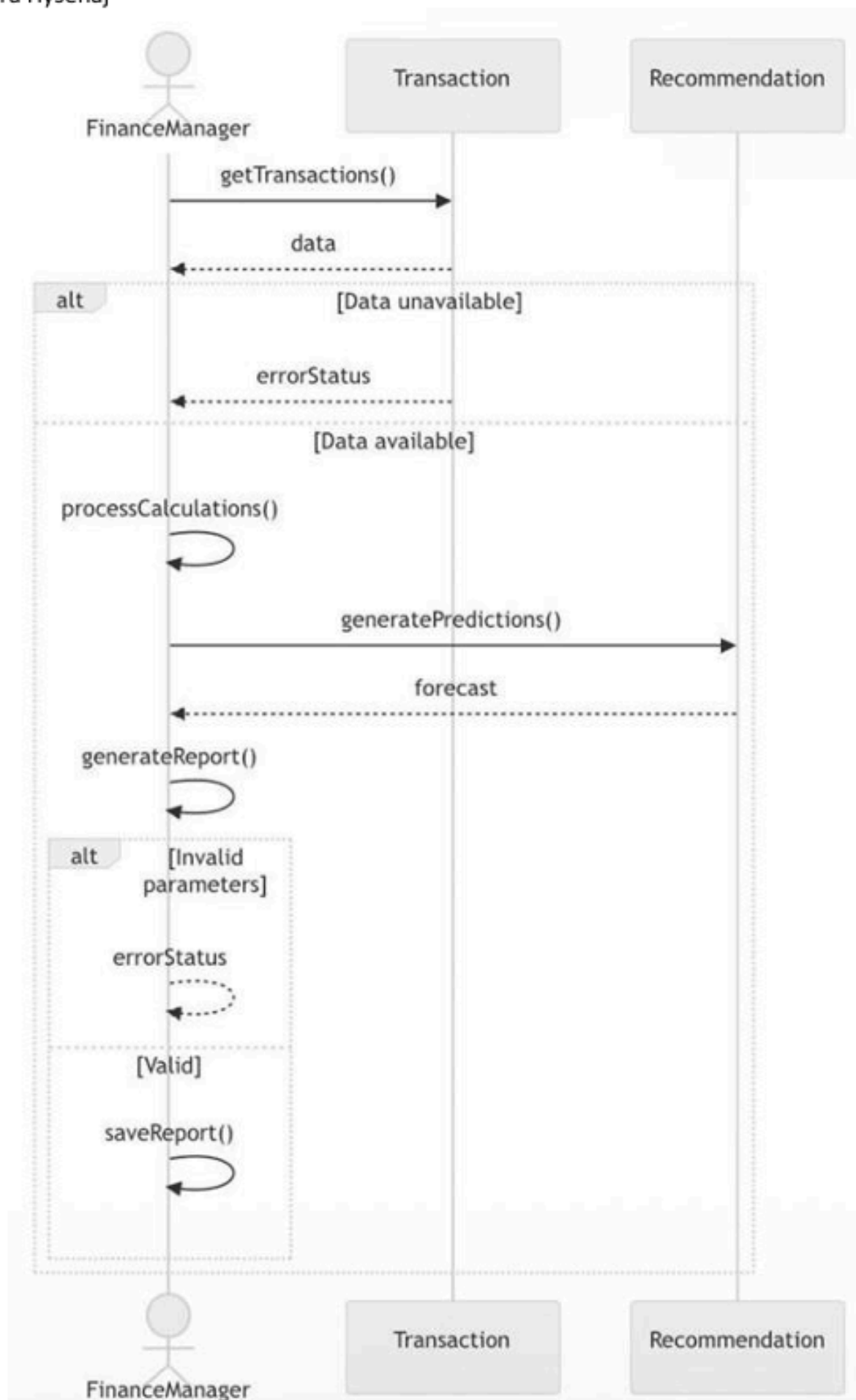


Sequence Diagram-UC-34- Billing / Refunds

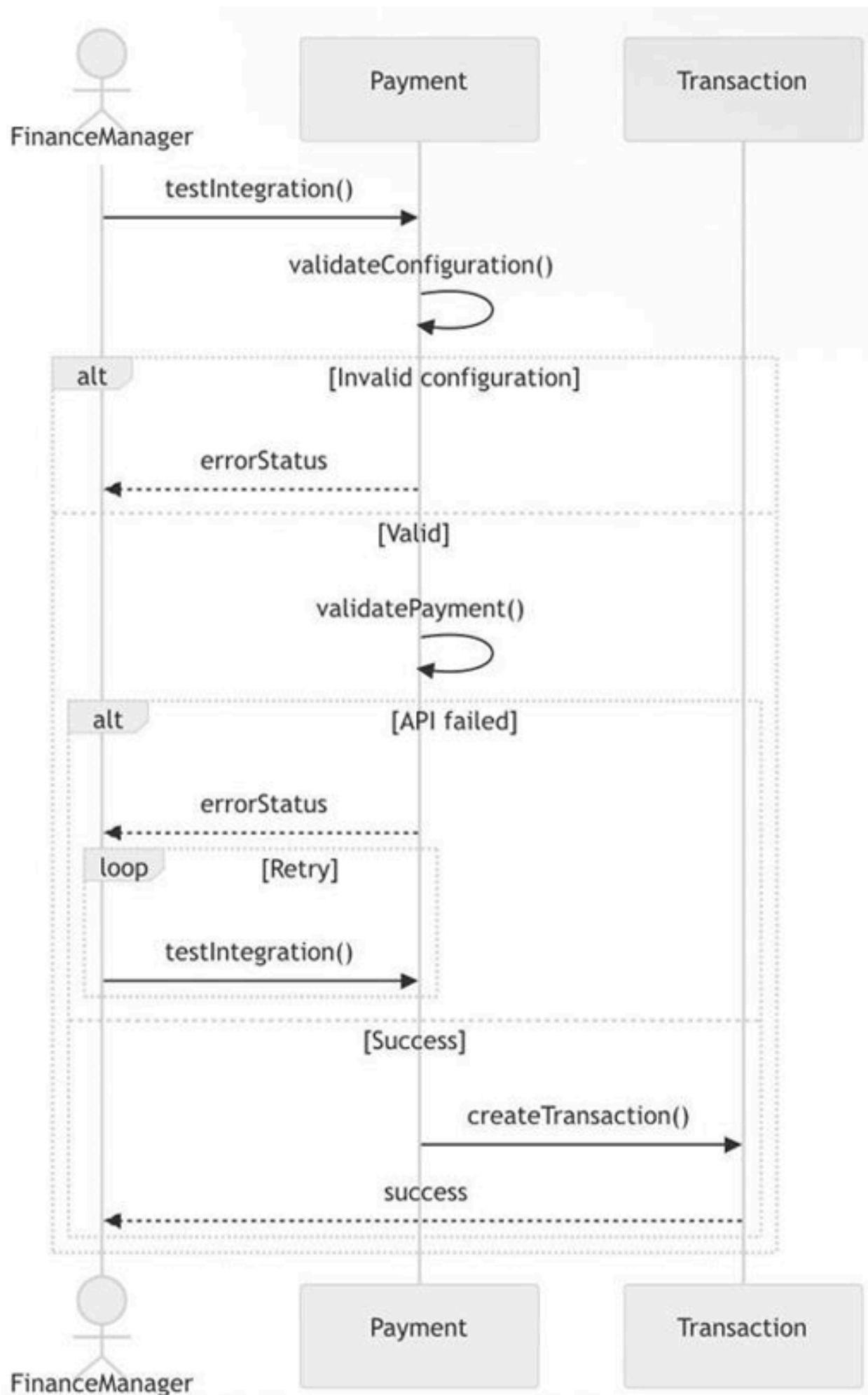


UC-35 - Generate Financial Reports and Forecasting

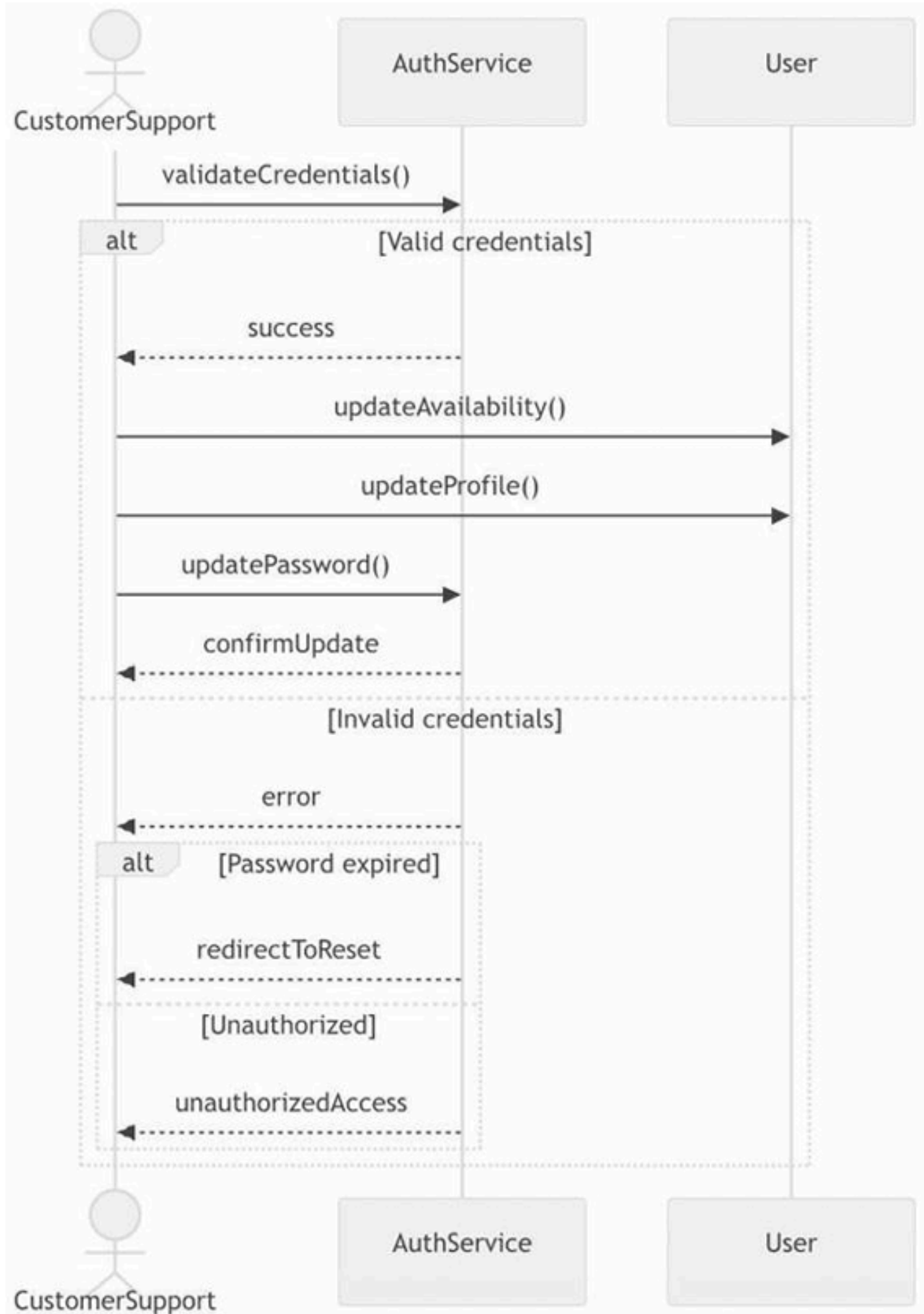
Author-Klara Hysenaj



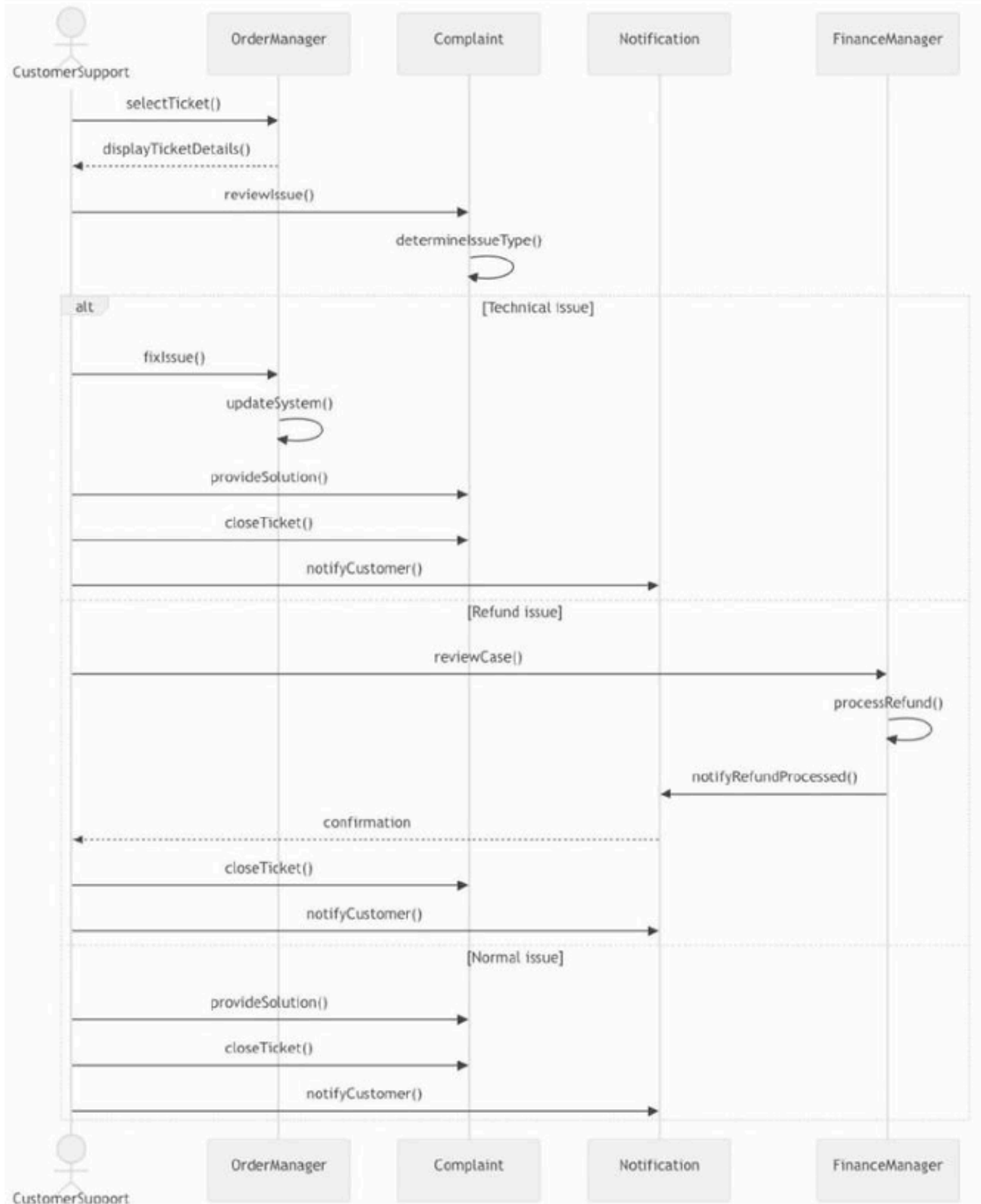
UC-36 - Manage Financial Operations



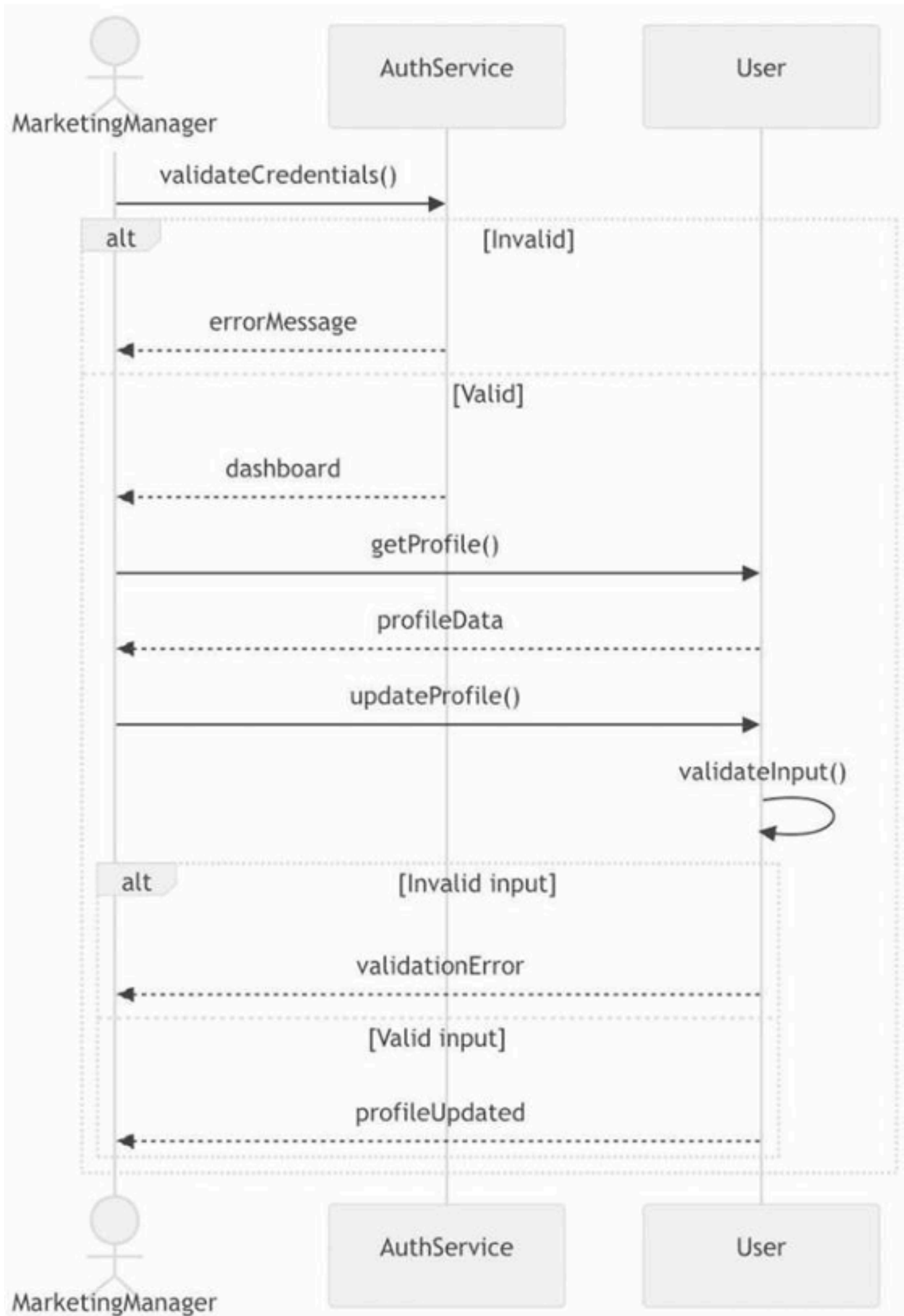
UC -37 - Customer Support Account Management



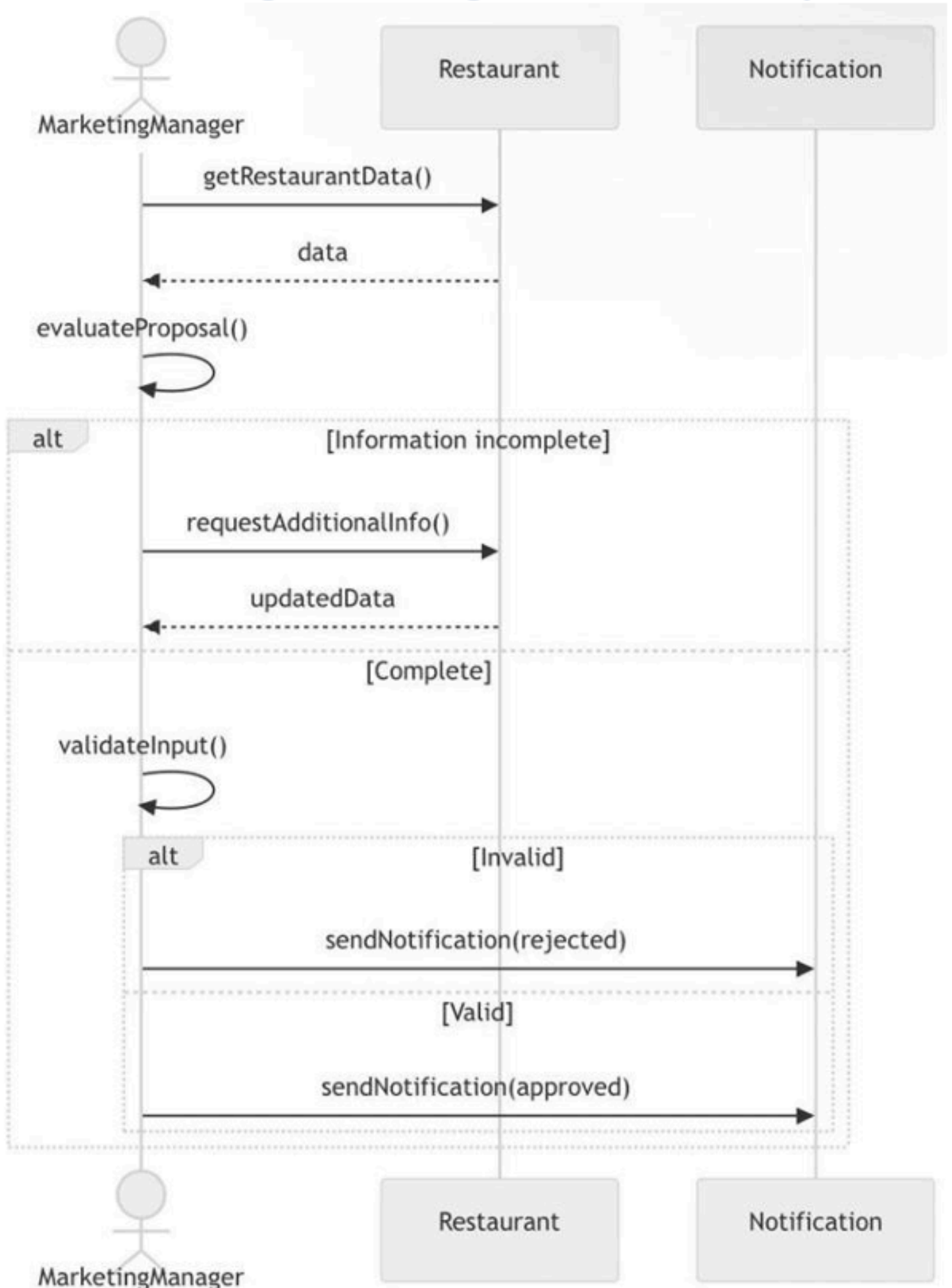
UC-38 - Manage User Support Operations



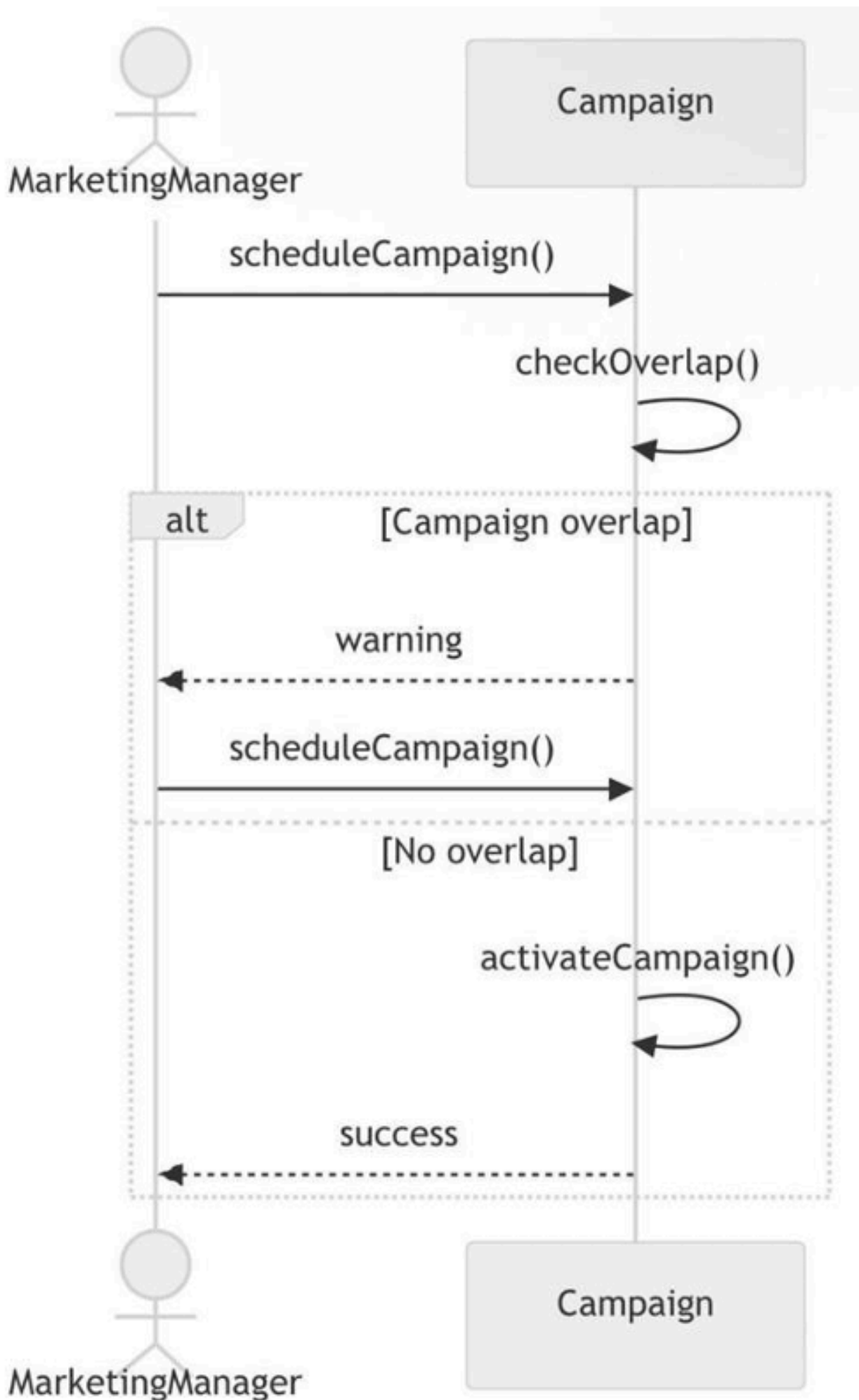
UC- 39-Manage Marketing Manager Account



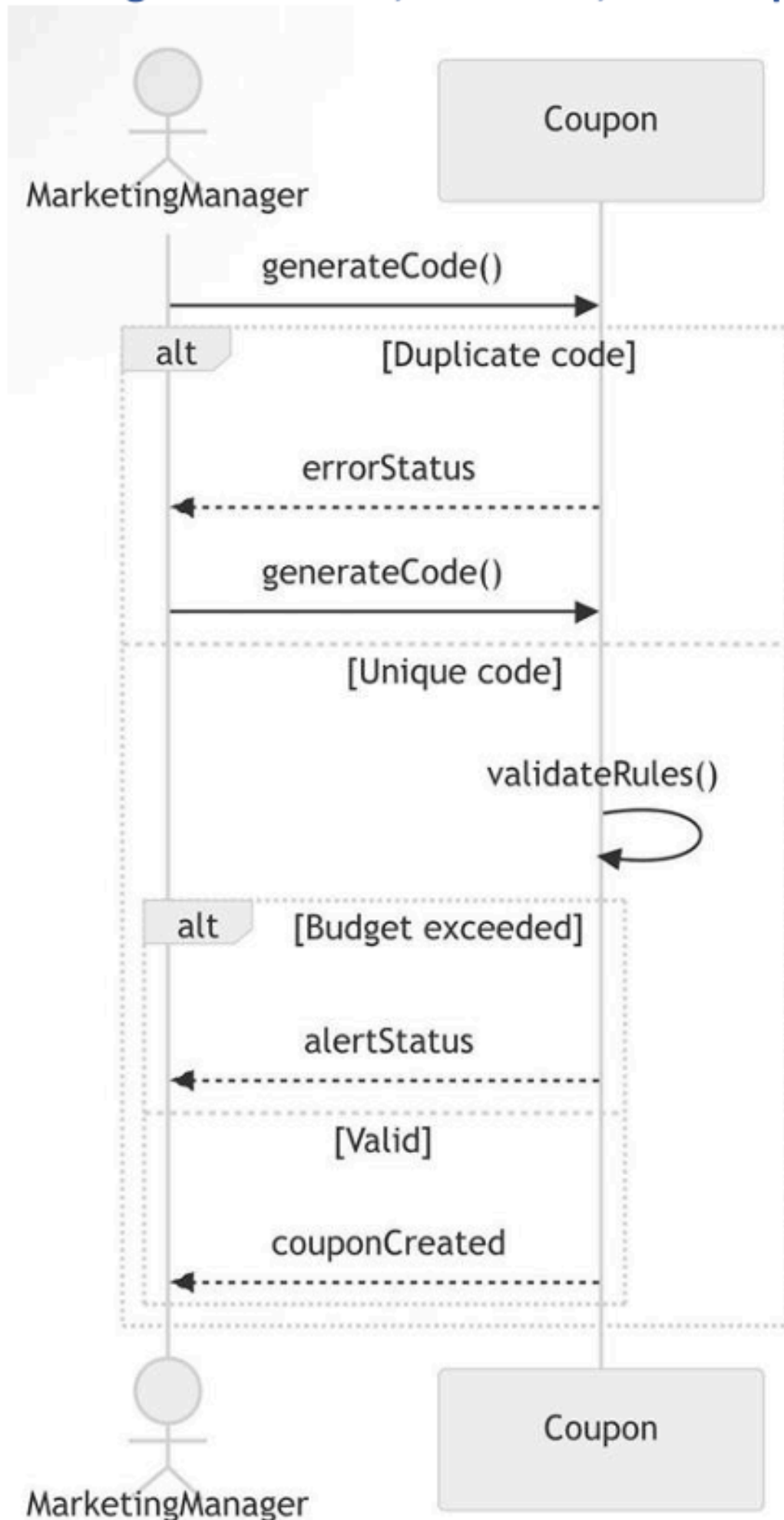
UC-40-ManageMarketing Collaboration Requests



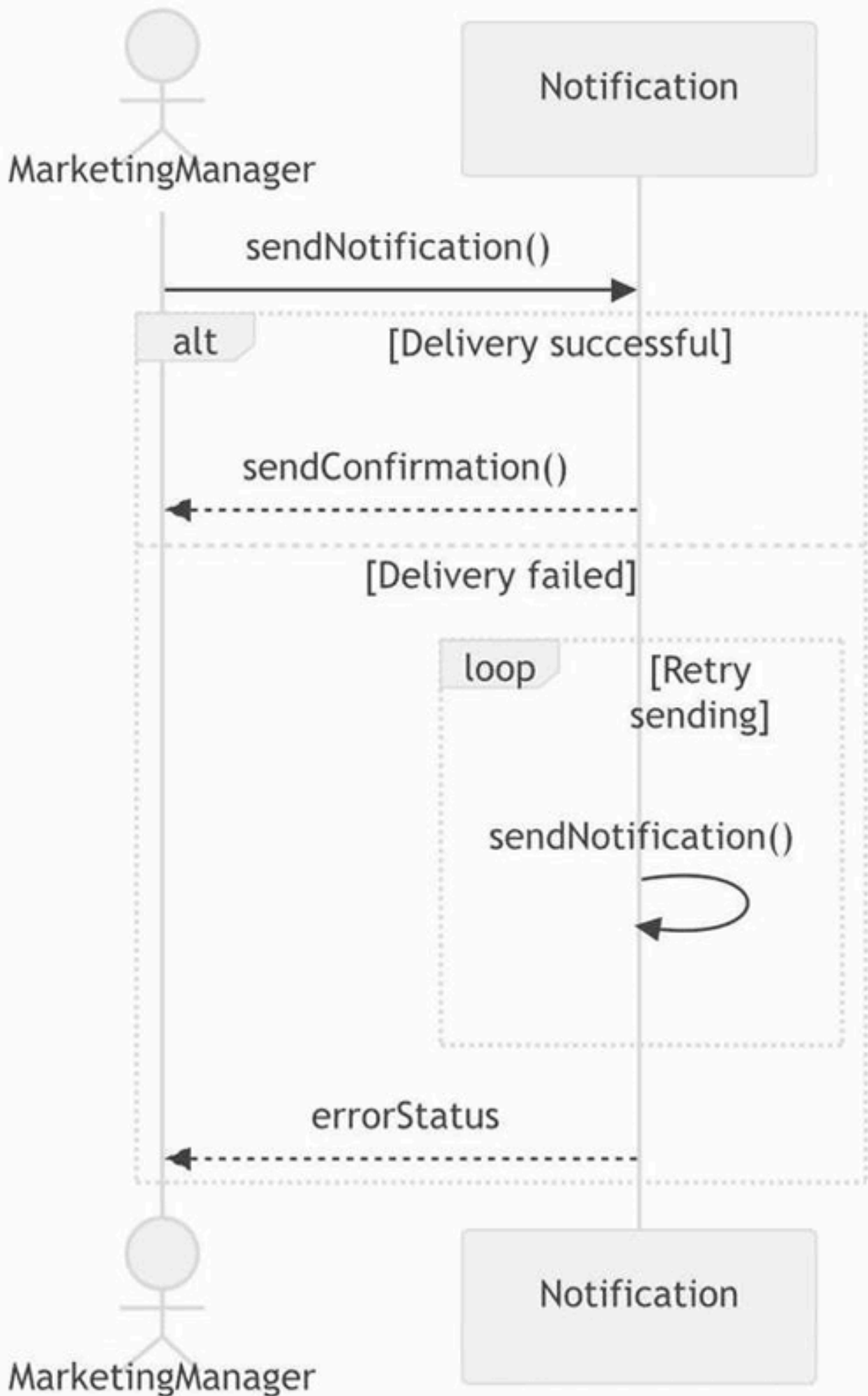
UC-41 – Manage Marketing Campaigns



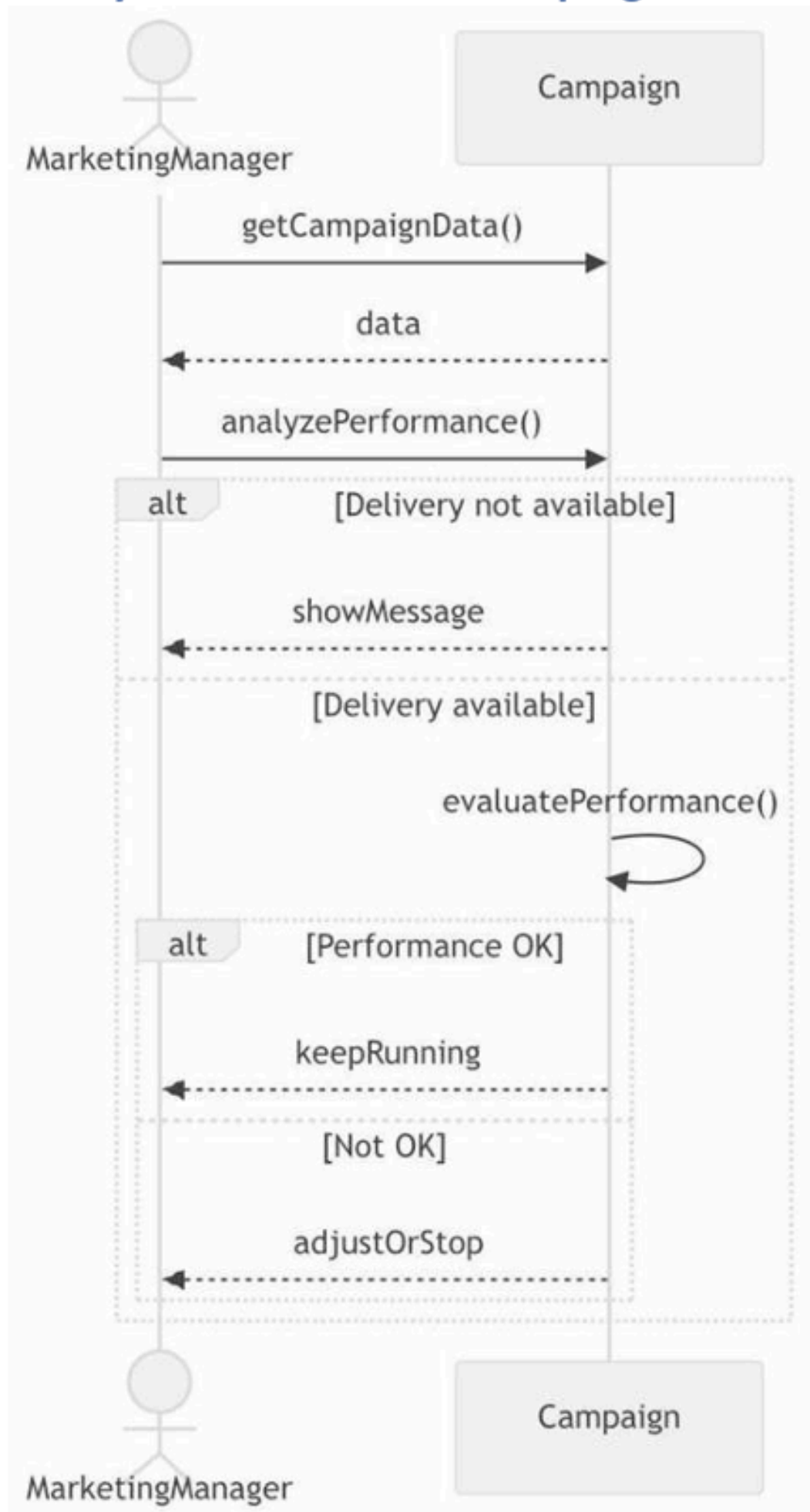
UC-42 – Manage Promotions, Discounts, and Coupon Codes



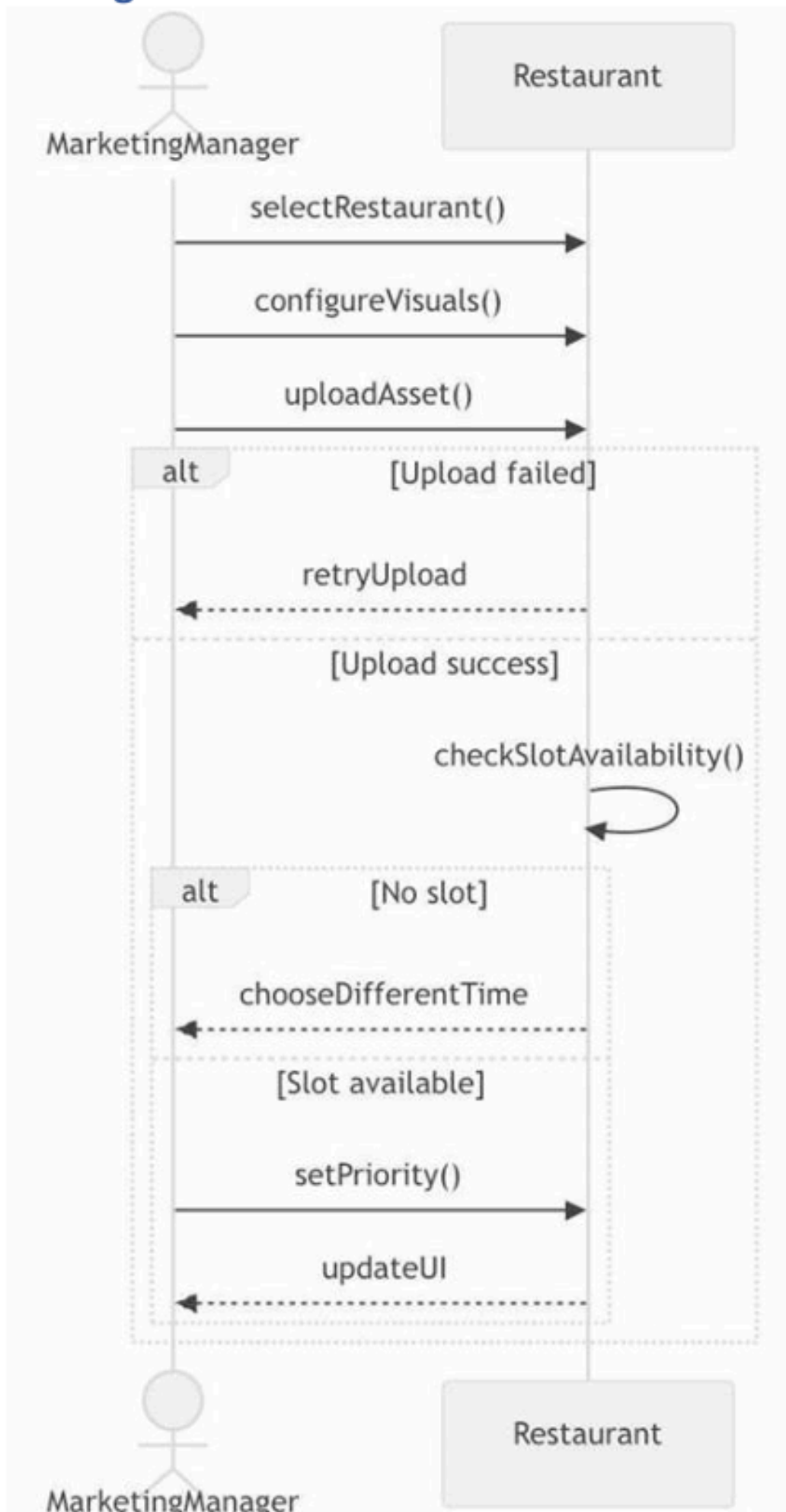
UC-43 – Manage Promotional Communication



UC-44 Analyze and Monitor Campaign Performance



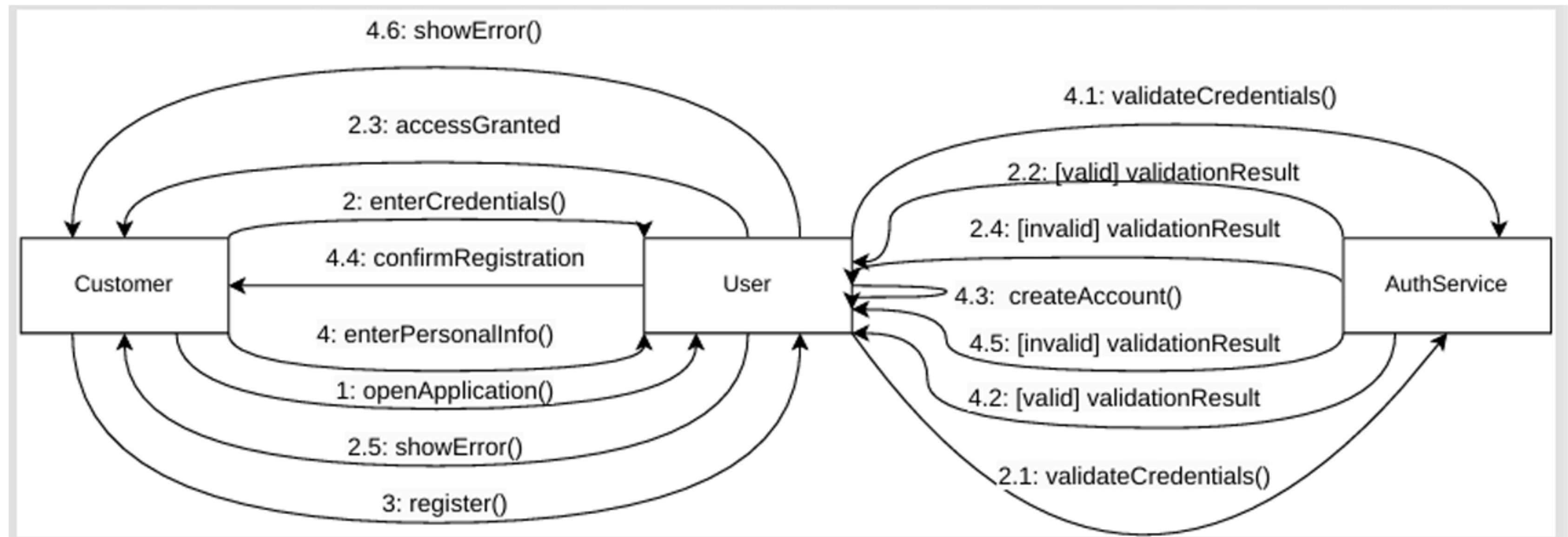
UC-45 – Manage Featured Restaurants and Advertisements



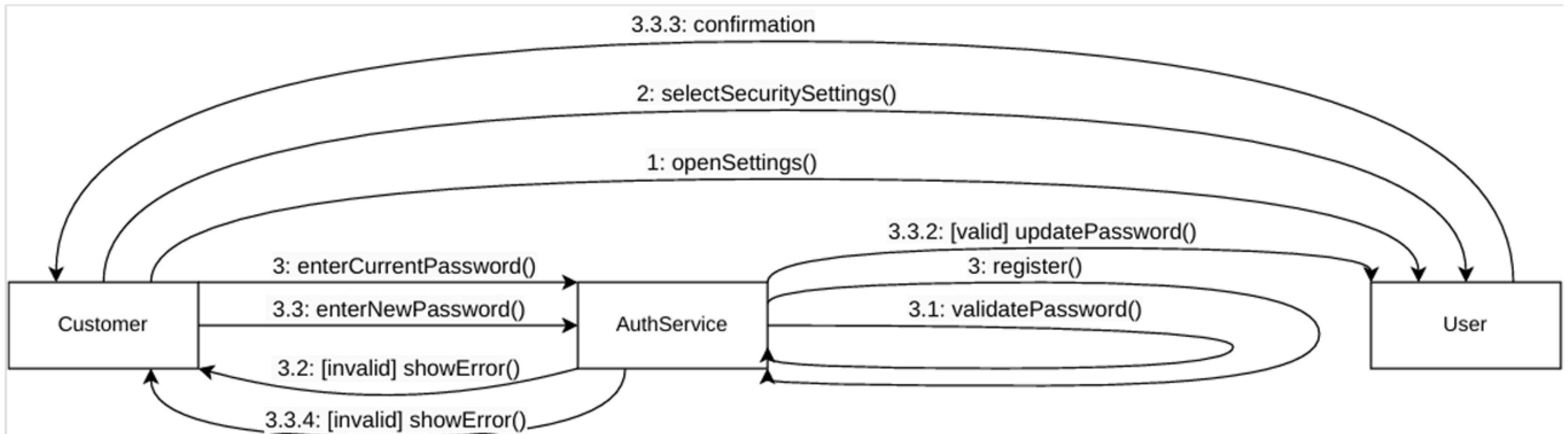
Collaboration diagrams

Author-Sofia Cala

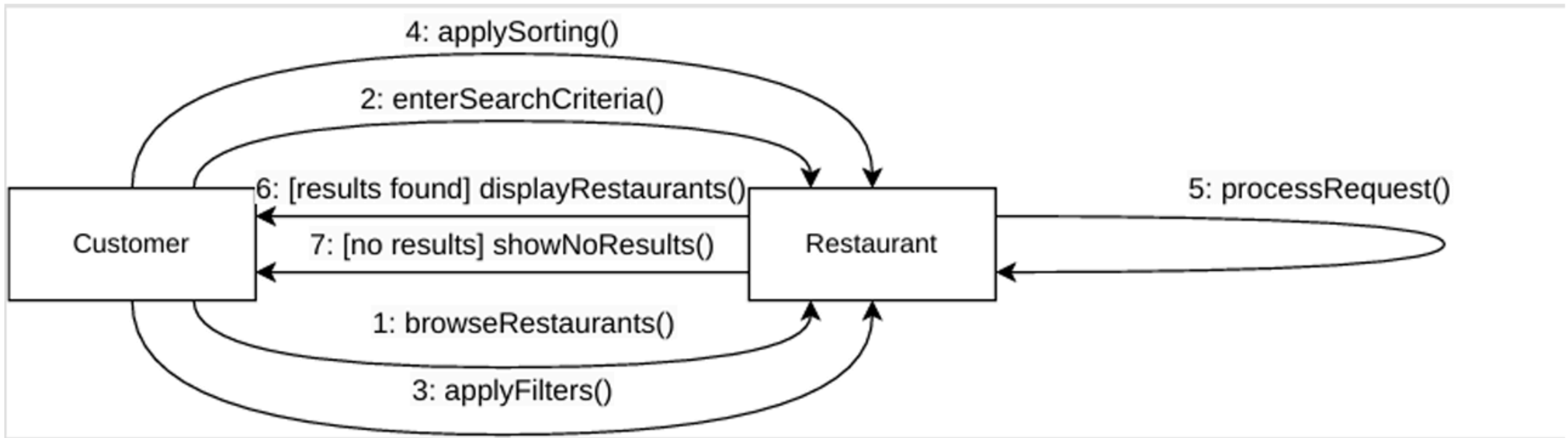
Collaboration Diagram-UC-01-Manage Customer Account



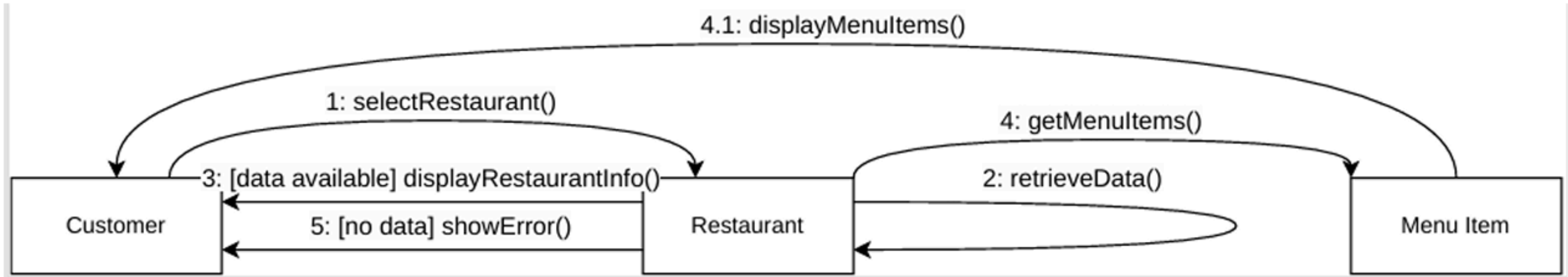
Collaboration Diagram-UC-02-Authentication&Security Settings



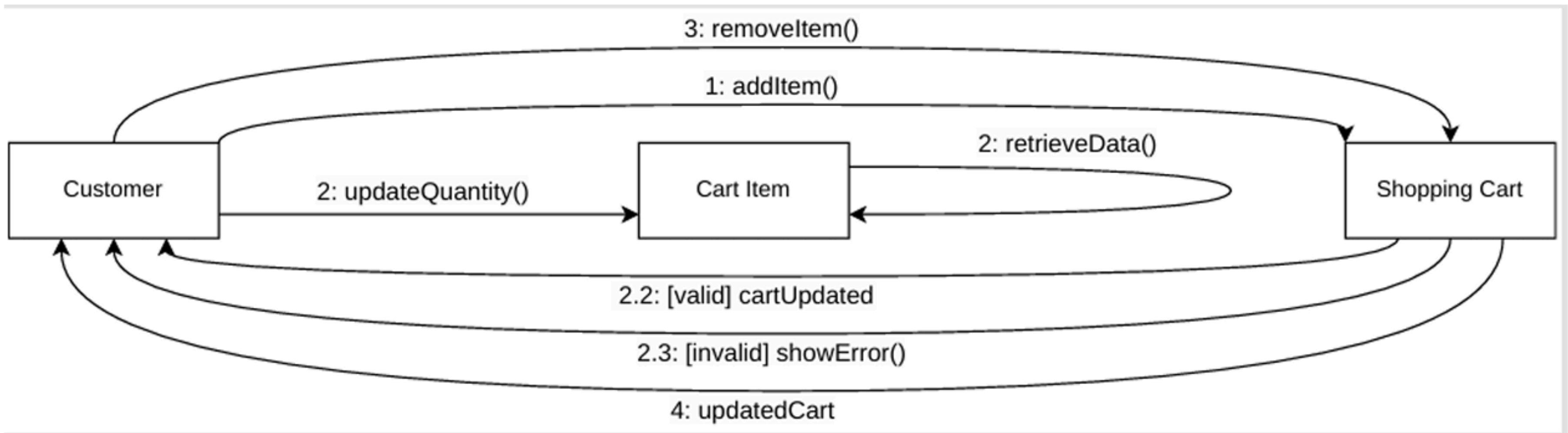
Collaboration Diagram-UC-03-Browse/Search/Filter Restaurant



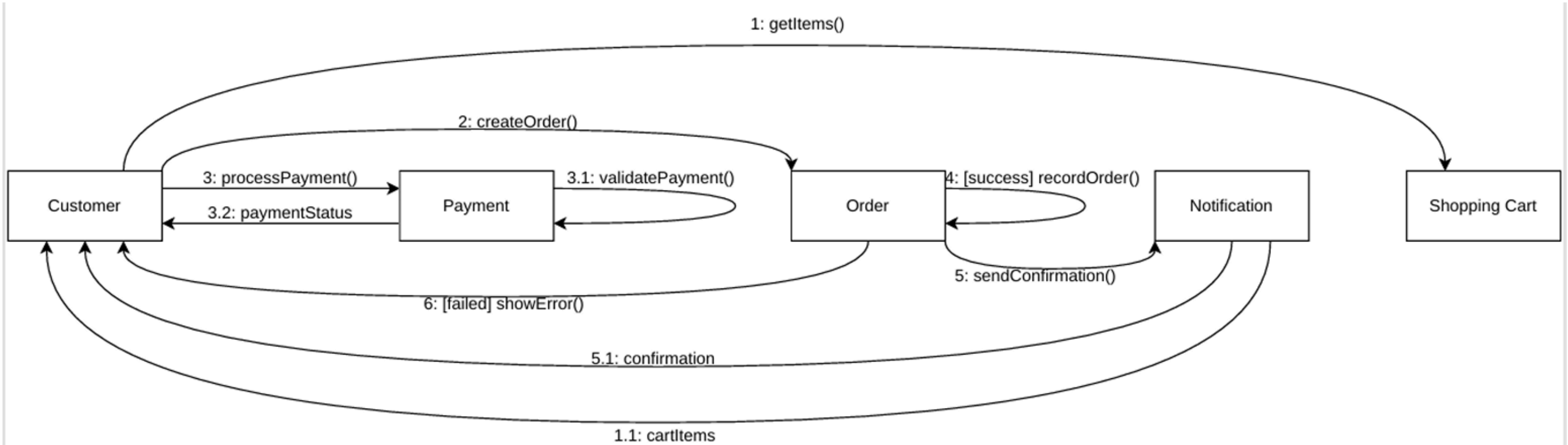
Collaboration Diagram-UC-04-View Restaurant & Food Details



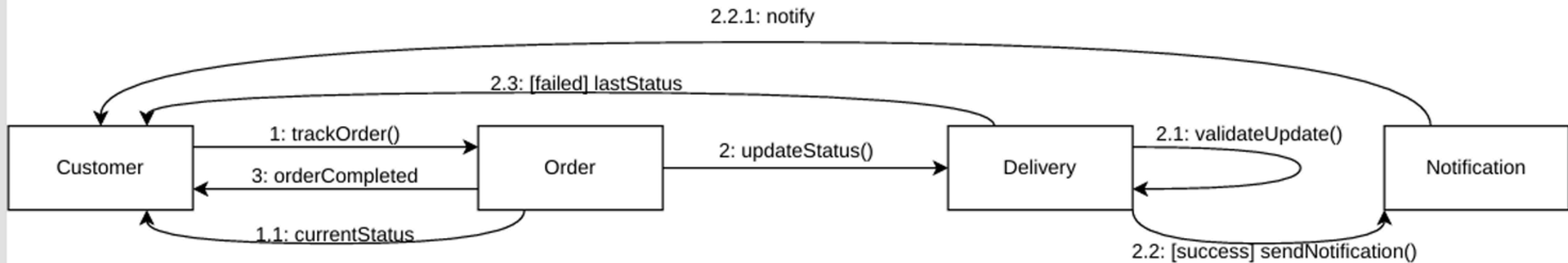
Collaboration Diagram-UC-05-Manage Shopping Cart



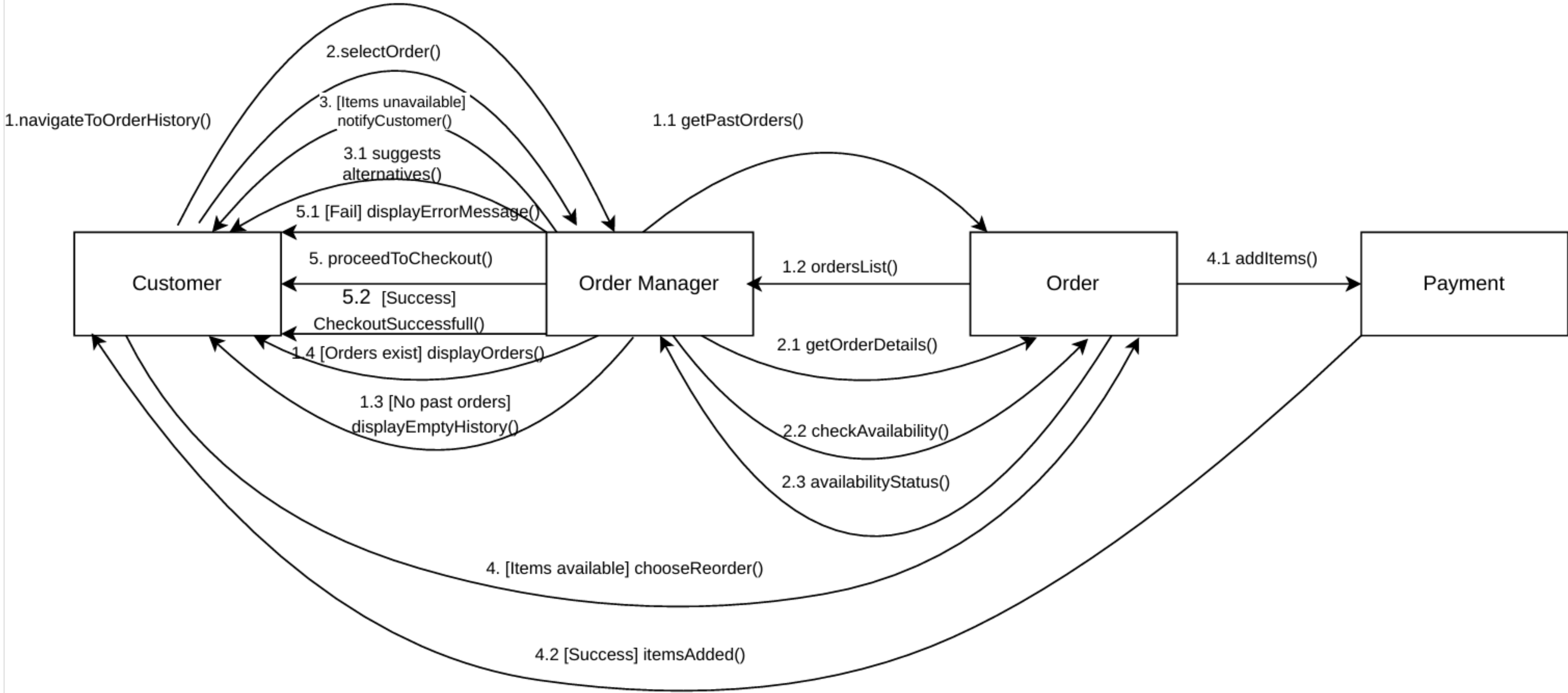
Collaboration Diagram-UC-06-Place Order



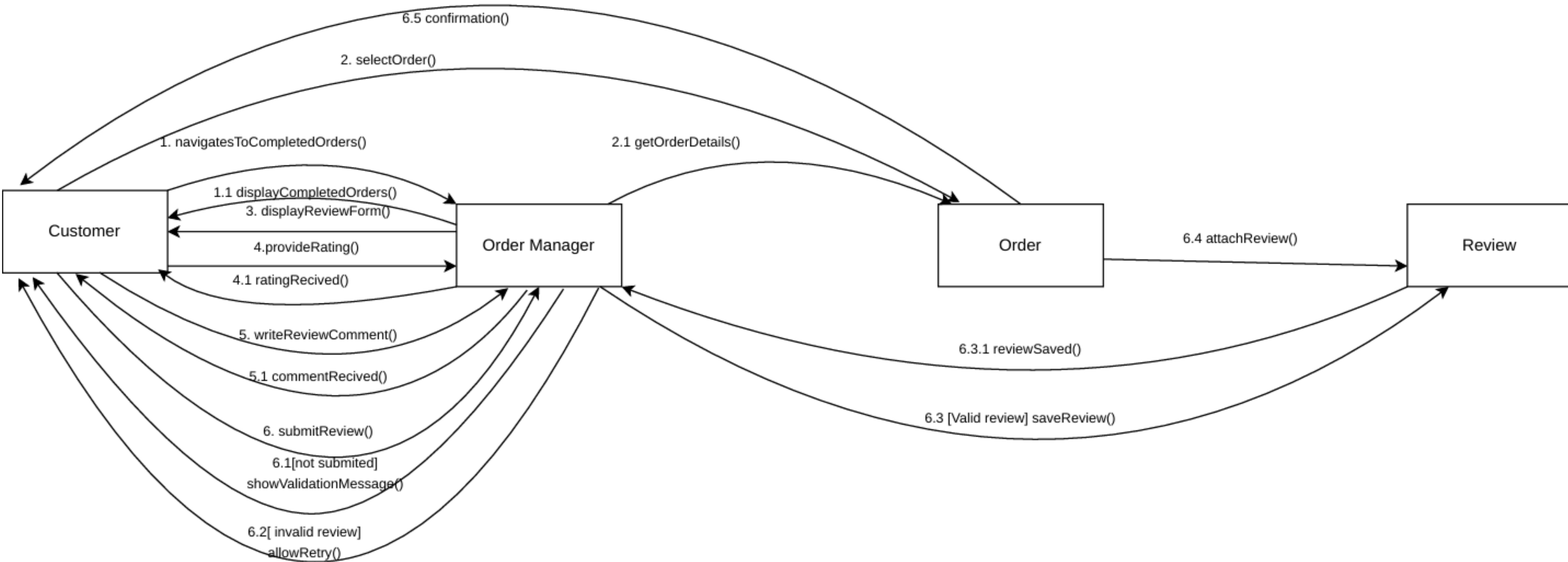
Collaboration Diagram-UC-07-Track Order & Notifications



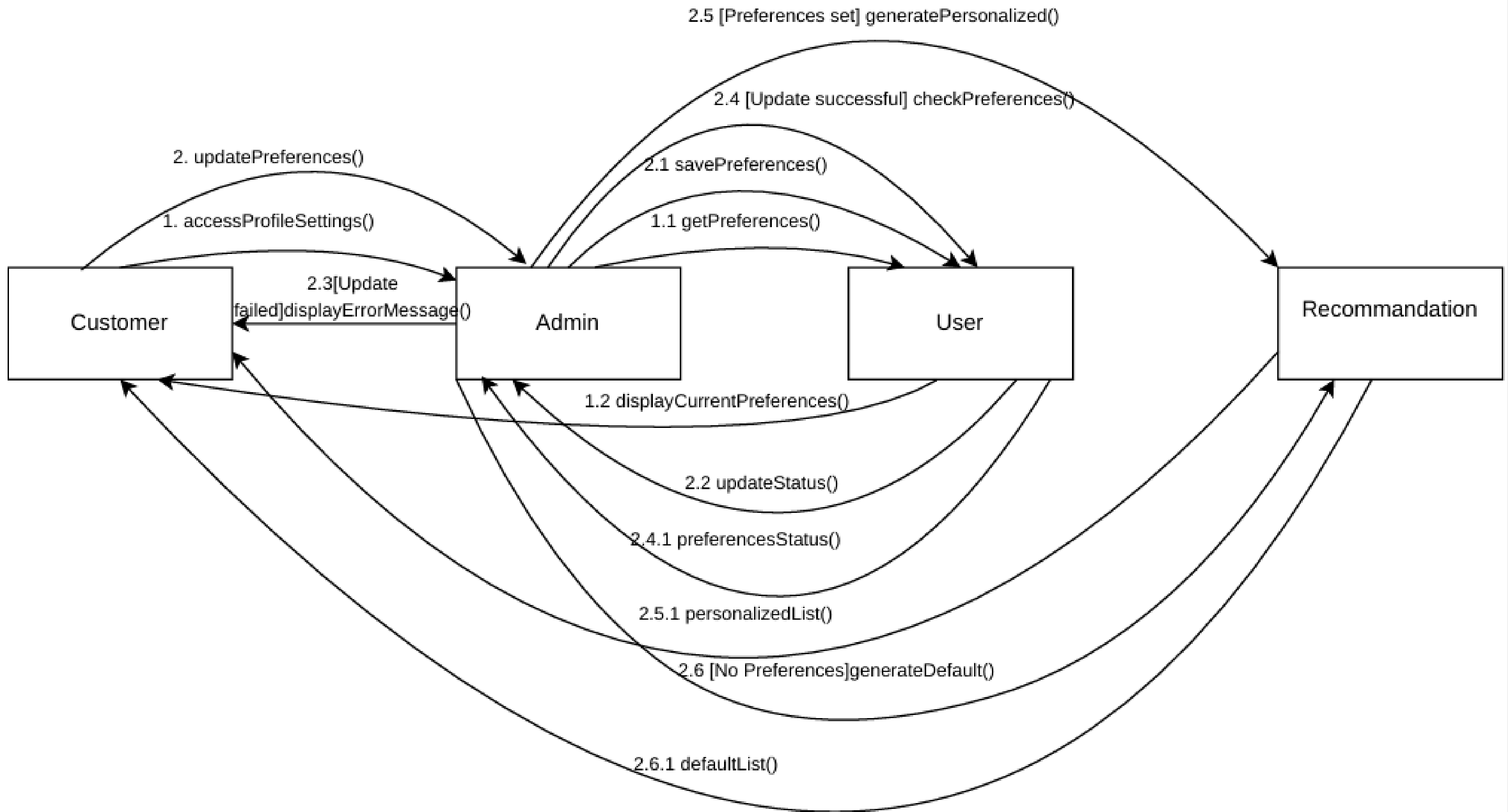
Collaboration Diagram 08-Xhesika Toska



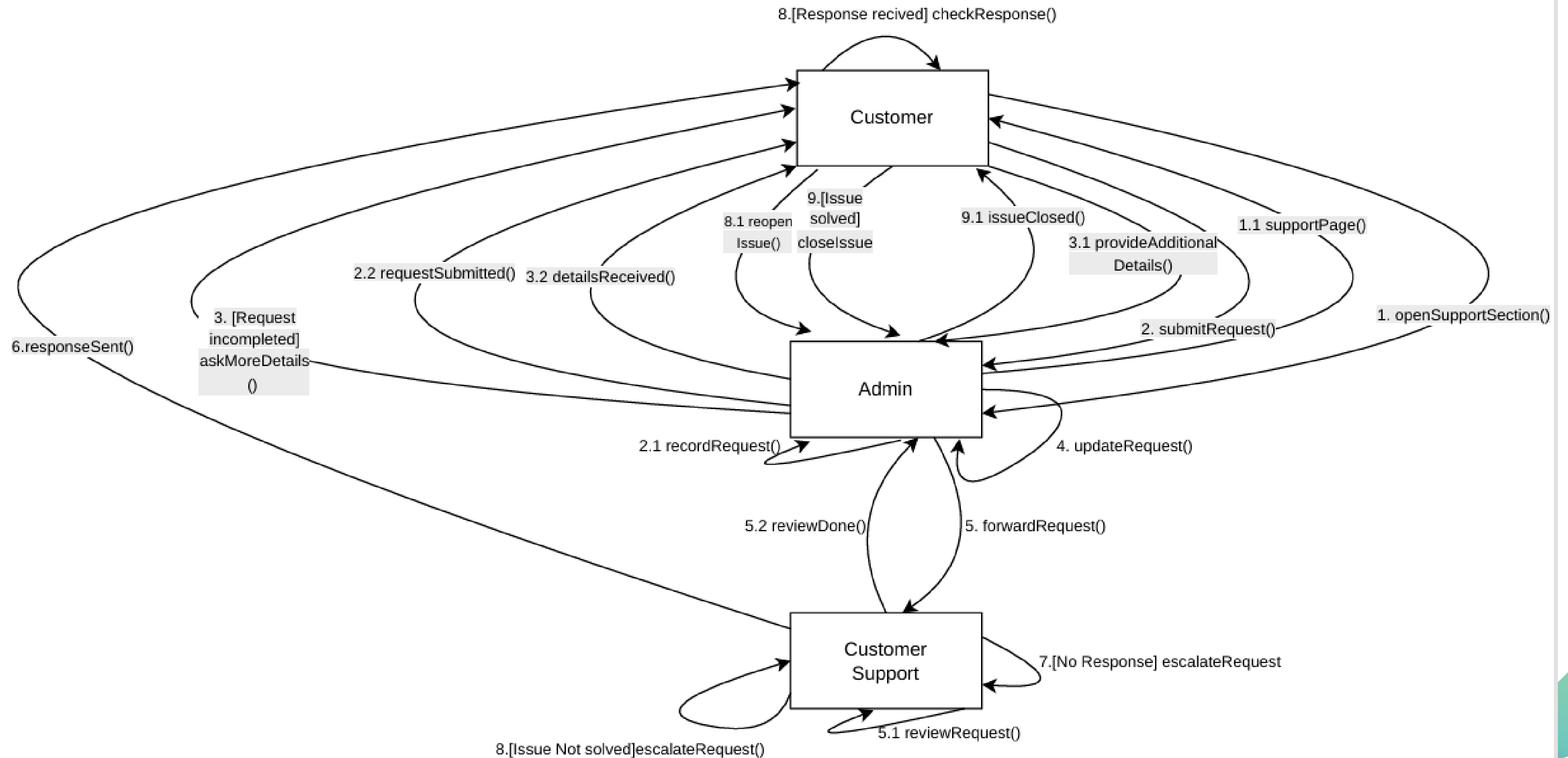
Collaboration Diagram 09



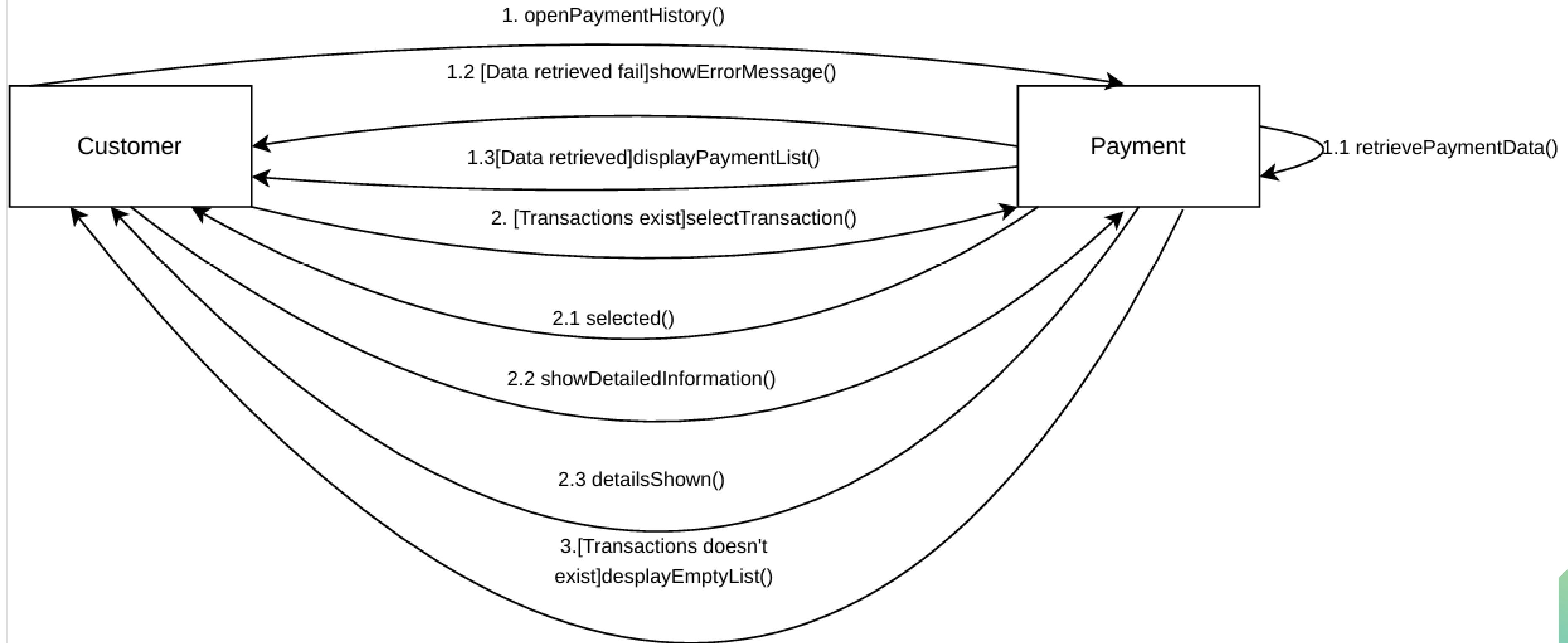
Collaboration Diagram 10



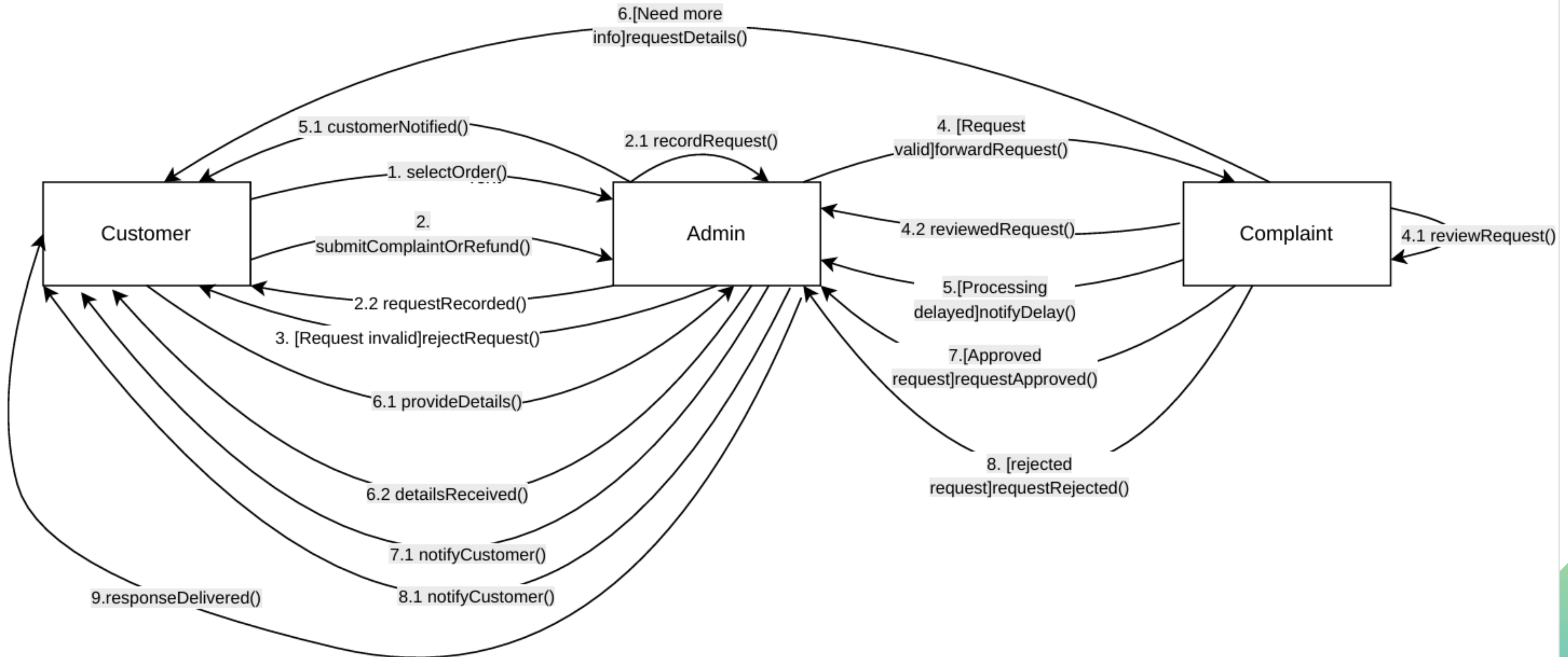
Collaboration Diagram 11



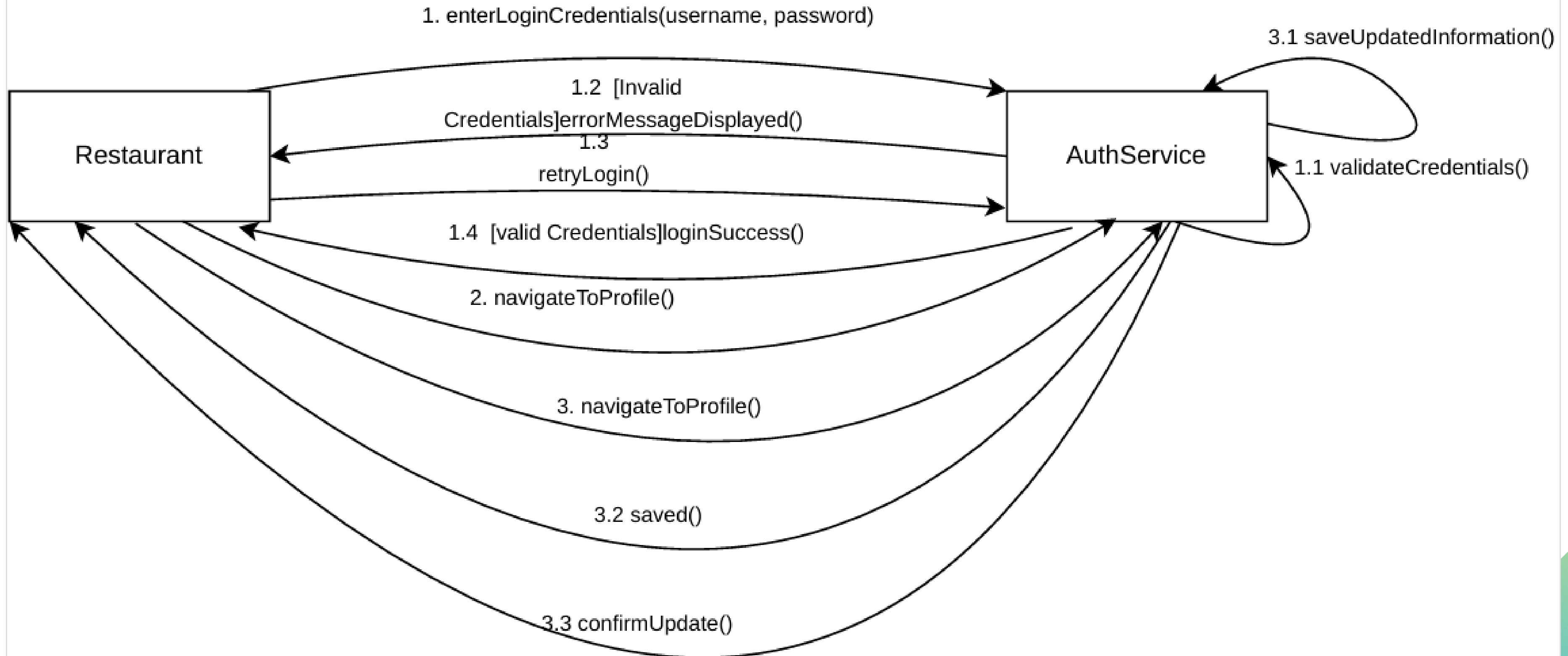
Collaboration Diagram 12



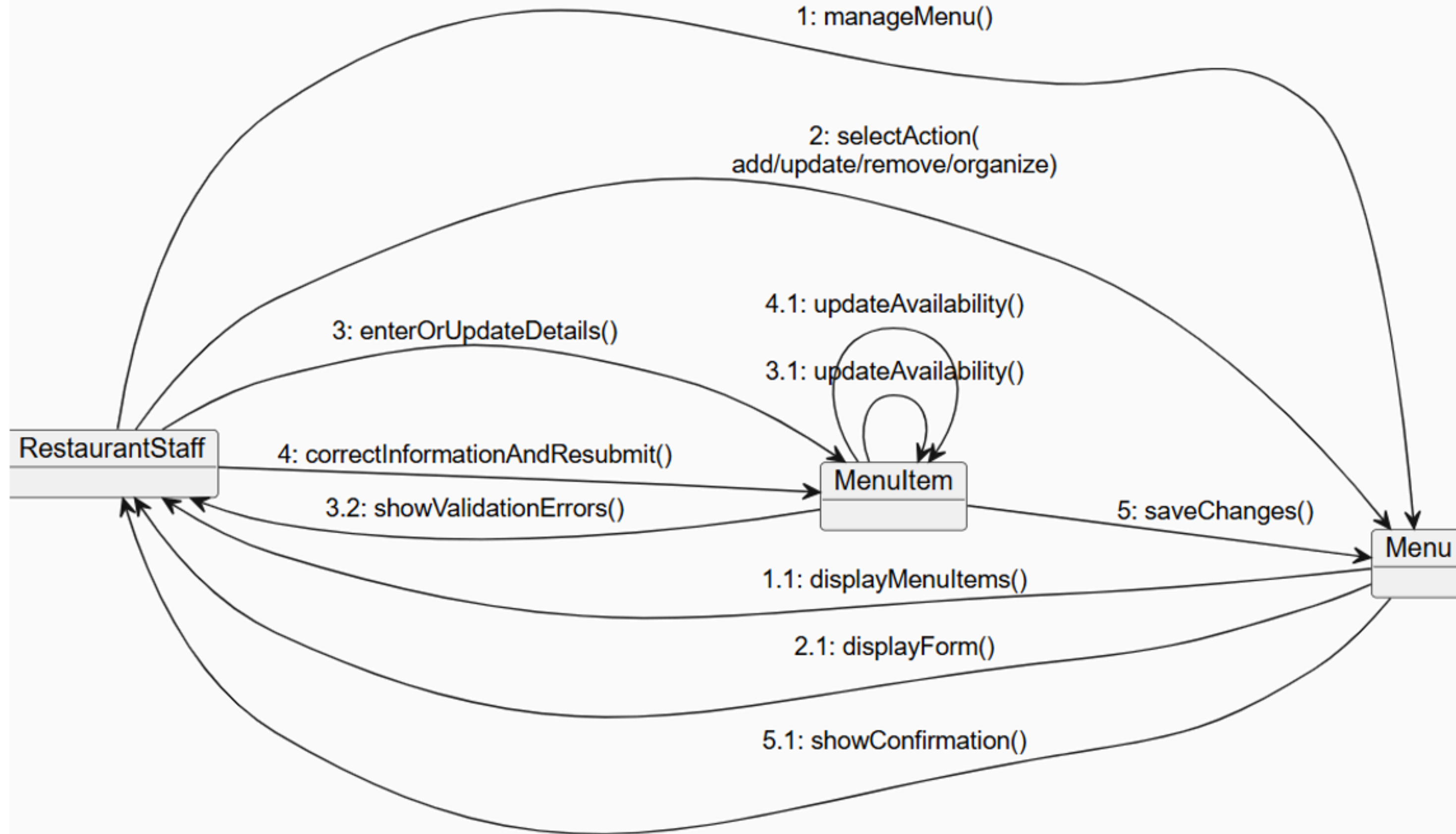
Collaboration Diagram 13



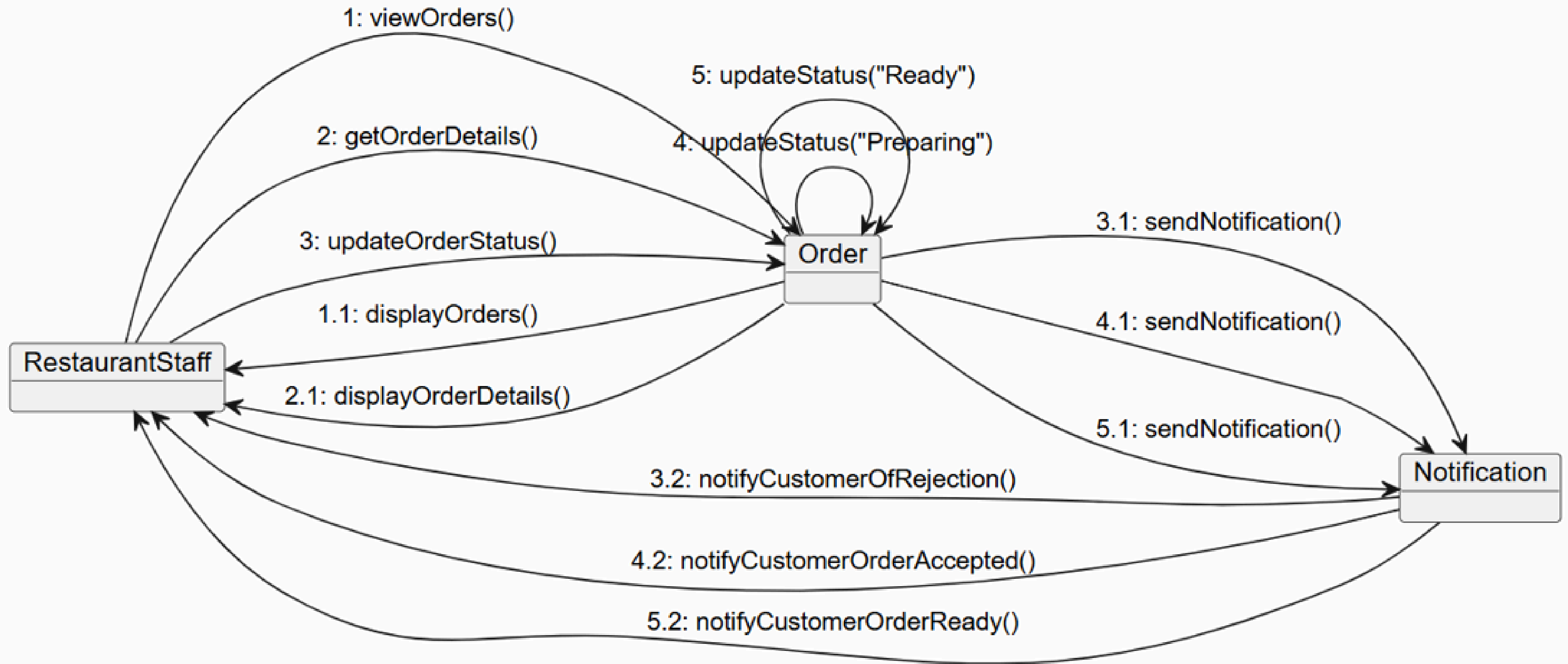
Collaboration Diagram 14



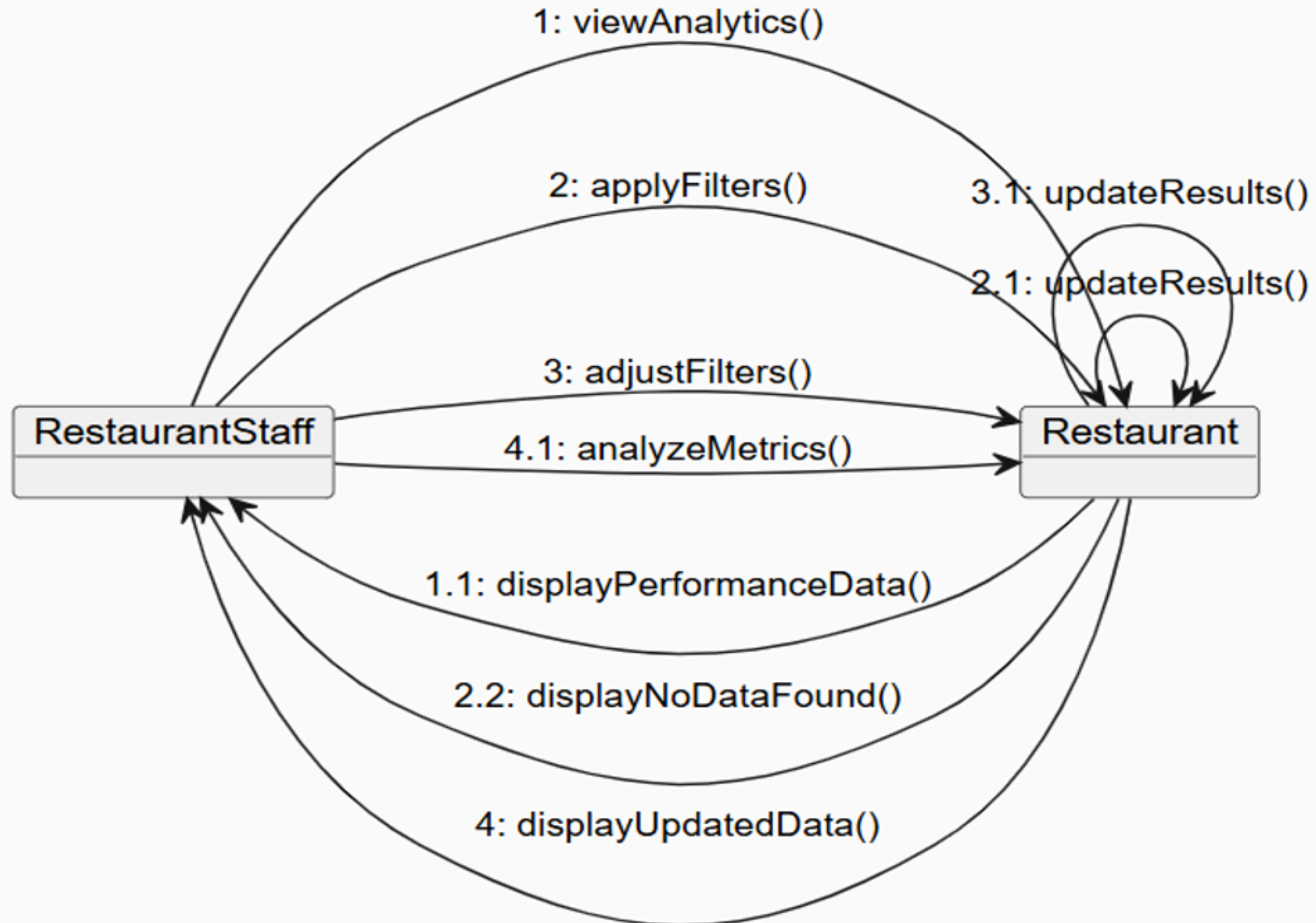
UC15-Restaurant staff manages menu items, categories, and food availability



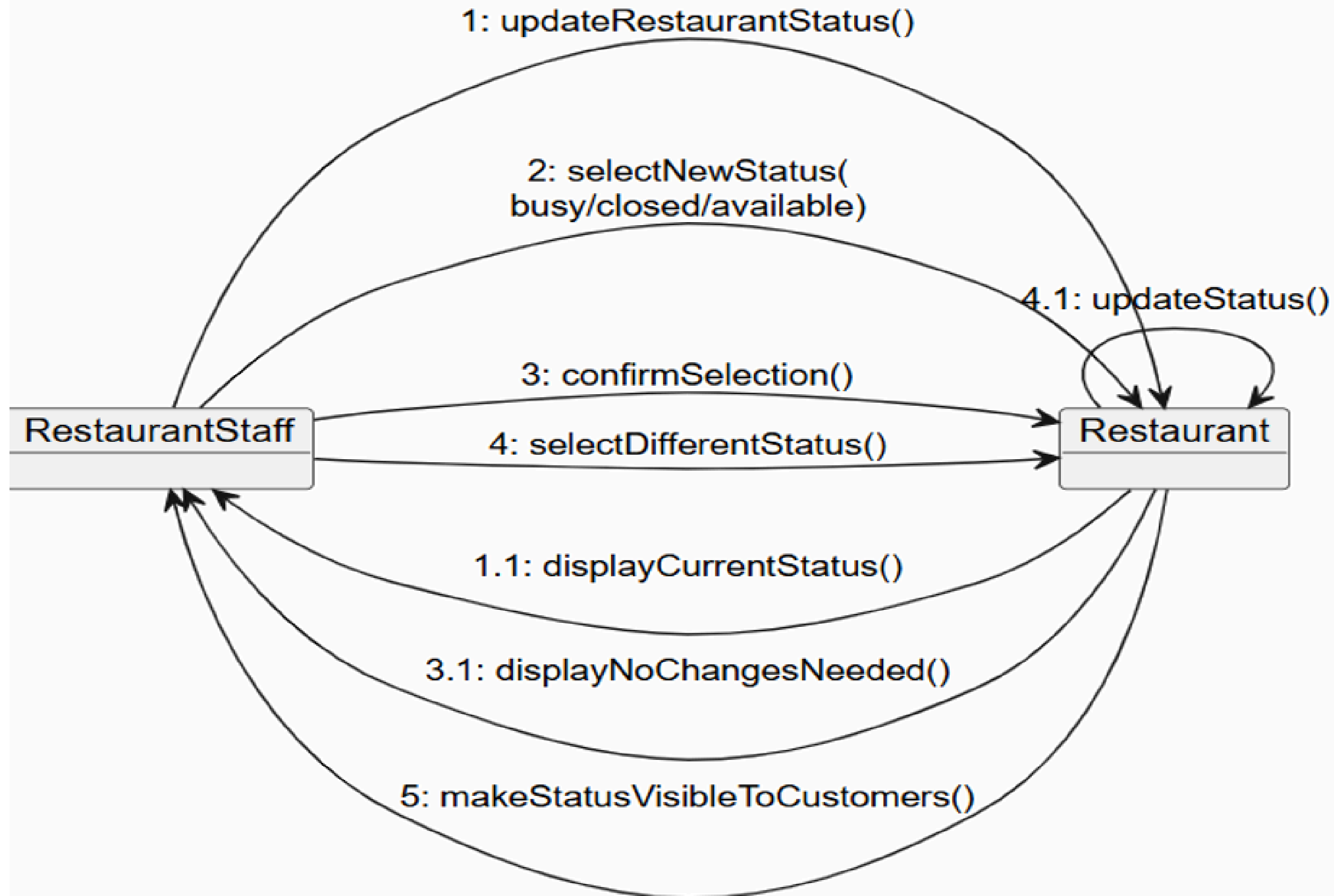
UC16 - Restaurant staff manages customer orders and updates order status



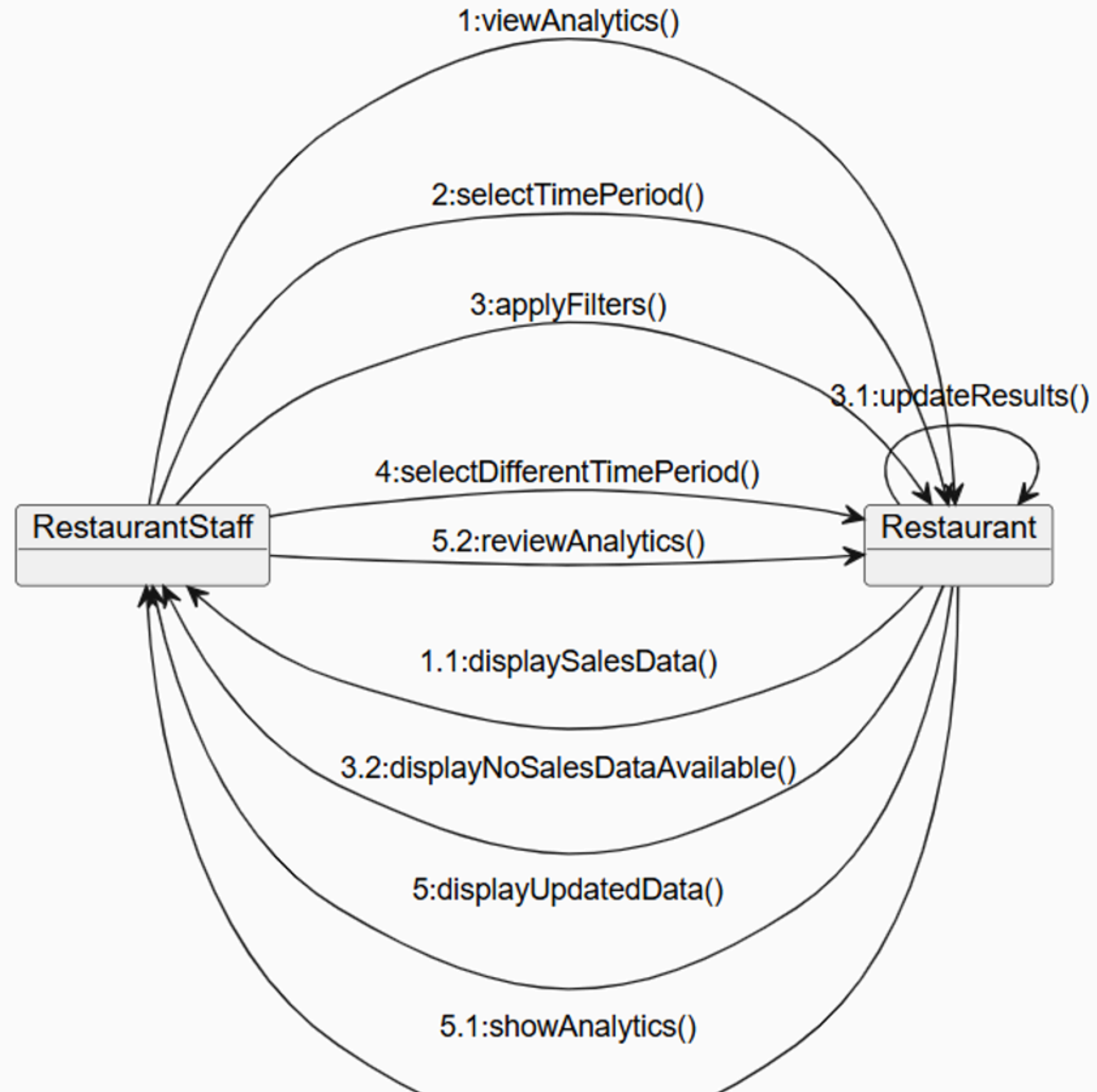
UC17-Restaurant staff monitors and analyzes restaurant performance



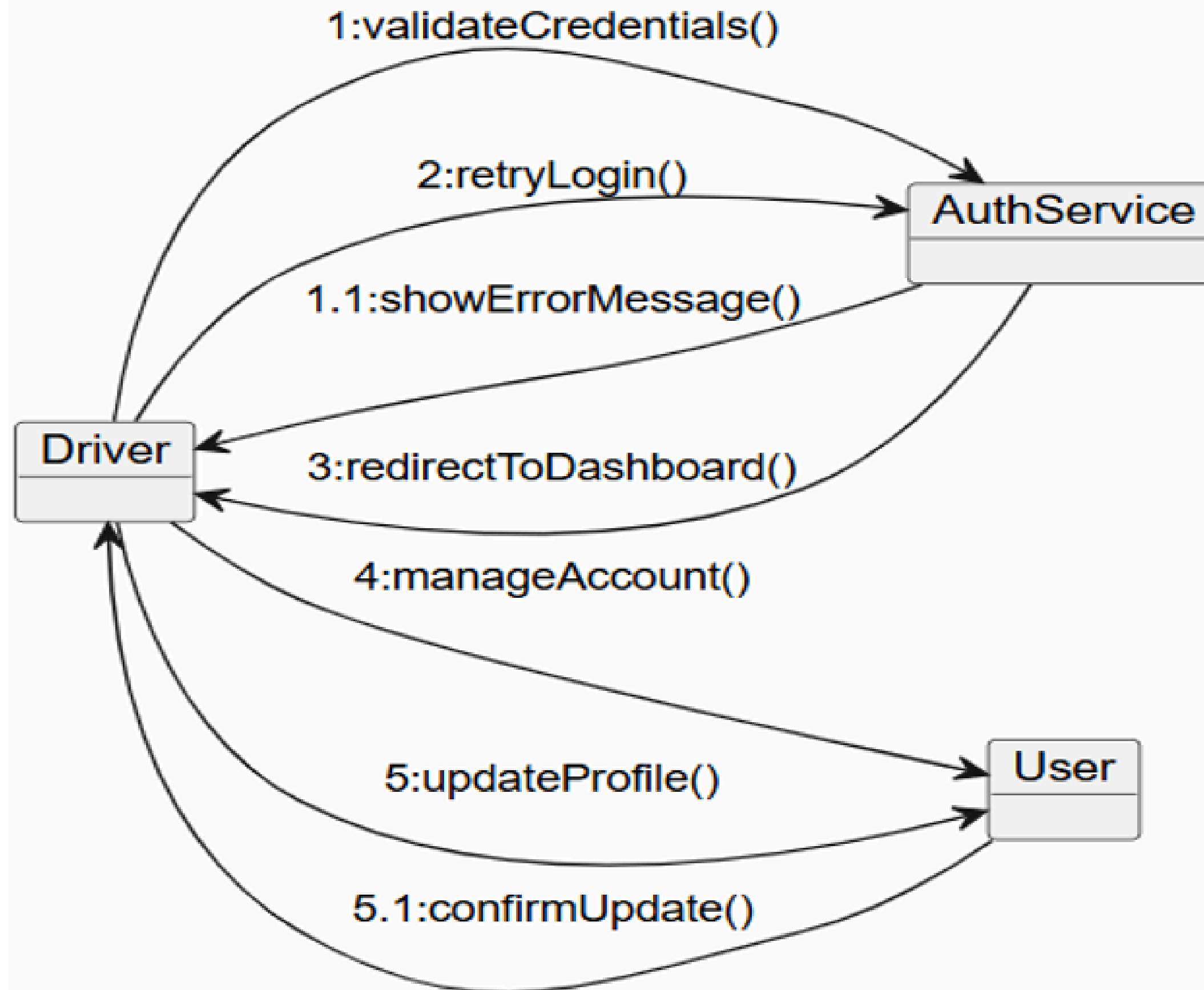
UC18-Restaurant staff manages temporary service interruptions



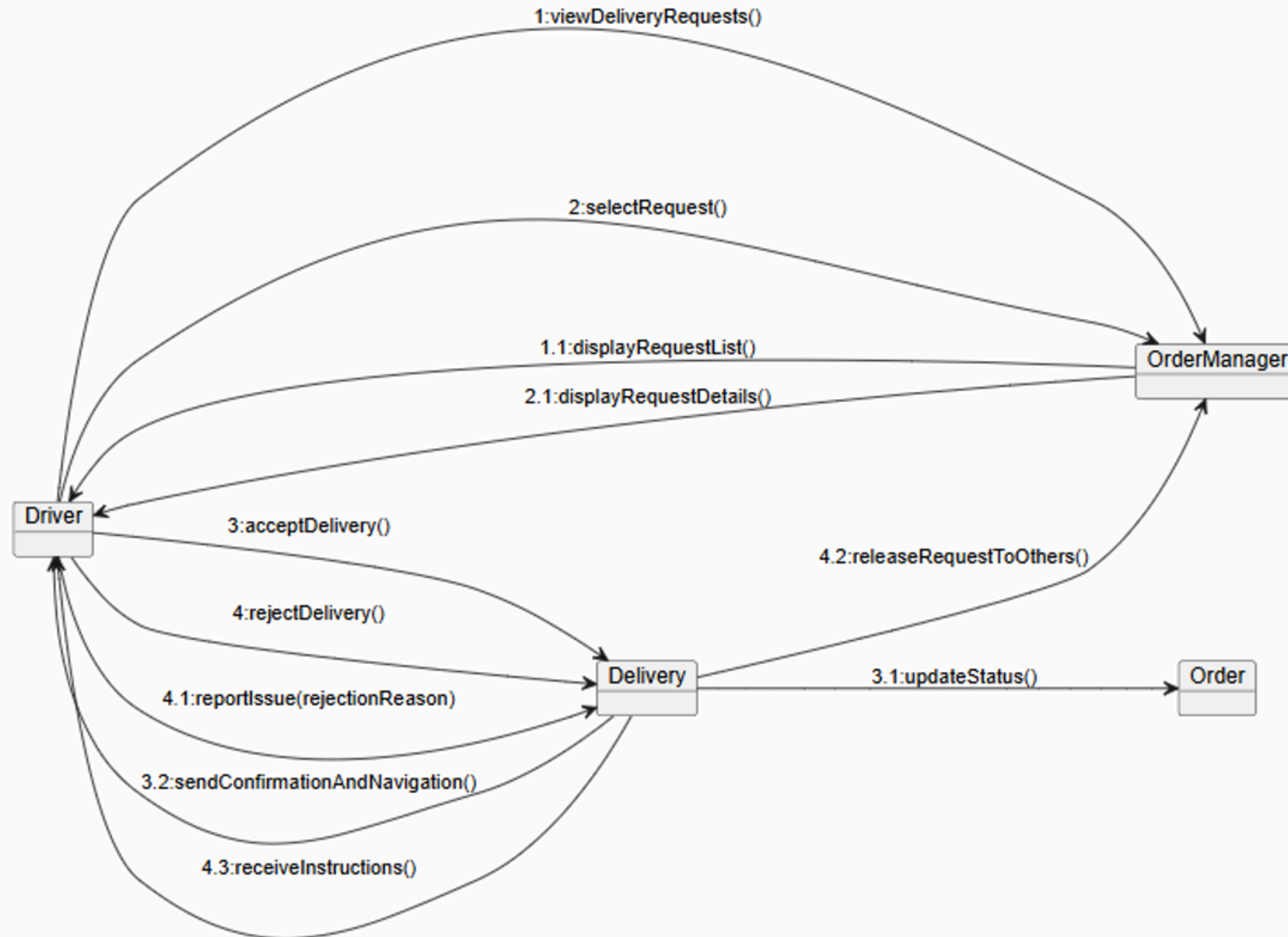
UC19-Restaurant staff views sales analytics



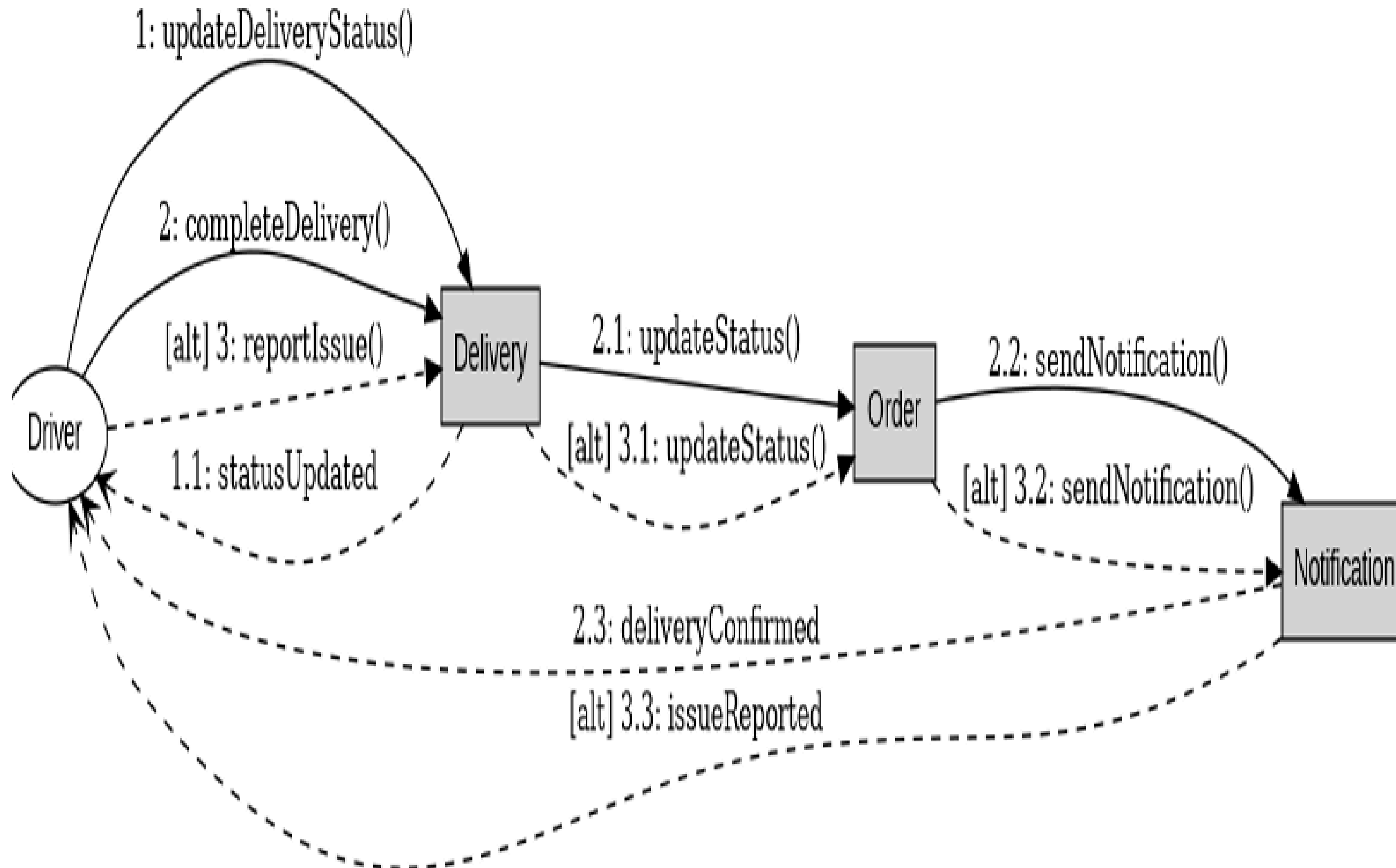
UC20-Delivery Driver Manages Account



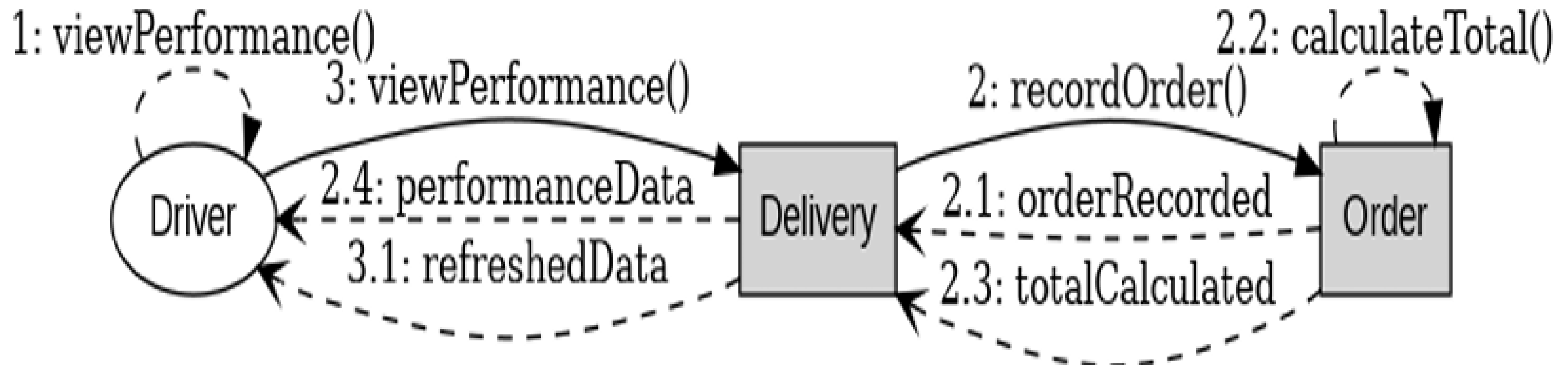
UC21-Delivery Driver Accepts or Rejects Delivery Requests



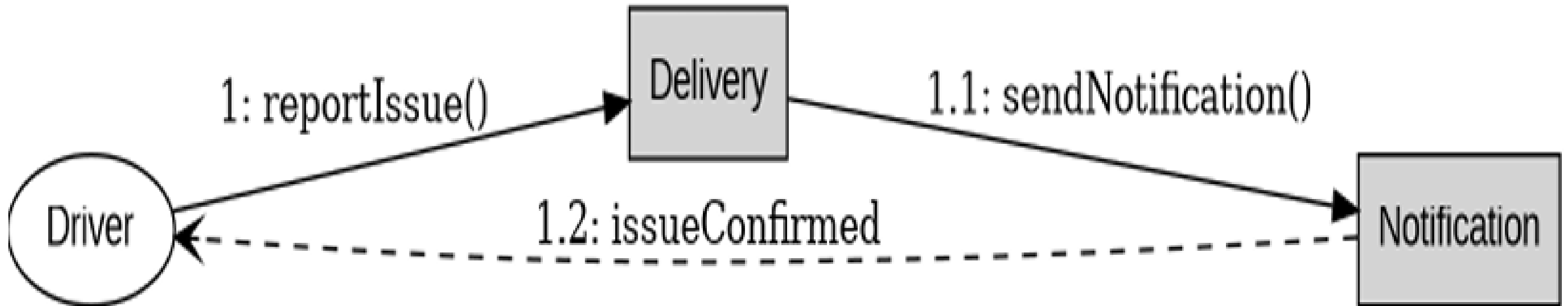
UC22 – Delivery Driver Completes Deliveries



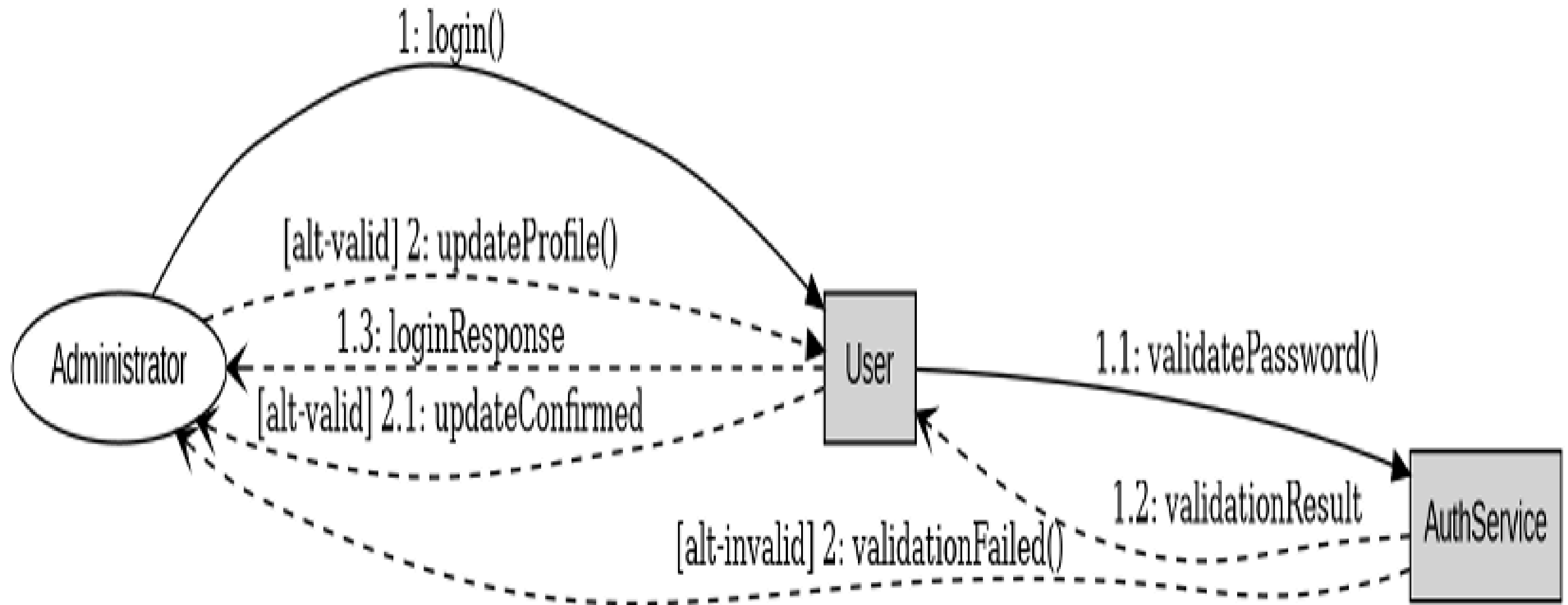
UC23 – Delivery Driver Monitors Personal Performance



UC24 – Delivery Driver Reports Issues During Delivery

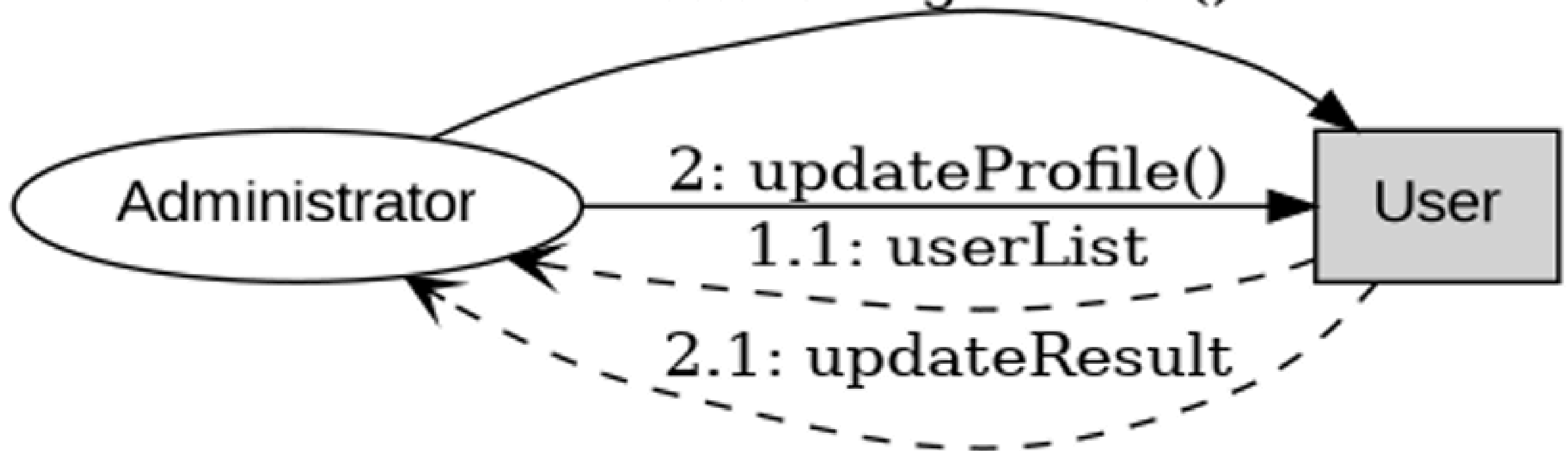


UC25 – Administrator Manages Account and Profile



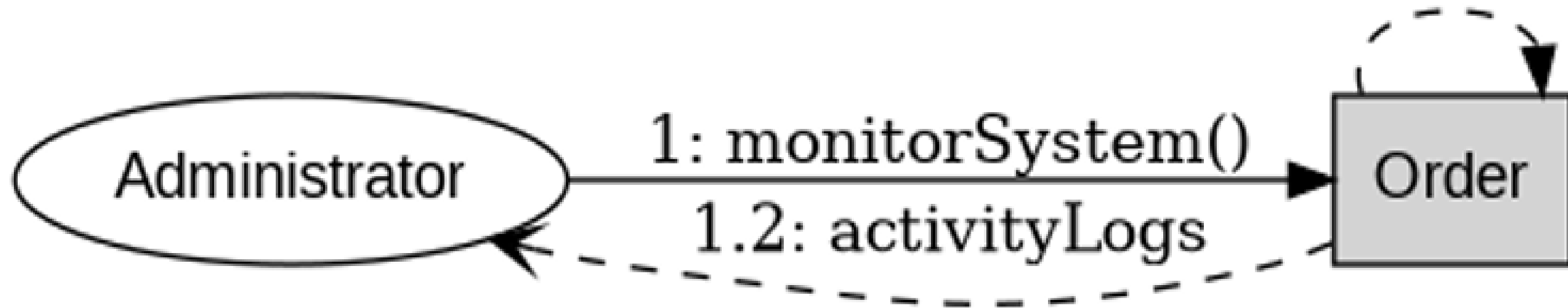
UC26 – Administrator Manages User Accounts

1: manageUsers()

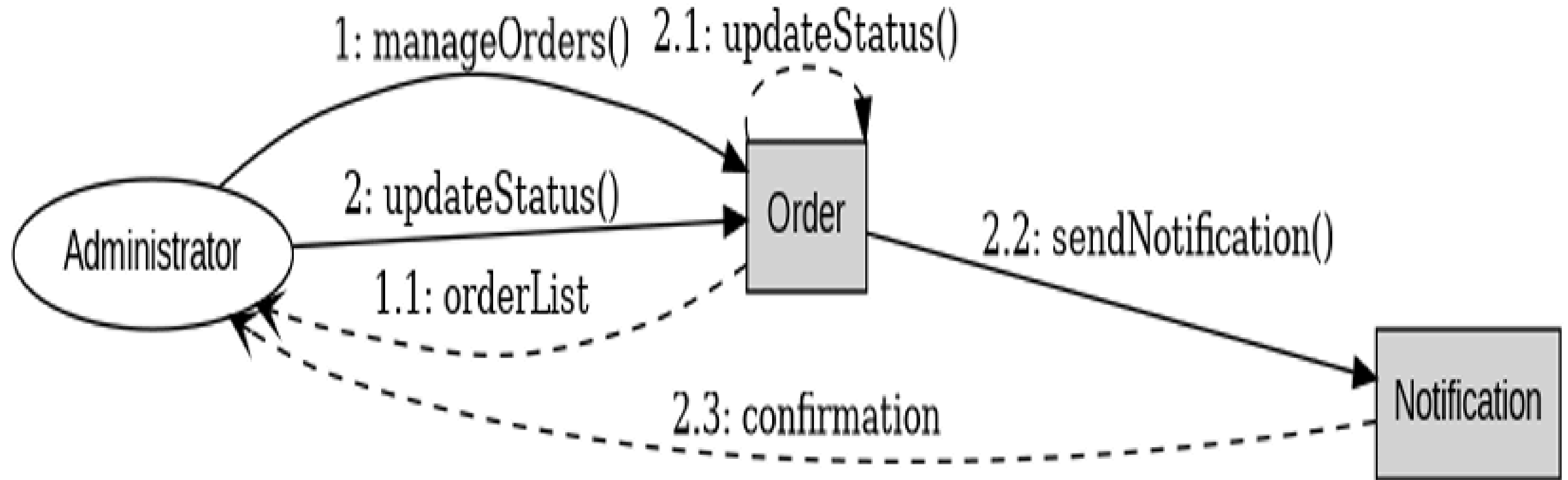


UC27 – Administrator Monitors System Activity

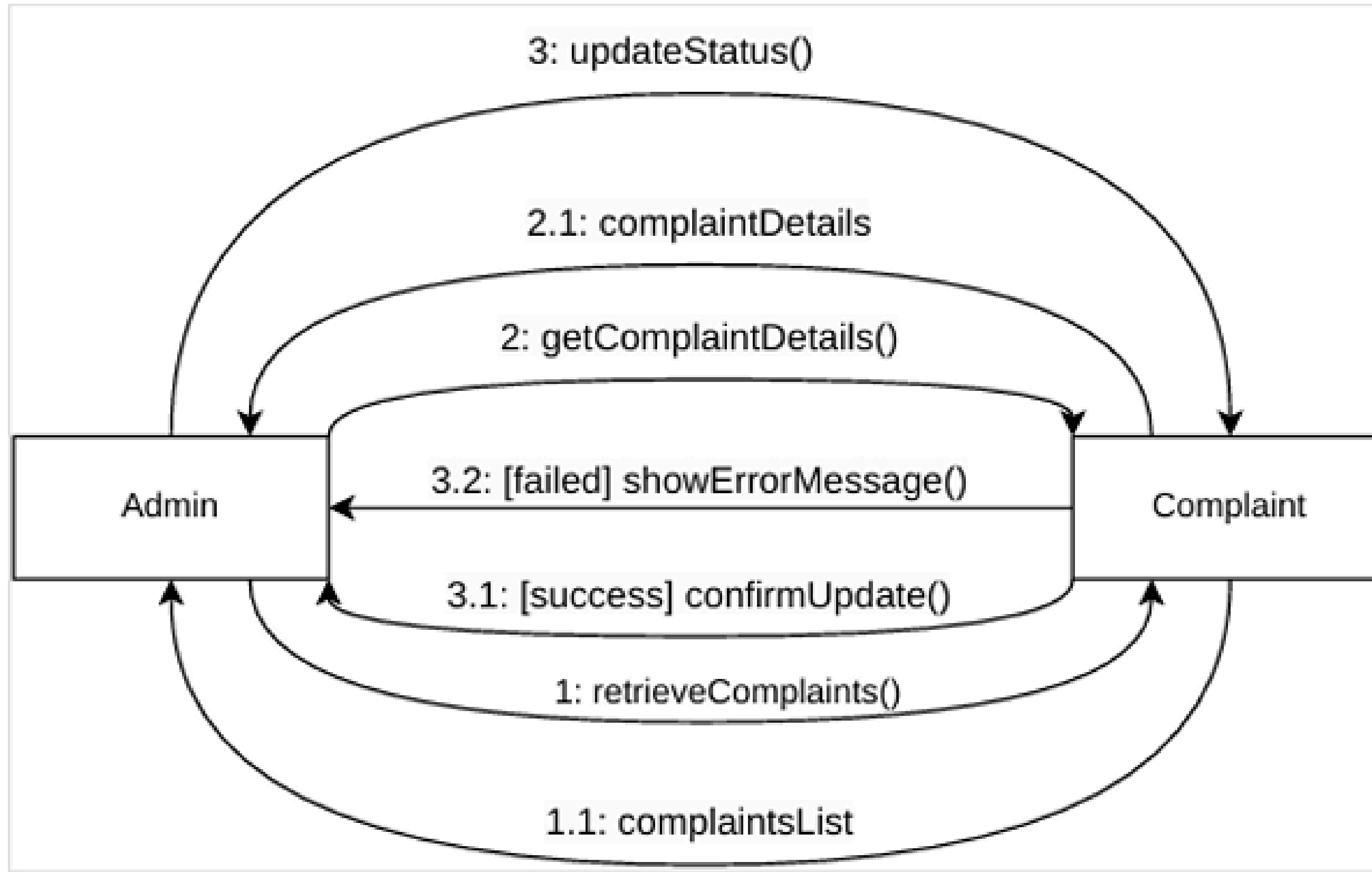
1.1: recordOrder()



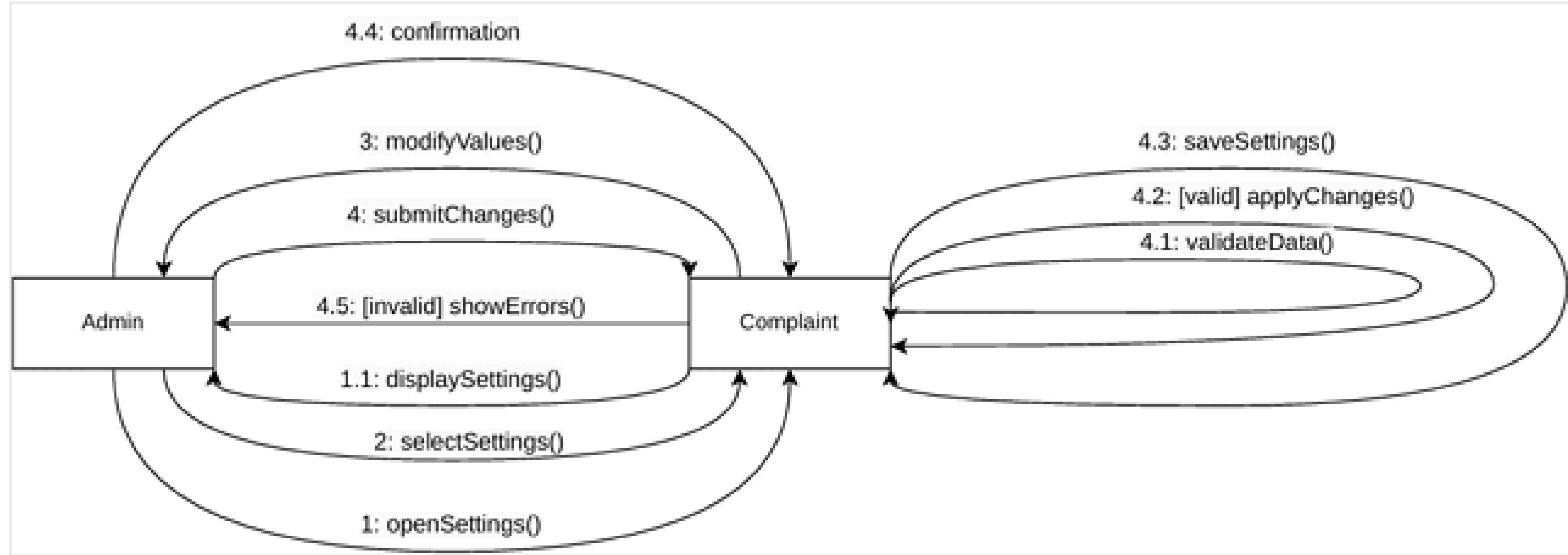
UC28 – Administrator Manages Orders



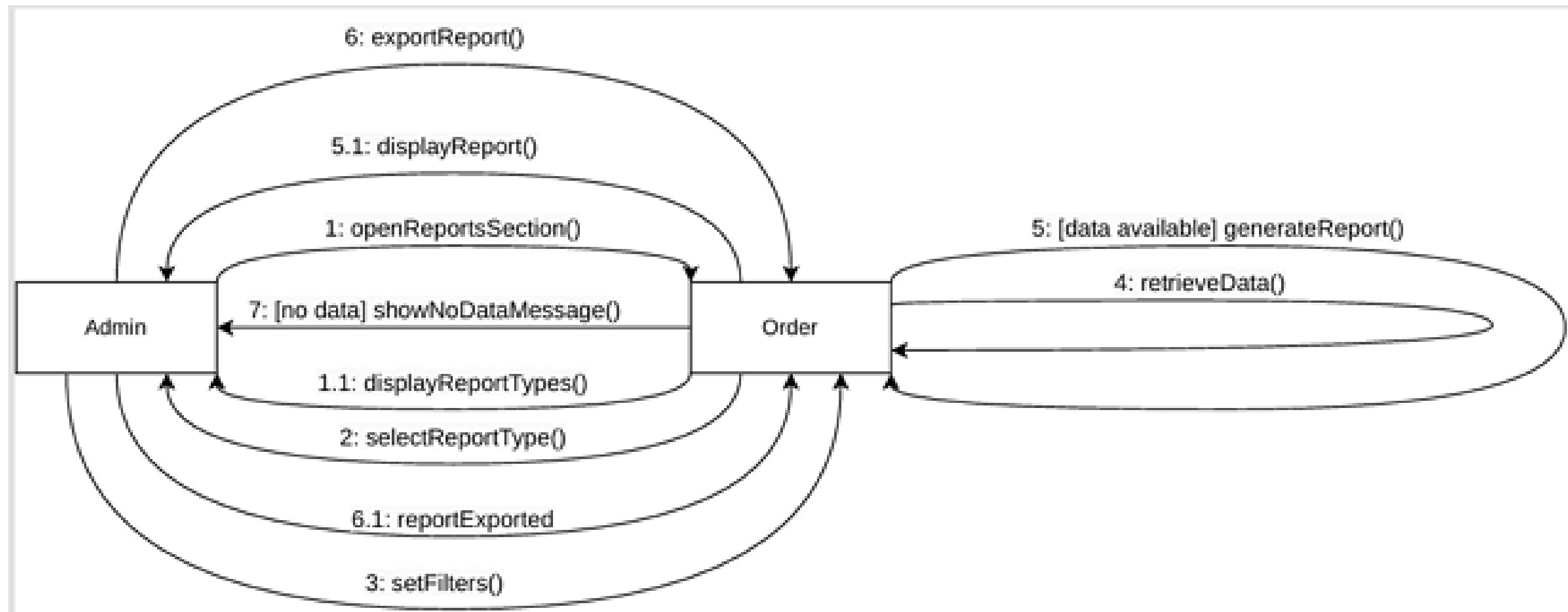
Collaboration Diagram-UC-29-Handle Complaints



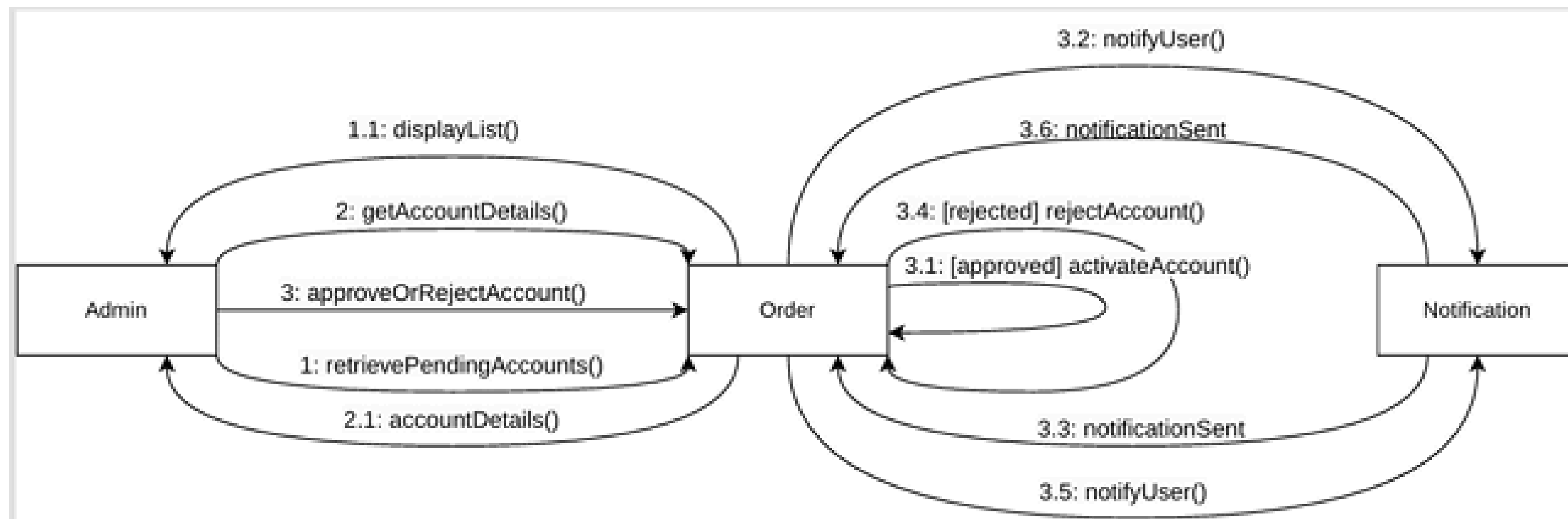
Collaboration Diagram-UC-30- Manage Settings



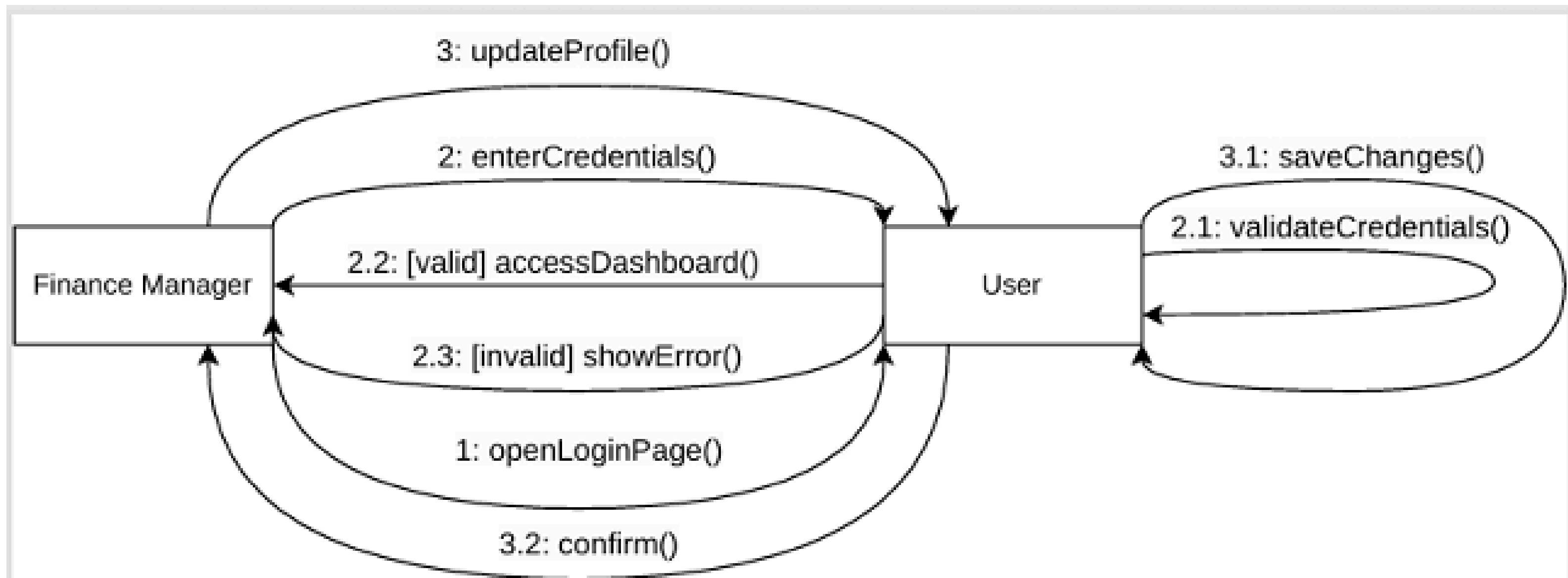
Collaboration Diagram-UC-31- Generate Reports



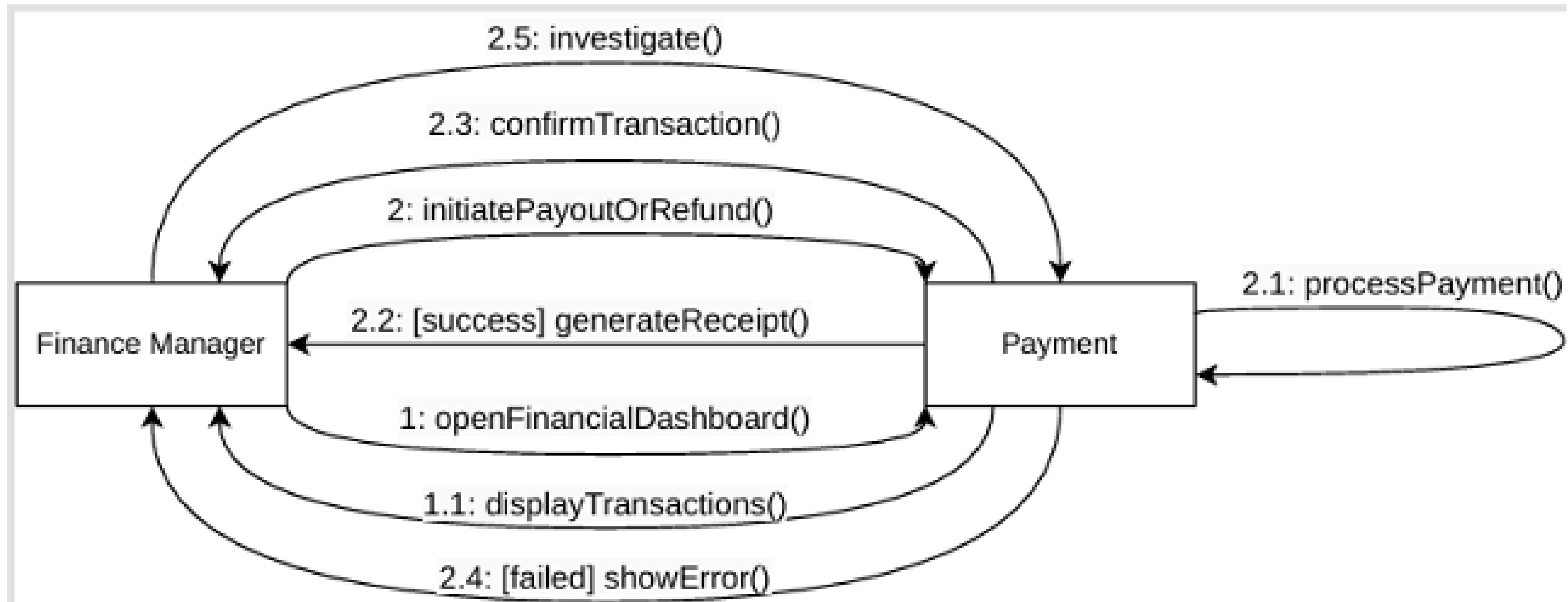
Collaboration Diagram-UC-32- Account Approvals



Collaboration Diagram-UC-33- Finance Manager Account

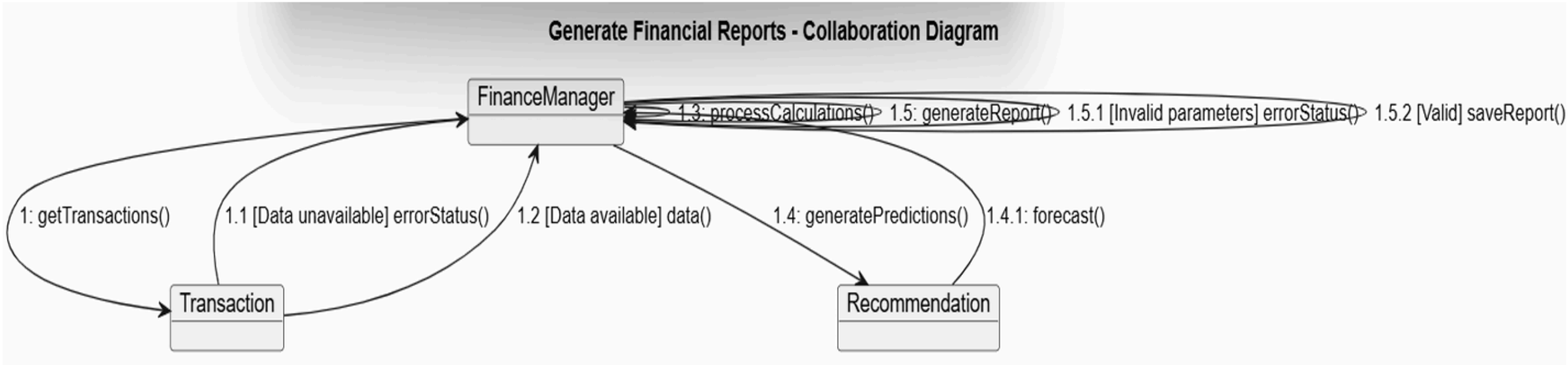


Collaboration Diagram-UC-34- Billing / Refunds



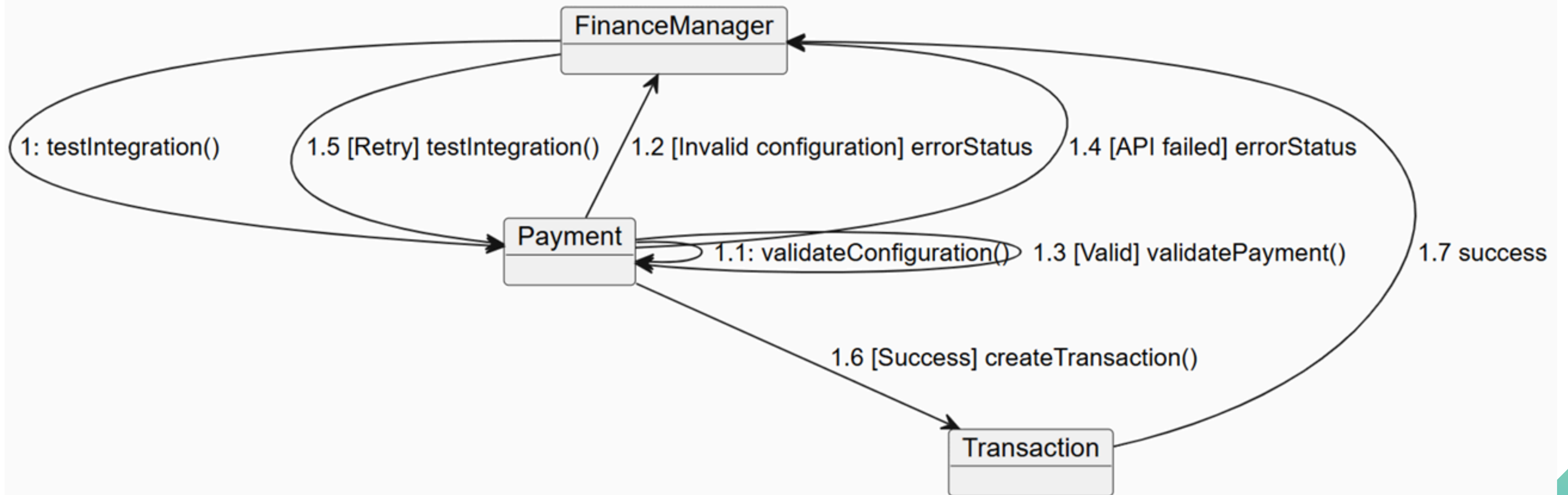
Author-Klara Hysenaj

UC 35 Generate Financial Reports and Forecasting

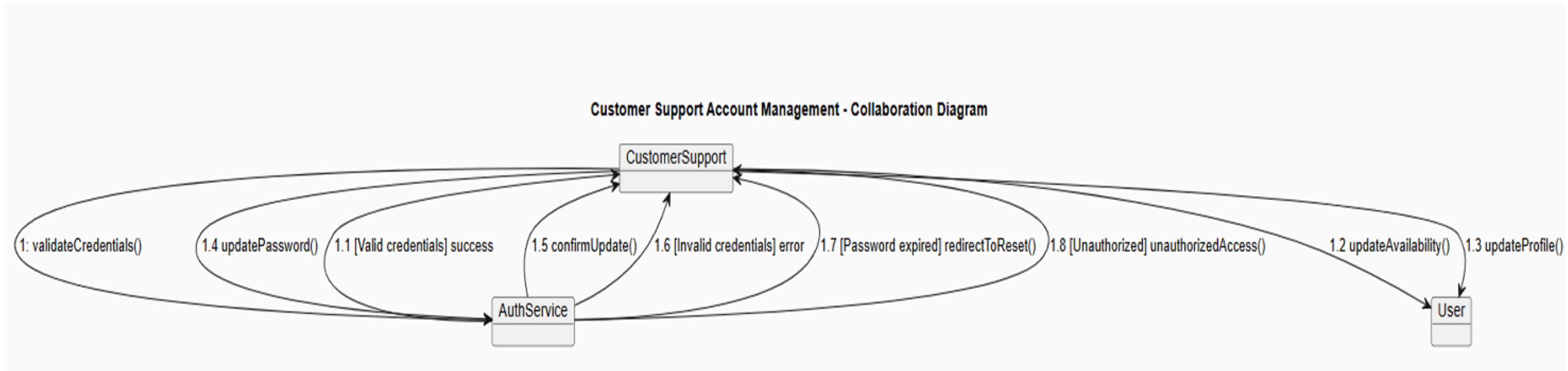


UC 36 Manage Financial Operations

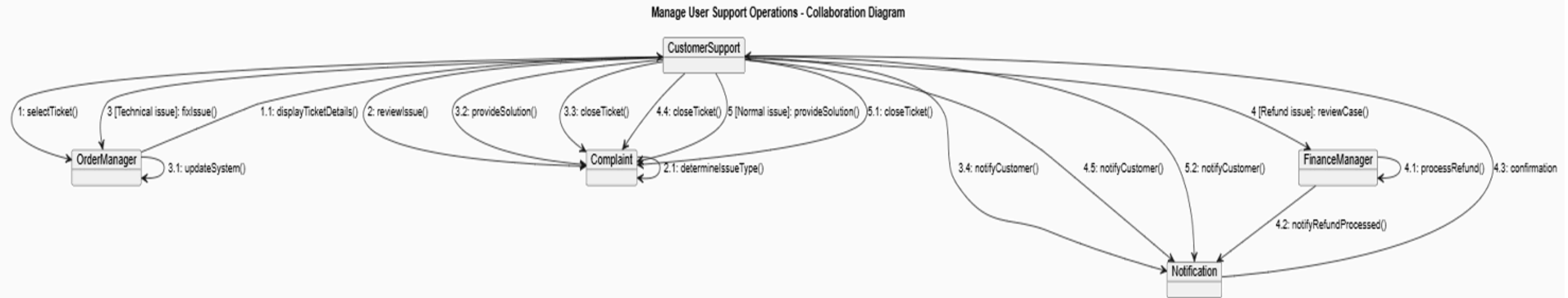
Manage Financial Operations - Collaboration Diagram



UC 37 Customer Support Account Management

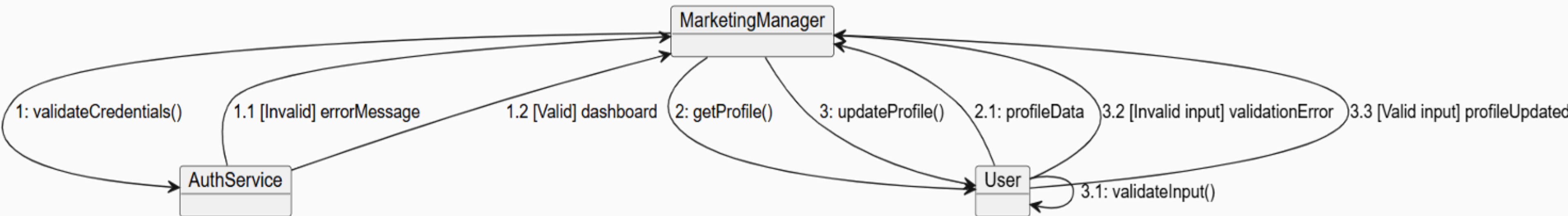


UC 38 Manage User Support Operations



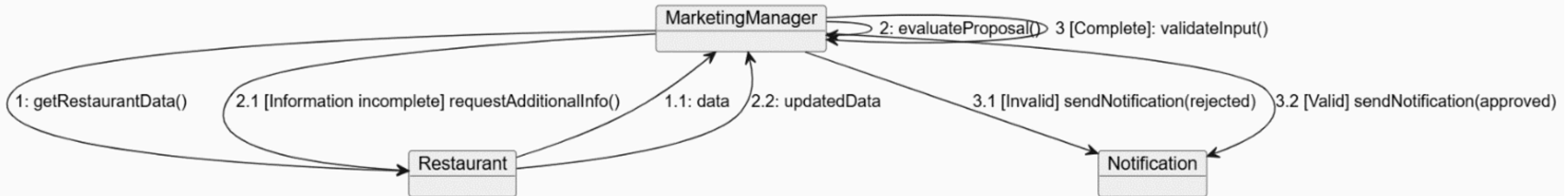
UC 39 Manage Marketing Manager Account

Manage Marketing Manager Account - Collaboration Diagram



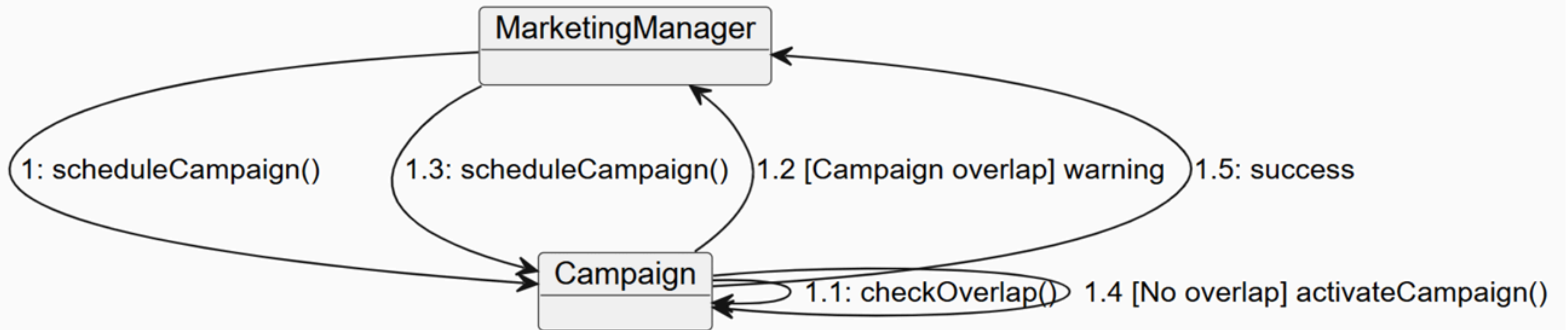
UC 40 Manage Marketing Collaboration Requests

Manage Collaboration Requests - Collaboration Diagram



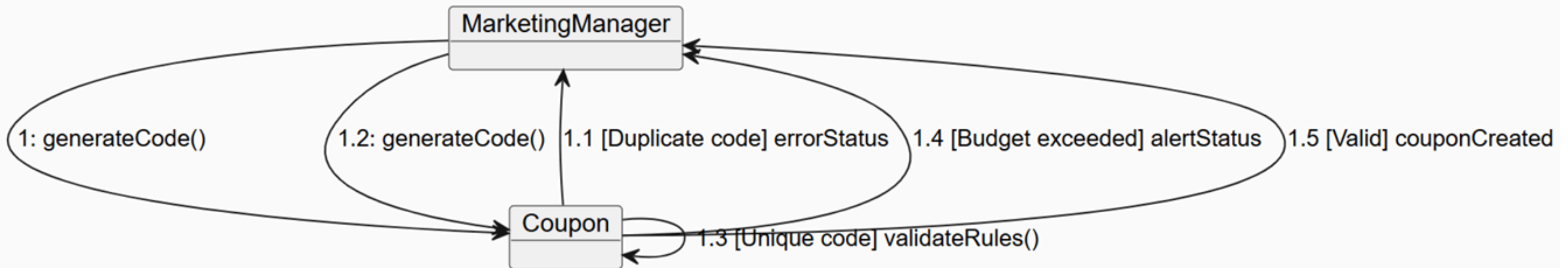
UC 41 Mange Marketing Campaigns

Manage Marketing Campaigns - Collaboration Diagram



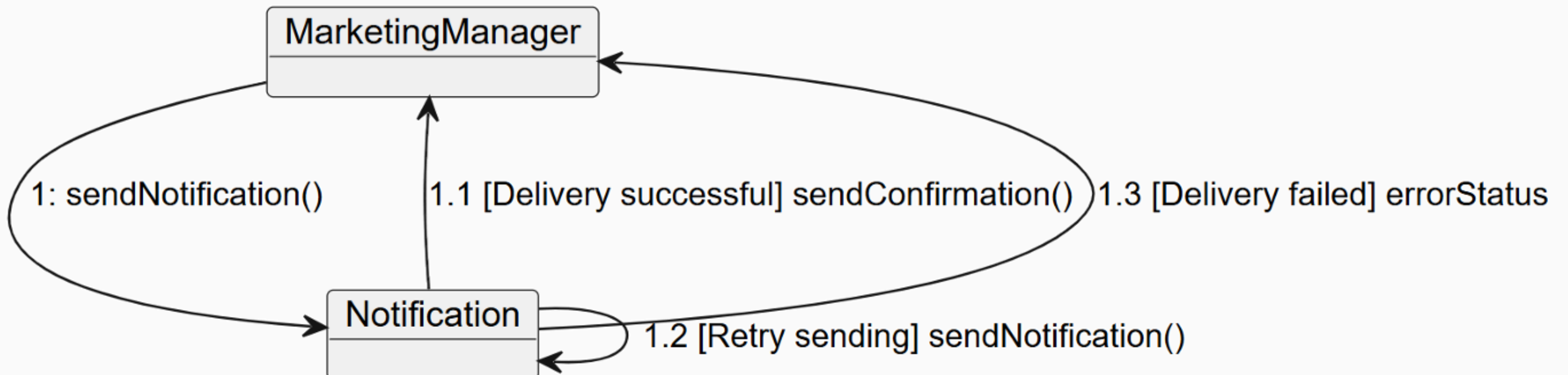
UC 42 Manage Promotions, Discounts and Coupon Codes

Manage Promotions, Discounts and Coupons - Collaboration Diagram



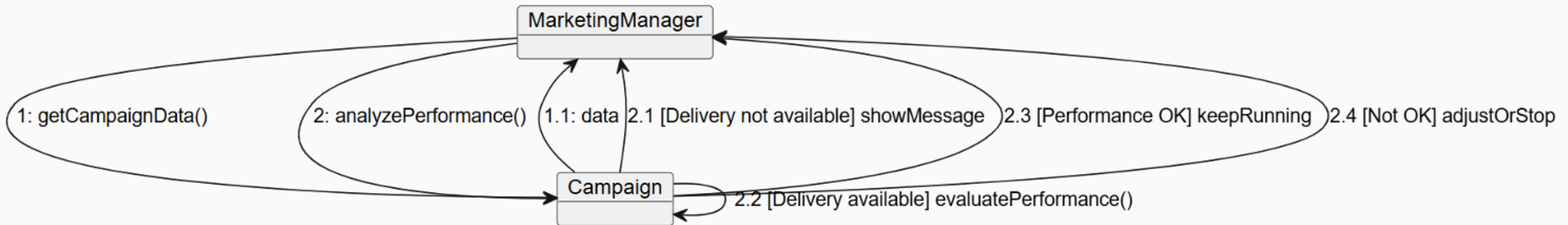
UC 43 Manage Promotional Communication

Manage Promotional Communication - Collaboration Diagram



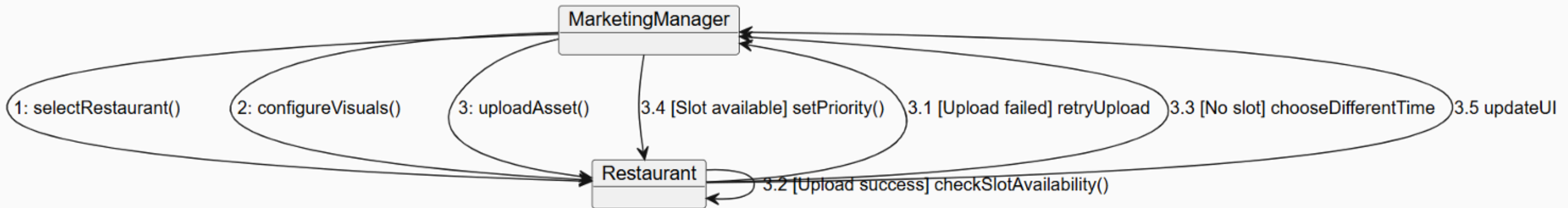
UC 44 Analyze&Monitor Campaign Performance

Analyze & Monitor Campaign Performance - Collaboration Diagram

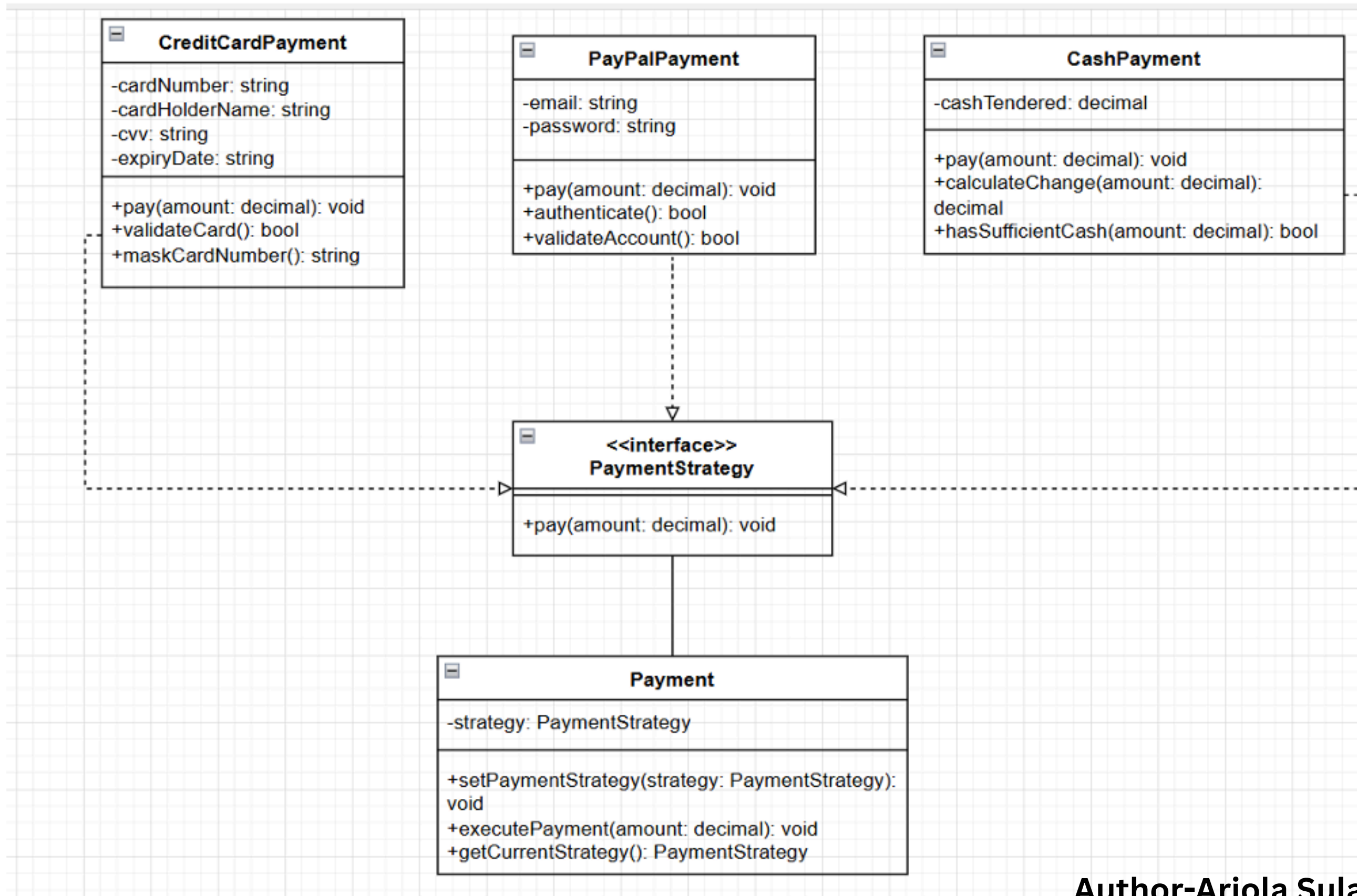


UC 45 Manage Featured Restaurants & Advertisements

Manage Featured Restaurants & Ads - Collaboration Diagram



Strategy Pattern



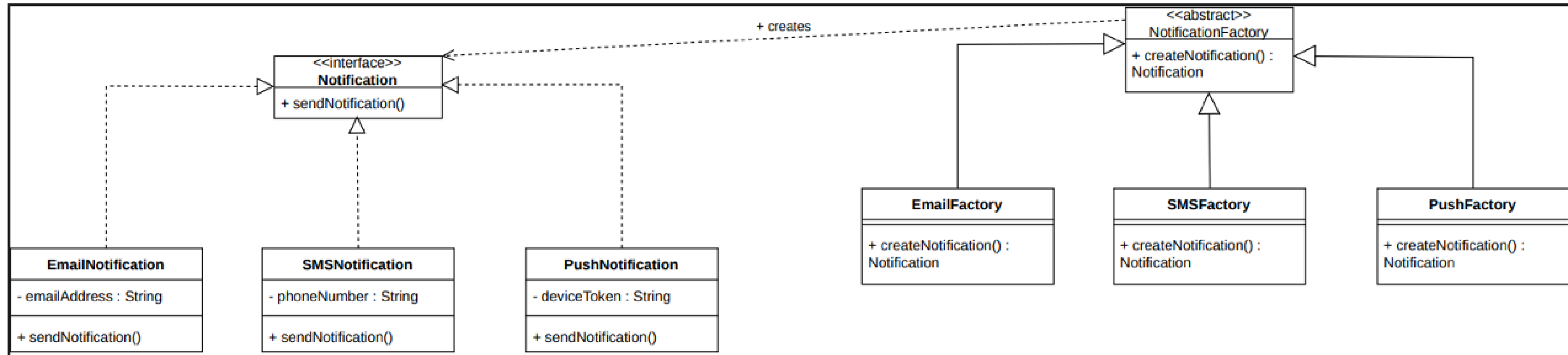
The Strategy Pattern helps the system use different payment methods without changing the main payment logic. In the food delivery system, this pattern is used to manage different payment methods such as Credit Card, PayPal, and Cash.

Justification

This pattern was chosen because the food delivery system supports multiple payment methods, including Credit Card, PayPal, and Cash, each with different payment processing logic. Using the Strategy Pattern allows each payment method to be implemented in a separate class while sharing the same interface. This makes the system more flexible, organized, and easier to maintain. It also avoids large if else statements inside the Payment class and allows new payment methods to be added in the future without modifying the existing code structure.

Design Patterns – Factory Method

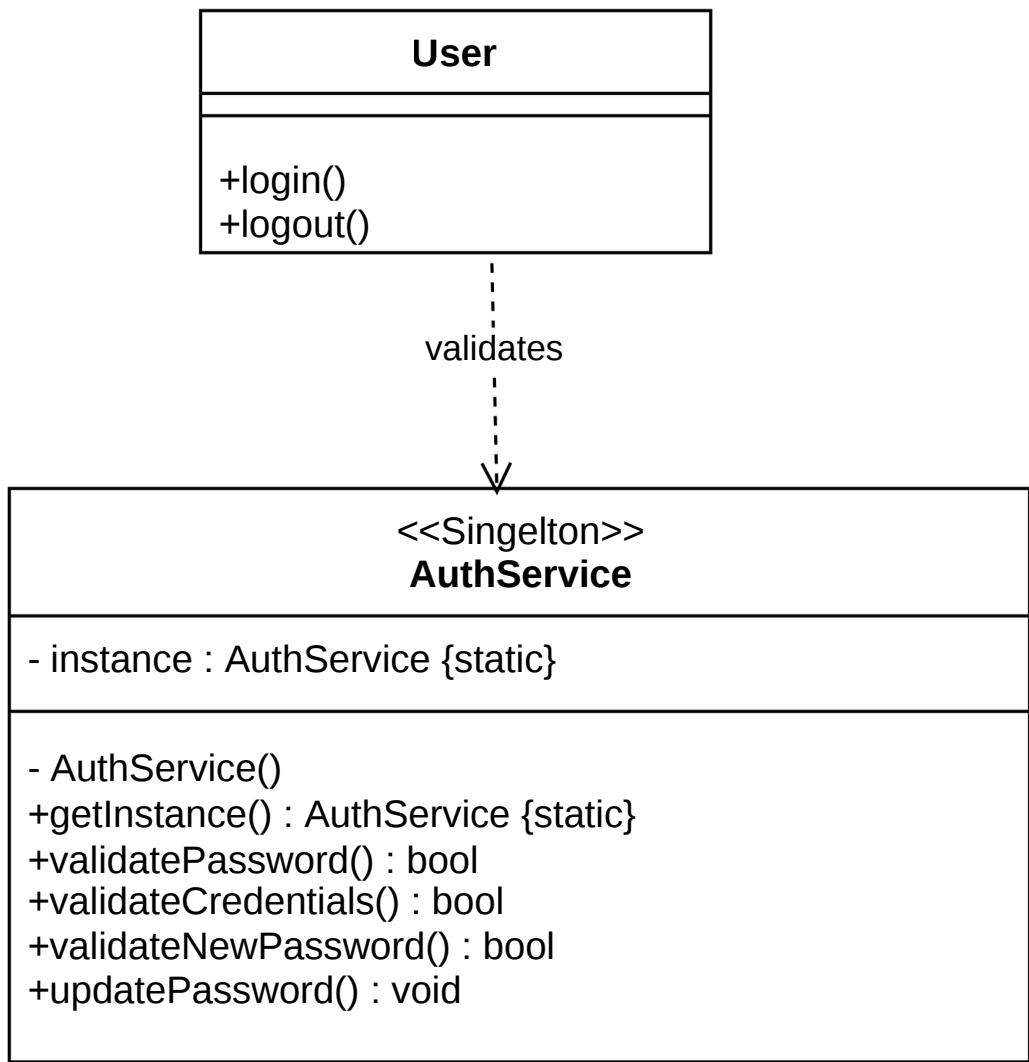
Author-Sofia Cala



Factory Method Pattern is used in the Food Ordering and Delivery System to dynamically create different types of notifications such as Email, SMS, and Push notifications. This pattern fits the system because the notification type depends on the communication method chosen by the user, while the system still needs a common and flexible way to create notification objects. By separating object creation from the business logic, the pattern improves scalability, flexibility, and maintainability.

Singleton Design Pattern – AuthService

Xhesika Toska



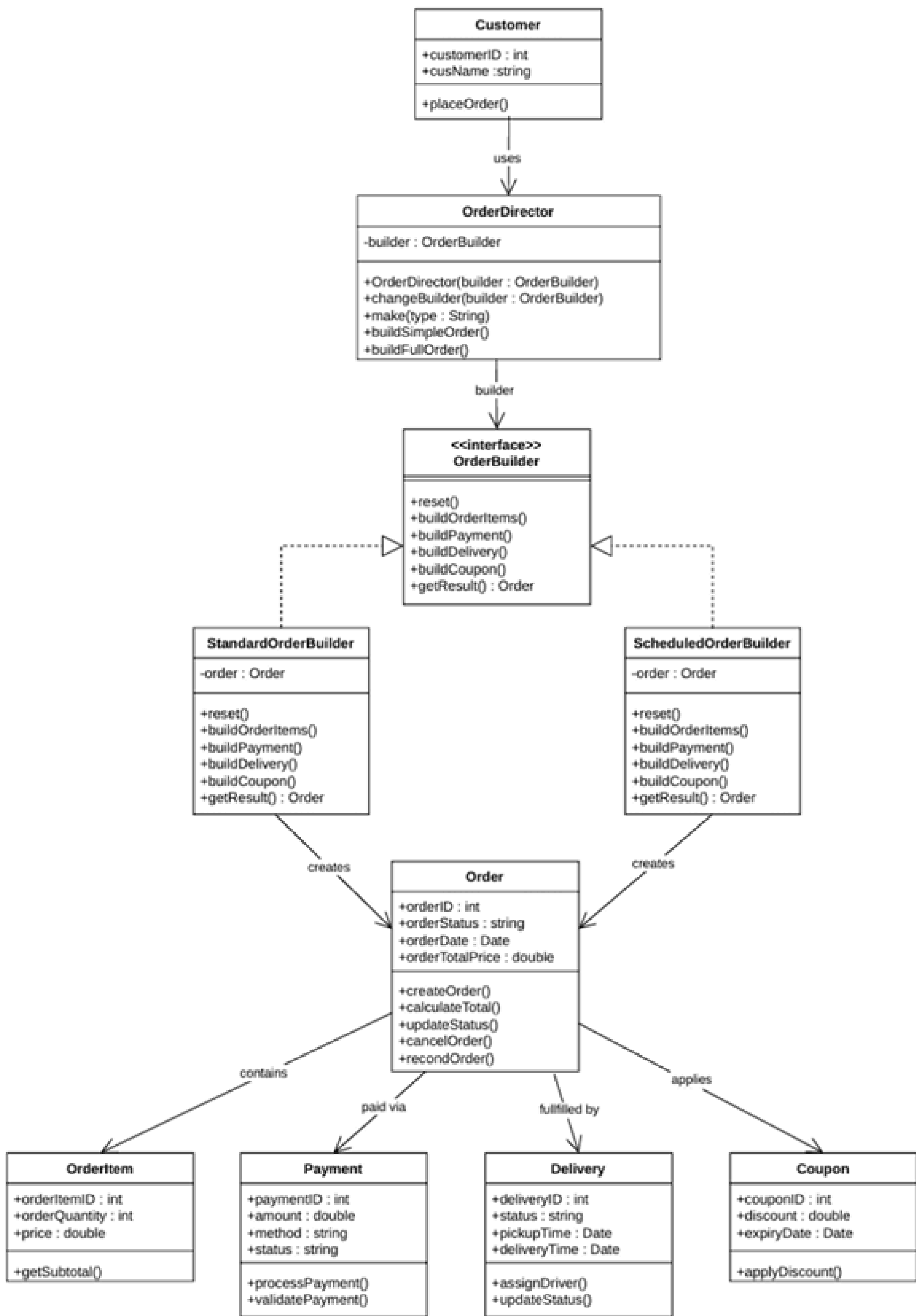
The Singleton Design Pattern is implemented in the AuthService class to ensure that only one instance of the authentication service exists throughout the entire system. This design pattern is useful for centralized functionalities such as user authentication, password validation, credential verification, and password updates.

In the UML diagram, the AuthService class contains a private static instance (instance : AuthService {static}), which stores the single object of the class. The private constructor (- AuthService()) prevents other classes from creating new objects directly. Instead, the system accesses the single instance through the public static method getInstance(), which always returns the same object. The User class interacts with AuthService through a dependency relationship (validates), meaning that users rely on this service for login validation and authentication-related operations. This implementation improves security, consistency, and resource management, since all authentication processes are handled by one centralized service instance.

Overall, the Singleton pattern ensures better control, maintainability, and efficiency in the authentication system.

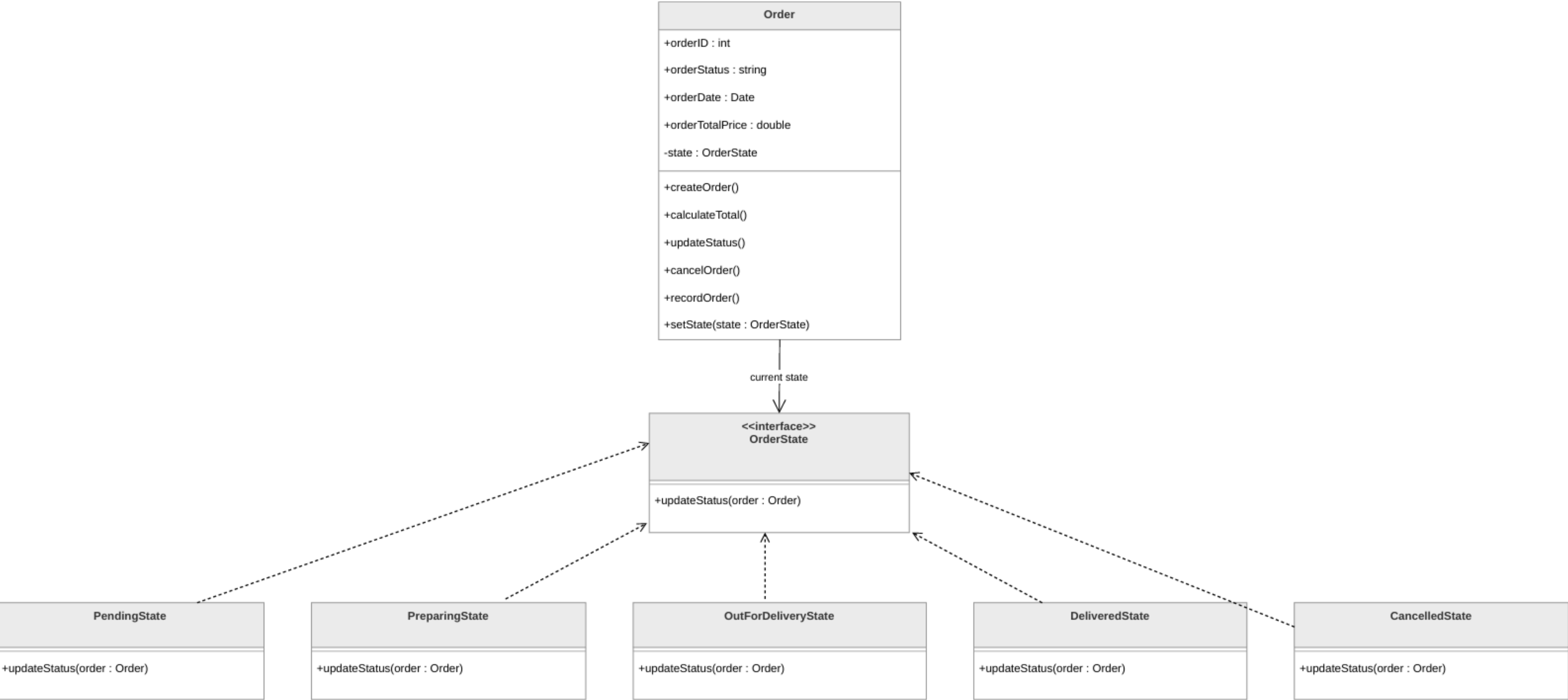
Builder Design Pattern- Order Creation

Author-Sofia Cala



The Builder Pattern is used in the Food Ordering and Delivery System to construct complex Order objects step by step, assembling different parts such as OrderItems, Payment, Delivery, and Coupon based on the order type chosen by the customer. This pattern fits the system because an order's composition varies depending on whether it is a standard or scheduled order, while the system still needs a consistent and flexible way to build these objects. By separating the construction logic from the client code the pattern improves scalability, flexibility, and maintainability.

State Design Pattern – Order Status



The State Design Pattern is implemented in the Order class to manage the different stages of an order in the food delivery system. Since an order changes its behavior depending on its current status (such as Pending, Preparing, Out for Delivery, Delivered, or Cancelled), the State pattern provides a structured and flexible way to handle these transitions. In this design, the Order class acts as the context class, containing a reference to the OrderState interface through the state attribute. Instead of using multiple if-else statements to check the order status, each state is represented by a separate class (PendingState, PreparingState, OutForDeliveryState, DeliveredState, and CancelledState). Each state class implements the updateStatus() behavior differently based on the current order phase. The OrderState interface defines the common behavior for all order states, while concrete state classes manage how the order behaves and transitions between statuses. This improves the system by making it more organized, scalable, and easier to maintain, especially when new order statuses need to be added in the future. Overall, the State pattern increases flexibility, readability, and maintainability by separating status-specific logic into independent classes instead of keeping all logic inside the Order class.